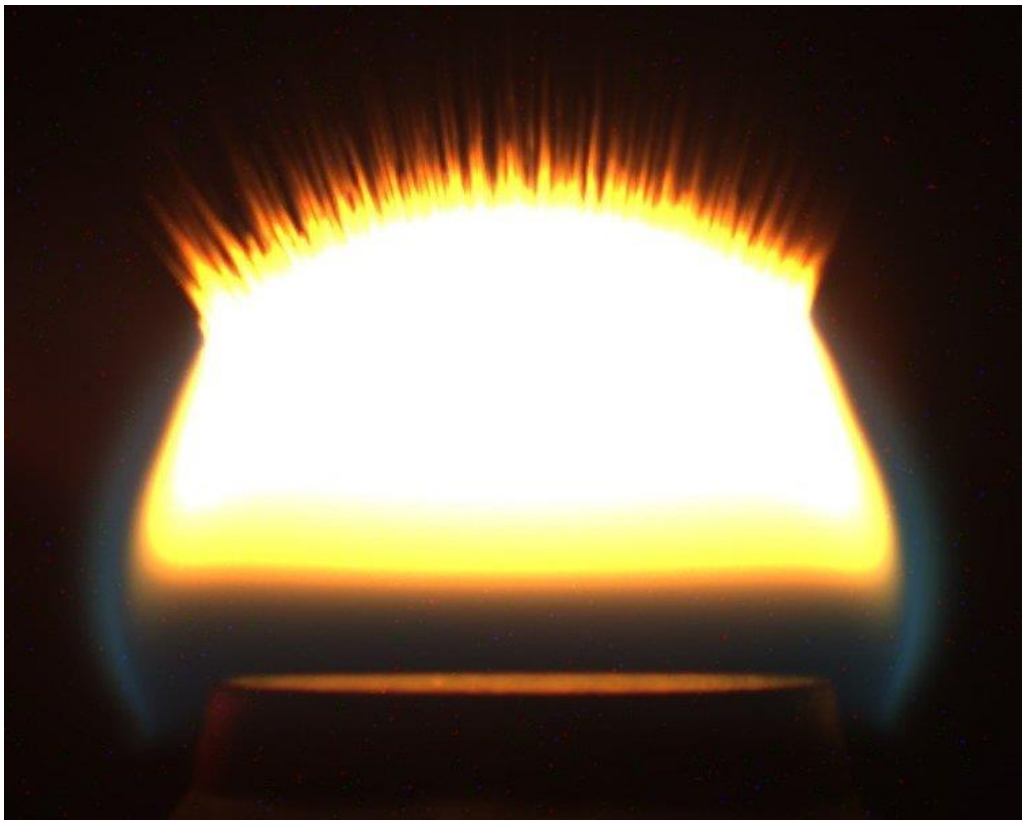
Advanced Combustion via Microgravity Experiments (ACME)
Burning Rate Emulator (BRE)

ACME/BRE Final Report (2022*)

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A BRE flame shortly after ignition, in which the burner outlet can also be seen.

**Data analysis and the publication of results is still underway, so it is expected that this report will be updated in the future with at least a revision of the list of papers and presentations.*

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1. Overview

The Burning Rate Emulator (BRE) experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments (ACME) project for which tests are conducted within the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

The experiment's Principal Investigator (PI) is Prof. James G. Quintiere of the Department of Fire Protection Engineering at the University of Maryland. The experiment has two Co-Investigators (Co-Is), namely Prof. Peter B. Sunderland, of the same U. Maryland department, and Dr. John L. de Ris (FM Global, retired). The ISS research also benefitted greatly from the work of Parham Dehghani, a doctoral student in the Fire Protection Engineering department.

Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is the Project Scientist (PS) for ACME including BRE and the rest of the ACME investigations. In preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University (CWRU) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team includes Drs. Uday G. Hegde and Vedha Nayagam who are also with CWRU.

The ISS operations were conducted in two separate rounds of testing during Increments 58-59 and 63-64. More specifically, the first round of testing was conducted from Feb. to April 2019, while the second-round testing occurred from Sept. 2020 to Jan. 2021. The first round of tests was limited to ethylene and the 25-mm BRE burner, while the second round also included tests with methane and the 50-mm burner (although not together).

2. Science Overview

A. Background

This background is the Executive Summary from the experiment's Science Requirements Document (SRD) of June 2013.

The goal of this research is to establish the burning conditions possible in a microgravity quiescent environment. The approach is to do this using a burner to emulate (BRE, Burning Rate Emulator) the burning for real condensed fuels, representative of common liquids and solids. This will be done for quasi-steady burning conditions, albeit ideal in some cases, but the BRE will range over a wide spectrum of real fuel properties. The key properties governing steady burning are (1) the heat of gasification, (2) the surface re-radiation heat flux, and (3) the heat of combustion. Flame radiation is also relevant in general and can be represented in terms of the laminar smoke point. Its importance in microgravity flames needs study, but the first three properties are key.

Currently NASA assesses the flammability hazard of materials for use in space flight by an upward flame spread "pass-fail" test (Test 1), and supplements with data from the Cone Calorimeter (Test 2). The latter test has the capability of measuring the three properties listed above. Hence, the successful application of the BRE in microgravity can enhance the usefulness of Test 2.

Fundamental and ad hoc testing in microgravity at quiescent conditions has shown that quasi-steady flames can be supported on candle-wicks, droplets, jets and, at very low ambient velocities, small plastic samples and a burner flame. The established flames tend to have spherical or semi-

ellipsoidal shapes. Our preliminary work burning ethanol on planar wicks of 1 × 4 cm suggest similar shapes in aircraft and drop tower tests. But this work showed instabilities in the drop tower, and needs further exploration. Theoretical analysis based on microgravity jet flame length correlations suggests that a 50 mm diameter planar material could support quasi-steady combustion in microgravity for some liquid and plastic fuels. The analysis indicates that burning depends on diameter, so several additional burner sizes will be needed to examine this effect.

Gas burners like the BRE have been used to successfully simulate burning of a wide range of fuels in terms of their *B* numbers and configurations over a range of Grashof numbers. We have successfully shown that for planar burning at different angle orientations that a burner can emulate the flame standoff distribution, and even match the onset of unsteadiness to turbulent flow. Theoretical analyses of such a problem are limited by the inability to accurately deal with flame radiation, turbulence and variable properties. A burner simulation can be far superior as it inherently contains these effects.

The construction of a 50 mm diameter prototype BRE burner with flame heat flux sensors has enabled us to emulate both a methanol and heptane pool fire of the 50 mm diameter, reproducing its color, flame character and shape, and heat of gasification for nearly the same surface temperatures. We show that the BRE is capable of displaying the burning conditions in terms of the fuel properties of heat of combustion, heat of gasification, and surface re-radiant heat flux (with some likely effect of smoke point). A preliminary 3-D plot of steady burning gives a flammability mapping in terms of these variables under conditions of gravity and atmospheric variations. A mapping in microgravity can demonstrate the burning conditions for a wide range of real materials in terms of these properties. In addition the effect of external radiation can indirectly be assessed. The 50 mm burner has been redesigned, along with a 25 mm burner, to reduce edge effects, provide improved heat flux sensors with temperatures, and cooling water has been dropped from the both the burner and the heat flux sensors in microgravity experiments.

The primary objective is to observe the burning behavior (including extinction) in long-duration microgravity for methane/nitrogen and ethylene/nitrogen mixtures on a porous circular burner of 25 and 50 mm diameter at various pressures, ambient oxygen, and CO2 (desired, to assess extinguishment). Ignition will not be directly studied in microgravity studies, as we believe normal-gravity experiments are sufficient for its study. Another objective is to obtain conditions that can be related to microgravity behavior of condensed fuels that embrace the scope of real liquid and solid fuels.

We think these results can establish fuel properties that directly relate to the fire hazard in spacecraft, and these properties can be derived from NASA Test 2. The BRE is an efficient, safe, and comprehensive technique for achieving these results.

This version of the Science Requirements Document has been refined since the Science Concept Review (Aug 28, 2012). The panel wrote at that time that they “strongly encourage continued development of the BRE experiment for implementation into the ACME and testing on the ISS.” We have tried to address most of their recommendations and suggestions. In addition, work has progressed to include the development of two papers, and the commencement of drop facility testing of the BRE. Those results have shown quasi-steady flames, and are very encouraging.

B. Rationale for Microgravity

This is the 'Justification for Extended Duration Microgravity' section (2.5) of the experiment's SRD from June 2013. Equations are not included here, but they can be found in the SRD.

Fire in a spacecraft microgravity environment is a likely event based on limited accident data, tests to burn materials, and research that has explored simple burning scenarios. Indeed, NASA screens the use of materials on spacecraft by its flammability Test 1.

Currently NASA Test 1 and Test 2 are incapable of directly analyzing material fire hazards in spacecraft. There is no predictive methodology to extrapolate from these tests. However, NASA Test 2, the Cone Calorimeter, is capable of determining heats of combustion and gasification, and surface reradiation in burning. Steady (or quasi-steady) burning rate data from the BRE in flight experiments will establish a flammability mapping in terms of these properties for microgravity. In addition, using Eq. (2), the results can establish the necessary heat flux condition for other fuels to burn in microgravity. The BRE is an efficient way to achieve quasi-steady burning. While microgravity tests of condensed fuels have been performed and are planned, they are limited in their ability to reveal sufficient information about the diverse condensed fuels that exist aboard the ISS. Moreover the time to achieve quasi-steady burning rate of solid fuels can be of the order of minutes as dictated by the time response of the solid. The BRE can be turned on and off quickly, and its quasi-steady condition is only dependent on the gas phase phenomenon. While this time is shorter, it can still be on the order of 10 seconds or more.

The proposed tests cannot be performed in parabolic aircraft owing to considerable g-jitter disturbances. While limited testing is possible in drop facilities, the characteristic times for the gas phase (approximately 20 s) and the burner surface temperature (approximately 40 s) preclude consideration of quasi-steady conditions. Preliminary testing in the NASA 5.2 s drop facility indicated the approach to quasi-steady burning, but longer tests are needed to confirm this tendency. The longer duration testing in the ISS with the efficient operation of the BRE can produce results only limited by the gas-phase response time.

C. Science Objectives

The proposed objectives text presented here is the 'Experimental Objectives' section (2.2) of the experiment's SRD from June 2013. ISS testing was not carried out to address the desired second objective, where NASA effectively limited this experiment to the first and third objectives.

The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows.

1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21 % oxygen: 10.2 psia, 26.5 %; and 8.2 psia, 34 % oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels.
2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired).
3. Consider the conditions of Objectives 1 – 2 for extinction limits.

D. Science Success Criteria

This text is from the BRE portion of the 'Success Criteria' section of the ACME Integrated Science Requirements Document (ISRD, ACME-SRD-001.B) from Sept. 2013, which is revision B of the version prepared for the Requirements Definition Review (RDR). Achievement of superior success, as defined below, was effectively ruled out by NASA because its plan didn't include the desired tests.

6.5.1. Minimum Success

- Obtain color video, burner surface temperature, and burner heat flux for two C2H4 flames that burn for at least 20 s and then extinguish.

6.5.2. High Success

All of the above plus:

- Obtain a flammability map for C2H4.

6.5.3. Complete Success

All of the above plus:

- Obtain a flammability map for CH4.
- Measure peak temperatures (using filament and/or soot pyrometry) as a function of time for at least 5 flames.
- Measure temperature distributions (using filament and/or soot pyrometry) at or near quasi-steady conditions for at least 5 flames.
- Measure temperature distributions (using filament pyrometry) near the extinction limit for at least 5 flames.
- Measure soot volume fraction profiles at (or near) quasi-steady conditions and near the extinction limit for at least 5 flames.

6.5.4. Superior Success

All of the above plus:

- Complete the desired parts of the test matrix.



3. Hardware Overview

A. General Description

This section is copied from section 2.A of the April 2021 version of 'ACME Engineering Implementation' which should be consulted for additional details. References to other sections within the boxed text of this section refer to the ACME Engineering Implementation document.

This section provides a general description a general description of the ACME engineering from the '3.0 Experimental' section of the following 2014 conference paper:

> *Dennis P. Stocker, Fumiaki Takahashi, J. Mark Hickman, and Andrew C. Suttles, "Gaseous Non-Premixed Flame Research Planned for the International Space Station," Spring Technical Meeting of the Central States Section of the Combustion Institute, Tulsa, OK, 16–18 March 2014.*

IMPORTANT: *The 2104 paper from which this section was extracted proceeded completion of the spaceflight hardware which was mostly launched in 2017. As a result, it does not accurately describe the final engineering in some regards. Therefore updates, clarifications, additional details, shortcomings, references, etc. have been included in boxed notes like this following the original description. As a minor note, the sizes of figures were adjusted in this report, but none were cropped.*

As previously discussed, the ACME experiments will be conducted within the Combustion Integrated Rack (CIR) using a chamber insert, depicted in Figure 2, which was designed to accommodate gas delivery to a variety of exchangeable burners. Besides the chamber insert, other ACME-unique hardware includes a high-definition color camera and an avionics package, which is for data acquisition and control.

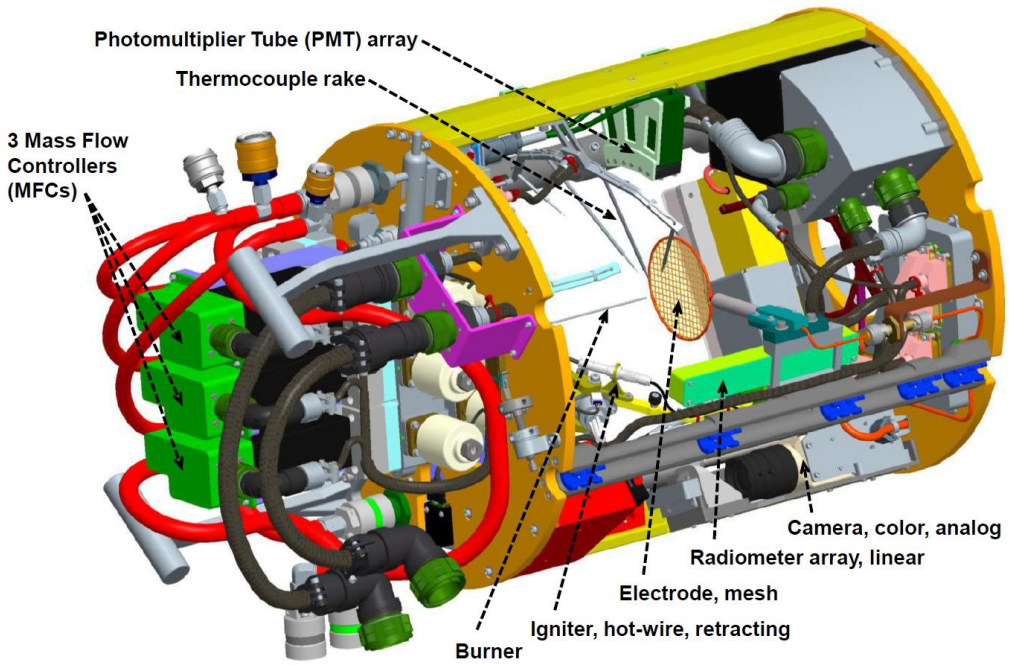


Figure 2: Design of the Advanced Combustion via Microgravity Experiments (ACME) chamber insert that will be used in the Combustion Integrated Rack (CIR) on ISS.

ACME's chamber insert, which is ~0.4 m in diameter, provides common mounting for a variety of burners on the chamber axis (but only one at a time) and includes a retracting resistive igniter. As depicted in Figure 3, there will initially be gas-jet burners (with inner diameters ranging from 0.4 to 3.2 mm), a small coflow burner (where the two concentric tubes have inner diameters of 2.1 and 25 mm), porous spherical burners (with diameters ranging from 6.35 to 12.7 mm), and two flat porous burners for the BRE experiment (with inner diameters of 25 and 50 mm). Most burners will be equipped with a surface thermocouple, and each BRE burner will have two heat flux transducers. The internal pressure of the porous burners will be measured to allow monitoring of the pressure differential across the outlet surface. The igniter will have a crew-exchangeable tip and arm (to accommodate the various burners) and its stepper-motor drive will allow for precise positioning.



Figure 3: ACME burner types, including from left to right, gas-jet burners, a co-flow burner, porous spherical burners, and flat porous burners for the BRE experiment.

Ultimately, the spherical burners were limited to a single diameter of 6.35 mm, and the fabricated gas-jet burners were limited to two sizes with nominal inner diameters of 1.3 and 2.1 mm. However, only the larger gas-jet burner was used during the spaceflight tests. The burners are addressed in detail in section 3.C.

Unfortunately, the spherical burners have not always yielded spherically symmetric flames, presumably as a result of (1) greater burner flow through the joint between the porous sphere and supply tube, and (2) heat loss to the support tube (encompassing the supply tube) causing local flame quenching.

Two types of igniter tips have been used during ACME testing. Most tests have been conducted using a loop igniter tip like those which had been successfully used in the droplet combustion experiments which preceded ACME research in the CIR. However, some tests were conducted with a coil igniter tip, where that included the second round testing for both the Burning Rate Emulator (BRE) and Structure and Response of Spherical Diffusion Flames (s-Flame) experiments, with the exception of the latter's first day testing. The CFI-G investigation will also use the coil igniter tip. For more information about the igniter tips, see section 3.D.

The chamber insert includes three mass flow controllers (visible on the left of Fig. 2) for the delivery of fuel, oxidizer or inert (e.g., helium or carbon dioxide), and nitrogen to the burner. The maximum fuel flow is 2 slpm (on a nitrogen basis), whereas much higher flows, e.g., 10 slpm (on a nitrogen basis), are possible for the oxidizer and nitrogen lines. The crew can exchange the mass flow controllers to provide appropriate ranges that maximize the flow accuracy. The gas delivery system enables on-orbit dilution of the fuel or oxidizer (but not both simultaneously), coflow, inverse flames (e.g., oxidizer delivery to the center of a coflow burner), and premixed flames,

although the latter are not planned for any of the current ACME experiments. Gases can also be delivered premixed to the space station where that is necessary in some circumstances, such as the blending of different fuel gases. CIR's standard bottle sizes are 1, 2.25, and 3.8 liters but ACME will sometimes use smaller bottles as a hazard control to limit the deficient reactant (e.g., fuel) that can be released into the combustion chamber.

Thus far, ACME tests were not conducted with carbon dioxide as a diluent. Furthermore, the fuel diluted with helium was premixed prior to launch rather than on orbit. Ultimately, oxidizer flows of up to 20 slpm (on a nitrogen basis) were possible, while the nitrogen flow was limited to a maximum of 5 slpm. ACME has not used gas bottles smaller than 1 liter as indicated above; the bottle fill pressure was instead limited as an additional hazard control.

ACME's primary flame diagnostics are a suite of imaging systems including five video cameras and two illumination systems, of which three cameras are part of the CIR facility and are already in use on orbit. All but one of the cameras is mounted outside of the chamber. Collectively, the imaging systems will enable color, CH*, and OH* imaging; pyrometry; measurement of the soot volume fraction; and operational support. However, it is unlikely that all functions would be exercised at any one time as will become apparent in the subsequent discussion.

Color imaging is provided by two cameras: a high-definition digital camera mounted outside of the chamber and a standard analog camera mounted on the chamber insert (with a turning mirror). Both cameras are part of the ACME hardware that is now in development. The analog camera signal will be downlinked in near real time to support the experiment operations. Light-emitting diodes within the chamber can be used to illuminate the igniter and burner to both support the operations and provide reference images. The 12-bit digital camera records at up to 30 full frames (1360x1024) per second, and is equipped with a motorized lens with independent control of the zoom, iris, and focus. The digital camera will be equipped with a filter assembly with four options, where selection of a clear "filter" allows for color imaging. Color imaging can also be carried out with a blue-green BG7 Schott glass filter, which balances the color planes to avoid saturation in the red while maintaining sensitivity in the blue and green. The natural flame color can be approximately reconstructed, but the primary purpose of the filter is pyrometry and measurement of the soot volume fraction, again via the techniques of [23-24]. Alternately, CH* imaging can be conducted with this camera because the filter assembly includes a 430-nm bandpass filter with a 10-nm full width at half-maximum transmission (FWHM). A 450-nm bandpass filter (again with a 10-nm FWHM) can be used with the CH* filter to effectively subtract interfering broadband soot emission. However, the filter assembly rotates slowly so such subtraction effectively requires a separate test with each filter and of course reproducible flame conditions.

The video from the analog operations (i.e., ops) camera, also referred to as the internal camera, is not archived in space. The video is instead recorded on the ground in the operations center. Its gain was sometimes adjusted for improved imaging of dim flames, e.g., approaching extinction.

The sensitivity of the color digital camera, often referenced as the ACME DataCam (i.e., Data Camera), to the dim microgravity flames was such that it was often used with long exposure times and thus low framing rates, for example on the order of 1 frame per second. The clear "filter" is of fused silica. Unfortunately, the thickness of the blue-green BG7 Schott glass filter and the added

antireflective (AR) coating caused too great a reduction in the red, so it does not balance the intensity of the three color planes as intended and thereby can limit the accuracy of temperature determination. Meanwhile, rather than the 450-nm bandpass filter described above, the high definition color camera was equipped with a bandpass filter for imaging formaldehyde emissions. For more information, see the Data Camera Characterization Report (ACME-RPT-010 Rev. A).

Imaging of the dim hydrogen flames in the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment was particularly challenging with these, if not all, cameras.

Pyrometry can alternately be accomplished with CIR's monochrome High-Bit Depth Multispectral (HiBMs) camera, equipped with a Liquid-Crystal Tunable Filter (LCTF), which allows for imaging at multiple spectra, but requires a quasi-steady flame for measurement at each desired band. This camera has a telecentric optical system and is recorded at 512x512 and 30 frames per second (fps). Although it has a 1024x1024 pixel array, binning has been necessary because of signal loss through the LCTF. Both soot pyrometry and Thin Filament Pyrometry (TFP) are possible, where ACME is equipped with a crew-replaceable array of 5 silicon carbide (SiC) fibers. The fibers are orthogonal to the flame axis and the array can be translated along the flame axis to the desired position downstream of the burner, e.g., based on the downlink of the analog video. During set up, the crew can manually orient the array so that it is orthogonal to either ACME's color digital camera or the HiBMs camera with the LCTF. An alternate fiber array including a platinum wire will be used for inflight characterization, which in combination with preflight calibration will enable determination of the soot volume fraction [24].

In contrast to the preceding paragraph, ACME did not use a HiBMs camera for pyrometry, but instead relied on use of ACME's color DataCam for such temperature measurements. For an example of how this can be done, see the techniques described in the following papers:

P.B. Kuhn, B. Ma, B.C. Connelly, M.D. Smooke, and M.B. Long, "Soot and Thin-filament Pyrometry Using a Color Digital Camera," Proceedings of the Combustion Institute, 33 (2011) 743-750.

B. Ma, M. B. Long, "Absolute light calibration using S-type thermocouples," Proceedings of the Combustion Institute, 34 (2) (2013) 3531-3539.

Thus far, TFP tests have been conducted with 14-micron fibers spaced apart by 5 mm for a total span of 20 mm (in the case of 5 intact ceramic fibers), neglecting optional translation of the array. Tests have been conducted using arrays with and without the platinum wire in the furthest downstream position. Tests have also been conducted with some of the fibers broken and subsequently removed, leaving only three fibers in the final testing for the BRE and Coflow Laminar Diffusion Flame (CLD Flame) experiments and at least the initial testing for the Cool Flames Investigation with Gases (CFI-G).

CIR has a second HiBMs camera, without a filter, that will be used in combination with CIR's illumination package for alternate measurements of the soot volume fraction. This will be accomplished via an extinction technique previously employed in microgravity [3,6,26], where the camera and the collimated light source (with its two fiber-coupled laser diodes) are positioned at opposite windows. The images are recorded at 1024x1024 and 30 fps. Thus far, the primary purpose of this combined system has been the measurement of the droplet diameter for determination of the droplet burning rate.

A HiBMs camera without the Liquid Crystal Tunable Filter (LCTF) has been used throughout much, but not all, of ACME's operations. In the first round of the CLD Flame experiment, it was used without a filter for flame imaging, but this proved to be largely unsuccessful because the camera is very insensitive to blue. Afterward, it was equipped with a red filter corresponding to the CIR illumination package for the rest of the ACME testing. Unfortunately, that filter is sufficiently broad that significant incandescence can pass through the filter limiting its utility for determinations of soot volume fraction via light extinction, e.g., through the technique described in the following papers:

D.L. Urban, Z.G. Yuan, P.B. Sunderland, K.C., Lin, Z. Dai, and G.M. Faeth, "Smoke-Point Properties of Nonbuoyant Round Laminar Jet Diffusion Flames," Proceedings of the Twenty-Eighth Symposium (International) on Combustion 28 (2000) 1965-1972.

F.J. Diez, C. Aalburg, P.B. Sunderland, D.L. Urban, Z.-G. Yuan, G.M. Faeth, "Soot properties of laminar jet diffusion flames in microgravity," Combustion and Flame, 156 (2009) 1514-1524.

P.S. Greenberg and J.C. Ku, "Soot Volume Fraction Imaging," Applied Optics, 36 (22) (1997) 5514-5522.

Nonetheless, soot volume fraction can also be determined from the color DataCam images.

During ACME operations, the HiBMs camera is typically archived at 15 fps because of data limits for transfer and downlink operations. And because operations are constrained by the data limits, image data from this camera are sometimes deleted without downlinking to enable additional testing. And in some testing, HiBMs images aren't even acquired given their lesser value.

Unfortunately, the HiBMs backlight illumination is significantly obstructed by the bracket for the electrode mesh further limiting its utility for the E-FIELD Flames experiment. This was a design trade-off necessitated primarily by ACME's large number of cameras and other optical instruments.

CIR's Low-Light-Level Ultraviolet (LLL-UV) package is a monochrome camera with an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging. Again as a result of the signal loss through the filter, binning is employed such that the OH* images are recorded at 512x512 and 30 fps.

During ACME operations, the LLL-UV camera has only been used for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

While not discussed in the preceding section, another intensified camera was used throughout ACME testing, although its image data was sometimes deleted or not even acquired to allow the performance of additional tests. This was the Cool Flames Investigation (CFI) camera which was generally configured with its CH filter, where an exception is some s-Flame testing with particularly dim flames. Due to design trade-offs, its Field of View (FOV) was partially obstructed by the arms of the Thin Filament Pyrometry (TFP) fiber array when it was moved inwards to the flame. More information about this camera can be found in its characterization report, CIR-RPT-4600.*

The ACME chamber insert will also be equipped with non-imaging optical diagnostics including both thermopile radiometers and Photomultiplier Tubes (PMTs). There is a crew-exchangeable linear array of radiometers of which there are two versions supporting the initial set of

experiments. The array has a fixed position which is parallel to the chamber axis and centered (along the chamber length) relative to the chamber windows and thus most of the other instrumentation. One array is equipped with a single wide-view radiometer with two narrow-view radiometers that are downstream and two more that are upstream, where all views are orthogonal to the chamber axis, as depicted in Figure 4. The narrow-view radiometers are intended to support studies of gas-jet flames, where ACME can accommodate shorter burners, making the "lower" detectors useful. The other array, also seen in Fig. 4, provides for three wide-view measurements of the flame, where one view is masked to nominally block radiant emission from the burner. It was specifically designed for the BRE experiment but will be a resource available to other ACME experiments. Given that the radiometer array can be exchanged by the crew, alternate arrays could be developed.

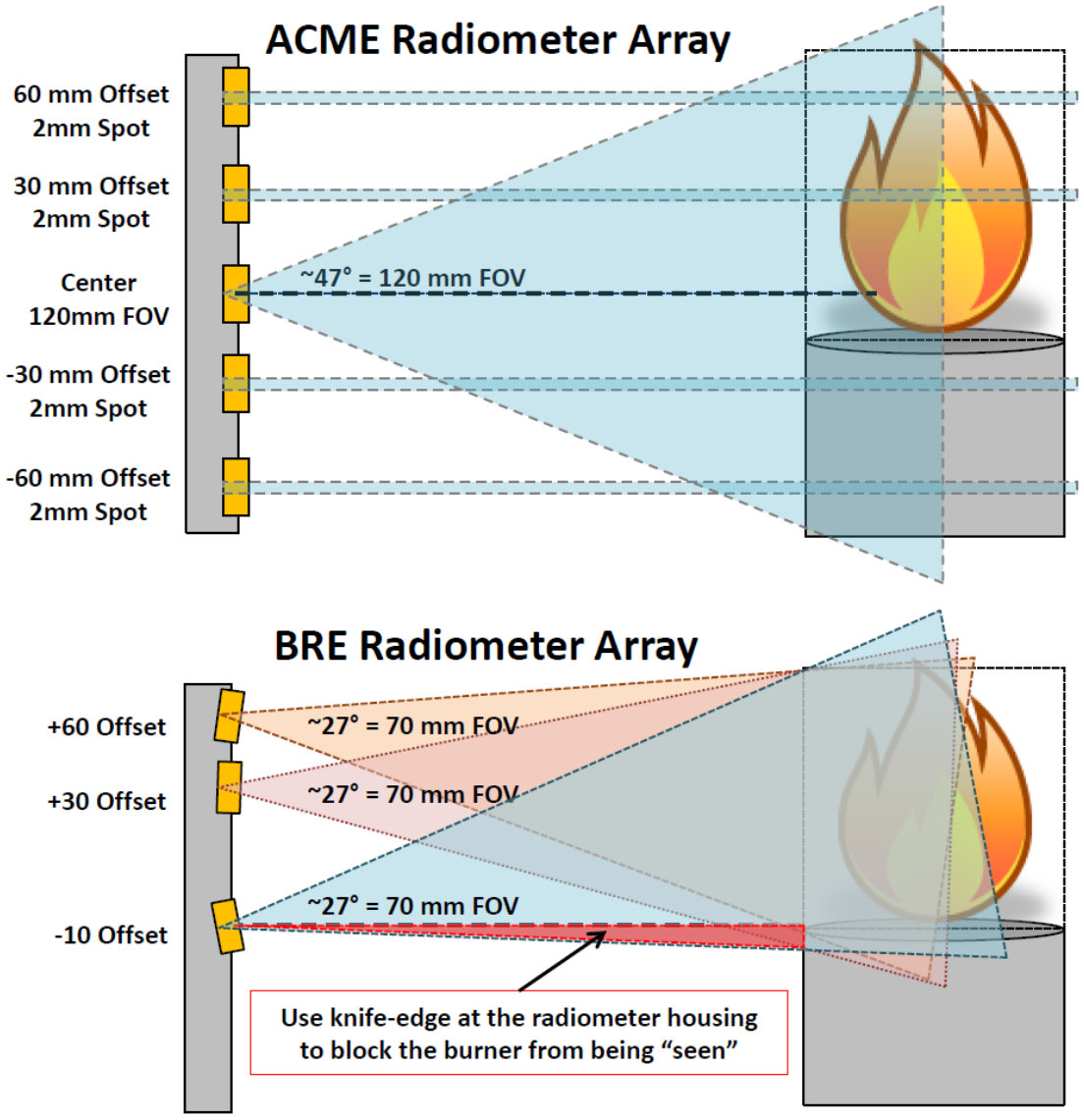


Figure 4: Crew-exchangeable radiometer arrays on the ACME chamber insert: (top) standard (ACME) array of 5 radiometers, and (bottom) array with three radiometers, designed for the BRE experiment, where the lowest detector is masked to avoid view of the burner.

A third radiometer array was developed for the CFI-G experiment, where it is important to understand that only a single radiometer array can be used at a time.

The narrow-view radiometers in the standard array are only fully sensitive in small spots, but they are each sensitive in a much greater Field of View (FOV) than represented in the preceding figure. They do not have lenses; their FOV is a result of an aperture where most of their FOV is within the penumbra which is only visible to a portion of the sensor. Figures representing the measured FOVs of the as-built standard and BRE radiometer arrays are found in section 3.F of this document.

Unfortunately, the radiometers in the BRE array typically saturated in the first round of BRE testing, so the standard array was used in its second round. However, those radiometers similarly saturated in many of round 2 tests. In this regard, each of ACME's BRE and standard radiometers have a fixed gain. Meanwhile, each sensor in the radiometer array developed for the CFI-G investigation has two fixed gains, i.e., a high gain and low gain.

For more information about the standard radiometer array used for most ACME testing, see the Radiometer Characterization Report (ACME-RPT-011). Meanwhile, you can learn about the other radiometer arrays from the BRE Radiometer Characterization Report (ACME-RPT-012) and the Cool Flames Radiometer Characterization Report (CIR-RPT-4600).

An additional wide-view radiometer, that is fixed and not crew exchangeable, is positioned downstream of the burner and flame at 30° from the chamber axis. This detector is primarily intended to support the spherical flame experiments and will have a quicker response than the more sensitive detectors that will used in the linear arrays. Like the radiometers on the two linear arrays, it will have a spectral range of nominally 0.2-11 micron and its field of view will be established by an appropriately selected aperture.

The FOV of the off-set radiometer is also shown in section 3.F. Learn more about this sensor from the Radiometer Characterization Report (ACME-RPT-011).

To detect the quenching of dim flames, the chamber insert will be equipped with a set of three PMTs. Each PMT will have a wide-angle view, a sight line that is orthogonal to the chamber axis, and will be centered (along the chamber length) relative to the chamber windows so that it is aligned with most of the other instrumentation. The PMTs' spectral range will be 230-700 nm, i.e., visible and near ultraviolet, but one will have a 310-nm bandpass filter and another will have a 430-nm bandpass filter (where both are with a 10-nm FWHM). The pair of detectors will enable measurement of OH* and CH* emission, respectively. The third PMT will be equipped with a neutral density filter to prevent saturation. Like many other components, the PMT array can be removed from the chamber insert by the crew.

Unlike the radiometers, each PMT had a readily adjustable gain. However, the PMT signals sometimes saturated when the gains were set too high. See the PMT Characterization Report (ACME-RPT-013) to learn more about the three Photomultiplier Tube (PMTs).

The chamber insert is also equipped with six far-field thermocouples to measure the spread of the thermal field in the spherical flame experiments. They are of limited utility for other experiments because they are positioned upstream of the burner tip. Therefore, they will

generally not be installed by the crew to reduce the chance of accidental damage. There will be two versions of a four-thermocouple rake to allow a choice of near and far measurements. Other arrays could be designed to support future experiments, but they cannot extend downstream of the burner tip without interfering with other (e.g., optical) measurements.

The two lone thermocouples, which are both at nominally 200 mm from the center of the spherical burner, have been installed throughout ACME testing. The tip of one was seen to be slightly bent early in testing, as discussed and depicted in section 3.G of this report.

Only the short (or far) version of the radiometer rake has been used thus far, where factors include the need for crew time to switch between rakes and the heat loss that could result from thermocouples in the long (close) rake. The short (far) rake's thermocouple tips should nominally be at 45, 60, 100, and 100 mm from the spherical burner center. For thermocouples at the same distance from the burner center, the angle between the rays from the burner center to the two thermocouples should be 90 degrees. The bead size of the thermocouples was not measured, but it is undoubtedly at least twice the size of the 0.2 mm diameter wire.

The two thermocouples closest to the burner outlets on each four-thermocouple rake are type B, while the others are type K.

Experiment-unique hardware is also mounted on the chamber insert. For example, to accommodate the E-FIELD Flames experiment, the insert is equipped with a high-voltage supply and a crew-removable electrode mesh which can be set between +10 and -10 kV. The circular mesh is orthogonal to the chamber axis and is positioned (during set up) at either ~35 or ~50 mm downstream of the insulated burner. The ion current between the burner and electrode mesh will be determined via measurement of the voltage drop across a known resistor and application of Ohm's Law.

Additional images of the chamber insert can be seen in subsequent sections of this report. The hardware specific to E-FIELD Flames is discussed further in section 2.B.

Additional supporting measurements will be made, where the chamber pressure is a simple example. The ACME experiments will use the CIR's gas chromatograph for pre- and post-test analysis of the chamber atmosphere, e.g., for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases. Acceleration measurements will be provided by the Space Acceleration Measurement System (SAMS) [27], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The CIR's Gas Chromatograph Instrumentation Package (GCIP) was not used for the CLD Flame or E-FIELD Flames investigations and thus far was only sometimes used during BRE, Flame Design, and s-Flame testing. Its use requires significant time and as such effectively reduces the number of tests that can be conducted. However, estimates of the chamber contents were computed assuming that the fuel in a normal flame test (for example) was either fully or not at all oxidized, e.g., for unsustained ignition in the latter case.

The CIR facility was equipped with a 3-axis SAMS sensor head and the acceleration measurements, along with relevant documentation, are accessible via the home page for the Principal Investigator

Microgravity Services (PIMS), https://gipoc.grc.nasa.gov/wp/pims/home/. To be clear, the acceleration measurements are not archived with the ACME experiment data in the Physical Sciences Informatics (PSI) website, www.nasa.gov/PSI.

All ACME experiments will be constant volume (i.e., rather than constant pressure) studies, where the chamber's free volume will be about 90 liters. Burner flows will generally be constrained to limit the pressure rise, and tests will typically be 0.5-2 minutes in duration.

The chamber's free volume is approximately 83 liters depending on the installed hardware.

The ACME experiments will be configured, but not conducted, by the ISS crew. Instead, the experiments will be commanded from the ground, specifically from the Telescience Support Center (TSC) at the NASA Glenn Research Center in Cleveland, Ohio. They can be conducted in a nominally automated mode using pre-programmed scripts. However, it will also be possible to effectively operate the experiments in a "manual" mode, where changes can be made in response to the analog video downlink and the numerical data available in the telemetry stream. For example, a flow rate could be adjusted to approach a stability or soot limit. Step and ramp changes can be made in the flow rate(s) or electric field potential. The TFP fiber array can be translated uniformly through a soot-free flame to get a continuum measurement of the temperature field. In summary, ACME's experiment control will allow for broad flexibility.

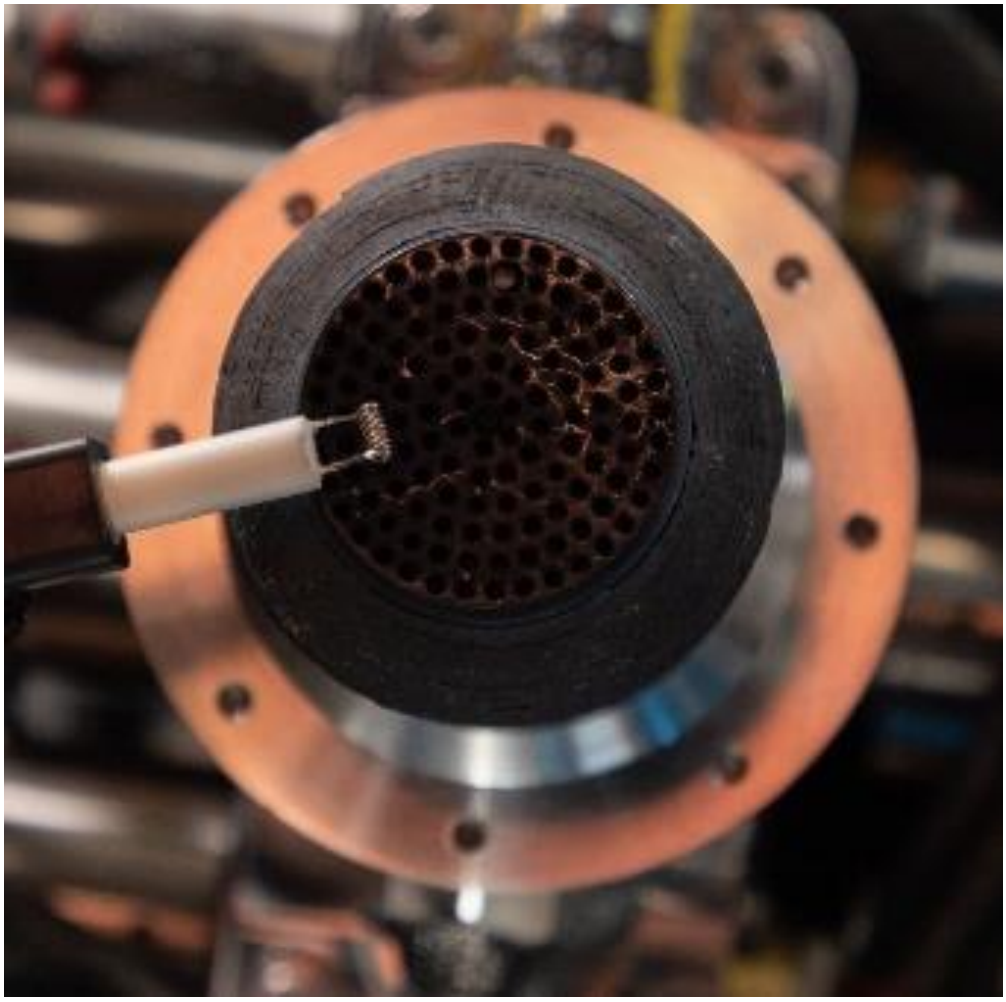
The TSC is now also referred to as the Glenn ISS Payload Operations Center (GIPOC). Thus far, the ACME tests have always been conducted in a nominally automated mode.

B. Hardware Shortcomings

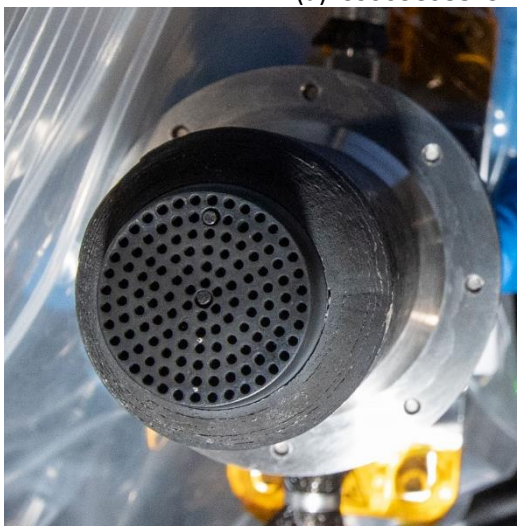
In terms of hardware, BRE's primary challenge was that the sensors for measuring the thermal radiation, which had fixed amplification, often saturated. Although the measurements were not documented as required to meet experiment success, they were considered sufficiently critical that BRE was equipped with its own radiometer array, featuring three thermopile detectors. Estimates had been made of the signal levels based on 5-second drop tests since the hot gases tend to blanket the flame in the absence of buoyancy and thereby remain within the view of the radiometers. As a result, the signal levels are strongly dependent on gravity. The estimates were not part of the science requirements and may not have been sufficiently tracked in the hardware development. Furthermore, the ISS tests lasted as long as 5 minutes, which is over four times the envisioned BRE test duration and 60 times the duration of the drop tests. With the fixed amplification of the radiometer signals, the data was frequently lost - other than the knowledge that the limits were exceeded. Saturation of the radiometers was partially mitigated in the round 2 testing through use of ACME's standard array instead of the one developed specifically for BRE which was used in the first round of testing.

The outlet surface of the 25-mm BRE burner (used in all but two of the BRE test days) became scratched and worn, as seen in Fig. 5(a), through ISS stowage and/or handling. This degradation presumably occurred following its removal after the first round of testing as seen in Fig 5(b). The scratches are not evident in Fig. 5(c) taken when the 25-mm burner was removed at the end of its use, but the outlet surface was clearly no longer uniformly black. These changes are important because of their effect on the absorptivity of the burner's copper outlet plate. Note that Figs. 5(b)

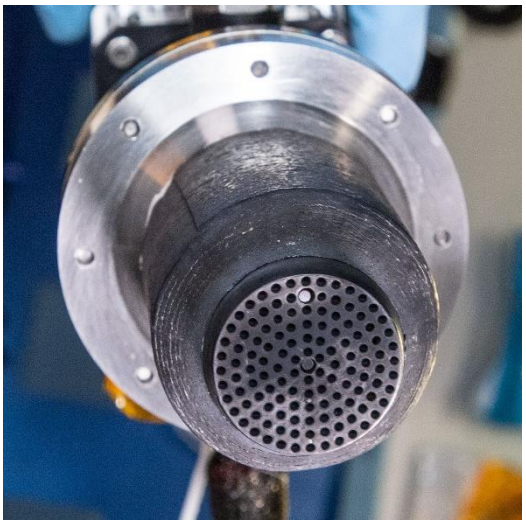
and (c) were rotated to nominally correspond to Fig. 5(a), i.e., with the offset heat flux transducer toward the image top and embedded thermocouples toward the bottom.



(a) ISS063e088291 detail, 14 Sept. 2020



(b) ISS059e017084 detail, 9 April 2019



(c) ISS064e026386 detail, 25 Jan. 2021

Figure 5. Changes in the appearance of the 25-mm BRE burner outlet

As discussed in the preceding section, the ACME DataCam is unfortunately not as sensitive as had been hoped to the dim microgravity flames. As a result, it was often used with a long exposure time limiting the framing rate, or a high gain amplifying noise in the image.

On 3 Sept. 2020 (20247), the first day of BRE's second round, the gas flow didn't always transition as to the desired flow rate as intended. The intended igniter arm was missing (but found before the next test day) and use of an alternate arm led to the concerns about ignition success and the decision to ignite at the same successful flow conditions and subsequently adjust the fuel flow rate to the desired test value. The first attempt was successful, but the transition time was then shortened to save fuel, but it was reduced to the degree that it unfortunately prevented the intended flow transition. Unfortunately, this shortcoming wasn't discovered until during a review of the data days later. As a result, the tests on that day were generally conducted at the same fuel flow rate, which was meant to be a variable. Therefore, a make-up test day was conducted on 28 Jan. 2021 (21028) to address the science loss.

More generally, ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel and nitrogen have unfortunately not always performed adequately, where the flow rates have not always been appropriately steady. Steps were made to mitigate such behavior, e.g., by zero-ing the MFCs prior to each test, but they were not completely successful. Furthermore, when an MFC is used at a small fraction of its maximum flow rate, the actual flow rate sometimes differs from that desired.

During testing on 23 Dec. 2020 (20358), it was discovered that the lowest of the 14-micron silicon carbide fibers on the Thin Filament Pyrometry (TFP) fiber array had broken. After some discussion, testing continued that day. Given that the translating array had five ceramic fibers, the loss of one was not a significant science loss. But it was a concern for crew safety. Recovery plans were developed and enacted on January 18 to remove the broken fragments from the harp-like array. Unfortunately, the next fiber was broken during the delicate operations where the fibers are only spaced 5 mm apart. As a result, the last two BRE test days were conducted with only three rather than five fibers in the TFP array. ACME does have alternate fiber arrays, but there were months of testing remaining and the decision was made to continue testing with only 3 fibers.

Ignition was not always successful. As an extreme example, it was discovered on 1 April 2019 (19091) that the hot-wire igniter tip had failed on the preceding test day, preventing further flame testing until it could be replaced. While such circumstances are uncommon for ACME operations, BRE tests were sometimes conducted under conditions, e.g., for igniter activation and deactivation, which led to momentary flashes without sustained ignition. Given that unplanned crew time is required to replace failed igniter tips, the BRE ops team limited - to varying degrees - the igniter voltage and the time that it was energized within the flame. The latter was known from earlier research in the Combustion Integrated Rack (CIR) to degrade the hot-wire igniter tip and hasten its demise. It is notable that the flashes of flame (across the burner surface) sometimes continued briefly after the igniter was deactivated and retracted. In one case on the first day of testing, there was excess fuel and a 'fireball' was ignited which promptly quenched. Ignition also sometimes failed due to a software limitation where the fuelFlowIgnitionDeltaMs parameter which specifies the relative timing of the fuel flow and igniter activation must be evenly divisible by 100 (ms). This limitation wasn't initially understood and the igniter sometimes didn't energize as a result.

While it rarely occurred, BRE tests sometimes automatically aborted (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time associated with the failed test attempt. Such aborts typically occur because of communication issues with an Image Processing and Storage Unit (IPSU).

In addition to this section (3), shortcomings are discussed in Appendices B to E of this report and the *ACME Engineering Implementation* document.

4. Preliminary Results

The burner has shown that quasi-steady burning is possible for real solids with a flat surface of 25 mm diameter. Up to now most flammability studies in microgravity have focused on flame spread with a wind for concurrent spread and at least a self-induced wind for opposed flow spread. We already knew that this would be possible from normal-gravity research. The burning of liquids, as in a wick of a candle flame and small droplets, has been shown to produce steady flames. However, many droplet flames were found to extinguish owing to radiative cooling of the flame. We see this as well in the extended ISS microgravity tests. However, this extinction was predominately seen for nominal ambient conditions of 21% oxygen and 14.7 psia. At conditions of 27% oxygen, 10.2 psia and 34% oxygen, 8.2 psia, the flames generally persisted for 3 minutes or more. These are all atmospheric conditions that NASA is considering for future spacecraft.

A. Changed Test Plans

While BRE's proposal included desired testing with carbon-dioxide-diluted atmospheres, NASA's plan for the ISS investigation did not include desired testing. Therefore, the exclusion of that testing was not a change in plans.

The most significant change was the use of ACME's standard radiometer array rather than the BRE radiometer array in BRE's second round of testing. This change was made because the BRE array was found to generally saturate during the ISS tests. The ACME array had a greater range but nonetheless still often saturated during testing.

Significant changes were made to the ignition procedure in the early first-round testing, both relative to the preceding ACME experiments and in the first few days of BRE testing. Compared to the other ACME burners (neglecting the outer plenum of the coflow burner), the BRE burners have a much greater internal volume. Therefore, the fuel flow was begun before the igniter was energized, where that period ranged from a few seconds to roughly 1 minute. The duration was typically about 15 and 30 seconds for the 25- and 50-mm BRE burners, respectively. In contrast, the fuel flow has commonly been started when the igniter is energized in the other ACME investigations.

Despite the duration of fuel flow prior to igniter activation, there was concern about understanding the composition of the gas initially flowing from the burner. Therefore, on the second BRE test day (20 Feb. 2019, i.e., 19051), a no-flame 'test' was run before each flame test in which the burner was flushed with nitrogen at 0.15 slpm for 15 seconds. A shortcoming of this measure was that the time between the nitrogen flushing and the start of the fuel flow was not

precisely controlled. Therefore, the software script used to run the BRE tests was subsequently altered to allow specification of a 'pre-flow' with a specified wait time between the pre-flow and the start of the gas flow for the flame. Although the script allowed for a pre-flow of nitrogen and/or fuel, only nitrogen was used for the pre-flow during BRE's ISS testing. The revised script was implemented on BRE's third test day (25 Feb. 2019, 19056) where a nitrogen flow rate of 0.15 slpm for 15 seconds continued to be used throughout the remainder of the first round.

A wait time of 2 minutes was used in the first two 19056 tests, while a 1-minute wait was used for the remainder of testing on that day. During the remaining round-one test days, the wait duration was consistently 50 seconds. With the script change, the basic non-imaging data file captured the data during both the pre-flow and in the flame test, but it is important to understand that data was not acquired throughout the wait duration. Therefore, one should pay close attention to the time stamp when reviewing BRE data prior to the start of the fuel flow because the listed time abruptly jumps.

Although the script wasn't significantly modified, there were changes to the ignition procedure in the second round of BRE tests. It was concluded that the nitrogen pre-flow wasn't needed for the first test of the day or the first test after the chamber was evacuated and refilled. The wait time between the pre-flow and the start of the fuel flow was also increased to 5 minutes throughout the second round, whereas it was 50 seconds in most of the first-round tests. This was done to ensure that the atmosphere in the chamber was quiescent prior to the start of the fuel flow. Pre-flow durations and flow rates were also somewhat different. In the tests on the first round-two test day (20247), which featured the 50-mm burner, the pre-flow was at ~0.5 slpm for 5 or 10 seconds. On the two subsequent test days (20261 and 20279), the pre-flow was at 0.19 slpm for 3.2 seconds. For the remaining test days (20339, 20346, 20352, 20358, 21020, and 21028), the pre-flow was instead 0.16 slpm for 5 seconds, where that includes the final test day which also featured the 50-mm burner. As such, the pre-flow differed significantly between the two test days with the larger BRE burner.

A few minor changes were made to the script after the addition of the pre-flow to flush the burner's internal volume:

~2020-03-17	flow ramp hold was fixed so that the hold values were achieved exactly
~2020-12-18	ignition detection was disabled prior to igniter arm retraction so that the retraction wasn't interrupted

The first change was implemented between the first and second rounds of testing, while the second was made during the second round. The final version of the script will be included in the upcoming *ACME/BRE Experimental Procedure* document.

While a minor factor, BRE made use of the coiled type of igniter tip in second-round testing as compared to the standard loop-type igniter used in the first round. BRE ignition seemed to require significant ignition energy and the coiled igniter design is believed to be more robust. The coiled igniter tip was developed as an alternate after ignition challenges with the Coflow Laminar Diffusion Flame (CLD Flame) experiment in late 2017. The development of the coiled igniter tips continued even though it was later determined that the igniter had not been properly aligned with the burner outlet in the early CLD Flame testing. The two igniter designs are discussed and depicted in the *ACME Engineering Implementation* document.

In contrast to BRE's success criteria (discussed in section 2.D), which identified the importance of

measuring the flame temperature, BRE's first round testing seldom made use of Thin Filament Pyrometry (TFP). In contrast, TFP data was acquired in a significant fraction of the second-round testing. In this regard, TFP use was important because the BRE flames quickly became non-sooty even with oxygen concentrations of up to 40% (by volume) so measurements couldn't be made via soot pyrometry.

Tests were conducted as planned at atmospheres being considered for future spacecraft, featuring elevated oxygen concentrations and reduced pressures (nominally maintaining the oxygen partial pressure), but more of the testing was conducted at roughly atmospheric pressure with elevated oxygen concentrations. The latter tests were carried out in a broader exploration of material flammability in microgravity.

B. Overview of Experiment Test Operations

While not yet prepared, additional details will be made available in the ACME/BRE Experimental Procedure document.

The narrative description of the experimental procedure is an excerpt from the 'Methods / Experimental' section of the following conference paper.

Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.

While the paper was written for the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment, much of the discussion applies to ACME in general, including the BRE investigation. The figure referenced in this excerpt is not included herein. Clarifications and additional information, not in the conference paper, are included in boxed notes like this.

The CIR experiments are configured, but not conducted, by the ISS crew. Thus far, ACME's activity has included the majority of the crew members, regardless of their nationalities. Most often the astronauts and cosmonauts exchange gas bottles, mass flow controllers, and igniter tips. In the latter case, the igniter tip exchange can occur after failure or proactively before failure when the crew already has changes to make changes in the interior of the insert. While these activities may seem mundane, the crew play a critical role in setting up the hardware for each experiment. As an extreme example, there were crew activities on 8 separate days in Sept.-Oct. 2017 to configure the CIR and ACME hardware for the first ACME experiment, namely the Coflow Laminar Diffusion Flame (CLD Flame) experiment. In contrast, only two crew days were required to reconfigure the hardware for the first E-FIELD Flames tests because of the initial use of the same coflow burner. Planning is critical because crew activity is scheduled three weeks in advance, for which detailed procedures must be prepared and uplinked to the ISS.

BRE has its own unique burners and it was the first ACME experiment conducted with large flames, as compared to those in the CLD Flame and E-FIELD Flames investigations. That required changes to some optical elements for BRE as well as subsequent ACME experiments. BRE was also the first experiment to use the CIR facility's Gas Chromatograph Instrumentation Package (GCIP). As a result, the initial BRE set-up included 9 days of crew activity, mostly by Canadian astronaut David Saint-Jacques, although that includes 4 days in which hardware was either gathered or stowed.

All CIR tests are remotely commanded from the ground from NASA's Glenn ISS Payload Operations Center (GIPOC) in Cleveland, Ohio. Typically, four to six people staff four consoles during the testing. The CIR hardware is controlled from the Rack Officer (RO) and Data Management Officer (DMO) consoles, and the ACME hardware is controlled from the Payload Developer (PD) console. Two shifts of personnel are scheduled to staff these three consoles on test days because they typically last 14 hours with the many pre- and post-testing activities. Given that the tests are remotely commanded, the operations are carried out in the day and evening in Cleveland, e.g., from 7 AM to 9 PM. A day's tests, i.e., with flames, are normally conducted in a period of 10 hours of less. Tests are normally conducted on three weekdays per two weeks. On intervening weekdays, the image data is downlinked and the crew often swap one or both gas bottles. Transferring and downlinking data from a day's tests also requires a 2-shift day for the RO and DMO consoles, where it is always necessary to verify that data was downlinked without errors and that data is not missing. Those two positions are also staffed to guide and oversee the crew activity. As a result, the CIR consoles are normally actively staffed on four weekdays per week. A team of four carry out the detailed planning for the CIR and ACME console staff, and the daily operational plans typically exceed 50 pages on days including testing.

The science console allows for review of the analog video and non-imaging telemetry (at 1 Hz) in near real time but doesn't allow for commanding. In the case of E-FIELD Flames, arrangements were made so that the analog video is streamed to a secure computer at UC Irvine to allow the research team to participate in test operations and to recommend adjustments to the day's planned parameters. That team does not see the telemetry during the testing, but is in communication via phone and e-mail with their NASA research partners at the GRC ISS Payload Operations Center (GIPOC). While there are many preparatory steps, the tests are ultimately conducted using an uploaded software script and parameter files. In advance of the test day, the UC Irvine science team plans the nearly 60 parameters in the voltage sweep script and specifies other parameters that are configured outside of the script, such as camera settings. It is also possible to effectively conduct tests in a 'manual' mode, where parameters are changed in response to the analog video downlink or telemetry. For example, the electric field potential can be adjusted to approach a stability or soot limit.

Provisions have been made to enable the investigators from all ACME experiments to remotely view the analog video while they communicated verbally with NASA science personnel in the operations center. Thus far, all ACME flame tests, including those for BRE, have been conducted in an automated manner using software scripts and extensive parameter files.

Typically, ten E-FIELD Flames tests are conducted per test day. Despite the short test durations, the low rate results from several unavoidable factors. After a test is conducted, the 125-Hz non-imaging data is downlinked for immediate review by the science team. A small sample of images are often downlinked to ensure that the camera exposures are appropriate, where the times for those images are typically selected based on non-imaging data to provide extreme examples. The review helps to verify parameters or plan for adjustments in subsequent tests, e.g., to select the voltage range after determining where the ion current saturates for the given flame conditions. After a test is conducted, a fan circulates the chamber atmosphere after which the chamber is vented (to space) back to atmospheric pressure, if needed. Midway through a day's E-FIELD Flames tests, the atmosphere is circulated for 15 minutes through a filter cartridge filled with molecular sieves to capture any soot and adsorb at least a portion of the combustion products, e.g., carbon dioxide, water vapor, and incomplete combustion products. The tests are only

conducted when both Ku and S band communication is available through NASA's Tracking and Data Relay Satellite (TDRS) System. Depending on the various orbits, there are periodic communication outages with a loss of signal (LOS) that can be a short handover (from one satellite to another) or can last as long as 30 minutes.

About eight flames were ignited per test day in BRE's first round, neglecting the day in which no tests were conducted because the igniter had unknowingly failed at the end of the preceding test day. The nominal durations were typically planned for 2 or 3 minutes but some tests were 5 minutes long. Meanwhile the second round of BRE had roughly 15 tests conducted per day, but the durations were nominally 2 minutes at most.

Unlike E-FIELD Flames, the BRE experiment also made use of the CIR's Gas Chromatograph Instrumentational Package (GCIP) to measure species concentrations of the atmosphere in the combustion chamber. There is no sampling probe and measurements are only made before, between, or after tests when the chamber atmosphere was 'stirred' through use of a fan. The GC measurements were made on roughly 60% of the test days with flames in the first round, specifically on 19039, 19063, 19066, 19070, 19087, and 19094. In the second round, such data was acquired on less than half of the test days, specifically on 20247, 20261, 20279, and 20352.

Like E-FIELD Flames, BRE made periodic use of the adsorber cartridge, for example between atmospheres and after GC measurements, if any, were made.

A variety of constraints limit the number of tests that can be performed, both overall and on a daily basis. The next combustion experiments in the CIR will feature solid fuels and focus on flame spread and material flammability. That follow-on plan limits the duration of ACME operations, where each ACME experiment is allotted four months for testing. Although it requires additional crew time, these allotments are split into two rounds so that the investigators have a break from testing for careful review of the early results so that they can maximize the scientific outcome of their second round. This paper reports on early results from the first round of E-FIELD Flames.

While nitrogen is available from the space station, the combustion testing is limited by the available gas bottles that were and are launched to support the tests. For E-FIELD Flames and most of ACME's experiments, the test matrices used to plan those gas resources were established in 2010. At that time, NASA limited the scope of the ACME tests matrices with the intent of focused experiments with limited objectives rather than comprehensive experiments. As such, the general target was 50 tests per experiment. For most ACME experiments, the limiting gas constraint is the oxygen/nitrogen bottles used to fill the chamber. This resource constraint is exacerbated by the CIR because it is equipped with a pair of redundant safety timers to limit the fuel that can flow into the combustion chamber (where the timers can be set for the oxidizer flow in the case of inverse flame testing). The timers can only be reset by venting the combustion chamber. Therefore, the timer limit combined with other constraints effectively corresponds to the day's maximum flow time for the fuel (or more generally, the limiting reactant). For E-FIELD Flames, the safety timers are set to the maximum value of 999 seconds, which corresponds to the 10 tests typically completed per day.

The 999-second limit on the valve safety timers has thus far been common for all ACME testing. However, the timers can be reset to 999 by nearly completely venting the chamber. Sufficient time

and gas resources are needed to vent and refill the chamber and still conduct tests afterward, but this was done on most BRE test days. The only exceptions were 19039, 19063, and 19091 where in the last case no flames were ignited because the igniter had unknowingly failed at the end of the prior test day.

Another constraint is the available time in a day, where the console personnel and ISS resources, such as the video downlink, are planned for a limited number of hours. When necessary for the research, NASA does have continuous 24-hour/day operations for some non-CIR experiments. But as discussed, the ACME tests are typically two minutes or less in duration, so the testing is almost always limited to the planned durations.

Data limits are yet another constraint, where the on-orbit hardware can store a limited amount of data. More importantly, 127,000 MB is the most that can be reasonably managed in a two-shift transfer and downlink day. Therefore, tests days are limited to this amount of data, where image data from select cameras is often archived at reduced rates to mitigate this challenge. Nonetheless, the limit sometimes forces decisions to abandon recorded images.

This detailed description of operations and hardware provides the context under which the ACME experiments are conducted. …

Meanwhile, Table 1 is a summary log of the flames ignited, including a limited overview of the test conditions, where it is an excerpt from this report's Appendix E. 30-Day Operational Accomplishments Report for ISS Increment 64. Note that it is comprehensive and includes the earlier testing conducted in the preceding increments. A minor correction was made where 19094 is now identified as April 4. It accidentally been identified as April 5 in the 30-day reports, although the testing did extend past midnight in Cleveland, OH so the final tests were conducted on April 5. Nonetheless, the BRE (and more generally ACME) test days and numbers are identified by the date on which the test day began, even if they occurred after midnight (whether on the ISS or in Cleveland).

Table 1. BRE log of flames ignited in Increments 58-59 (round 1) and 63-64 (round 2)

Test Day[1]	Year	Month	Date	Burner Size[2]	Fuel bottle	Oxygen conc.[3]	Flames ignited[4]
First round of tests, with the 25-mm burner, from Feb.-April 2019							
19039	2019	Feb.	8	25-mm	100% C2H4	21%, 26.5%, 40%	10
19051			20			26.5%, 34%	8
19056			25		50% C2H4	21%, 26.5%, 34%, 40%	8
19063		March	4			26.5%, 40%	6
19066			7			34%	6
19070			11			26.5%, 34%	7
19087			28		100% C2H4	26.5%, 34%	8
19091		April	1			21%	0
19094			4			21%, 26.5%, 34%	9
Second round of tests, with the 25-and 50-mm burners, from Sept. 2020 to Jan. 2021							
20247	2020	Sept.	3	50-mm	100% C2H4	25%, 34%, 40%	13
20261			17	25-mm		26.5%, 34%	13
20279		Oct.	5		100% CH4	25%, 34%, 40%	21
20339		Dec.	4			25%, 40%	18
20346			11		50% C2H4		12
20352			17		100% C2H4	34%, 40%	10
20358			23			25%, 34%, 40%	17
21020	2021	Jan.	20			21%, 26.5%, 34%	15
21028			28	50-mm		25%, 34%, 40%	13

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Burner size** refers to the diameter of the BRE burner's perforated circular outlet.
[3]**Oxygen conc.** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations for tests that ignited are listed. *The chamber pressures were also varied but are not identified in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

C. Summary of Preliminary Results

Figure 6 shows a typical test of ignition and flame growth. Although extinction occurred at 45 s with the flame still growing, the heat flux was nearly steady by 30 s. Therefore, a heat of gasification (L) could be estimated. This was done by using the arithmetic average of the center and offset heat flux sensor as the net average heat flux.

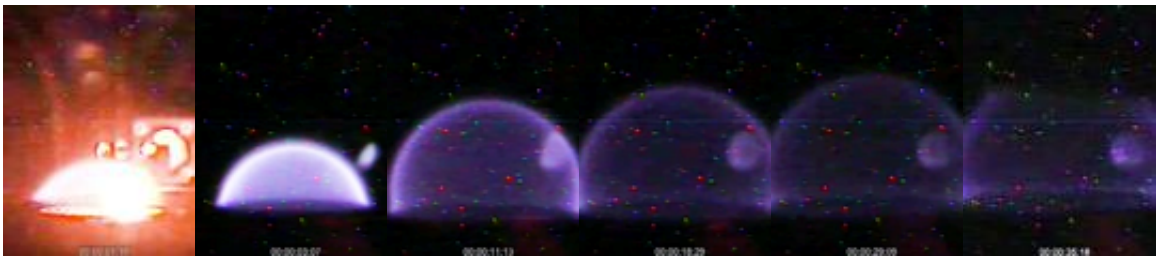


Figure 6. Flame growth from ignition to self-extinction in test 19056F1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 4.55 g/m^{2}s, 27% O_2, 10.2 psia, L = 0.66 kJ/g, extinction at 45 s.

Figures 7 through 10 illustrate results for nominal atmospheric conditions of 26.5% oxygen and 0.7 atm (10.2 psia), and 34% oxygen and 0.56 atm (8.2 psia), respectively with two figures each. Both use 50% mixtures of nitrogen and ethylene giving a fuel mixture heat of combustion of 24 kJ/g. Both tests reached quasi-steady conditions and were terminated by shutting off the fuel. The plots of transient heat flux shown by Figures 8 and 10, show the condensation effect removed after 80 s and 40 s respectively and arithmetic average of the two heat flux gauge measurements at the end of each test is recorded as steady surface heat flux which are used to obtain heats of gasification.



3 s 10 s 52 s 91 s 125 s 155 s 178 s

Figure 7. Flame growth from ignition to steady state in test 19070K1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 2.3 g/m^{2}s, 26.5% O_2, 10.2 psia, L = 3.0 kJ/g, steady.

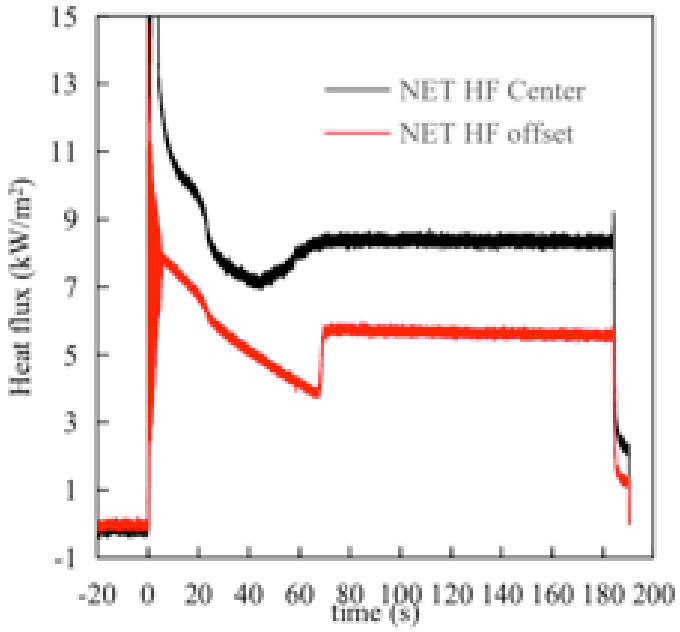


Figure 8. Measured heat flux in test 19070K1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 2.3 g/m^{2}s, 26.5% O2, 10.2 psia, L = 3.0 kJ/g, steady.

In Figures 9 and 11 the flame reaches a quasi-steady state in about 80 seconds as seen by the flame shape and the heat flux data. This emulation corresponds to a material with a heat of combustion of about 24 kJ/g and a heat of gasification of about 3 kJ/g. As such it could represent PMMA as a material solid. This is a profound result indicating a common plastic could steadily burn under these NASA proposed atmospheric conditions.



0.2 s 2 s 6 s 17 s 64 s 172 s

Figure 9. Flame growth from ignition to steady state in test 19070D1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Figure 9 shows steady results for a NASA proposed nominal atmosphere of 34% oxygen and 0.56 atm. After about 100 s the flame appears to be steady. The heat of gasification here is estimated as 2.42 kJ/g.

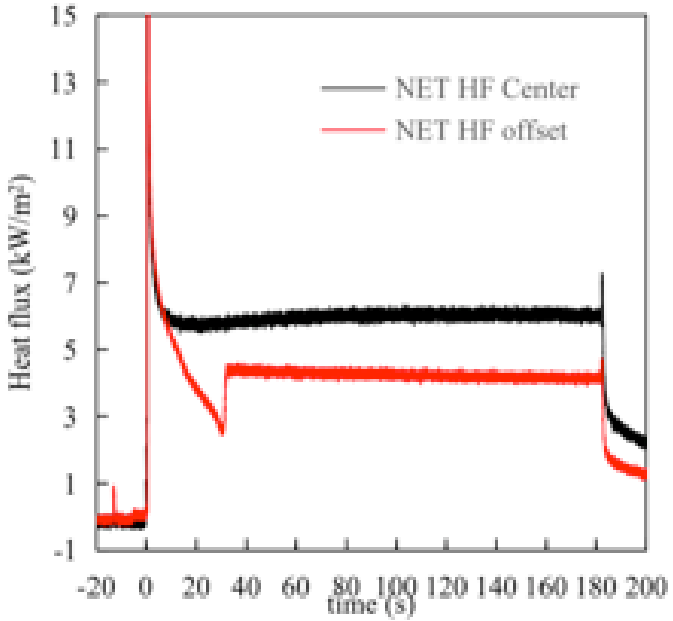


Figure 10. Measured heat flux in test 19070D1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Radiative loss fraction from the flames are obtained using the radiometer's measurements. Unfortunately, the data set is limited as the BRE sensors were saturated for several tests. The +60 mm and the offset radiometers were used to obtain the radiative loss fraction as these two sensors provided data in multiple cases while the other two radiometers were saturated. The transient radiative loss fraction for 19066D5 and 19070L1 are shown in Figure 10 for both radiometers, which show good agreement with each other. Table 3 provides information about the test conditions. As BRE radiometers were out of range for many tests of round 1, BRE radiometer array was replaced with ACME array to possibly prevent sensors from saturation. The offset radiometer data has been analyzed and the results are presented in Fig. 11.

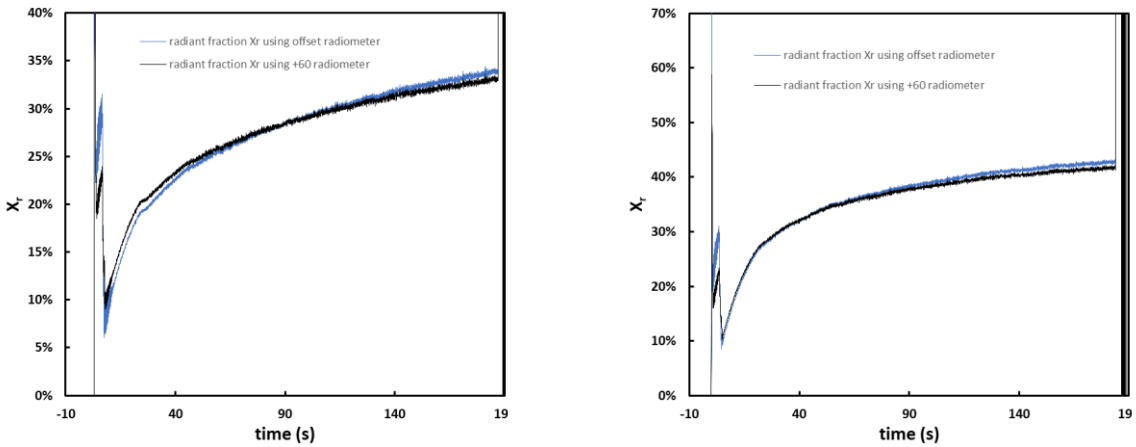


Figure 11. Transient radiative loss fraction for tests 19066D5 (a) and 19070L1 (b).

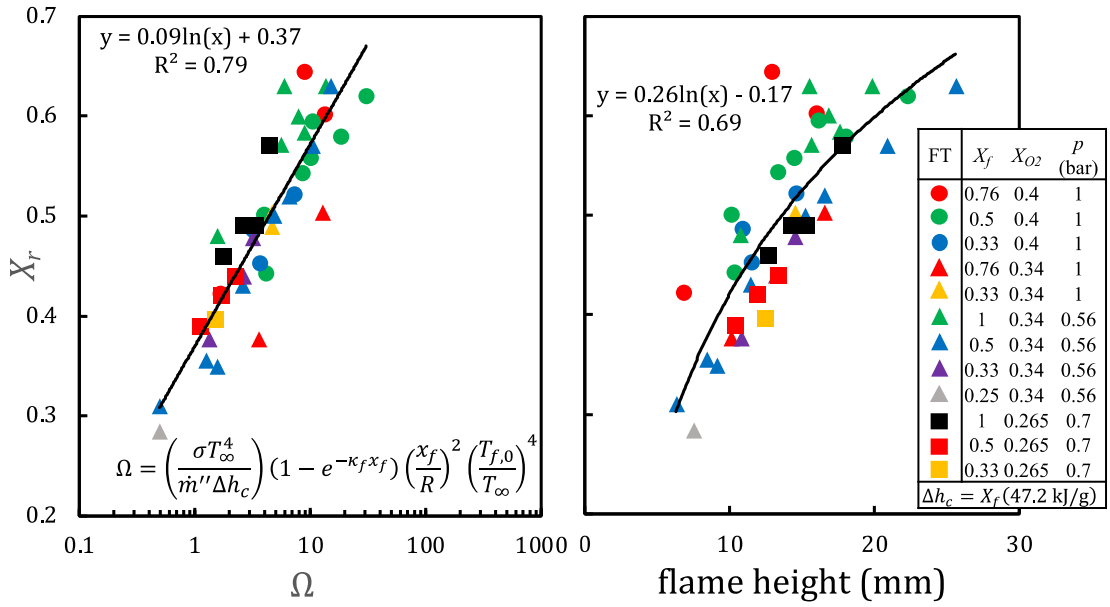


Figure 12. Flame radiant fraction vs. mass flux (a) and flame height (b) for all steady end-point results.

Flame heights at the end of the tests are measured from the image data. The radiative loss fraction is correlated with flame height as shown in Fig 12 (b). As more design parameters are included in the radiation analysis (i.e., oxygen mole fraction, pressure, and mass flux), more accurate empirical relationship was found between radiative loss fraction and a dimensionless parameter Ω. The results are shown in Figure 12. Radiative loss fraction was calculated for all tests using the parameters shown in Fig. 12 (a) and the results were compared to the measured values; see Fig. 13.

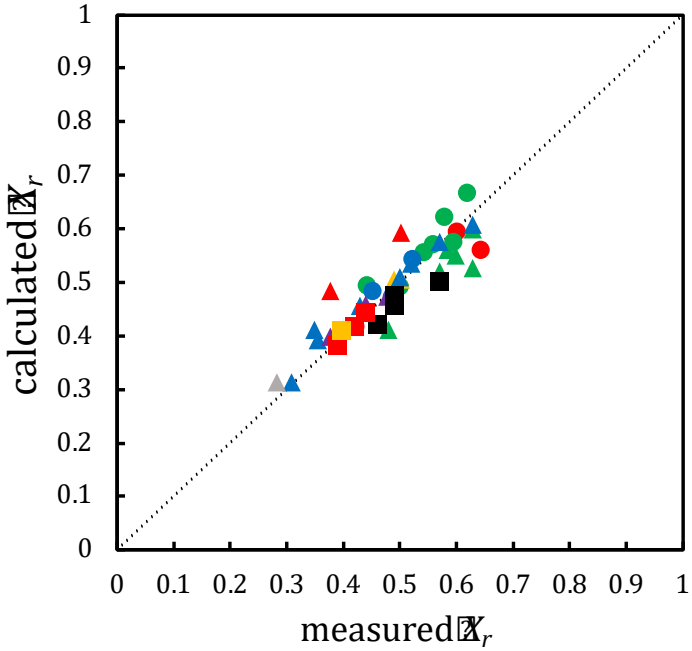


Figure 13. Radiative heat flux on the burner surface obtained by the View Factor method for nominal ambient conditions used for round 1 BRE tests on the ISS.

The experimental flammability maps are shown in Figures 14 and 15. Both plots indicate a decreasing, inverse asymptotic-like curve that approaches zero, for heat of gasification with mass flux. It is surprising that the data cluster about a nearly singular trend. However, both plots show the heat of gasification is up to 20% higher for the lower heat of combustion, 23.6 kJ/g. Our experiments in the University of Maryland lab have shown, with the BRE in 1-g, that similar results for propane-nitrogen mixtures have the opposite behavior with heats of gasification decreasing with heat of combustion for the same mass flux. But again, the data mostly cluster about one trend line except for very low heats of combustion. However, the 1-g corresponding heat of combustion data shift far to the right and down from the microgravity data of Figures 14 and 15. For example, the 1-g data would indicate ($\dot{m}$", L) points of about 30 g/m^{2}s, 2 kJ/g and 10, 0.5. Moreover, our previous work in a 5- s drop facility suggested similar data trends but with an apparent vertical asymptote at about 3 g/m^{2}s, rather than 1 as shown here. Obviously, the current data at durations of minutes are more accurate than those limited to 5 s. However, it has been shown that the ISS heat flux data even at self-extinction are theoretically nearly steady. Figures 14 and 15 include the heats of gasification derived from the self-extinguished endpoint data as well. Remarkably, the self-extinguished not-so-steady data follow the same trend as the steady data.

Let us interpret Figures 14 and 15 in terms of the flammability of real materials in microgravity. Typical melting plastics have heats of gasification ranging from about 2 to 3 kJ/g, with effective heats of gasification for charring plastics above 4 kJ/g. Some selected literature values for materials with heats of combustion in the BRE range are listed in Table 2. The accuracy of the calorimeter heat flux is found to be less than the two sensors, thus, we relied on the arithmetic average of the two heat flux sensors data for obtaining the flammability map. However, same results are shown considering the calorimeter data. Figure 14 would suggest that polyethylene (PE) and polypropylene (PP) would burn in microgravity at about 3 g/m^{2}s. Nylon and PMMA would burn at about 2.5 g/m^{2}s, and the phenolic glass fiber composite would burn at about 1 g/m^{2}s.

Table 2. Materials close to BRE heats of combustion and corresponding heats of gasification

Material	***LHV*, kJ/g**	***L*, kJ/g**
PE	43.4	1.9
PP	44	2.0
Nylon	28.8	1.5, 2.4
PMMA	24.2	1.6, 2.3
Phenolic glass fibers	22.0	7.3

BRE tests in orbit were finished with 196 successful ignitions. The measured heat flux and the end point data of the long duration tests (i.e., the tests that fuel was terminated as opposed to the tests that self-extinguished) were used to obtain heats of gasification. The measured heat flux for all quasi steady tests are shown in Fig. 14. The heats of gasification of these data from all the tests vs. fuel mixture mass flux are shown in Fig. 15. One important fire safety consideration, based on the findings of this study, is that solid materials with heats of combustion in the range 23.6 to 47.2 kJ/g and heats of gasification from about 1 to 10 kJ/g can burn in microgravity. The NASA alternative flammability Test 2 [1] using the Cone Calorimeter can measure these material properties. As this study develops a more complete understanding, NASA may wish to put more emphasis into Test 2.

Regression analysis was performed on the experimental data recorded from the quasi-steady tests and the burning rate vs. parameter μ is shown in Fig. 16. The symbol μ indicates the importance of each design parameter. As shown, the oxygen mole fraction, radiant fraction, and heat of gasification are the factors with largest exponents. The larger the exponent, the greater its effect on the burning rate.

Moreover, the flammability map based on our theoretical analysis is shown in Figure 17. It shows similar asymptotic trend and clearly indicates lower pressure and higher oxygen concentration as favorable conditions for long duration burning.

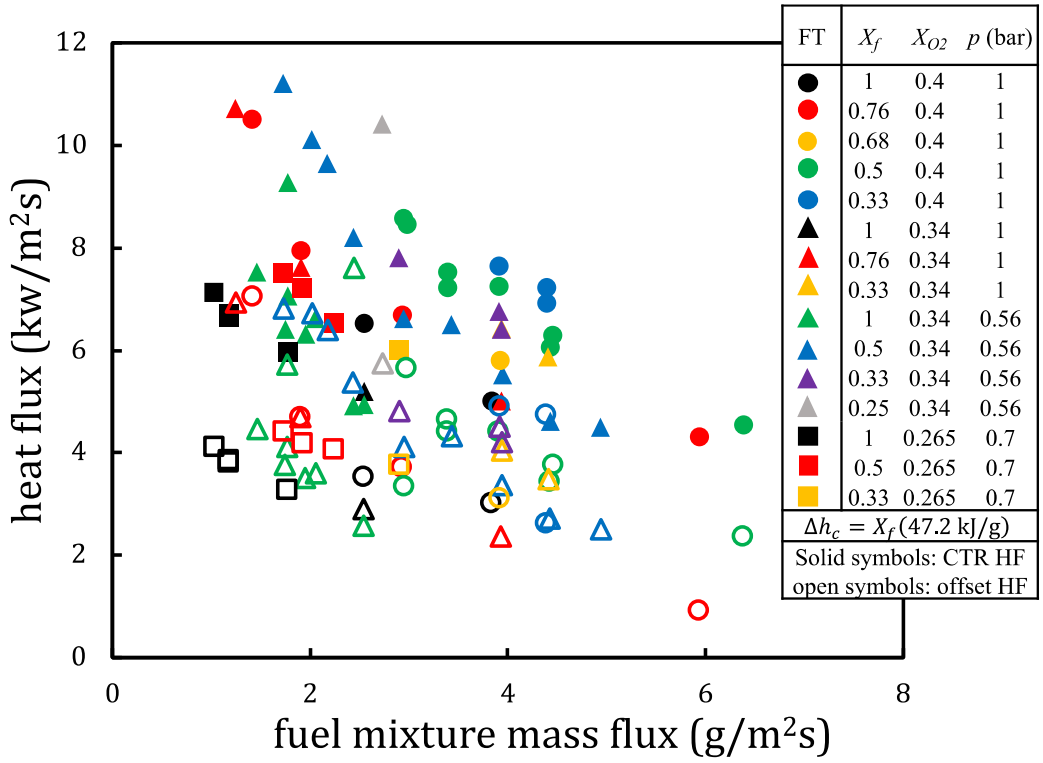


Figure 14. Round 1 result for the emulation of 25-mm flat materials using the arithmetic average of the two local heat flux measurements.

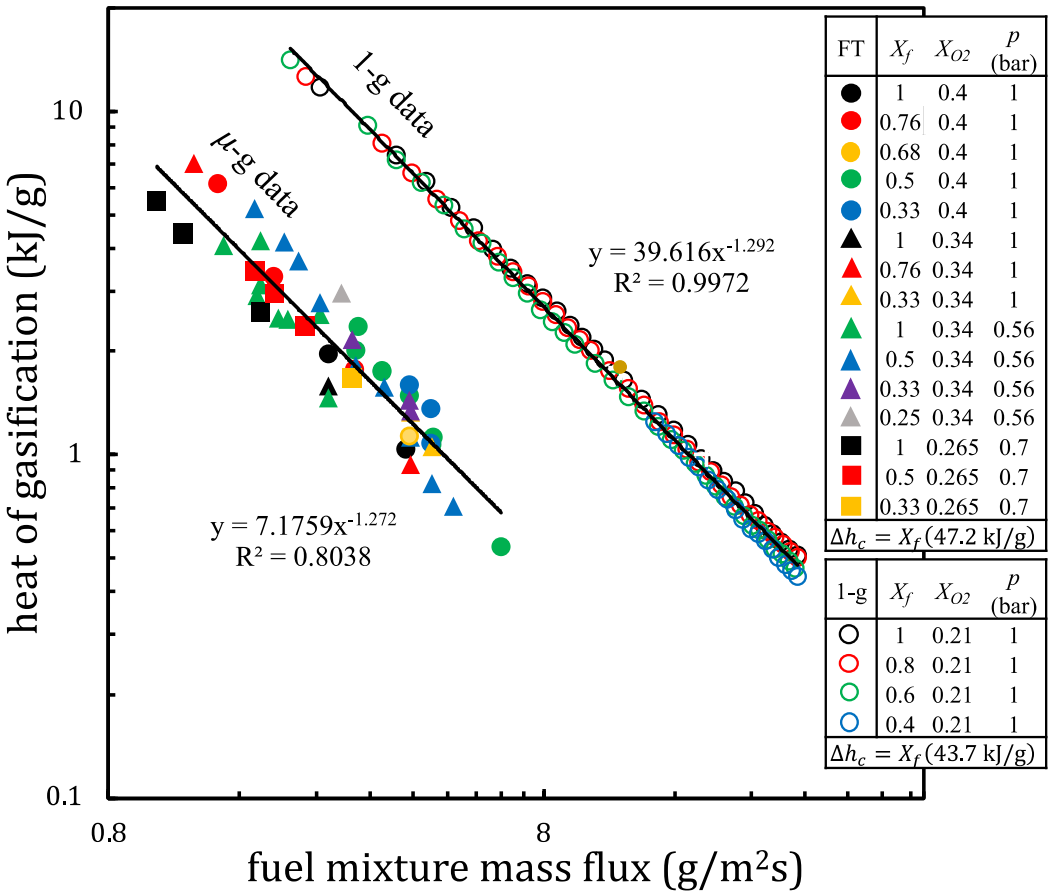


Figure 15. Preliminary flammability map based on the arithmetic average of the two heat flux sensors data.

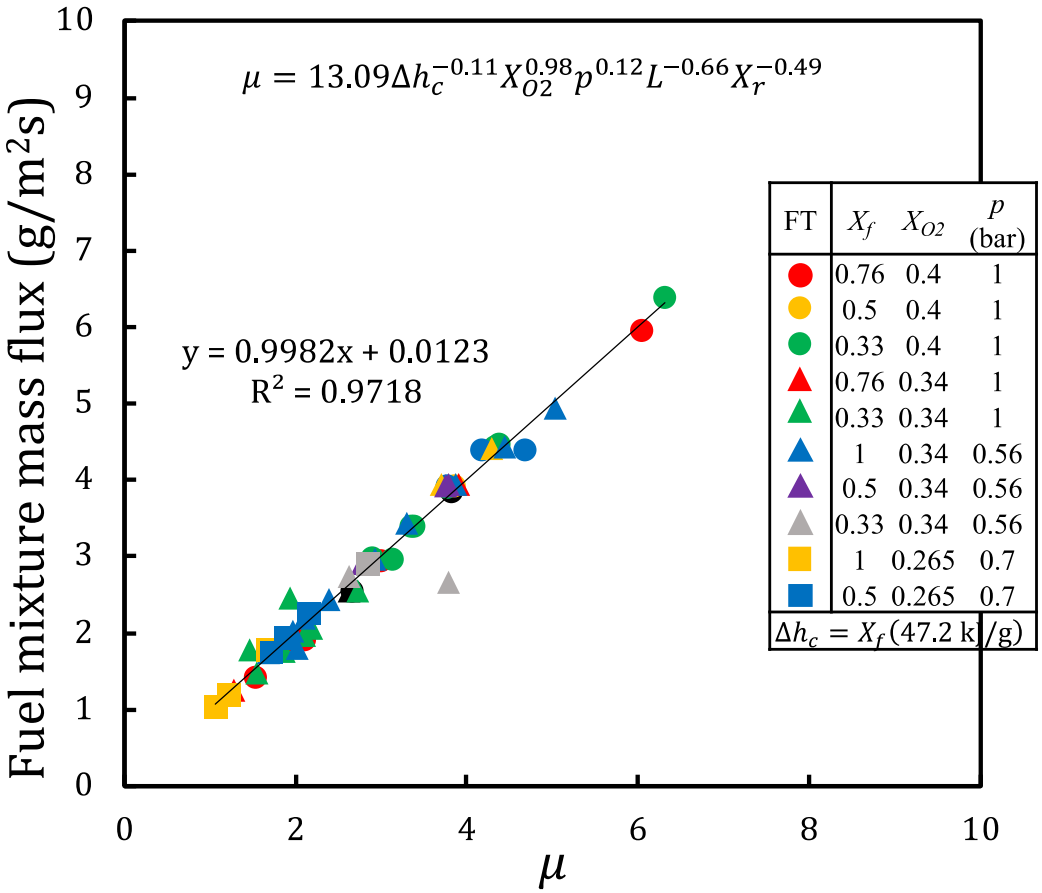


Figure 16. Heats of gasification calculated from the corrected calorimeter measurements vs. fuel mixture mass flux.

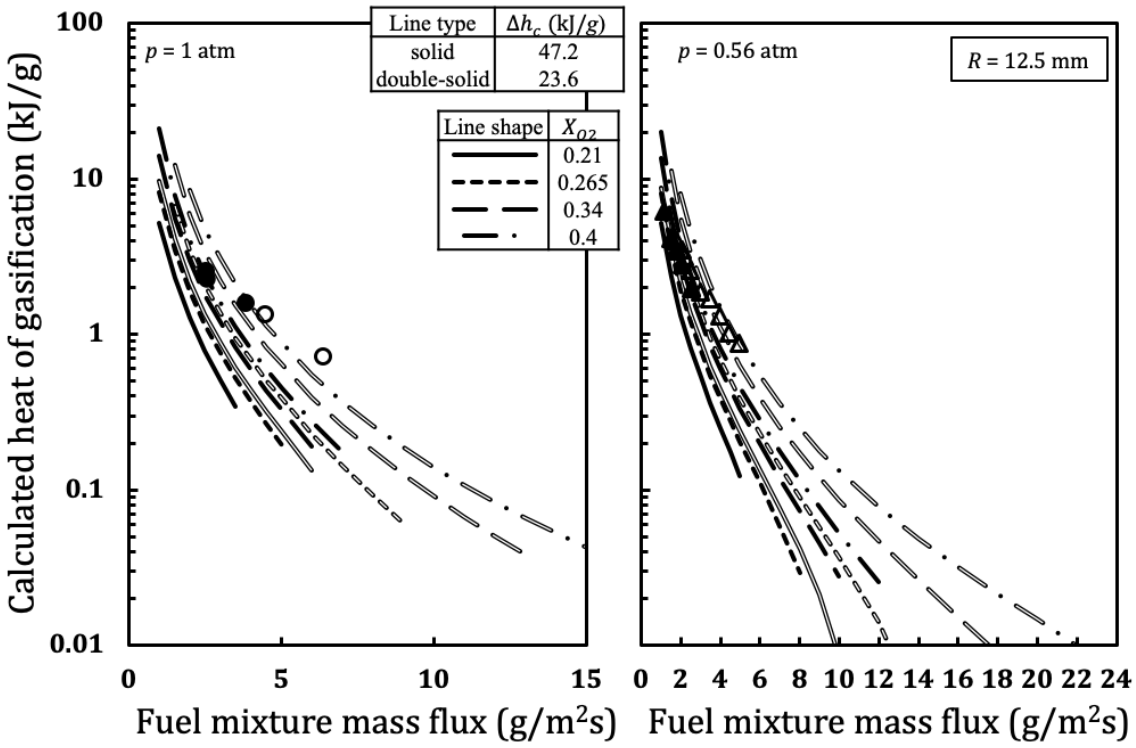


Figure 17. Calculated heats of gasification vs. mass flux for different oxygen mole fractions and p = 1 atm (left) p = 0.56 atm (right).

The imaging data from ops camera and ACME camera are used to create videos and image sequences for the tests. Many of the first tests with the 50-mm BRE burner were not at the intended flow rate. A make-up day was carried out to accomplish that day's scientific objectives. The plan for round 2's second day contained a combination of pure and diluted ethylene at 3 different ambient conditions. Pure methane was the fuel used for testing on the second round's third test day at 3 different nominal ambient conditions. It was observed that methane was hard to ignite compared to ethylene and flames tend to self-extinguish faster even at a high oxygen concentration.

Our preliminary analysis on the calculated vs. measured heats of gasification led to the comparison plot shown in Figure 18 which contains all the tests with quasi-steady data.

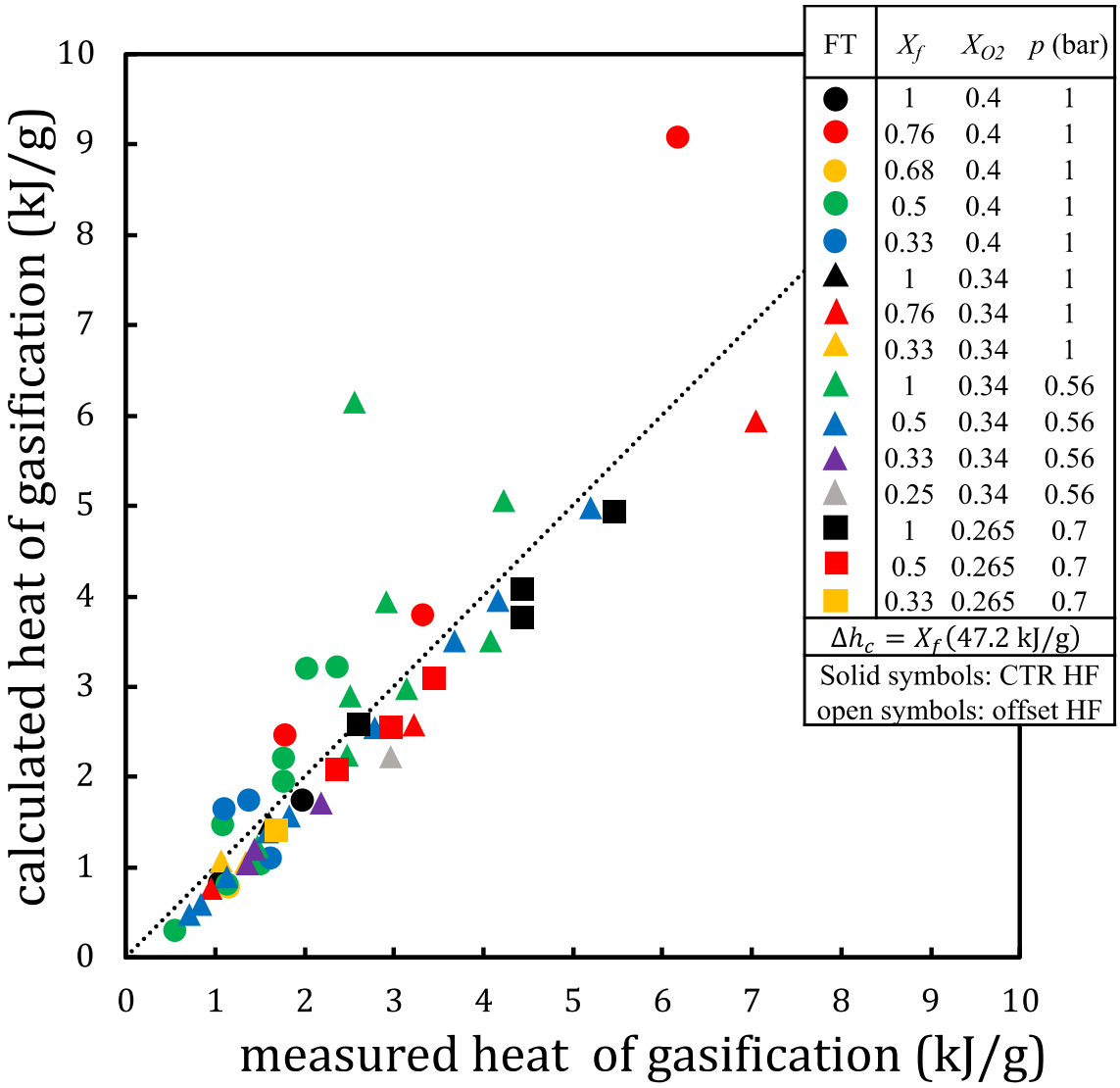


Figure 18. Convective heat flux on the burner surface obtained by the View Factor method for nominal ambient conditions used for round 1 BRE tests on the ISS.

As the two major factors used to emulate real fuel are heats of gasification and heats of combustion, the plot of the former parameter vs. the latter was developed to identify a region for possibility of flammability that changes with ambient oxygen mole fraction. Thus, oxygen mole fraction will be included as a third dimension to make the plot more clear, but this is still in progress. The preliminary results for all quasi-steady data points are shown in Fig. 19.

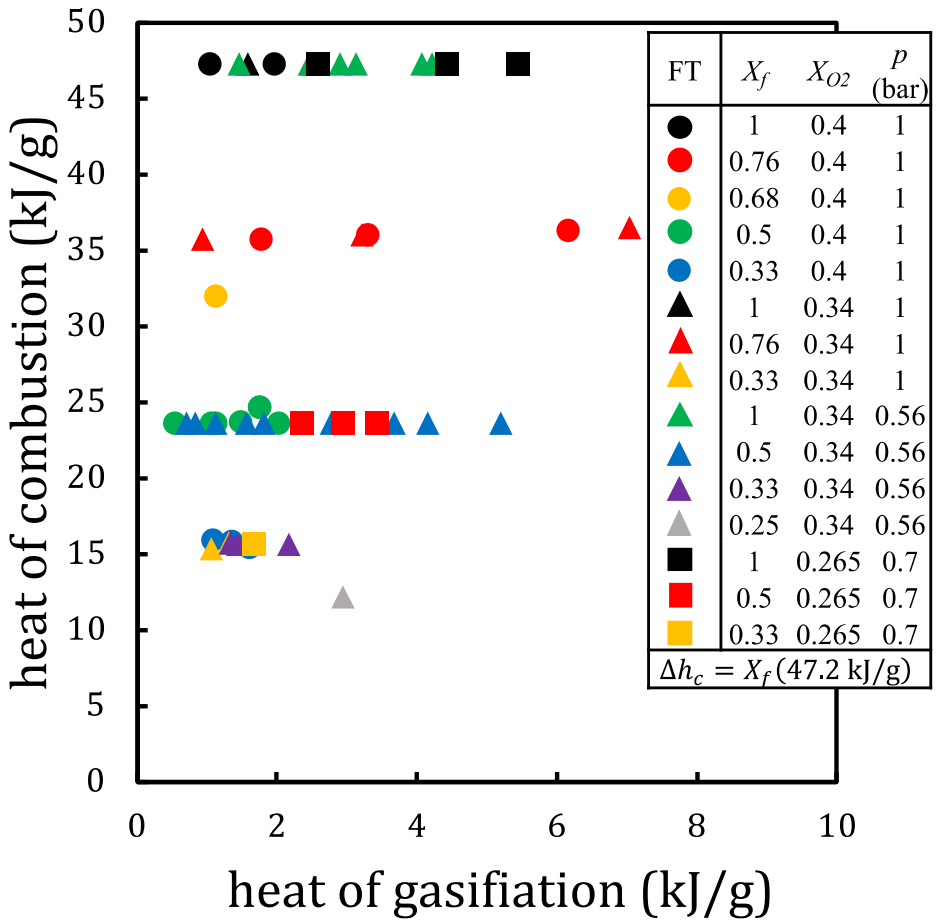


Figure 19. Radiation fraction vs. fuel mixture mass flux.

Thin Filament Pyrometry technique was used during many tests of round 2. The technique of converting data to peak flame temperature at fiber locations was previously developed and published in ref. [2].

Four flames were analyzed to obtain transient flame temperature [3]. The results are shown in Figs. 20 to 23.

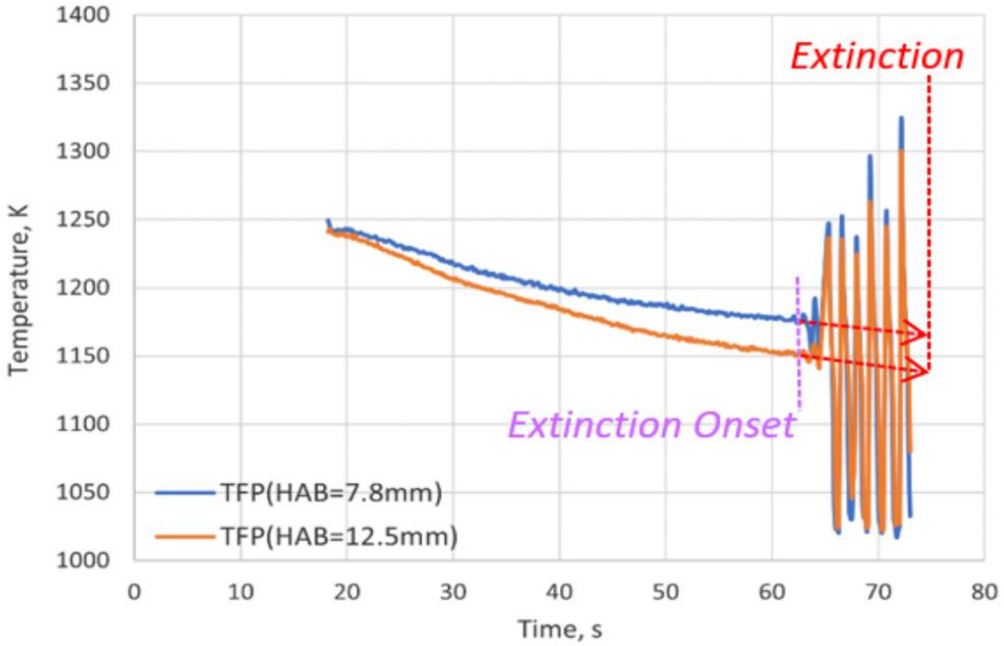


Figure 20. TFP measurements for test 21020D1.

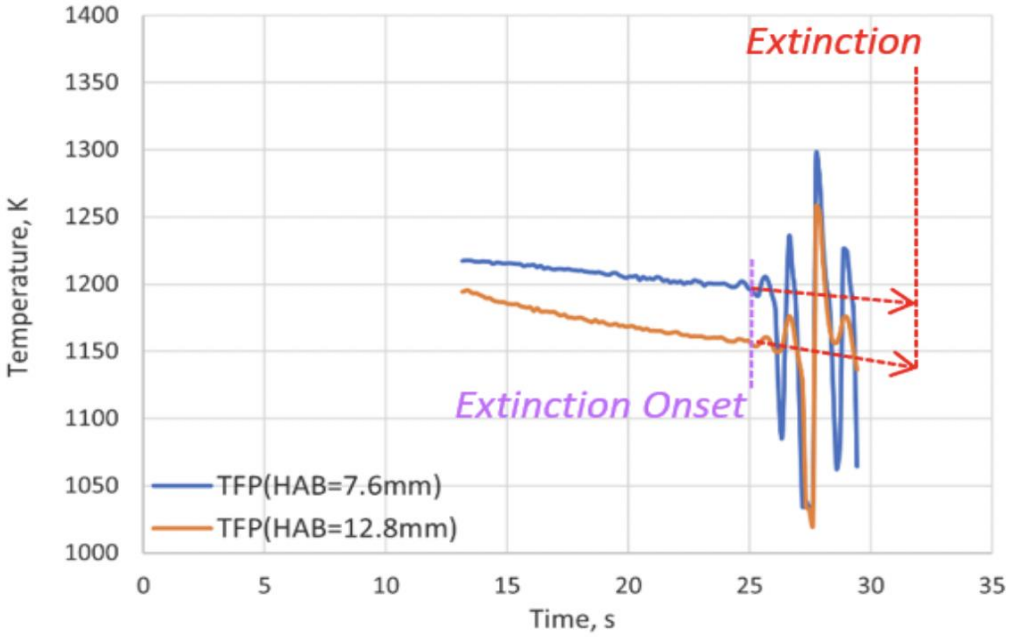


Figure 21. TFP measurements for test 20358G2.

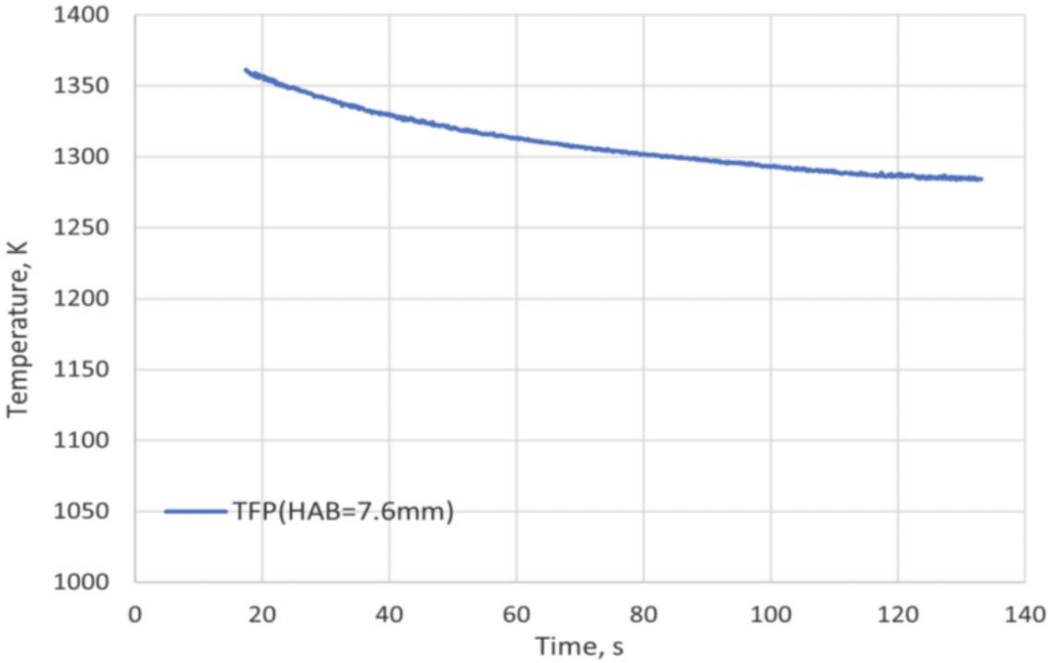


Figure 22. TFP measurements for test 21020A1.

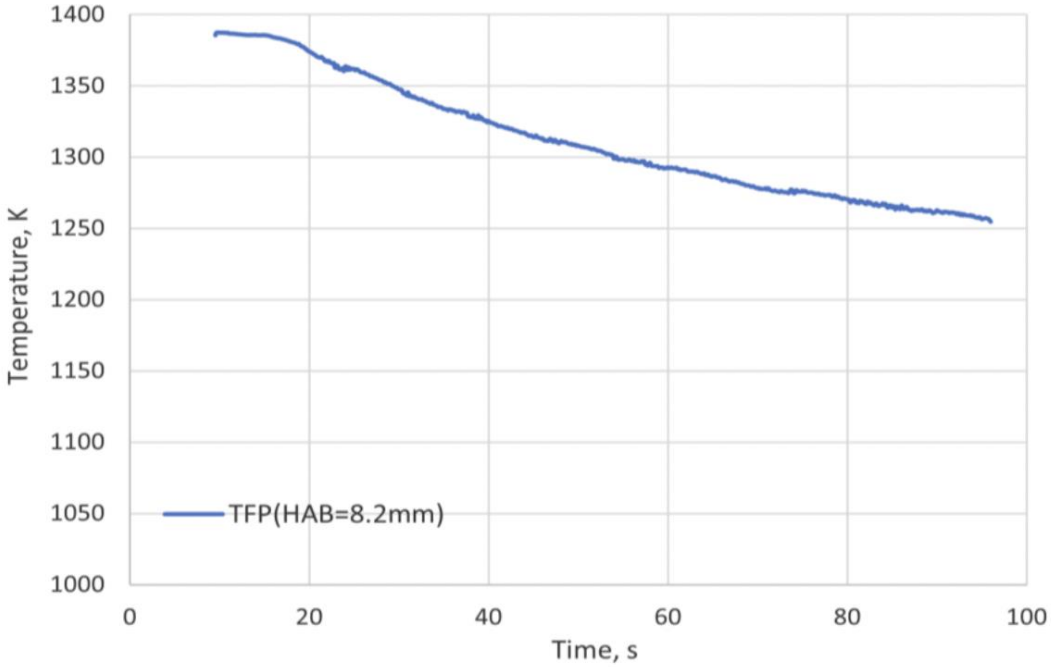


Figure 23. TFP measurements for test 20346A5.

Measured oscillation onset and extinction times of all BRE flames were analyzed which resulted in a modeling of these timings as a function of Reynolds number, ambient oxygen mole fraction. The empirical coefficient and exponents (i.e., A) were found by fitting the modeled vs. measured data [4]. The results are shown in Figs. 24 and 25.

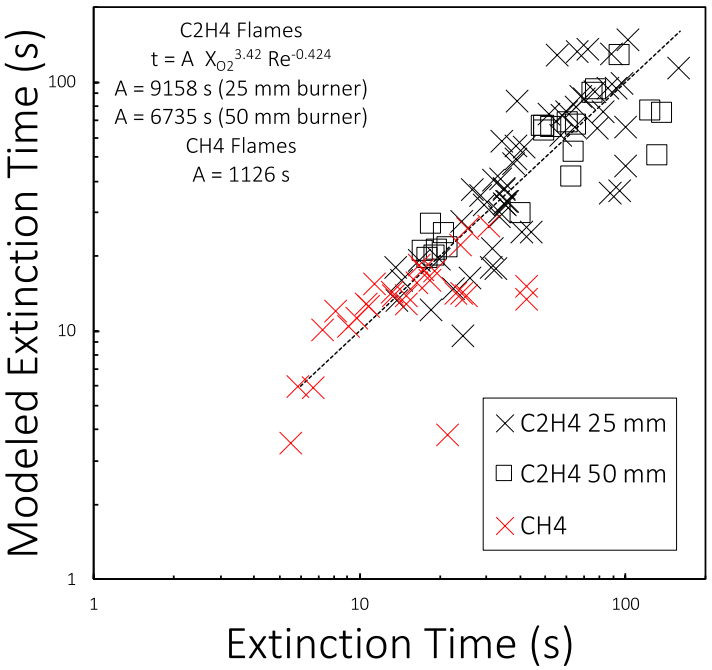


Figure 24. TFP measurements for test 20346A5.

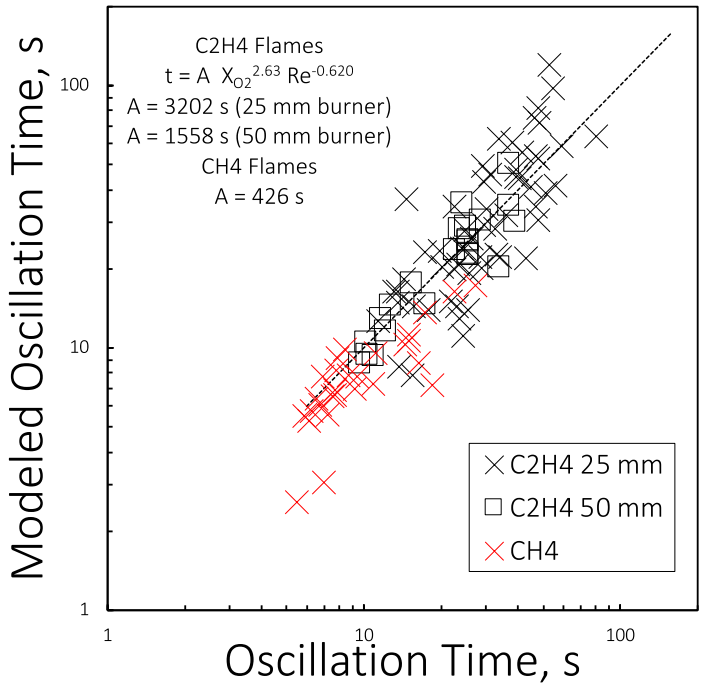


Figure 25. TFP measurements for test 20346A5

Section 4.C references

1. NASA-STD-6001B. Flammability, Offgassing, and Compatibility Requirements and Test Procedures (2011). https://standards.nasa.gov/standard/nasa/nasa-std-6001
2. P.H. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, and R.L. Axelbaum, "First Observations of Long Duration Microgravity Spherical Diffusion Flames Aboard the International Space Station," *Combustion and Flame,* 229 (2021) 111373. https://doi.org/10.1016/j.combustflame.2021.02.019
3. Venzon, R., 2021, "Thin-Filament Pyrometry and Analysis of Extinction Onset Temperature for Ethylene Flames on the Burning Emulator in the International Space Station," University of Edinburgh.
4. A. Wright, P. Dehghani, P. Sunderland, and J. de Ris, "Using A Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station," poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.

D. Surprises and New Discoveries

Initial surprises and discoveries include:

(1) It is found out that quasi-steady burning on a flat surface, representing a pool fire, can exist in microgravity. Ambient oxygen mole fraction plays the most significant role in this regard. The quasi-steady flames were observed for nominal ambient oxygen mole fractions of 0.265 and above.

(2) Although ambient pressure has a weak effect on the quality of burning, it affects radiative heat loss by changing the partial pressures of radiating species, i.e., H2O and CO2.

(3) Methane nitrogen-generated flames self-extinguished within a minute for all ambient conditions. That indicates that the burn time depends on the fuel chemistry.

(4) Ethylene flames, which tend to soot under normal gravity conditions, did not soot under microgravity conditions. Sooting only occurred upon ignition and disappeared quickly after ignition. Nitrogen dilution reduced the amount of soot.

(5) Radiative loss fraction for the flames in microgravity can reach about 60% compared to 30% for the normal gravity flames.

(6) TFP measurements of four flames confirmed that critical temperature range for extinction is 1100 to 1200 K. This is confirmed by numerical simulations.

(7) A It was shown that the BRE can emulate several liquids and solids including Acetone, Ethanol, Methanol, PMMA, Nylon, Phenolic glass fiber, etc. as demonstrated by a flammability diagram, as a plot of heats of gasification vs. the mass burning rate, based on the steady average surface heat flux.

(8) For the first time, two limits of extinction were obtained on a plot of heat of combustion vs. heats of gasification. The plot shows the flammability region for each nominal ambient oxygen mole fraction with radiative extinction for large flames and another extinction limit for small flames.

E. Accomplishment of Approved Objectives

Two of the three approved objectives were accomplished through the ISS experiment, where NASA did not intend to address the desired second objective through the investigation.

The objectives are presented here with brief summaries of the findings; readers are encouraged to refer to section 4.C for more information.

1. *Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25- and 50-mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21 % oxygen; 10.2 psia, 26.5 %; and 8.2 psia, 34 % oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels.*

   An elevated oxygen concentration in the ambient enhanced flammability and quasi-steady flames for nominal oxygen mole fractions of 0.265 and above. Pressure had a minimal effect on the quality of burning. The microgravity flames corresponded to the combustion of a range of liquid and solid fuels.

2. *Consider the conditions of Objective 1 for ambient of O2/CO2 (desired).*

   As planned, this objective was not investigated in the conducted tests. It would be a good candidate for further exploration in a follow-on experiment.

3. *Consider the conditions of Objectives 1 – 2 for extinction limits.*

   All flames ignited in normal atmosphere self-extinguished within 40 s. In contrast, oxygen mole fractions of 0.265 and above resulted in quasi-steady flames. Large flames extinguished due to radiative heat loss, while small flames quenched at a second limit.

F. Additional Objectives Achieved

An empirical correlation for the radiative loss fraction was obtained through a dimensional analysis of the radiation data. The correlation was used to predict the flames' radiative loss fractions from their mass burning rate, oxygen mole fraction in the ambient, chamber pressure, and heat of combustion. This result was used to correct the heat of combustion in the analytical solution for these flames.

G. Impacts, Benefits, and Products

BRE yielded important data relevant to spacecraft fire safety. Quasi-steady burning corresponding to various condensed-phase materials (such as PMMA and Nylon) was observed in microgravity despite the absence of air flow, where the cessation of ventilation is an established step in putting out spacecraft fires. The ongoing combustion without air flow was notably seen in elevated-oxygen atmospheres which are being considered for use in future spacecraft, whereas such flames in a normal atmosphere of 21% oxygen self-extinguished in a short time.

While BRE's results are still preliminary, they suggest that NASA may want to increase use of its alternative flammability Test 2, with the cone calorimeter, where that test can measure material properties corresponding to the microgravity flames observed with the BRE burner.

H. Patents

Thus far, no patents have been filed from the BRE research.

5. Follow-On Experiments

A study of the burning behavior in an ambient of O2/CO2 (the desired objective) is a good candidate for a follow-on experiment because of the utility of CO2 in spacecraft fire extinguishers. We would also like to expand the testing to include propane and propylene to emulate additional condensed fuels. Furthermore, the addition of external radiation to the experimental apparatus would improve the simulation of real fires.

Appendix A. Abbreviations and Acronyms

#####	ACME days are identified by year (2 digits) and the day of the year (3 digits, 001-366)
#####x#	ACME tests are identified by day (5 digits; see above), a letter, and a 1-digit number
0-g	microgravity
1-g	normal gravity (on the surface of the Earth)
ACME	Advanced Combustion via Microgravity Experiments (*CIR project*)
ADP	Acceptance Data Package
ASGSR	American Society for Gravitational and Space Research
AVP	avionics package
BRE	Burning Rate Emulator (*an ACME experiment*)
C_2H_4	ethylene
C_3H_8	propane
CH_2O	formaldehyde
CASIS	Center for the Advancement of Science in Space
CCD	Charge-Coupled Device (*such as a camera sensor*)
CFI	Cool Flames Investigation (*an MDCA experiment preceding ACME in the CIR*)
CFI-G	Cool Flames Investigation with Gases (*an ACME experiment*)
CH_4	methane
CIA	Chamber Insert Assembly (*inserted into the CIR combustion chamber*)
CIR	Combustion Integrated Rack (*ISS facility in which ACME experiments are conducted*)
CLD Flame	Coflow Laminar Diffusion Flame (*an ACME experiments*)
CM	Combustion Module (*space shuttle precursor to the CIR*)
cm	centimeter
CO_2	carn dioxide
Co-I	Co-Investigator
CRS	Commercial Resupply Service (*a designation of ISS cargo missions)*
CSA	Canadian Space Agency
CSV	comma-separated values
CWRU	Case Western Reserve University
Da	Damköhler number
DataCam	Data Camera (*ACME's primary camera which was characterized for pyrometry, etc.)*
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames (*an ACME experiment*)
EMI	electromagnetic interference
EMS	Experiment Mounting Structure
ERD	Engineering Requirements Document
ESA	European Space Agency
Fiji	Fiji Is Just ImageJ (*a version of ImageJ preloaded with many plug-in features*)
Flame Design	Flame Design (*an ACME experiment*)
FOMA	Fuel and Oxidizer Management Assembly (*CIR's gas management subsystem*)
FOV	field of view
fps	frames per second
GC	Gas Chromatograph (*see GCIP*)
GCIP	Gas Chromatograph Instrumentation Package (*for measuring CIR chamber contents*)
GigE	Gigabit Ethernet (*an interface standard for high performance cameras*)
GIPOC	Glenn ISS Payload Operations Center (*also known as the TSC*)
GMT	Greenwich Mean Time (*where most references in this report are the day*)
H_2	hydrogen

HiBMs	High Bit-depth Multi-spectral camera (*developed for the CIR facility*)
HOBJ	Halcon object (*binary uncompressed format; used for the DataCam*)
HV	high voltage
HVPS	High-Voltage Power Supply
ICDERS	International Colloquium on the Dynamics of Explosions and Reactive Systems
ID	inner diameter (*e.g., of burner tube*)
ImageJ	Image Processing and Analysis in Java (*open-source image processing software*)
IPSU	Image Processing and Storage Unit (*CIR device for recording images on ISS)*
IPSU-G	IPSU for GigE cameras (*including the DataCam and CFI intensified camera*)
ISRD	Integrated Science Requirements Document (*for a set of experiments*)
ISS	International Space Station
JAXA	Japan Aerospace Exploration Agency
kPa	kilopascal
kV	kilovolt
LLL-UV	Low Light Level Ultraviolet camera ((*developed for the CIR facility*)
LSP	Laminar Soot Processes (*a shuttle study conducted in the Combustion Module*)
MDCA	Multi-user Droplet Combustion Apparatus (*a CIR chamber insert*)
MFC	Mass Flow Controller
Mist	Water-Mist Fire Suppression Experiment (*a Combustion Module experiment*)
MIUL	Materials Identification/Usage List
mm	millimeter
N_2	nitrogen
NSF	National Science Foundation
NTSC	National Television System Committee
O_2	oxygen
OMA2	Optical Multi-channel Analyzer 2 (*open-source software for image processing)*
ops	operations
PARIS	Passive Rack Isolation System (*with which the CIR is equipped, but not used for ACME*)
PI	Principal Investigator
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (*where ACME is equipped with three PMT modules*)
PS	Project Scientist
PSI	Physical Science Informatics System
psia	pounds per square inch
PT	pressure transducer
QD	quick disconnect
qty	quantity
RSA	Russian Space Agency, i.e., Roscosmos
s	second
SAMS	Space Acceleration Measurement System
SDP	Safety Data Package
s-Flame	Structure and Response of Spherical Diffusion Flames (*an ACME experiment*)
SiC	Silicon Carbide (*material of the TFP fibers*)
slpm	standard liters per minute (*a measure of the mass or molar flow rate of a gas*)
SOFBALL	Structure Of Flame Balls At Low Lewis-number (*a Combustion Module study*)
SoFIE	Solid Fuel Ignition and Extinction (*project to follow ACME in the CIR facility*)
SORD	Science Operations Requirements Document (*an ACME supplement to the ISRD*)
SRD	Science Requirements Document

TDRS	Tracking and Data Relay Satellite (System)
TFP	Thin Filament Pyrometry (*a method for gas-phase temperature measurements*)
TIFF or TIF	Tagged Image File Format (*for the CFI intensified, HiBMs, and LLL-UV cameras*)
TM	thermistor
TS	Transport Stream (*a video format*)
TSC	Telescience Support Center (*also known as the GIPOC*)
UV	ultraviolet
Ze	Zeldovich number
zero-g	zero gravity, i.e., microgravity
ZIN	ZIN Technologies, Inc. (*ACME engineering and operations provider*)
µA	microamp

While this list of abbreviations and acronyms is long, it is not comprehensive. Contact ACME Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) for additional clarification.

Appendix B. 30-Day Operational Accomplishments Report for ISS Increments 57 and 58

31 May 2019

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 57 and 58

ACME Advanced Combustion via Microgravity Experiments
BRE Burning Rate Emulator

BRE is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation will be carried out in two separate rounds, where the first round began in Increment 58, so there is nothing to be reported for Increment 57.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for BRE and ACME information.

Principal Investigator (PI):	Prof. James G. Quintiere, University of Maryland, College Park, MD
Co-Investigator (Co-I):	Prof. Peter B. Sunderland, U. of Maryland, College Park, MD
Co-Investigator (Co-I):	Dr. John L. de Ris, FM Global (retired)
Collaborator:	Prof. Alexander Yu. Snegirev, Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia
Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Uday G. Hegde, Case Western Reserve University, Cleveland, OH

I. Introduction

The Burning Rate Emulator (BRE) experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. BRE is focused on spacecraft fire prevention. More specifically, BRE's goals are to improve our fundamental understanding of materials flammability, such as extinction behavior and the conditions needed for sustained combustion, and to assess the relevance of existing flammability test methods for low and partial-gravity environments. A flat porous burner fed with gaseous fuel simulates the burning of solid and liquid fuels, where measurements are made of the thermal feedback (i.e., to the burner) upon which the vaporization of condensed-phase fuels is dependent. A small number of gaseous fuels (including mixtures with inert gases) are used to simulate the burning of fuels such as paper, plastic, and alcohol by matching properties including the heats of combustion and gasification, the surface temperature, and smoke point. The experiment's results can establish fuel properties that

directly relate to fire hazards in spacecraft, and if effective in microgravity could provide an efficient, safe, and comprehensive technique for assessing flammability and selecting fire-resistant spacecraft materials.

BRE's specific experimental objectives are as follows, where it must be emphasized that NASA does not currently plan to carry out the desired tests for objective B.

A. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21 % oxygen; 10.2 psia, 26.5 %; and 8.2 psia, 34 % oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels.
B. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired).
C. Consider the conditions of Objectives 1 – 2 for extinction limits.

The observations will enable quantification of the microgravity behavior of condensed fuels (i.e., liquids and non-charring plastics).

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flame, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

BRE and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in

response to the analog video downlink and the numerical data available in the telemetry stream. For example, gas flow rate(s) can be adjusted after quasi-steady data has been collected. But testing has not been conducted in a 'manual' mode thus far for BRE or ACME in general.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

The first round of BRE tests began in February 2019, where a total of 45 flames were ignited during Increment 58. That round was completed in April 2019, where the latter portion will be covered in the upcoming Increment 59-60 report. Meanwhile, the second round of BRE testing is expected to begin near the end of Increment 60 and continue into Increment 61. The 'interruption' in the BRE testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

All of the first round of tests were conducted with the 25-mm diameter BRE burner while burning ethylene (C2H4) - sometimes diluted with nitrogen. The tests were carried out at a range of chamber pressures and oxygen concentrations, generally representative of possible crew environments in future spacecraft or habitats. More specifically, testing was especially focused on atmospheric conditions at (1) 14.7 psia, 21 % oxygen; (2) 10.2 psia, 26.5 %; and (3) 8.2 psia, 34 % oxygen. In all cases, the atmosphere was diluted with nitrogen and any incomplete combustion products (e.g., water vapor and carbon dioxide) from preceding tests.

BRE does not have living subjects like a life science experiment, but the tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C2H4), is highly sooting because of its carbon double bond. Tests are conducted with pure fuel and fuel mixed with nitrogen, where the fuel and nitrogen was premixed on the ground prior to launch for some cases. But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations.

There are no samples (e.g., of soot) collected for BRE, where all of the ISS results are electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for BRE during Increment 58, including a limited overview of the test conditions, is found in Table 1.

Table 1.

BRE log of flames ignited in Increment 58 (*when BRE testing began*)							
Test Day[1]	**Year**	**Month**	**Date**	**Burner Size[2]**	**Fuel bottle**	**Oxygen conc.[3]**	**Flames ignited[4]**
First round of tests, with the 25-mm burner, which began in Feb. 2019 (Increment 58)							
19039	2019	2	8	25-mm	100% C2H4	21%, 26.5%, 40%	10
19051			20			26.5%, 34%	8
19056			25		50% C2H4	21%, 26.5%, 34%, 40%	8
19063		3	4			26.5%, 40%	6
19066			7			34%	6
19070			11			26.5%, 34%	7

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Burner size** refers to the diameter of the BRE burner's perforated circular outlet.
[3]**Oxygen conc.** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations for tests that ignited are listed. *The chamber pressures were also varied but are not identified in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

There were no significant anomalies hindering BRE operations during Increment 58.

2. Any other difficulties in obtaining scientific data

BRE's primary problem was that the sensors for measuring the thermal radiation often saturated. Although the measurements were not required to meet experiment success, they were considered sufficiently critical that BRE was equipped with its own radiometer array, featuring three thermopile detectors. Estimates had been made of the signal levels based on 5-second drop tests since the hot gases tend to blanket the flame in the absence of buoyancy and thereby remain within the view of the radiometers. As a result, the signal levels are strongly dependent on gravity. The estimates were not part of the science requirements and may not have been sufficiently tracked in the hardware development. Furthermore, the ISS tests lasted as long as 5 minutes, which is over four times the envisioned BRE test duration and 60

times the duration of the drop tests. With the fixed amplification of the radiometer signals, the data was frequently lost - other than the knowledge that the limits were exceeded. An engineering review failed to identify an acceptable solution and BRE testing continued unchanged.

From the BRE PI Jim Quintiere:
The radiometry array proved problematic as its sensors were mostly out of range and quickly became saturated. However, for some of the small size flames, these data were complete. Those data will be vital to assessing the role of flame and combustion product radiation in the destiny of the flames.

Ignition was not always successful, where BRE tests were sometimes conducted under conditions, e.g., for igniter activation and deactivation, which led to momentary flashes without sustained ignition. Given that unplanned crew time is required to replace failed igniter tips, the BRE ops team limited - to varying degrees - the igniter voltage and the time that it was energized within the flame. The latter was known from earlier research in the Combustion Integrated Rack (CIR) to degrade the hot-wire igniter tip and hasten its demise. It is notable that the flashes of flame (across the burner surface) sometimes continued briefly after the igniter was deactivated and retracted. In one case on the first day of testing, there was excess fuel and a 'fireball' was ignited which promptly quenched. Ignition also sometimes failed due to a software limitation where the fuelFlowIgnitionDeltaMs parameter which specifies the relative timing of the fuel flow and igniter activation must be evenly divisible by 100 (ms). This limitation wasn't initially understood and the igniter didn't energize as a result.

From the BRE PI Jim Quintiere:
It was realized that the igniter for the burner gases could only be applied for up to 4 seconds, on insertion and retraction. This produced an unanticipated problem, as the time for the burner gases to reach the igniter could be 10s of seconds depending on the set flow rate. Also, after the first test, the burner would contain that fuel mixture minus diffusion with the chamber. Thus the burner would need to be flushed with nitrogen before the second fuel mixture was initiated.

Intense communication with the NASA team proceeded the day before the first test. The gas volume of the burner along with the feed pipe volume was supplied by NASA. Accordingly ground experiments at Maryland were conducted with an identical NASA BRE burner along with its piping volume from the flow valve. These experiments and theory were used to develop an algorithm for determining the relative timing of the igniter and gas flow activation that will result in ignition.

Day 1 was conducted with rough estimates made on the fly, and fortunately nearly all of the tests led to a satisfactory ignition. By Day 2 an algorithm was established

and successfully used for the remainder of the testing. This algorithm will need to be refined and applied for the larger 50-mm diameter burner that will be used in Round 2.

A document on the development of the algorithm and its refinement for the larger burner is being prepared and will be shared with the NASA team to facilitate Round 2.

The second day of flame testing was delayed to allow for the development of a new ignition procedure, where concerns arose that the mixture of gas flowing from the burner wasn't sufficiently well understood, i.e., in terms of concentrations. This concern had to do with the burner's large internal volume relative to that of the burners used for the other ACME experiments. To be clear, this issue had to do with the analysis of results rather than the achievement of ignition. Re-planning was also necessary early in BRE testing because the test durations were often much longer than had earlier been envisioned by the investigators. While these factors didn't directly interfere with the acquisition of test data, they did slow progress.

While it rarely occurred, BRE tests sometimes automatically aborted (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time associated with the failed test attempt. Such aborts typically occur because of communication issues with an Image Processing and Storage Unit (IPSU).

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel and nitrogen have unfortunately not always performed adequately, where the flow rates have not always been appropriately steady. The resulting effects on the flame behavior are tentatively assumed to be minor, but this still needs to be assessed.

C. Completeness/Quality of Data

The number of flames ignited significantly exceeds the planned tests – under the tested conditions (25-mm burner and ethylene as fuel) - which were specified as required in the ACME Integrated Science Requirements Document (ISRD). A total of 25 such tests were required, while 45 flames were ignited in the Increment 58 portion of the experiment's first round. As such, roughly 80% more flames were ignited than had been planned even though the round hadn't been completed.

It should be understood that a complete set of image data is not necessarily available for all tests. Recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

From BRE Co-I Peter Sunderland:
The video and image data of the testing produced satisfactory data to display the nature of the burning sequence from ignition to self-extinction or fuel termination. However, the image quality is less than was hoped for. The ACME ops camera provides video, but only offers 0.3 megapixels of resolution, and for many flames it is too zoomed out. The GigE camera has poor sensitivity in the blue, and after the ignition transient all our flames have been blue. This camera only offers 1.4 megapixel of resolution and requires about 1 s exposures for blue flames, which is far too slow to be considered video. The images provided by both cameras are, according to UMD marketing staff, at too low resolution to be used in press releases.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

See discussion in III.D in regard to fouling or degradation of the paint on the BRE burner's thermopile sensors.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and BRE's first round (which ended in Increment 59) is considered to be successful despite the frequent saturation of the radiometer data discussed in section III.B.2.

D. Status of Data Analysis **(submitted by BRE PI Jim Quintiere)**

All of Round 1 tests have had the thermopile sensor data reduced using their original calibrations. As presented immediately below, the data must be continually processed to address the effect of the combustion product water and any paint degradation.

The primary measurements are used, by two techniques, to obtain the net heat flux from the flame to the burner surface. In the steady state, this net heat flux divided by the mass flow rate per unit area of the fuel mixture gives the heat of gasification of the emulated fuel. The key hypothesis is that the heat of combustion, smoke point, and heat of gasification of the fuel mixture will represent a real solid fuel (or liquid fuel absorbed in an inert matrix). Thus achieved steady burning would represent the potential for real solid fuels to achieve steady burning in microgravity. Other than a candle flame, this has not been shown before.

The first technique uses two thermopile heat flux gages embedded in the burner surface. One gage is at the center the other is offset from the center. The resulting data suffered from two possible effects. The first effect is the condensation of water vapor. As the sensors are not cooled, but rise with temperature as the flame grows, water from its combustion products can condense and then evaporate from the sensor. This occurs, as the dew point for combustion products is about 60 – 70 ^{o}C, and the sensor begins at ambient temperature and can achieve temperatures over 100 ^{o}C. Thus condensation can occur at temperatures below the dew point that leads to added heat to the sensor. Above the dew point, the condensed water will evaporate and reduce the heat to the sensor. Once the sensor is dry, it will only measure the heat flux from the flame gases as intended.

We have duplicated this condensation effect in 1-g laboratory experiments with our duplicative NASA BRE burner. The figure below (Fig. 1) shows an illustration of the condensation and evaporation effect in the ISS tests. The offset sensor in red shows a clear jump increase when the water fully evaporates. More analysis is needed to properly correct for this effect. Fortunately by the end of the test shown here (and for all of the other data), this water effect is gone.

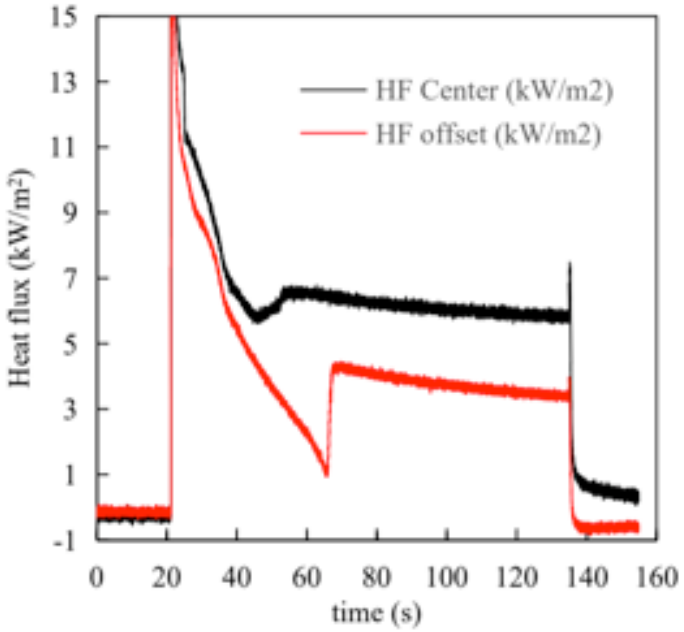


Figure 1. Typical net heat flux of thermopile sensors

The second effect on the thermopile sensors is that the heat may have fouled their paint. This could then affect their calibration. Figure 2 shows an enlargement of the burner after Round 1 tests were concluded. It can be seen that the paint on the sensors is degraded. This was the original manufacturer's paint, instead of that used for the copper top. The copper paint appears fine. We are working with the thermopile manufacturer to explore the effect of heat damage on the calibration of the thermopile sensors. This will give us a way to determine a quantitative change in the calibration.

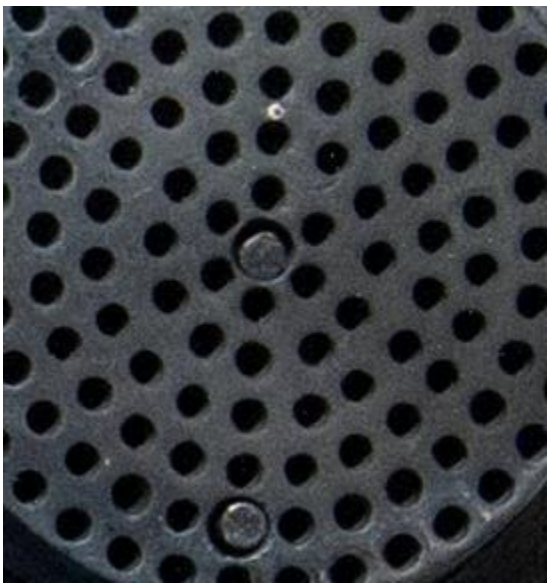


Figure 2. Paint degradation on thermopile sensors

The second technique to determine the heat flux is the copper calorimeter forming the top of the burner. The temperature rise of the copper is key to that determination. The next figure (Fig. 3) shows that typically the copper temperature is nearly uniform and smooth in time. It does not appear that condensation or evaporation is having any effect on its temperature behavior. So we believe we are justified in using this measurement to predict the overall average heat flux from the flame to the burner. However, we do need to evaluate any possible effect of the combustion product water. In addition, these calorimeter data are not fully processed yet because we are refining our methodology for its calibration. If the thermopile data cannot be completely used with good accuracy, we have the calorimeter to fall back on.

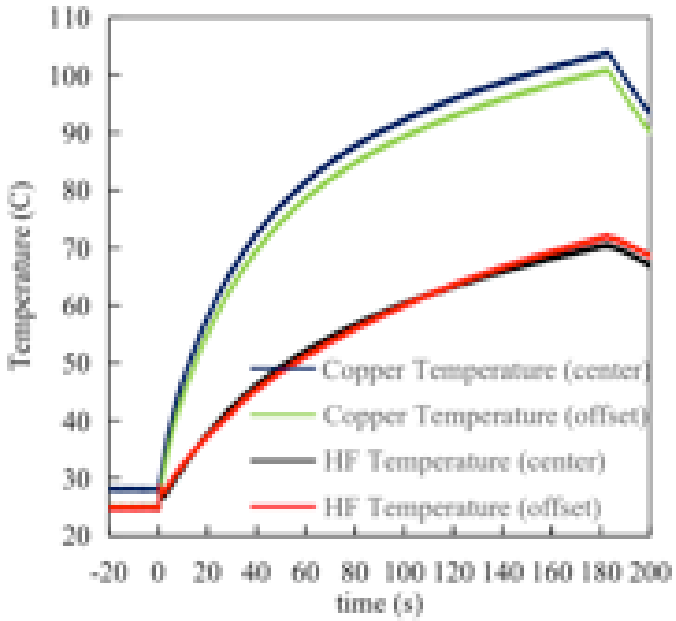


Figure 3. Copper and heat flux sensor temperatures, Day 6, D1

Preliminary analysis of the radiometer sensor data has indicated its consistent ability to measure the flame overall radiation heat loss fraction. The method to determine the radiative heat flux component to the burner is more complex, and it will rely on theoretical modeling. This will take some time.

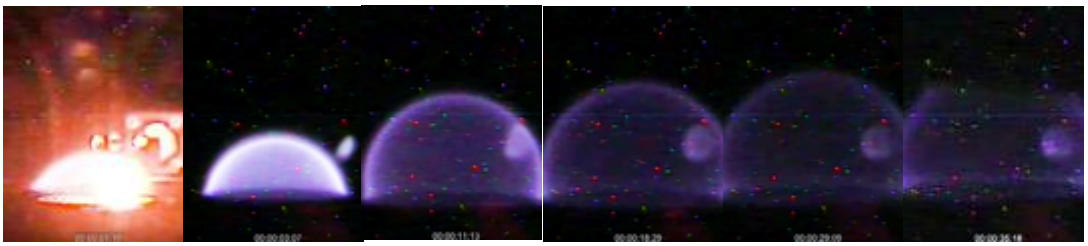
The video images of all the flames have been compiled, and are being quantified in term of shape and dimensions.

Alexander Snegirev, our Russian partner, has attempted a numerical simulation of at least one of the ISS BRE tests. That work is still in progress. It should be remarked his student won the Student Prize for Best Presentation at the 9th International Seminar on Fire and Explosion Hazards, St. Petersburg, April 2019 for simulating our 5-s BRE drop tests.

E. Preliminary Research Findings (submitted by BRE PI Jim Quintiere)

Despite having much data to digest and completely analyze, there is a profound conclusion from these tests. The burner has definitively shown that steady burning is possible for real solids with a flat surface of 25 mm diameter. Up to now most flammability studies in microgravity have focused on flame spread with a wind for concurrent spread and at least a self-induced wind for opposed flow spread. We already knew that this would be possible from 1-g research. The burning of liquids, as in a wick of a candle flame and small droplets, has been shown to produce steady flames. However, in the case of droplet burning cool flames could sustain normal burning, and normal burning would extinguish. The extinction has been shown to result from the radiative cooling of the flame. We see this as well in the extended ISS microgravity tests. However this extinction was predominately seen for atmospheric conditions of 21 %

oxygen and 14.7 psia. At conditions of 27 % oxygen, 10.2 psia and 34 % oxygen, 8.2 psia, the flames mostly persisted for 2 minutes or more as steady. These are all atmospheric conditions that NASA may use. Some examples will be shown to illustrate results over these atmospheres.



Ethylene/N_2, Δh_c = 22 kJ/g, 4.55 g/m^{2}s, 27% O2, 10.2 psia, L = 0.66 kJ/g, extinction @ 45 s

Figure 4. Day 3, Test F1. Flame growth from ignition to self-extinction

Figure 4 shows a typical test of ignition and flame growth. Although extinction occurred at 45 s with the flame still growing, the heat flux was nearly steady by 30 s. Therefore a heat of gasification (L) could be estimated. This was done in a preliminary estimation by using the center heat flux sensor as the net average heat flux. If the heating is just due to diffusion, theory indicates that the average heat flux is double the center value. Hence such an estimated heat of gasification could be low by 50 %. On the other hand, if the sensor is degraded in its calibration and radiation is a significant component of the heat flux, then the estimate will also be low. Hence, using the center heat flux sensor directly gives a conservatively low estimate of the heat of gasification in the emulation of real solids. This will be significant when the totality of results for the 25 mm flat burning surface is display over its spectrum of conservatively low values for L.

Figures 5 and 6 illustrate results for atmospheric conditions of 26.5 % oxygen and 0.7 atm (10.2 psia), and 34 % oxygen and 0.56 atm (8.2 psia), respectively. Both use 50 % mixtures of nitrogen and ethylene giving a fuel mixture heat of combustion of 21.8 kJ/g. Both of these tests reached steady conditions and were terminated by shutting off the fuel.



3 s 10 s 52 s 91 s 125 s 155 s 178 s

Ethylene/N_2, 22 kJ/g, 2.3 g/m^{2}s, L = 3.0 kJ/g, steady

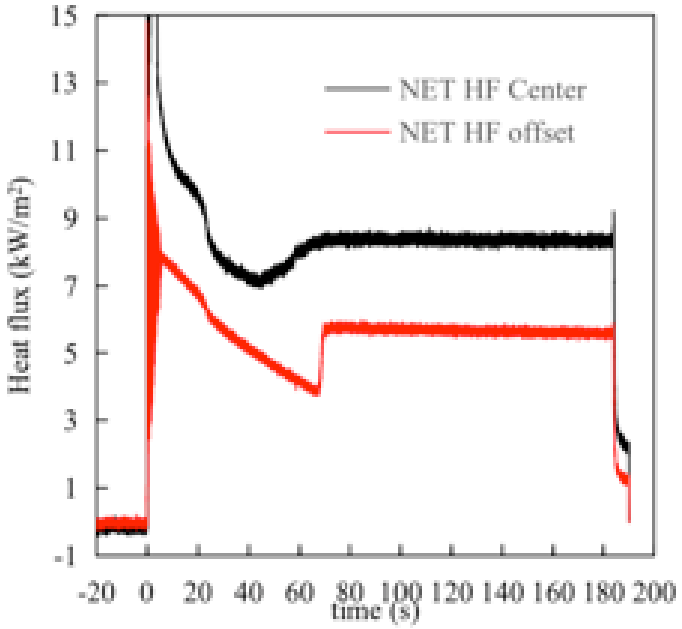
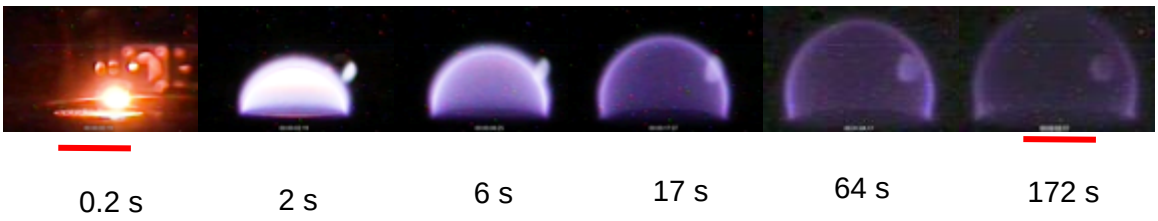


Figure 5. Day 6 H1. Flame growth and heat flux

In Figure 5 the flame reaches a steady state in about 80 seconds as seen by the flame shape and the heat flux data. This emulation corresponds to a material with a heat of combustion of about 22 kJ/g and a heat of gasification of about 3 kJ/g. As such it could represent PMMA as a material solid. This is a profound result indicating a common plastic could steadily burn under these NASA proposed atmospheric conditions. Let us examine another case.



Ethylene/N_2, 22 kJ/g, 5.0 g/m^{2}s, 34 % O2, 8.2 psia, *L* = 0.8 kJ/g, steady

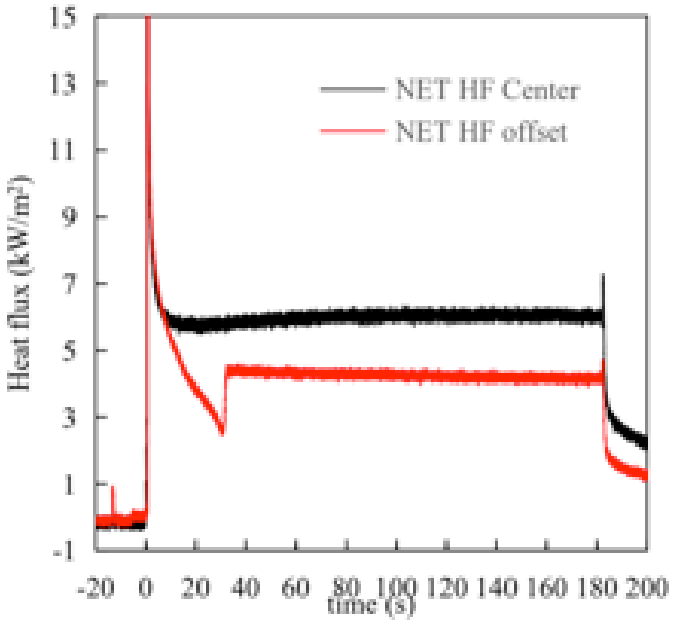


Figure 6. Day 6 D1 Flame growth and heat flux

Figure 6 shows steady results for a NASDA proposed atmosphere of 34 % oxygen and 0.56 atm. After about 100 s the flame appears to be steady. The heat of gasification here is estimated as 0.8 kJ/g. It again suggests that common plastics could steadily burn under these conditions.

Let us take a preliminary view of all the tests at atmospheres ranging from 21 % oxygen and 1 to 0.56 atm with fuel mixtures of about 22 and 44 kJ/g heats of combustion. With only discriminating by oxygen concentrations, Figure 7 shows a dramatic result. The fuel mixture mass flux is plotted against the estimated value of the center heat flux reading at the end of the test, as it appeared steady. As stated in the figure all of the flames at 21 % oxygen led to self-extinction in less than about 25 s, most of the flames at 27 % oxygen and all of the flames at 34 % oxygen reached steady burning. As stated earlier, the estimate for the heat of gasification could be low in this preliminary presentation; however, the range of *L* values displayed in Figure 7 clearly shows that it includes nearly

all combustible solid and liquid materials. This result has a profound implication for fire safety in spaceflight and planetary human missions.

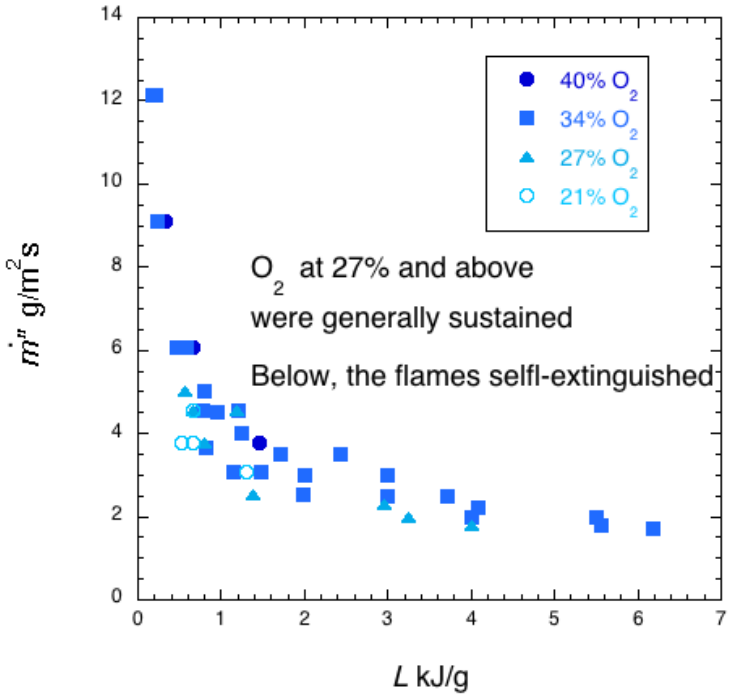


Figure 7. Round 1 results for the emulation of 25 mm flat materials

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Videos describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Videos related to BRE and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

The results of this study will have a substantial impact on the methodology used by NASA to assess the fire safety of materials in the future. [From BRE PI Jim Quintiere]

Appendix C. 30-Day Operational Accomplishments Report for ISS Increments 59 and 60

17 December 2019

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 59 and 60

ACME Advanced Combustion via Microgravity Experiments

BRE Burning Rate Emulator

BRE is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation will be carried out in two separate rounds, where the first round began in Increment 58 and was completed in Increment 59, so there is nothing to be reported for Increment 60. The second round of testing will be conducted in Increments 61 and/or 62.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for BRE and ACME information.

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Co-Investigator (Co-I):	Dr. John L. de Ris, FM Global (retired)
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Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Uday G. Hegde, Case Western Reserve University, Cleveland, OH

I. Introduction

The Burning Rate Emulator (BRE) experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. BRE is focused on spacecraft fire prevention. More specifically, BRE's goals are to improve our fundamental understanding of materials flammability, such as extinction behavior and the conditions needed for sustained combustion, and to assess the relevance of existing flammability test methods for low and partial-gravity environments. A flat porous burner fed with gaseous fuel simulates the burning of solid and liquid fuels, where measurements are made of the thermal feedback (i.e., to the burner) upon which the vaporization of condensed-phase fuels is dependent. A small number of gaseous fuels (including mixtures with inert gases) are used to simulate the burning of fuels such as paper, plastic, and

alcohol by matching properties including the heats of combustion and gasification, the surface temperature, and smoke point. The experiment's results can establish fuel properties that directly relate to fire hazards in spacecraft, and if effective in microgravity could provide an efficient, safe, and comprehensive technique for assessing flammability and selecting fire-resistant spacecraft materials.

BRE's specific experimental objectives are as follows, where it must be emphasized that NASA does not currently plan to carry out the desired tests for objective B.

D. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21 % oxygen; 10.2 psia, 26.5 %; and 8.2 psia, 34 % oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels.
E. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired).
F. Consider the conditions of Objectives 1 – 2 for extinction limits.

The observations will enable quantification of the microgravity behavior of condensed fuels (i.e., liquids and non-charring plastics).

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flame, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

BRE and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the operations center at the NASA Glenn Research Center. The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, gas flow rate(s) can be adjusted after quasi-steady data has been collected. But testing has not been conducted in a 'manual' mode thus far for BRE or ACME in general.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

The first round of BRE tests began in February 2019, where a total of 45 flames were ignited during Increment 58. That round was completed in April 2019, during the beginning of Increment 59, when an additional 17 flames were ignited. As such, a total of 62 flames were ignited in the first round of testing. Meanwhile, the second round of BRE testing is expected to begin near the end of Increment 61 and continue into Increment 62. The 'interruption' in the BRE testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

All of the first round of tests were conducted with the 25-mm diameter BRE burner while burning ethylene (C2H4) - sometimes diluted with nitrogen. The tests were carried out at a range of chamber pressures and oxygen concentrations, generally representative of possible crew environments in future spacecraft or habitats. More specifically, testing was especially focused on atmospheric conditions at (1) 14.7 psia, 21% oxygen; (2) 10.2 psia, 26.5% oxygen; and (3) 8.2 psia, 34% oxygen. In all cases, the atmosphere was diluted with nitrogen and any incomplete combustion products (e.g., water vapor and carbon dioxide) from preceding tests. An exception is tests conducted after the chamber atmosphere was vented and refilled, such as the first test of the day. But the atmosphere was typically refreshed once during each of testing.

BRE does not have living subjects like a life science experiment. The tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C2H4), is normally highly sooting

because of its carbon double bond. Tests are conducted with pure fuel and fuel mixed with nitrogen, where the fuel and nitrogen was premixed on the ground prior to launch for some cases. But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations.

There are no samples (e.g., of soot) collected for BRE, where all of the ISS results are electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for BRE during the first round of testing - in Increments 58 and 59 - is found in Table 1. The table includes a limited overview of the test conditions. For completeness, a row was included for testing on April 1 (GMT 091) even though it was quickly discovered that the hot-wire igniter tip had failed at the end of the preceding test day. As a result, flames were not ignited until a few days later after astronaut Nick Hague had replaced the igniter tip.

Table 1. BRE log of flames ignited in Increments 58 and 59
(*during BRE's first round of tests*)

Test Day[1]	**Year**	**Month**	**Date**	**Burner Size[2]**	**Fuel bottle**	**Oxygen conc.[3]**	**Flames ignited[4]**
First round of tests, with the 25-mm burner, which began in Feb. 2019 (Increment 58)							
19039	2019	Feb.	8	25-mm	100% C2H4	21%, 26.5%, 40%	10
19051			20			26.5%, 34%	8
19056			25		50% C2H4	21%, 26.5%, 34%, 40%	8
19063		March	4			26.5%, 40%	6
19066			7			34%	6
19070			11			26.5%, 34%	7
19087			28		100% C2H4	26.5%, 34%	8
19091		April	1			21%	0
19094			5			21%, 26.5%, 34%	9

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Burner size** refers to the diameter of the BRE burner's perforated circular outlet.
[3]**Oxygen conc.** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations for tests that ignited are listed. *The chamber pressures were also varied but are not identified in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

There were no significant anomalies hindering BRE operations during Increment 59, but there was one in Increment 58 during the preparations for BRE testing began (that was mistakenly not addressed in the 30-day report including that increment). It was discovered on Feb. 4 (GMT 035) that the pressure was low in a check gas bottle for the CIR's Gas Chromatograph Instrumentation Package (GCIP). The bottle had been on the space station for 9 years, and the low pressure was attributed to a nominal leak rate. The check gas was for calibration of the GCIP measurements and not needed for regular operations which proceeded unhindered. The bottle was replaced on June 13 to enable the necessary calibration.

2. Any other difficulties in obtaining scientific data

BRE's primary problem was that the sensors for measuring the thermal radiation often saturated. Although the measurements were not required to meet experiment success, they were considered sufficiently critical that BRE was equipped with its own radiometer array, featuring three thermopile detectors. Estimates had been made of the signal levels based on 5-second drop tests since the hot gases tend to blanket the flame in the absence of buoyancy and thereby remain within the view of the radiometers. As a result, the signal levels are strongly dependent on gravity. The estimates were not part of the science requirements and may not have been sufficiently tracked in the hardware development. Furthermore, the ISS tests lasted as long as 5 minutes, which is over four times the envisioned BRE test duration and 60 times the duration of the drop tests. With the fixed amplification of the radiometer signals, the data was frequently lost - other than the knowledge that the limits were exceeded. An engineering review failed to identify an acceptable solution and BRE testing continued unchanged.

> ***From the BRE PI Jim Quintiere:***
> The radiometry array proved problematic as its sensors were mostly out of range and quickly became saturated. However, for some of the small size flames, these data were complete. Those data will be vital to assessing the role of flame and combustion product radiation.

Ignition was not always successful. As previously discussed, it was discovered on April 1 (GMT 091) that the hot-wire igniter tip had failed on the preceding test day, preventing further flame testing until it could be replaced. While such circumstances are uncommon for ACME operations, BRE tests were sometimes conducted under conditions, e.g., for igniter activation and deactivation, which led to momentary flashes without sustained ignition. Given that unplanned crew time is required to replace failed igniter tips, the BRE ops team limited - to varying degrees - the igniter voltage and the time that it was energized within the flame. The latter was known from earlier research in the Combustion Integrated Rack (CIR) to degrade the hot-wire

igniter tip and hasten its demise. It is notable that the flashes of flame (across the burner surface) sometimes continued briefly after the igniter was deactivated and retracted. In one case on the first day of testing, there was excess fuel and a 'fireball' was ignited which promptly quenched. Ignition also sometimes failed due to a software limitation where the fuelFlowIgnitionDeltaMs parameter which specifies the relative timing of the fuel flow and igniter activation must be evenly divisible by 100 (ms). This limitation wasn't initially understood and the igniter didn't energize as a result.

From the BRE PI Jim Quintiere:
It was realized that the igniter for the burner gases could only be applied for up to 4 seconds, on insertion and retraction. This produced an unanticipated problem, as the time for the burner gases to reach the igniter could be tens of seconds depending on the flow rate. Also, after the first test, the burner would contain that fuel mixture mixed with the chamber gas. Thus the burner would need to be flushed with nitrogen before the second fuel mixture was initiated.

Intense communication with the NASA team proceeded the day before the first test. The gas volume of the burner along with the feed pipe volume was supplied by NASA. Accordingly ground experiments at UMD were conducted with an identical NASA BRE burner along with its piping volume from the flow valve. These experiments and theory were used to develop an algorithm for determining the relative timing of the igniter and gas flow activation that will result in ignition.

Day 1 was conducted with rough estimates made on the fly, and fortunately nearly all of the tests led to a satisfactory ignition. By Day 2 an algorithm was established and successfully used for the remainder of the testing. This algorithm will need to be refined and applied for the larger 50 mm diameter burner that will be used in Round 2.

A document on the development of the algorithm and its refinement for the larger burner is being prepared and will be shared with the NASA team to facilitate round 2.

The second day of flame testing was delayed to allow for the development of a new ignition procedure, where concerns arose that the mixture of gas flowing from the burner wasn't sufficiently well understood, i.e., in terms of concentrations. This concern had to do with the burner's large internal volume relative to that of the burners used for the other ACME experiments. To be clear, this issue had to do with the analysis of results rather than the achievement of ignition. Re-planning was also necessary early in BRE testing because the test durations were often much longer than had earlier been envisioned by the investigators. While these factors didn't directly interfere with the acquisition of test data, they did slow progress.

While it rarely occurred, BRE tests sometimes automatically aborted (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time associated with the failed test attempt. Such aborts typically occur because of communication issues with an Image Processing and Storage Unit (IPSU).

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel and nitrogen have unfortunately not always performed adequately, where the flow rates have not always been appropriately steady. The resulting effects on the flame behavior are tentatively assumed to be minor, but this still needs to be assessed.

C. Completeness/Quality of Data

The number of flames ignited in Increments 58 and 59 significantly exceeds the planned tests – under the tested conditions (25 mm burner and ethylene as fuel) - which were specified as required in the ACME Integrated Science Requirements Document (ISRD). A total of 25 such tests were required, while 62 flames with those conditions were ignited. As such, more than twice as many flames were ignited as had been planned.

It should be understood that a complete set of image data is not necessarily available for all tests. Recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done to maximize the scientific return of each day of testing within the project constraints.

From BRE Co-I Peter Sunderland:
The video and image data of the testing produced satisfactory data to display the nature of the burning sequence from ignition to self-extinction or fuel termination. However, the image quality is less than was hoped for. The ACME ops camera provides video, but only offers 0.3 megapixels of resolution, and for many flames it is too zoomed out. The GigE camera has poor sensitivity in the blue, and after the ignition transient all our flames have been blue. This camera only offers 1.4 megapixels of resolution and requires about 1 s exposures for blue flames, which is far too slow to be considered video. The images provided by both cameras are, according to UMD marketing staff, at too low resolution to be used in press releases. In round 2 we will increase the gain on the GigE camera, an option we were unaware of in round 1.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

See discussion in III.D in regard to fouling or degradation of the paint on the BRE burner's heat flux sensors.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and BRE's first round, completed in Increment 59, is considered to be successful despite the frequent saturation of the radiometer data discussed in section III.B.2.

D. Status of Data Analysis **(submitted by BRE PI Jim Quintiere and doctoral student Parham Dehghani)**

All measured transient raw data (e.g., heat flux sensors, thermocouples, radiometers, mass flow meters and chamber pressure sensor) have been converted to engineering units and plotted using the calibrations provided in ACME's Software Interface Design Description (ACME-DOC-010 Rev. D).

All of the round 1 tests have had the heat flux sensor data reduced using their original calibrations. As presented below, the data is being processed to address the effect of the combustion product water and paint degradation.

The primary measurements are used, by two techniques, to obtain the net heat flux from the flame to the burner surface. In the steady state, this net heat flux divided by the mass flow rate per unit area of the fuel mixture gives the heat of gasification of the emulated fuel. The key hypothesis is that the heat of combustion, smoke point, and heat of gasification of the fuel mixture will represent a real solid fuel (or liquid fuel absorbed in an inert matrix). Thus, achieved steady burning would represent the potential for real solid fuels to achieve steady burning in microgravity. Other than for a candle flame, this has not been shown before.

The first technique uses two thermopile heat flux gages embedded in the burner surface. One gage is at the center the other is offset from the center. The resulting data suffered from two possible effects. The first effect is the condensation of water vapor. As the sensors are not cooled, but rise with temperature as the flame grows, water from its combustion products can condense and then evaporate from the sensor. This occurs, as the dew point for combustion products is about 60-70°C, and the sensor begins at ambient temperature and can achieve temperatures over 100°C. Thus, condensation can occur at temperatures below the dew point that leads to added heat to the sensor. Above the dew point, the condensed water will evaporate and reduce the heat to the sensor.

When the sensor is dry and above 100°C, it will only measure the heat flux from the flame gases as intended.

We have duplicated this condensation effect in 1-g laboratory experiments with our duplicative NASA BRE burner. Fig. 1 illustrates the condensation and evaporation effect in the ISS tests. The offset sensor in red shows a clear jump increase when the water fully evaporates. More analysis is needed to properly correct for this effect. Fortunately, by the end of the test shown here (and for all of the other data), this water effect is gone.

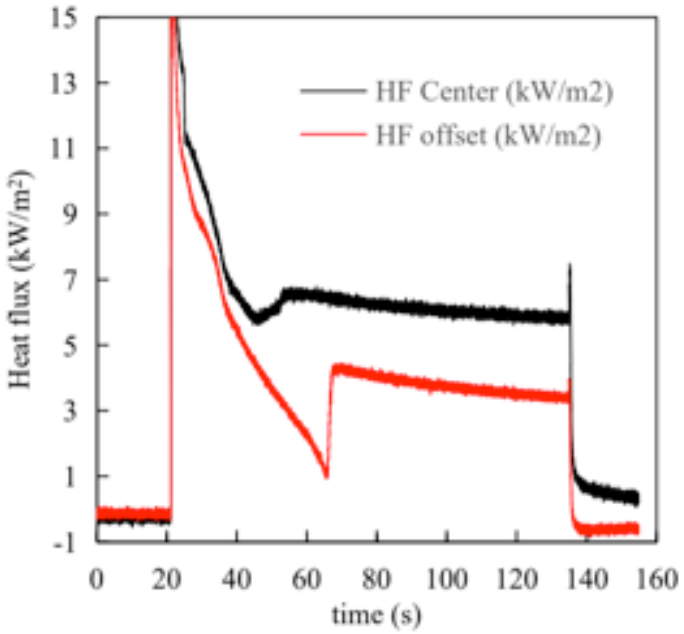


Figure 1. Typical net heat flux of the heat flux sensors.

The second effect on the heat flux sensors is that the heat may have fouled their paint. This could then affect their calibration. Figure 2 shows an image of the ISS burner after round 1 tests were concluded. It can be seen that the paint on the sensors is degraded. This was the original manufacturer's paint, instead of that used for the copper top. The copper paint appears fine. We are working with the heat flux gage manufacturer to explore the effect of heat damage on the calibration of the sensors. This will give us a way to determine a quantitative change in the calibration.



Figure 2. Paint degradation on heat flux sensors.

The second technique to determine the heat flux is the copper calorimeter forming the top of the burner. The temperature rise of the copper is key to that determination. Figure 3 shows that typically the copper temperature is nearly uniform and smooth in time. It does not appear that condensation or evaporation is having any effect on its temperature behavior. So, we believe we are justified in using this measurement to predict the overall average heat flux from the flame to the burner. However, we do need to evaluate any possible effect of the combustion product water. In addition, these calorimeter data are not fully processed yet because we are refining our methodology for its calibration. If the heat flux data cannot be completely used with good accuracy, we have the calorimeter to fall back on.

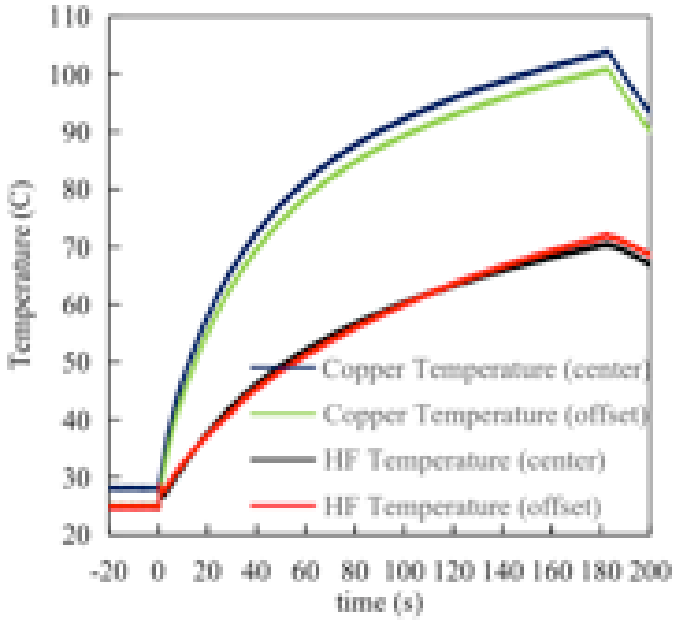


Figure 3. Copper and heat flux sensor temperatures in test 19070D1.

The calorimeter required a new calibration to account for heat being absorbed through the bevel edge of copper plate. So, control surfaces are now adjusted (i.e., the edge and burner back surface area, A_{back}) and then optimized for new set of calibration constants which are specific to the burner used in ISS experiments (i.e., mc_{Cu}, h_{back}, h_{rod}, and e_0). The calorimeter represents the average heat flux to the surface and is being used to obtain heats of gasification. Figure 4 shows the transient plots of test 19066D5. The test conditions for the plots used in this report are presented in Table 2.

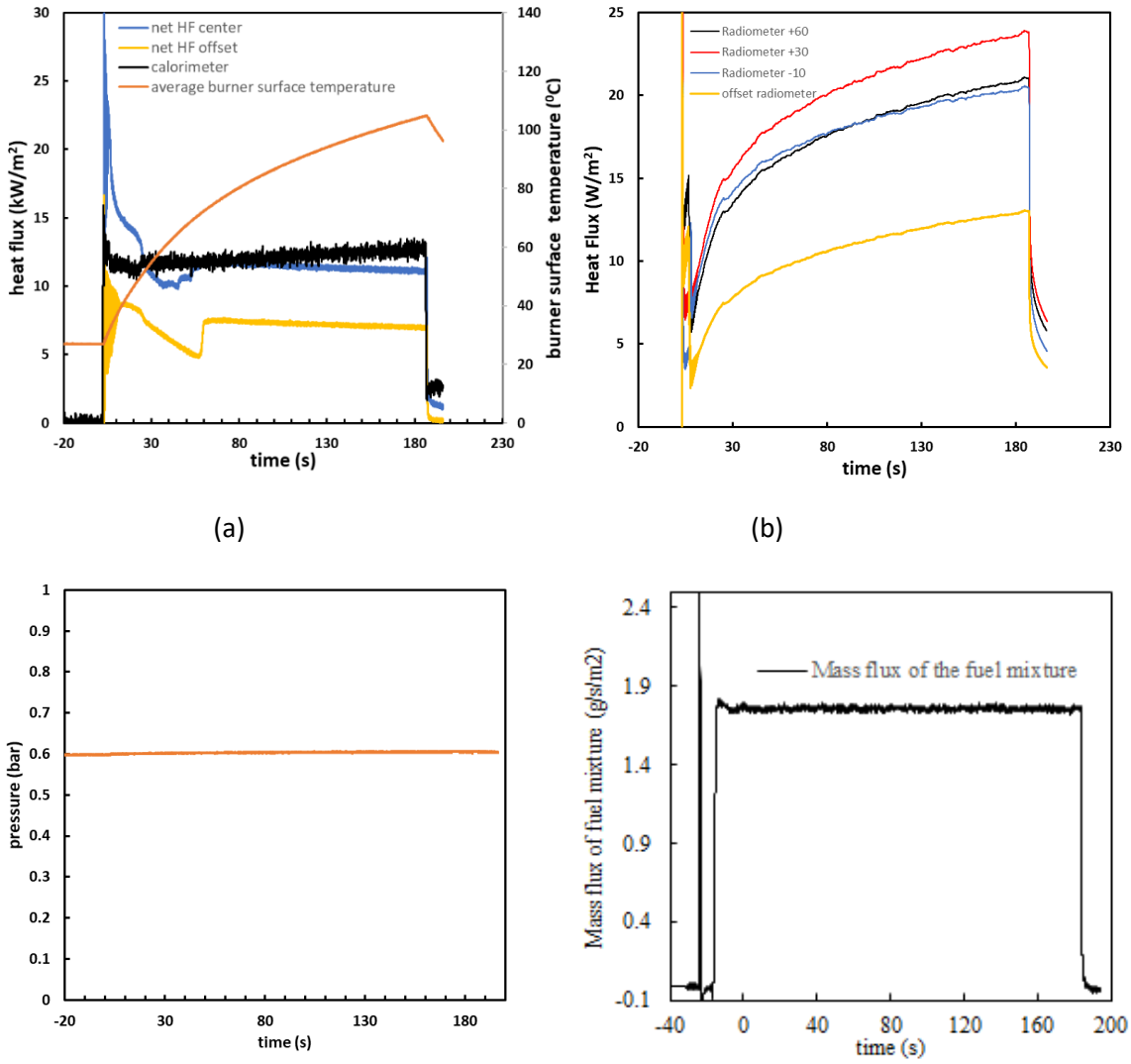


Figure 4. Average burner surface temperature and measured heat flux to the burner surface calorimeter (a), radiometer measurements (b), measured chamber pressure (c), and measured burner mass flux (d).

Table 2. Test conditions of 19066D5 and 19070L1.

Test day	test point	Burner Diameter, mm	X_{O2}	Chamber Pressure, bar	X_{fuel}	measured mass flux, g/m^2-s
Mar. 7	19066D5	25	34%	0.56	0.5	1.76
Mar. 11	19070L1	25	27%	0.7	0.5	1.74

Preliminary analysis of the radiometer sensor data has indicated its consistent ability to measure the flame overall radiation heat loss fraction. The method to determine the radiative heat flux component to the burner is more complex, and it will rely on theoretical modeling. This will take some time.

The video images of all the flames have been compiled and are being quantified in terms of flame shapes.

Alexander Snegirev, our Russian partner, has attempted a numerical simulation of at least one of the ISS BRE tests. That work is still in progress. It should be remarked his student won the Student Prize for Best Presentation at the 9th International Seminar on Fire and Explosion Hazards held in St. Petersburg, Russia in April 2019 for simulating our 5-s BRE drop tests.

E. Preliminary Research Findings **(submitted by BRE PI Jim Quintiere and doctoral student Parham Dehghani)**

Despite having much data to digest and analyze, there is a profound conclusion from these tests. The burner has definitively shown that steady burning is possible for real solids with a flat surface of 25 mm diameter. Up to now most flammability studies in microgravity have focused on flame spread with a wind for concurrent spread and at least a self-induced wind for opposed flow spread. We already knew that this would be possible from normal-gravity research. The burning of liquids, as in a wick of a candle flame and small droplets, has been shown to produce steady flames. However, many droplet flames were found to extinguish owing to radiative cooling of the flame. We see this as well in the extended ISS microgravity tests. However, this extinction was predominately seen for atmospheric conditions of 21% oxygen and 14.7 psia. At conditions of 27% oxygen, 10.2 psia and 34% oxygen, 8.2 psia, the flames generally persisted for 2 minutes or more. These are all atmospheric conditions that NASA is considering for future spacecraft. Some examples are shown below to illustrate these results.

Figure 5 shows a typical test of ignition and flame growth. Although extinction occurred at 45 s with the flame still growing, the heat flux was nearly steady by 30 s. Therefore, a heat of gasification (L) could be estimated. This was done in a preliminary estimation by using the center heat flux sensor as the net average heat flux. If the heating is just due to diffusion, theory indicates that the average heat flux is double the center value. Hence such an estimated heat of gasification could be low by 50%. On the other hand, if the sensor is degraded in its calibration and radiation is a significant component of the heat flux, then the estimate will also be low. Hence, using the center heat flux sensor directly gives a conservatively low estimate of the heat of gasification in the emulation of real solids. This will be significant when the totality of results for the 25 mm flat burning surface is displayed over its spectrum of conservatively low values for L.

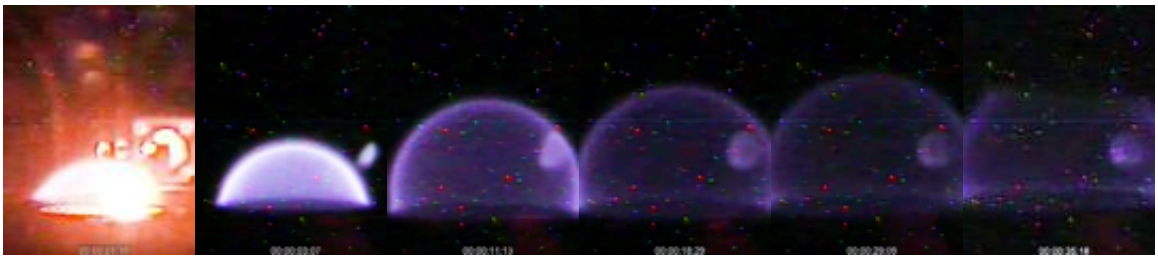
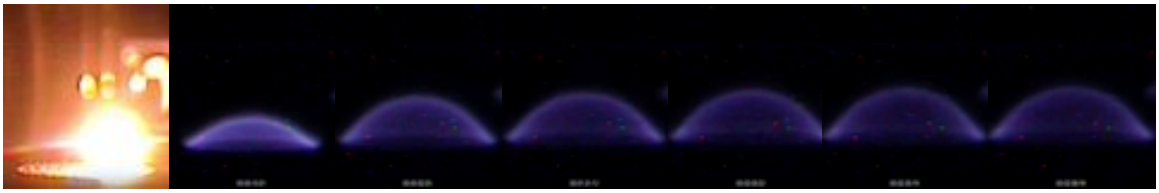


Figure 5. Flame growth from ignition to self-extinction in test 19056F1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 4.55 g/m^{2}s, 27% O_2, 10.2 psia, L = 0.66 kJ/g, extinction at 45 s.

Figures 6 and 7 illustrate results for atmospheric conditions of 26.5% oxygen and 0.7 atm (10.2 psia), and 34% oxygen and 0.56 atm (8.2 psia), respectively. Both use 50% mixtures of nitrogen and ethylene giving a fuel mixture heat of combustion of 21.8 kJ/g. Both of these tests reached steady conditions and were terminated by shutting off the fuel.



3 s 10 s 52 s 91 s 125 s 155 s 178 s

Figure 6. Flame growth from ignition to steady state in test 19070K1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 2.3 g/m^{2}s, 26.5% O_2, 10.2 psia, L = 3.0 kJ/g, steady.

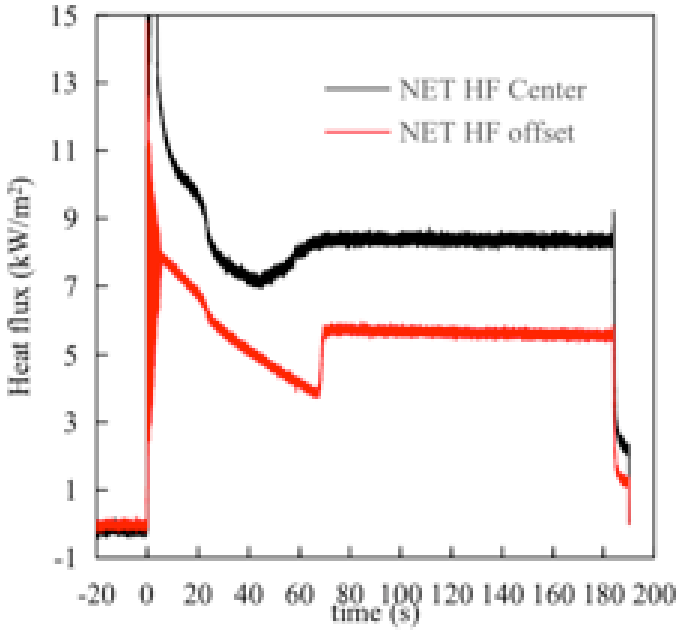


Figure 7. Measured heat flux in test 19070K1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 2.3 g/m^{2}s, 26.5% O2, 10.2 psia, L = 3.0 kJ/g, steady.

In Figure 8 the flame reaches a steady state in about 80 seconds as seen by the flame shape and the heat flux data. This emulation corresponds to a material with a heat of combustion of about 22 kJ/g and a heat of gasification of about 3 kJ/g. As such it could represent PMMA as a material solid. This is a profound result indicating a common plastic could steadily burn under these NASA proposed atmospheric conditions. Let us examine another case.



Figure 8. Flame growth from ignition to steady state in test 19070D1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Figure 9 shows steady results for a NASA proposed atmosphere of 34% oxygen and 0.56 atm. After about 100 s the flame appears to be steady. The heat of gasification here is estimated as 2.42 kJ/g.

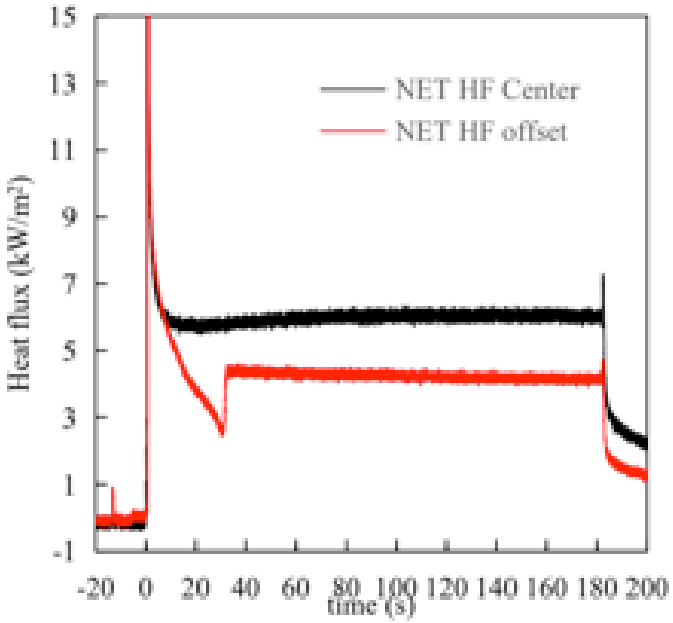


Figure 9. Measured heat flux in test 19070D1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Let us take a preliminary view of all the tests at atmospheres ranging from 21% oxygen and 1 to 0.56 atm with fuel mixtures of about 22 and 44 kJ/g heats of combustion. With only discriminating by oxygen concentrations, Figure 10 shows a dramatic result. The fuel mixture mass flux is plotted against the estimated value of the center heat flux reading at the end of the test, as it appeared steady. As seen in the figure, all of the flames at 21 % oxygen led to self-extinction in less than about 25 s, most of the flames at 27 % oxygen and all of the flames at 34% oxygen reached steady burning. As stated earlier, the

estimate for the heat of gasification could be low in this preliminary presentation; however, the range of *L* values displayed in Figure 10 and 14(a). clearly show that it includes nearly all combustible solid and liquid materials. This result has a profound implication for fire safety in spaceflight and planetary human missions.

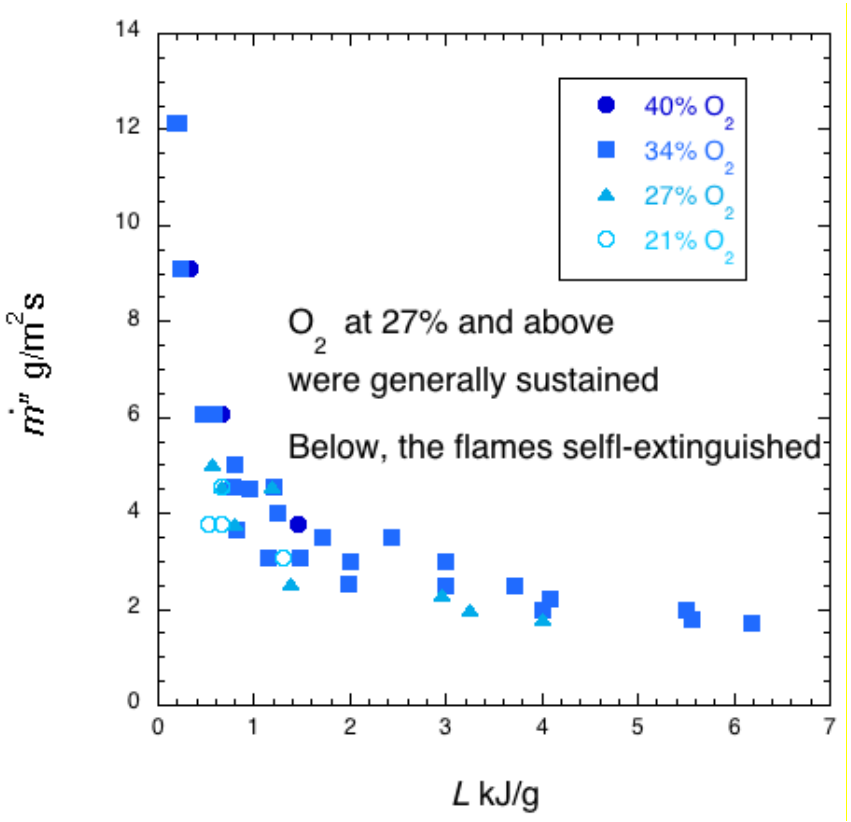


Figure 10. Round 1 results for the emulation of 25-mm flat materials using the center heat flux.

Radiative loss fraction from the flames are obtained using the radiometer's measurements. Unfortunately, the data set is limited as the BRE sensors were saturated for several tests. The +60 mm and the offset radiometers were used to obtain the radiative loss fraction as these two sensors provided data in multiple cases while the other two radiometers were saturated. The transient radiative loss fraction for 19066D5 and 19070L1 are shown in Figure 11. Radiative loss fractions obtained from both radiometers show good agreement for all studied cases and it is shown for 2 steady tests in Figure 11. Table 2 provides information about the test conditions.

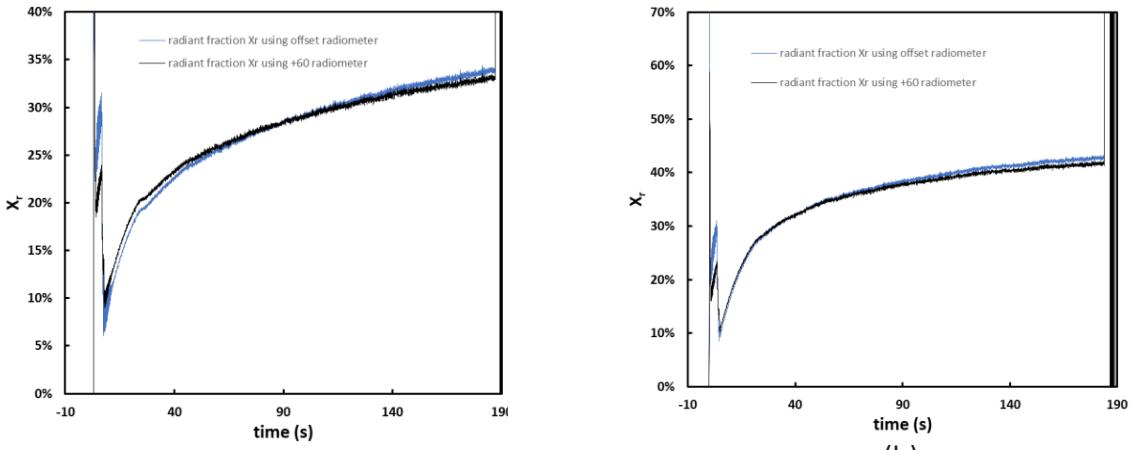


Figure 11. Transient radiative loss fraction for tests 19066D5 (a) and 19070L1 (b).

Flame heights and flame base diameter at the end of the test are measured from the image data. The radiative loss fraction is correlated with these two quantities and are shown in figure 12.

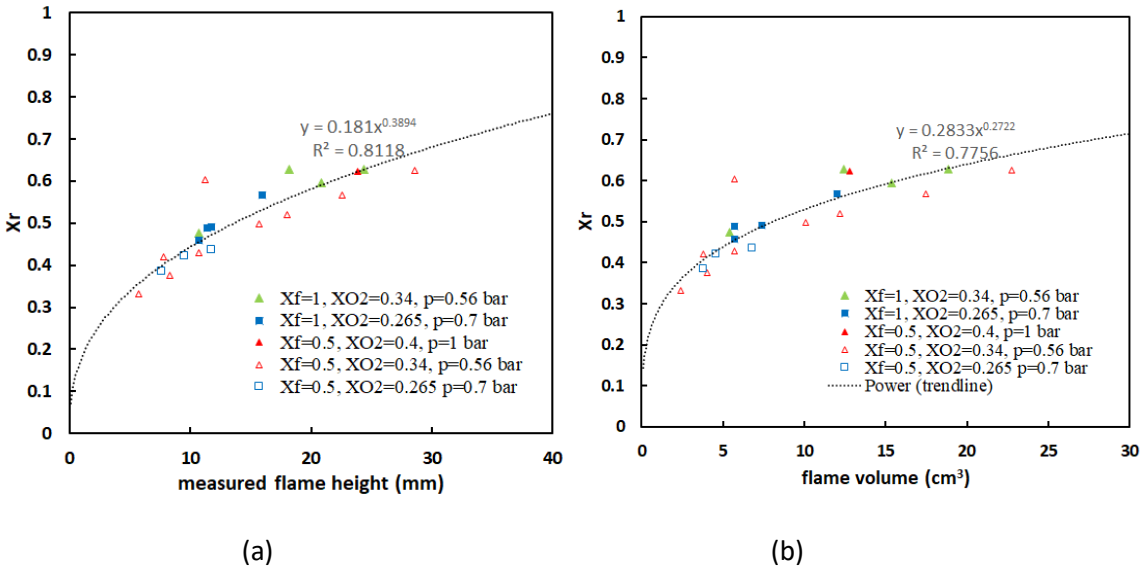
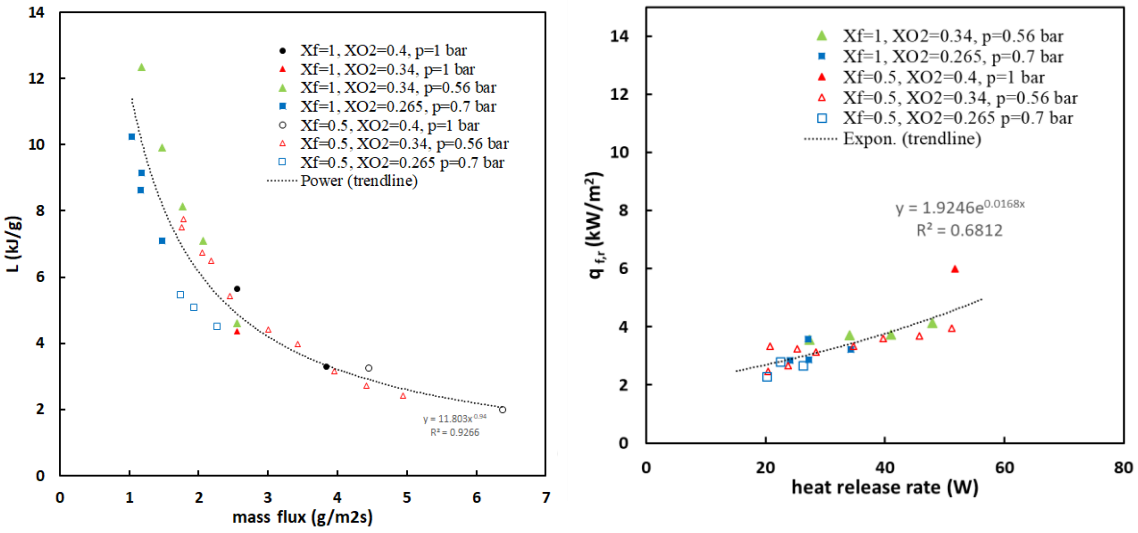


Figure 12. Radiative loss fraction vs. measured flame height (a) and flame volume (b).

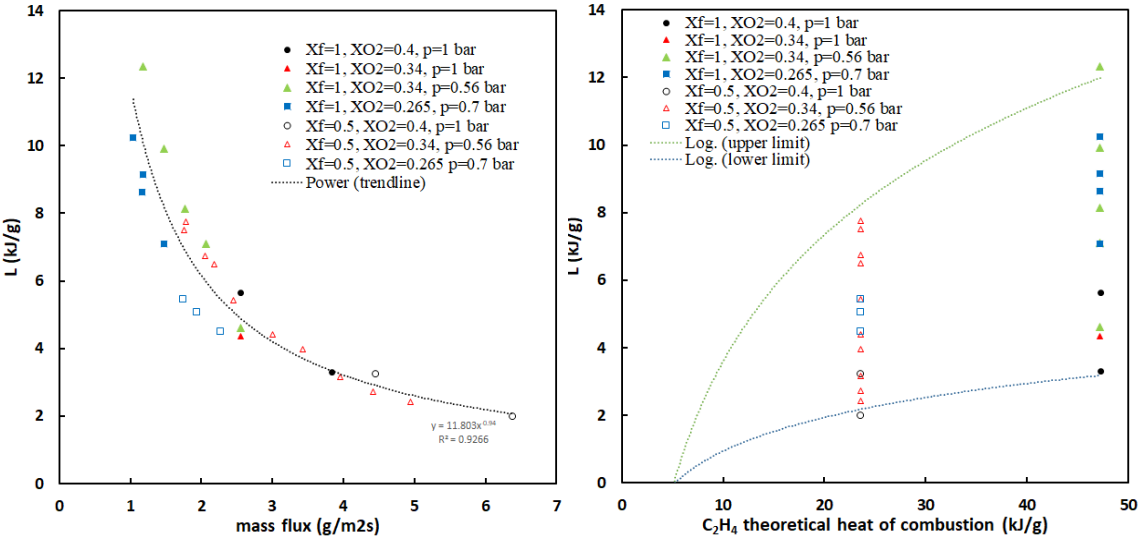
Two methods are proposed to obtain radiative heat flux on the burner surface. The first method is called 'Mean Beam Length' method and it is assumed that the radiative gasses are distributed homogeneously throughout the flame envelope. The second method is called the 'View Factor' method and it assumes that the flame is a thin sheet where all the radiation is generated at this sheet. Radiation view factor of the surface of the burner with respect to the flame sheet is then calculated which leads to obtaining the radiative heat flux on the burner surface. Figure 13 shows the radiative heat flux on the burner surface obtained by mentioned methods.



(a) (b)

Figure 13. Radiative heat flux on the burner surface obtained by the Mean Beam Length method (a) and View Factor method (b).

The heats of gasification obtained by the calorimeter are plotted against burner mass flux and heat of combustion. Figure 14 shows the flammability of materials in microgravity. It shows that solids like plastics and wood can burn steadily under NASA standard atmospheres. The flammability region shown in Figure 14(b) clearly shows that steady flames could exist for oxygen concentrations higher than 27%.



(a) (b)

Figure 14. Burning rate map (a) and flammability map (b).

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Videos related to BRE and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

The results of this study will have a substantial impact on the methodology used by NASA to assess the fire safety of materials in the future. [From BRE PI Jim Quintiere]

Appendix D. 30-Day Operational Accomplishments Report for ISS Increment 63

12 December2020

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increment 63

ACME Advanced Combustion via Microgravity Experiments
BRE Burning Rate Emulator

BRE is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation is being carried out in two separate rounds, where the first round was conducted during Increments 58 and 59 and the second round began in Increment 63 and should be completed in Increment 64.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for BRE and ACME information.

Principal Investigator (PI):	Prof. James G. Quintiere, University of Maryland, College Park, MD
Co-Investigator (Co-I):	Prof. Peter B. Sunderland, U. of Maryland, College Park, MD
Co-Investigator (Co-I):	Dr. John L. de Ris, FM Global (retired)
Collaborator:	Prof. Alexander Yu. Snegirev, Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia
Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Uday G. Hegde, Case Western Reserve University, Cleveland, OH

I. Introduction

The Burning Rate Emulator (BRE) experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. BRE is focused on spacecraft fire prevention. More specifically, BRE's goals are to improve our fundamental understanding of materials flammability, such as extinction behavior and the conditions needed for sustained combustion, and to assess the relevance of existing flammability test methods for low and partial-gravity environments. A flat porous burner fed with gaseous fuel simulates the burning of solid and liquid fuels, where measurements are made of the thermal feedback (i.e., to the burner) upon which the vaporization of condensed-phase fuels is dependent. A small number of gaseous fuels (including mixtures with inert gases) are used to simulate the burning of fuels such as paper, plastic, and alcohol by matching properties including the heats of combustion and gasification, the surface

temperature, and smoke point. The experiment's results can establish fuel properties that directly relate to fire hazards in spacecraft, and if effective in microgravity could provide an efficient, safe, and comprehensive technique for assessing flammability and selecting fire-resistant spacecraft materials.

BRE's specific experimental objectives are as follows, where it must be emphasized that NASA does not currently plan to carry out the desired tests for objective B.

G. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21 % oxygen; 10.2 psia, 26.5 %; and 8.2 psia, 34 % oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels.
H. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired).
I. Consider the conditions of Objectives 1 – 2 for extinction limits.

The observations will enable quantification of the microgravity behavior of condensed fuels (i.e., liquids and non-charring plastics).

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flame, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

BRE and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the operations center at the NASA Glenn Research Center. The science portion of the testing is typically conducted in

a nominally automated mode using pre-programmed scripts. However, it could alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, gas flow rate(s) can be adjusted after quasi-steady data has been collected. But testing has not been conducted in a 'manual' mode thus far for BRE or ACME in general.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

The first round of BRE tests began in February 2019, during Increment 58, when a total of 45 flames were ignited. That round was completed in April 2019, during Increment 59, when an additional 17 flames were ignited. As such, a total of 62 flames were ignited in the first round of testing. Meanwhile, the BRE's second-round flame testing began in Sept. 2020, during Increment 63, during which 47 flames were ignited. The second and final round of tests should be completed during Increment 64.

The 'interruption' in the BRE testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome. However, the break was longer than had been planned because of combination of hardware issues and limited crew availability. As a result, BRE's second round was rescheduled to occur after both rounds of the preceding ACME experiment, Structure and Response of Spherical Diffusion Flames (s-Flame), was completed, where that occurred in July 2020. A factor in that decision was the delayed start of commercial crew launches and understanding that crew availability would be limited until such crew launches entered an operational, rather than demonstration, phase.

All first-round tests were conducted with the 25-mm diameter BRE burner while burning ethylene (C2H4) - sometimes diluted with nitrogen. The tests were carried out at a range of chamber pressures and oxygen concentrations, generally representative of possible crew environments in future spacecraft or habitats. More specifically, testing was especially focused on atmospheric conditions at (1) 14.7 psia, 21% oxygen; (2) 10.2 psia, 26.5% oxygen; and (3) 8.2 psia, 34% oxygen. In all cases, the atmosphere was diluted with nitrogen and any incomplete combustion products (e.g., water vapor and carbon dioxide) from preceding tests. An exception is tests conducted after the chamber atmosphere was

vented and refilled, such as the first test of the day. But during tests days, the atmosphere was typically refreshed once or twice between sets of tests.

Second round testing is being carried out including both BRE burner diameters, i.e., 25 and 50 mm, and on some days with methane (CH4) – again sometimes diluted with nitrogen. Therefore, the second round of tests features a broader range of conditions than investigated during the first round.

BRE does not have living subjects like a life science experiment. The tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C2H4), is normally highly sooting because of its carbon double bond. Tests are conducted with pure fuel and fuel mixed with nitrogen, where the fuel and nitrogen were premixed on the ground prior to launch for some cases. But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations.

There are no samples (e.g., of soot) collected for BRE, where all ISS results are electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for BRE during Increments 58 and 59 (the first round of tests) and Increment 63 (the start of second round) is found in Table 1. The table includes a limited overview of the test conditions. For completeness, a row was included for testing on April 1 (GMT 091) even though it was quickly discovered that the hot-wire igniter tip had failed at the end of the preceding test day. As a result, flames were not ignited until a few days later after astronaut Nick Hague had replaced the igniter tip.

Table 1. BRE log of flames ignited in Increments 58-59 (round 1) and 63 (round 2 start)

Test Day[1]	Year	Month	Date	Burner Size[2]	Fuel bottle	Oxygen conc.[3]	Flames ignited[4]
First round of tests, with the 25-mm burner, from Feb.-April 2019							
19039	2019	Feb.	8	25-mm	100% C2H4	21%, 26.5%, 40%	10
19051			20			26.5%, 34%	8
19056			25		50% C2H4	21%, 26.5%, 34%, 40%	8
19063		March	4			26.5%, 40%	6
19066			7			34%	6
19070			11			26.5%, 34%	7
19087			28		100% C2H4	26.5%, 34%	8
19091		April	1			21%	0
19094			5			21%, 26.5%, 34%	9
Second round of tests, with the 50- and 25-mm burners, beginning Sept. 2020							
20247	2020	Sept.	3	50-mm	100% C2H4	25%, 34%, 40%	13
20261			17	25-mm		26.5%, 34%	13
20279		Oct.	5		100% CH4	25%, 34%, 40%	21

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Burner size** refers to the diameter of the BRE burner's perforated circular outlet.
[3]**Oxygen conc.** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations for tests that ignited are listed. *The chamber pressures were also varied but are not identified in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

There were no significant anomalies hindering BRE operations during Increments 59 and 63, but there was one in Increment 58 during the preparations for BRE testing (that was mistakenly not addressed in the 30-day report including that increment). It was discovered on Feb. 4 (GMT 035) that the pressure was low in a check gas bottle for the CIR's Gas Chromatograph Instrumentation Package (GCIP). The bottle had been on the space station for 9 years, and the low pressure was attributed to a nominal leak rate. The check gas was for calibration of the GCIP measurements and not needed for regular operations which proceeded unhindered. The bottle was replaced on June 13 to enable the necessary calibration.

2. Any other difficulties in obtaining scientific data

BRE's primary problem was that the sensors for measuring the thermal radiation often saturated. Although the measurements were not required to meet experiment success, they were considered sufficiently critical that BRE was equipped with its own radiometer array, featuring three thermopile detectors. Estimates had been made of the signal levels based on 5-second drop tests since the hot gases tend to blanket the flame in the absence of buoyancy and thereby remain within the view of the radiometers. As a result, the signal levels are strongly dependent on gravity. The estimates were not part of the science requirements and may not have been sufficiently tracked in the hardware development. Furthermore, the ISS tests lasted as long as 5 minutes, which is over four times the envisioned BRE test duration and 60 times the duration of the drop tests. With the fixed amplification of the radiometer signals, the data was frequently lost - other than the knowledge that the limits were exceeded. An engineering review failed to identify an acceptable solution and BRE testing continued unchanged.

From the BRE PI Jim Quintiere:
The radiometry array proved problematic as its sensors were mostly out of range and quickly became saturated. However, for some of the small size flames, these data were complete. Those data will be vital to assessing the role of flame and combustion product radiation.

Ignition was not always successful. As previously discussed, it was discovered on 1 April 2019 (GMT 091) that the hot-wire igniter tip had failed on the preceding test day, preventing further flame testing until it could be replaced. While such circumstances are uncommon for ACME operations, BRE tests were sometimes conducted under conditions, e.g., for igniter activation and deactivation, which led to momentary flashes without sustained ignition. Given that unplanned crew time is required to replace failed igniter tips, the BRE ops team limited - to varying degrees - the igniter voltage and the time that it was energized within the flame. The latter was known from earlier research in the Combustion Integrated Rack (CIR) to degrade the hot-wire igniter tip and hasten its demise. It is notable that the flashes of flame (across the burner surface) sometimes continued briefly after the igniter was deactivated and retracted. In one case on the first day of testing, there was excess fuel and a 'fireball' was ignited which promptly quenched. Ignition also sometimes failed due to a software limitation where the fuelFlowIgnitionDeltaMs parameter which specifies the relative timing of the fuel flow and igniter activation must be evenly divisible by 100 (ms). This limitation wasn't initially understood and the igniter didn't energize as a result.

From the BRE PI Jim Quintiere:
It was realized that the igniter for the burner gases could only be applied for up to 4 seconds, on insertion and retraction. This produced an unanticipated problem, as the time for the burner gases to reach the igniter could be tens of seconds depending on the flow rate. Also, after the first test, the burner would contain that fuel mixture mixed with the chamber gas. Thus the burner would need to be flushed with nitrogen before the second fuel mixture was initiated.

Intense communication with the NASA team proceeded the day before the first test. The gas volume of the burner along with the feed pipe volume was supplied by NASA. Accordingly ground experiments at UMD were conducted with an identical NASA BRE burner along with its piping volume from the flow valve. These experiments and theory were used to develop an algorithm for determining the relative timing of the igniter and gas flow activation that will result in ignition.

Day 1 [of the first round] was conducted with rough estimates made on the fly, and fortunately nearly all of the tests led to a satisfactory ignition. By Day 2 an algorithm was established and successfully used for the remainder of the testing. This algorithm will need to be refined and applied for the larger 50 mm diameter burner that will be used in Round 2.

A document on the development of the algorithm and its refinement for the larger burner [was] prepared and … shared with the NASA team to facilitate [ignition in] round 2.

The second day of round-1 flame testing was delayed to allow for the development of a new ignition procedure, where concerns arose that the mixture of gas flowing from the burner wasn't sufficiently well understood, i.e., in terms of concentrations. This concern had to do with the burner's large internal volume relative to that of the burners used for the other ACME experiments. To be clear, this issue had to do with the analysis of results rather than the achievement of ignition. Re-planning was also necessary early in BRE testing because the test durations were often much longer than had earlier been envisioned by the investigators. While these factors didn't directly interfere with the acquisition of test data, they did slow progress.

While it rarely occurred, BRE tests sometimes automatically aborted (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time associated with the failed test attempt. Such aborts typically occur because of communication issues with an Image Processing and Storage Unit (IPSU).

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel and nitrogen have unfortunately not always performed adequately, where the flow rates have not

always been appropriately steady. The resulting effects on the flame behavior are tentatively assumed to be minor, but this still needs to be assessed.

While it didn't prevent the acquisition of scientific data, the intended igniter arm, which establishes the relative positioning of the igniter tip and burner, was not found for the round-2 hardware set-up. Therefore, an alternate arm, which positioned the igniter further from the burner than intended, was used for the 3 Sept. 2020 testing. Fortunately, the flames were still reliably ignited and the proper igniter arm was found and installed prior to the subsequent test day.

However, it was determined that the 3 Sept. 2020 science objectives were not accomplished because the fuel didn't always transition to the intended test flow rate. As a result of concerns about the ignition success because of the use of the alternate igniter arm, the decision was made to repeatedly ignite the flame at the same fuel flow rate and then subsequently adjust it to the desired test flow rate. The first attempt was successful, but the transition time was shortened to save fuel, etc. which unfortunately prevented the flow transition. In other words, the transition time was too short where that wasn't discovered until review of the data days later. As a result, the tests were generally conducted at the same fuel flow rate, where that was meant to be a variable. As a result, the data currently doesn't allow a full assessment of the effect of the burner size on the flammability. Therefore, a make-up test day in Increment 64 has been planned to address this shortcoming.

C. Completeness/Quality of Data

The number of flames ignited in Increments 58 and 59 significantly exceeds the planned tests – under the tested conditions (25 mm burner and ethylene as fuel) - which were specified as required in the ACME Integrated Science Requirements Document (ISRD). A total of 25 such tests were required, while 62 flames with those conditions were ignited. As such, more than twice as many flames were ignited as had been planned.

An additional 48 tests were specified as required in the ISRD to fully accomplish the BRE science objectives, i.e., where a total of 73 test points were required. While there were issues with the Increment 63 testing, e.g., discussed in section B, 47 flames were ignited in Sept.-Oct. 2020, where 6 additional test days are currently planned for Increment 64 (including the make-up day previously discussed). As such, it is expected that far more tests will be conducted than had been originally planned for BRE's second round.

It should be understood that a complete set of image data is not necessarily available for all tests. Recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done to maximize the scientific return of each day of testing within the project constraints.

From BRE Co-I Peter Sunderland:
The video and image data of the testing produced satisfactory data to display the nature of the burning sequence from ignition to self-extinction or fuel termination. However, the image quality is less than was hoped for. The ACME ops camera provides video, but only offers 0.3 megapixels of resolution, and for many flames it is too zoomed out. The GigE camera has poor sensitivity in the blue, and after the ignition transient all our flames have been blue. This camera only offers 1.4 megapixels of resolution and requires about 1 s exposures for blue flames, which is far too slow to be considered video. The images provided by both cameras are, according to UMD marketing staff, at too low resolution to be used in press releases. In round 2 we will increase the gain on the GigE camera, an option we were unaware of in round 1.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

See the discussion in III.D regarding fouling or degradation of the paint on the BRE burner's heat flux sensors.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). Note that although the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS) it was not utilized for BRE (or any ACME) testing because of the crew time needed to enable and disable the isolation before and after testing.

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and BRE's first round is considered to be successful despite the frequent saturation of the radiometer data discussed in section III.B.2.

D. Status of Data Analysis **(submitted by BRE PI Jim Quintiere and doctoral student Parham Dehghani)**

All measured transient raw data (e.g., heat flux sensors, thermocouples, radiometers, mass flow meters and chamber pressure sensor) have been converted to engineering units and plotted using the calibrations provided in ACME's Software Interface Design Description (ACME-DOC-010 Rev. D).

All of the round 1 tests have had the heat flux sensor data reduced using their original calibrations. As presented below, the data is being processed to address the effect of the combustion product water condensation on the sensor surfaces.

The primary measurements are used, by two techniques, to obtain the net heat flux from the flame to the burner surface. In the steady state, this net heat flux divided by the mass flow rate per unit area of the fuel mixture gives the heat of gasification of the emulated fuel. The key hypothesis is that the heat of combustion, smoke point, and heat of gasification of the fuel mixture will represent a real solid fuel (or liquid fuel absorbed in an inert matrix). Thus, achieved steady burning would represent the potential for real solid fuels to achieve steady burning in microgravity. Other than for a candle flame, this has not been shown before.

The first technique uses two thermopile heat flux gages embedded in the burner surface. One gage is at the center the other is offset from the center (r = 9.5 mm). As the sensors are not cooled, but rise with temperature as the flame grows, water from its combustion products can condense and then evaporate from the sensor. This occurs, as the dew point for combustion products is about 60-70°C, and the sensor begins at ambient temperature and can achieve temperatures over 100°C. Thus, condensation can occur at temperatures below the dew point that leads to added heat to the sensor. Above the dew point, the condensed water will evaporate and reduce the heat to the sensor. When the sensor is dry and above 100°C, it will only measure the heat flux from the flame gases as intended.

We have duplicated this condensation effect in 1-g laboratory experiments with the duplicative NASA BRE burner. Fig. 1 illustrates the condensation and evaporation effect in the ISS tests. The offset sensor in red shows a clear jump increase when the water fully evaporates. More analysis is needed to properly correct for this effect. Fortunately, by the end of the test shown here (and for all of the other data), this water effect is gone.

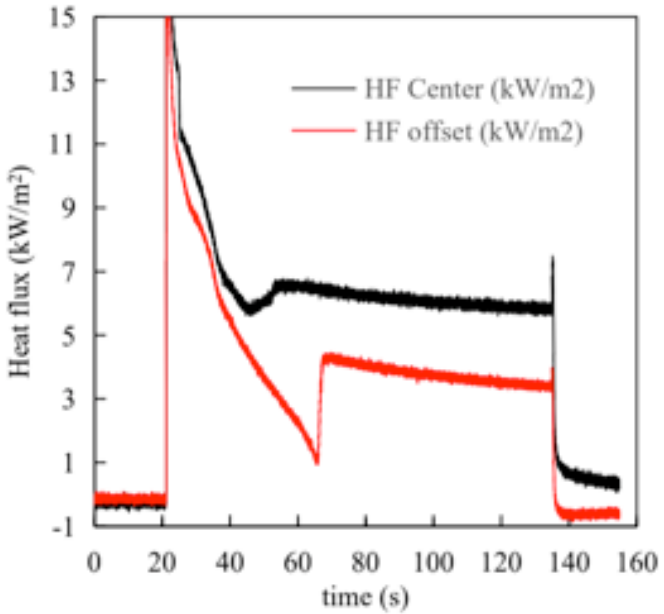


Figure 1. Typical net heat flux of the heat flux sensors.

The second effect on the heat flux sensors is that the heat may have fouled their paint. This could then affect their calibration. Figure 2 shows an image of the ISS burner after round 1 tests were concluded. It can be seen that the paint on the sensors is degraded. This was the original manufacturer's paint, instead of that used for the copper top. The copper paint appears fine. We worked with the heat flux gage manufacturer to explore the effect of heat damage on the calibration of the sensors. The results showed that the calibration was stable and didn't change during the process of heating and cooling.



Figure 2. Paint degradation on heat flux sensors.

The second technique to determine the heat flux is the copper calorimeter forming the top of the burner. The temperature rise of the copper is key to that determination. Figure 3 shows that typically the copper temperature is nearly uniform and smooth in time. It does not appear that condensation or evaporation is having any effect on its temperature behavior. So, we believe we are justified in using this measurement to predict the overall average heat flux from the flame to the burner.

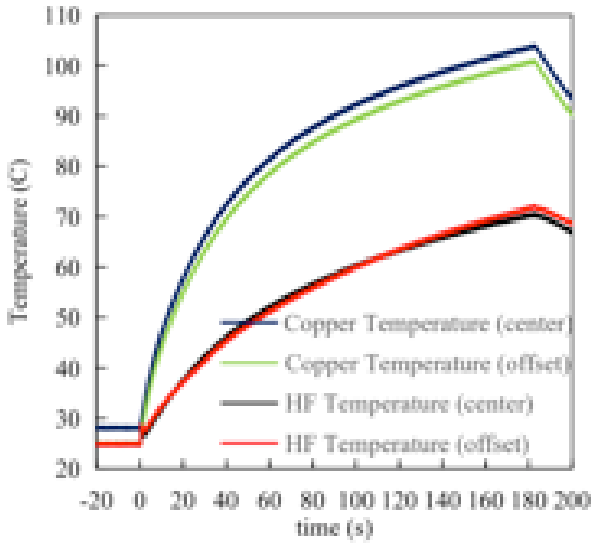


Figure 3. Copper and heat flux sensor temperatures in test 19070D1.

The calorimeter required a new calibration to account for heat being absorbed through the bevel edge of copper plate. So, control surfaces are now adjusted (i.e., the edge and burner back surface area, A_{back}) and then optimized for new set of calibration constants which are specific to the burner used in ISS tests (i.e., mc_{Cu}, h_{back}, h_{rod}, and e_0). The calorimeter represents the average heat flux to the surface and is being used to obtain heats of gasification. Figure 4 shows the transient plots of test 19066D5. The test conditions for the plots used in this report are presented in Table 2. We found out that there is significant heat absorption by the bevel edge of the burner that induces error to the calorimeter heat flux. A theoretical analysis was implied to estimate the radiative, convective and re-radiation heat flux to and from the top surface and the bevel edge of the burner which helped in estimation of the error induced to the calorimeter data.

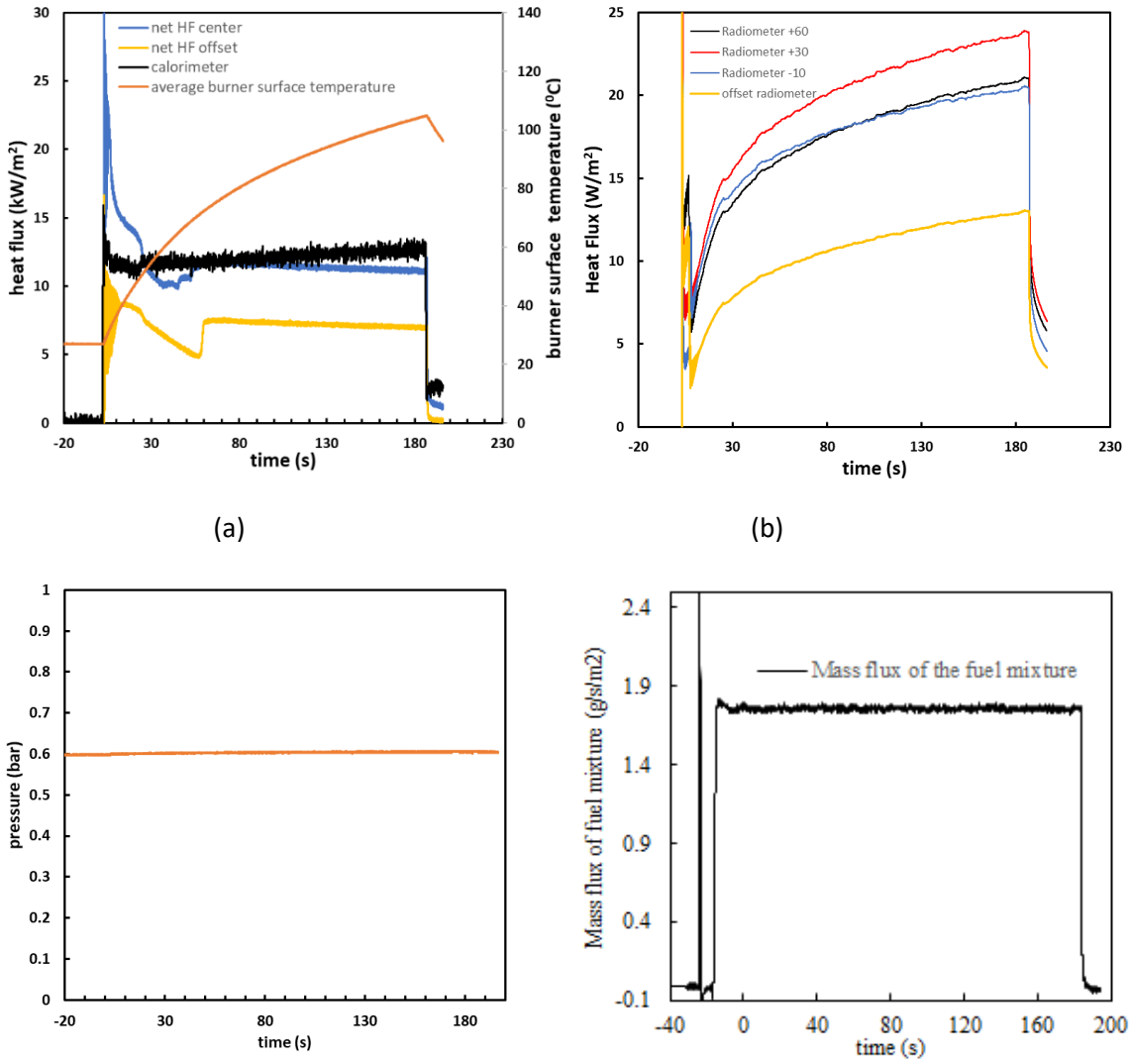


Figure 4. Average burner surface temperature and measured heat flux to the burner surface calorimeter (a), radiometer measurements (b), measured chamber pressure (c), and measured burner mass flux (d).

Table 2. Nominal test conditions of 19066D5 and 19070L1.

Test day	test point	Burner Diameter, mm	X_{O2}	Chamber Pressure, bar	X_{fuel}	measured mass flux, g/m^2-s
Mar. 7	19066D5	25	34%	0.56	0.5	1.76
Mar. 11	19070L1	25	27%	0.7	0.5	1.74

Our analysis of the radiometer sensor data has indicated its consistent ability to measure the flame overall radiation heat loss fraction. The method to determine the radiative heat flux component to the burner is more complex, and it is based on the geometrical details of the radiometers and burner surface relative distance and orientation. Also, the burner surface will re-radiate heat that impinges the radiometer and it needs to be corrected. All of the available radiation data are processed for the first round of testing and an empirical radiation correlation with flame height is found.

The video images of all the flames have been compiled and are being quantified in terms of flame shapes. Also, a summary booklet of all of the key findings along with heat flux and surface temperature data for each test was compiled.

Alexander Snegirev, our Russian partner, has attempted a numerical simulation of at least one of the ISS BRE tests. That work is still in progress. It should be remarked his student won the Student Prize for Best Presentation at the 9th International Seminar on Fire and Explosion Hazards held in St. Petersburg, Russia in April 2019 for simulating our 5-s BRE drop tests.

E. Preliminary Research Findings **(submitted by BRE PI Jim Quintiere and doctoral student Parham Dehghani)**

The burner has shown that quasi-steady burning is possible for real solids with a flat surface of 25 mm diameter. Up to now most flammability studies in microgravity have focused on flame spread with a wind for concurrent spread and at least a self-induced wind for opposed flow spread. We already knew that this would be possible from normal-gravity research. The burning of liquids, as in a wick of a candle flame and small droplets, has been shown to produce steady flames. However, many droplet flames were found to extinguish owing to radiative cooling of the flame. We see this as well in the extended ISS microgravity tests. However, this extinction was predominately seen for nominal ambient conditions of 21% oxygen and 14.7 psia. At conditions of 27% oxygen, 10.2 psia and 34% oxygen, 8.2 psia, the flames generally persisted for 3 minutes or more. These are all atmospheric conditions that NASA is considering for future spacecraft

Figure 5 shows a typical test of ignition and flame growth. Although extinction occurred at 45 s with the flame still growing, the heat flux was nearly steady by 30 s. Therefore, a heat of gasification (L) could be estimated. This was done by using the arithmetic average of the center and offset heat flux sensor as the net average heat flux.

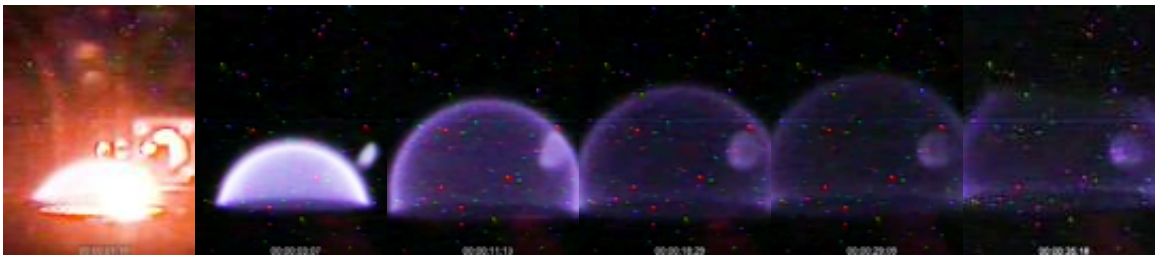


Figure 5. Flame growth from ignition to self-extinction in test 19056F1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 4.55 g/m^{2}s, 27% O_2, 10.2 psia, L = 0.66 kJ/g, extinction at 45 s.

Figures 6 through 9 illustrate results for nominal atmospheric conditions of 26.5% oxygen and 0.7 atm (10.2 psia), and 34% oxygen and 0.56 atm (8.2 psia), respectively with two figures each. Both use 50% mixtures of nitrogen and ethylene giving a fuel mixture heat of combustion of 24 kJ/g. Both of these tests reached quasi-steady conditions and were terminated by shutting off the fuel. The plots of transient heat flux shown by Figures 7 and 9, show the condensation effect removed after 80 s and 40 s respectively and arithmetic average of the two heat flux gauge measurements at the end of each test is recorded as steady surface heat flux which are used to obtain heats of gasification.



3 s 10 s 52 s 91 s 125 s 155 s 178 s

Figure 6. Flame growth from ignition to steady state in test 19070K1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 2.3 g/m^{2}s, 26.5% O_2, 10.2 psia, L = 3.0 kJ/g, steady.

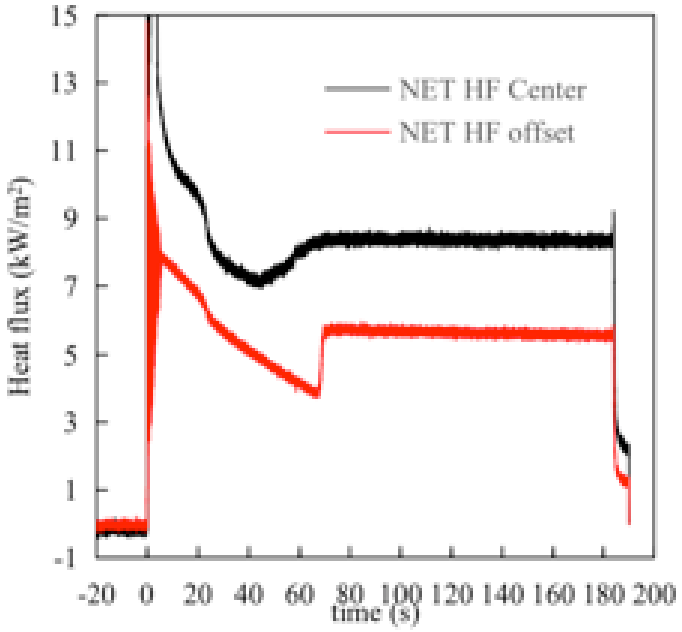


Figure 7. Measured heat flux in test 19070K1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 2.3 g/m^{2}s, 26.5% O2, 10.2 psia, L = 3.0 kJ/g, steady.

In Figures 8 and 9 the flame reaches a quasi-steady state in about 80 seconds as seen by the flame shape and the heat flux data. This emulation corresponds to a material with a heat of combustion of about 24 kJ/g and a heat of gasification of about 3 kJ/g. As such it could represent PMMA as a material solid. This is a profound result indicating a common plastic could steadily burn under these NASA proposed atmospheric conditions.



0.2 s 2 s 6 s 17 s 64 s 172 s

Figure 8. Flame growth from ignition to steady state in test 19070D1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Figure 9 shows steady results for a NASA proposed nominal atmosphere of 34% oxygen and 0.56 atm. After about 100 s the flame appears to be steady. The heat of gasification here is estimated as 2.42 kJ/g.

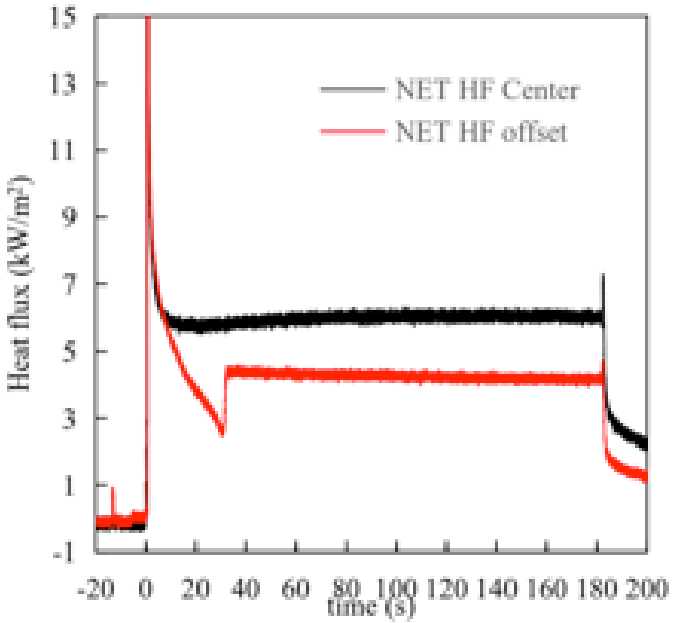


Figure 9. Measured heat flux in test 19070D1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Radiative loss fraction from the flames are obtained using the radiometer's measurements. Unfortunately, the data set is limited as the BRE sensors were saturated for several tests. The +60 mm and the offset radiometers were used to obtain the radiative loss fraction as these two sensors provided data in multiple cases while the other two radiometers were saturated. The transient radiative loss fraction for 19066D5 and 19070L1 are shown in Figure 10 for both radiometers, which show good agreement with each other. Table 2 provides information about the test conditions.

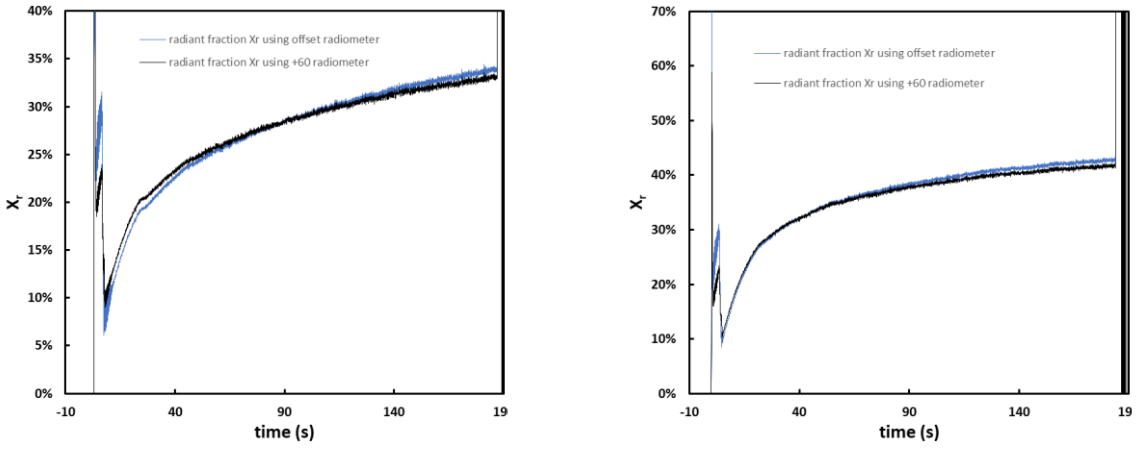


Figure 10. Transient radiative loss fraction for tests 19066D5 (a) and 19070L1 (b).

Flame heights at the end of the tests are measured from the image data. The radiative loss fraction is correlated with flame height and fuel mixture mass flux. The results from first round tests are shown in Figure 11.

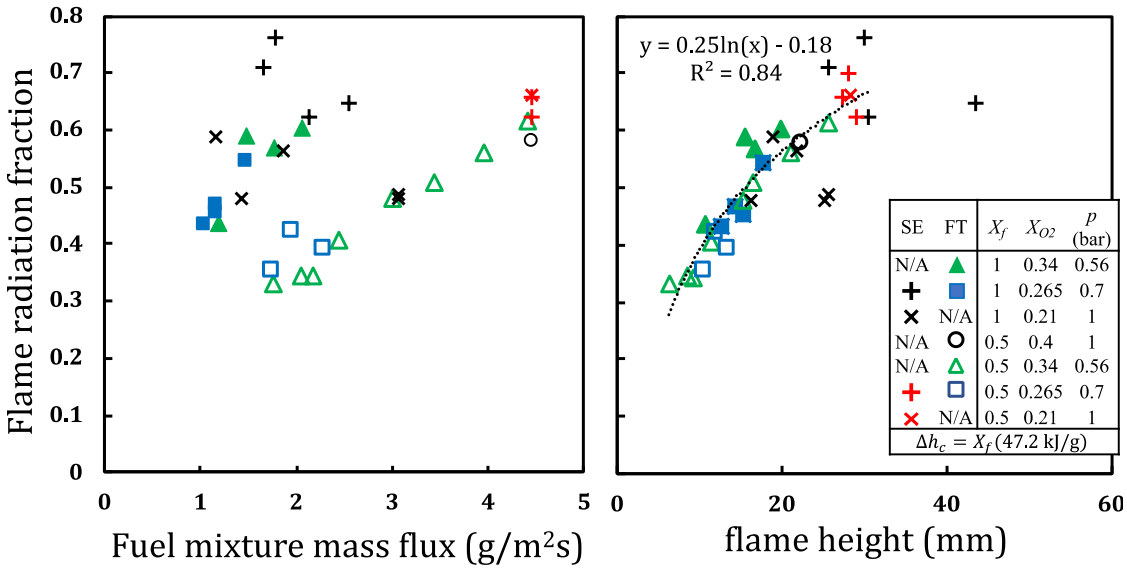


Figure 11. Flame radiant fraction vs. mass flux (a) and flame height (b) for all steady end-point results.

A simple analysis based on geometrical view factor of surface with respect to the flame sheet, allowed us to estimate the fraction of the total available heat that impinges surface. Also, the analytical model (the Ellipsoidal model) based on classical diffusion theory was used to estimate the convective heat flux to the surface which are shown by Figures 12 and 13, respectively. Adding up the convective, radiative and re-radiation from surface results in total average heat flux that gives theoretical heats of gasification when divided by fuel mixture mass flux.

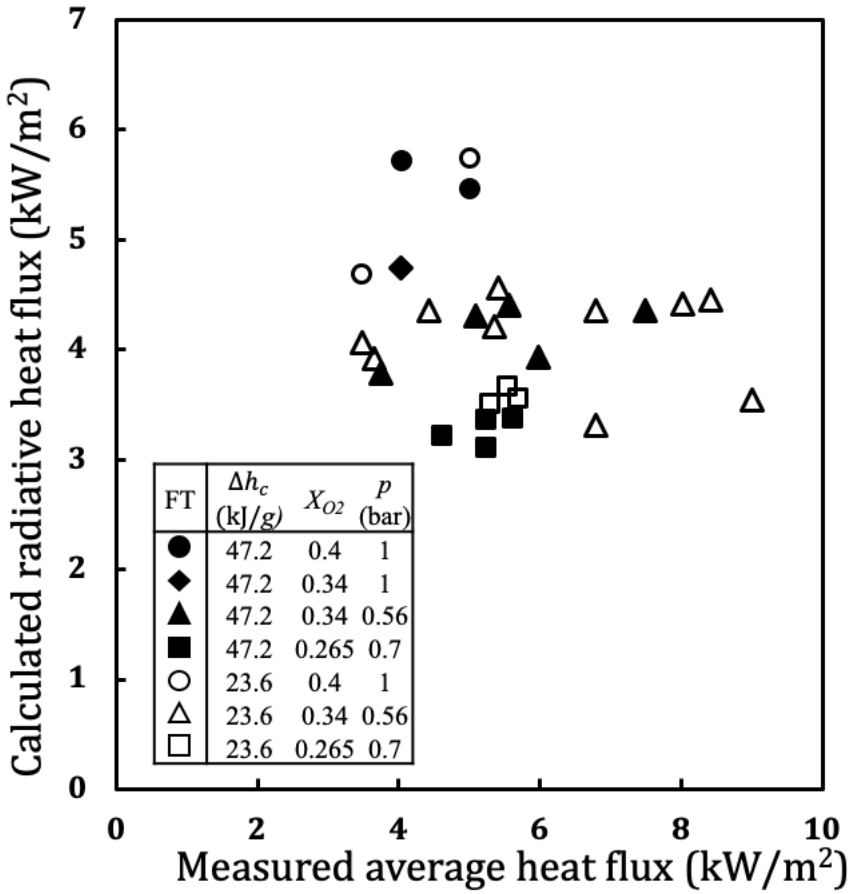


Figure 12. Radiative heat flux on the burner surface obtained by the View Factor method for nominal ambient conditions used for round 1 BRE tests on the ISS.

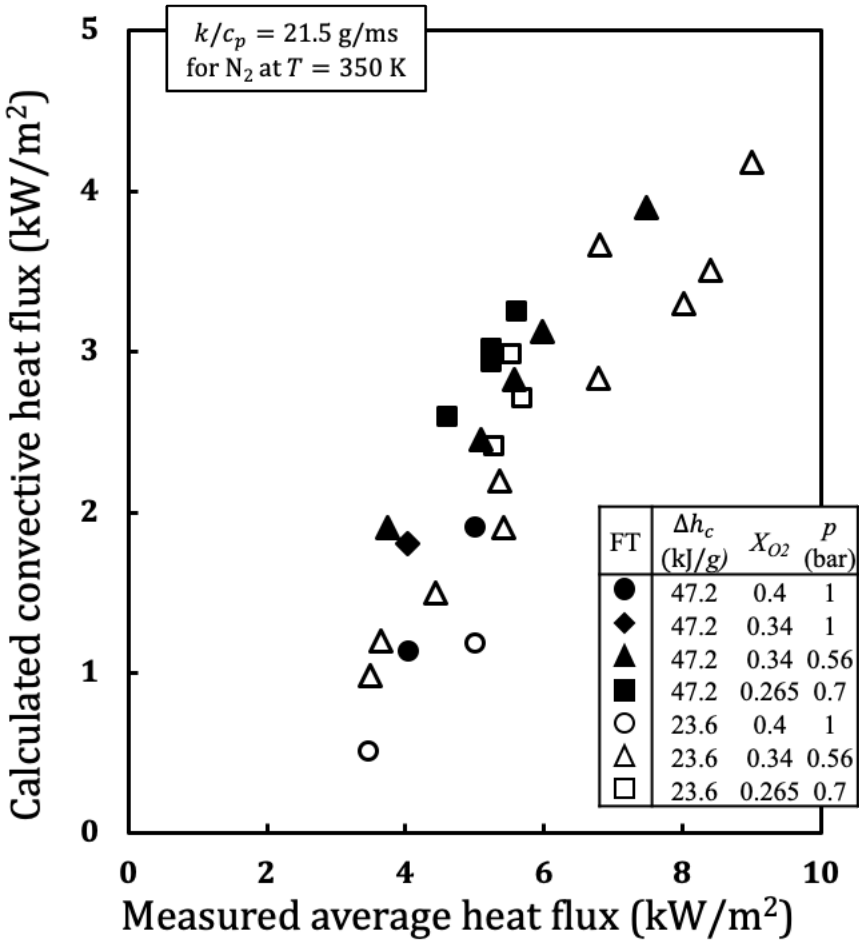


Figure 13. Convective heat flux on the burner surface obtained by the View Factor method for nominal ambient conditions used for round 1 BRE tests on the ISS.

The experimental flammability maps are shown in Figures 14 and 15. Both plots indicate a decreasing, inverse asymptotic-like curve that approaches zero, for heat of gasification with mass flux. It is surprising that the data cluster about a nearly singular trend. However, both plots show the heat of gasification is up to 20 % higher for the lower heat of combustion 23.6 kJ/g. Our experiments in University of Maryland lab has shown, with the BRE in 1-g, that similar results for propane-nitrogen mixtures have the opposite behavior with heats of gasification decreasing with heat of combustion for the same mass flux. But again, the data mostly cluster about one trend line except for very low heats of combustion. However, the 1-g corresponding heat of combustion data shift far to the right and down from the microgravity data of Figures 14 and 15. For example, the 1-g data would indicate ($\dot{m}$", L) points of about 30 g/m^{2}s, 2 kJ/g and 10, 0.5. Moreover, our previous work in a 5- s drop facility suggested similar data trends but with an apparent vertical asymptote at about 3 g/m^{2}s, rather than 1 as shown here. Obviously, the current data at durations of minutes are more accurate than those limited to 5 s. However, it has been shown that the ISS heat flux data even at self-extinction are theoretically nearly steady. Figures 14 and 15 include the heats of gasification derived from the self-

extinguished endpoint data as well. Remarkably, the self-extinguished not-so-steady data follow the same trend as the steady data.

Let us interpret Figures 14 and 15 in terms of the flammability of real materials in microgravity. Typical melting plastics have heats of gasification ranging from about 2 to 3 kJ/g, with effective heats of gasification for charring plastics above 4 kJ/g. Some selected literature values for materials with heats of combustion in the BRE range are listed in Table 3. The accuracy of the calorimeter heat flux is found to be less than the two sensors, thus, we relied on the arithmetic average of the two heat flux sensors data for obtaining the flammability map. However, same results are shown considering the calorimeter data. Figure 14. would suggest that polyethylene (PE) and polypropylene (PP) would burn in microgravity at about 3 g/m^{2}s. Nylon and PMMA would burn at about 2.5 g/m^{2}s, and the phenolic glass fiber composite would burn at about 1 g/m^{2}s.

Table 1. Materials close to BRE heats of combustion and corresponding heats of gasification

Material	*LHV*, kJ/g	*L*, kJ/g
PE	43.4	1.9
PP	44	2.0
Nylon	28.8	1.5, 2.4
PMMA	24.2	1.6, 2.3
Phenolic glass fibers	22.0	7.3

One important fire safety consideration, based on the findings of this study, is that solid materials with heats of combustion in the range 23.6 to 47.2 kJ/g and heats of gasification from about 1 to 10 kJ/g can burn in microgravity. The NASA alternative flammability Test 2 [1] using the Cone Calorimeter can measure these material properties. As this study develops a more complete understanding, NASA may wish to put more emphasis into Test 2.

Moreover, the flammability map based on our theoretical analysis is shown in Figure 16. It shows similar asymptotic trend and clearly indicates lower pressure and higher oxygen concentration as favorable conditions for long duration burning.

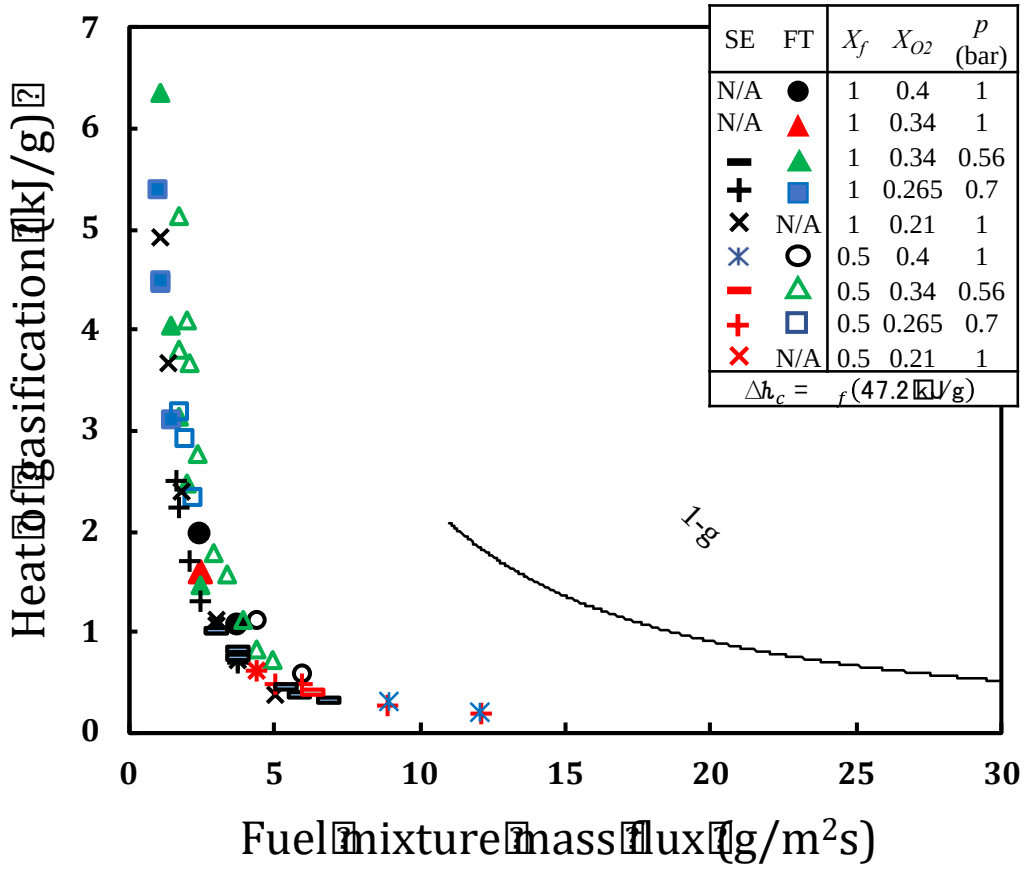


Figure 14. Round 1 results for the emulation of 25-mm flat materials using the arithmetic average of the two local heat flux measurements.

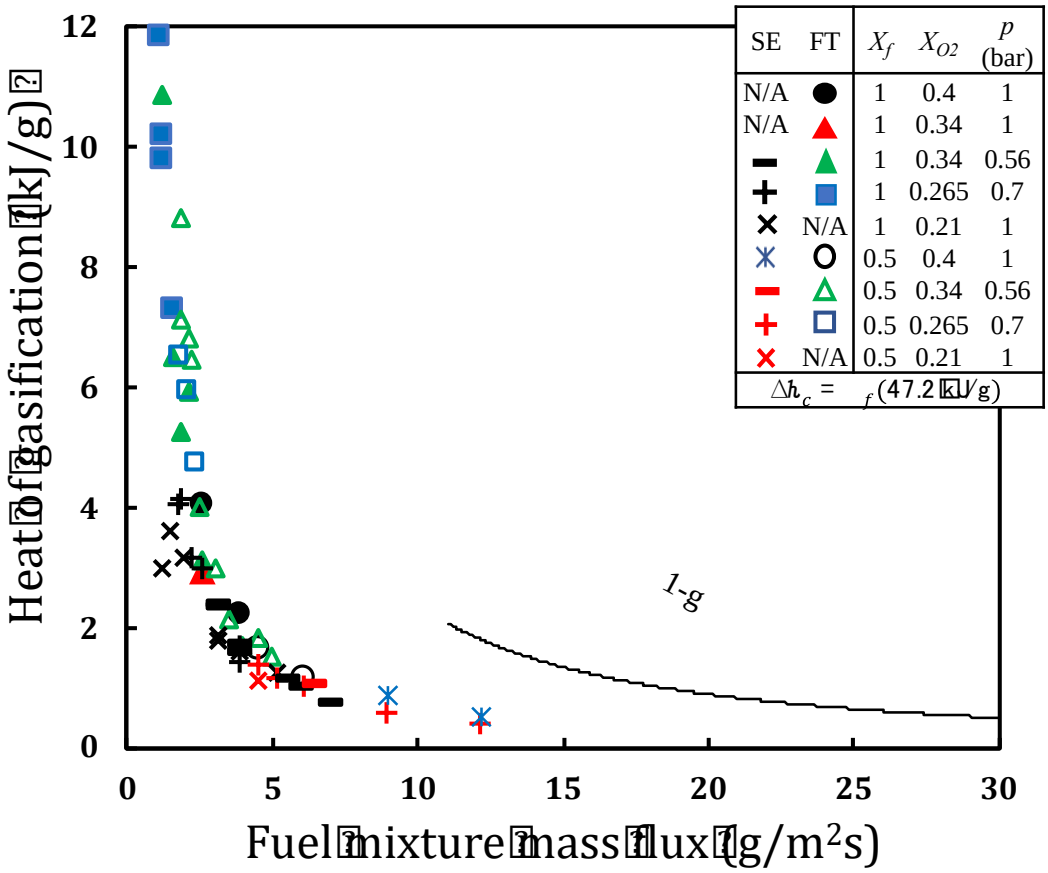


Figure 15. Heats of gasification calculated from the corrected calorimeter measurements vs. fuel mixture mass flux.

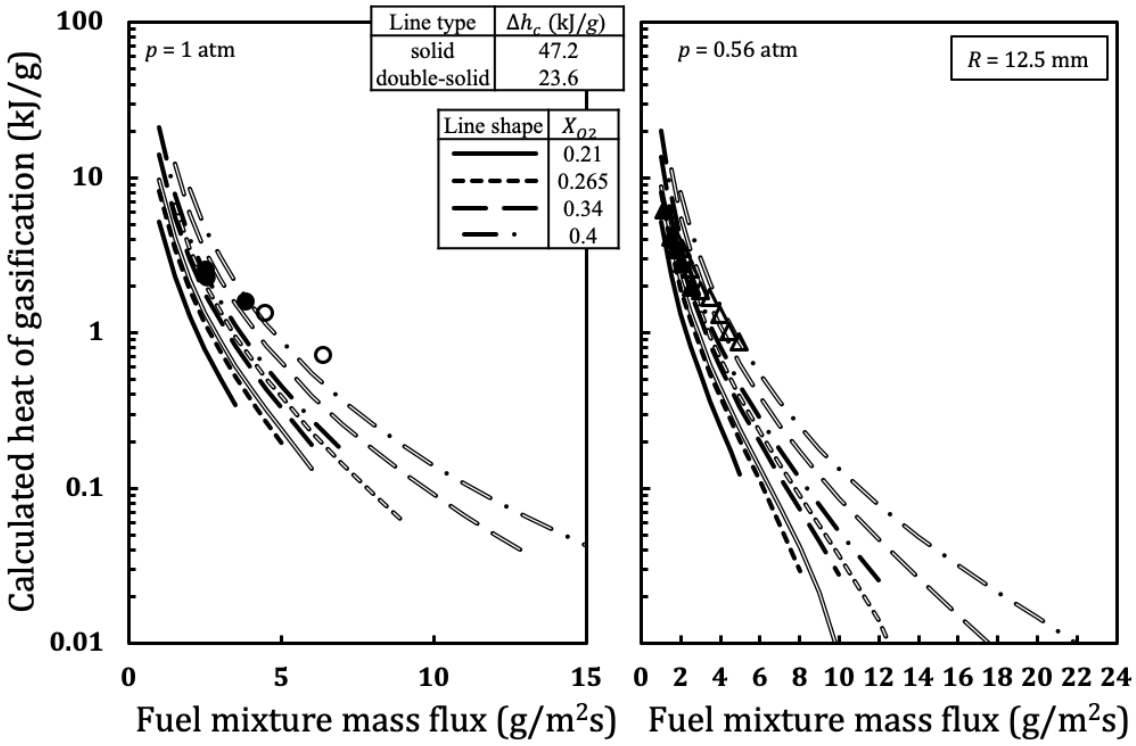


Figure 16. Calculated heats of gasification vs. mass flux for different oxygen mole fractions and p = 1 atm (left) p = 0.56 atm (right).

The imaging data from ops camera and ACME camera are used to create videos and image sequences for all of the tests taken so far. Preliminary analysis on the first three days of the second round of ISS testing data are performed. Many of the first day tests with the 50-mm BRE burner were not at the intended flow rate. A make-up day will be carried out to accomplish the day's scientific objectives. The second day plan contained a combination of pure and diluted ethylene at 3 different ambient conditions. Pure methane was the fuel used for testing on the third day at 3 different nominal ambient conditions. It was observed that methane was hard to ignite compared to ethylene and flames tend to self-extinguish faster even at a high oxygen concentration.

A new method is used to observe the asymptotic behavior of the heat flux data. The inverse of the burn-time was used as the abscissa and the transient data from the two heat flux sensors are shown in Figure 17 for one of the tests.

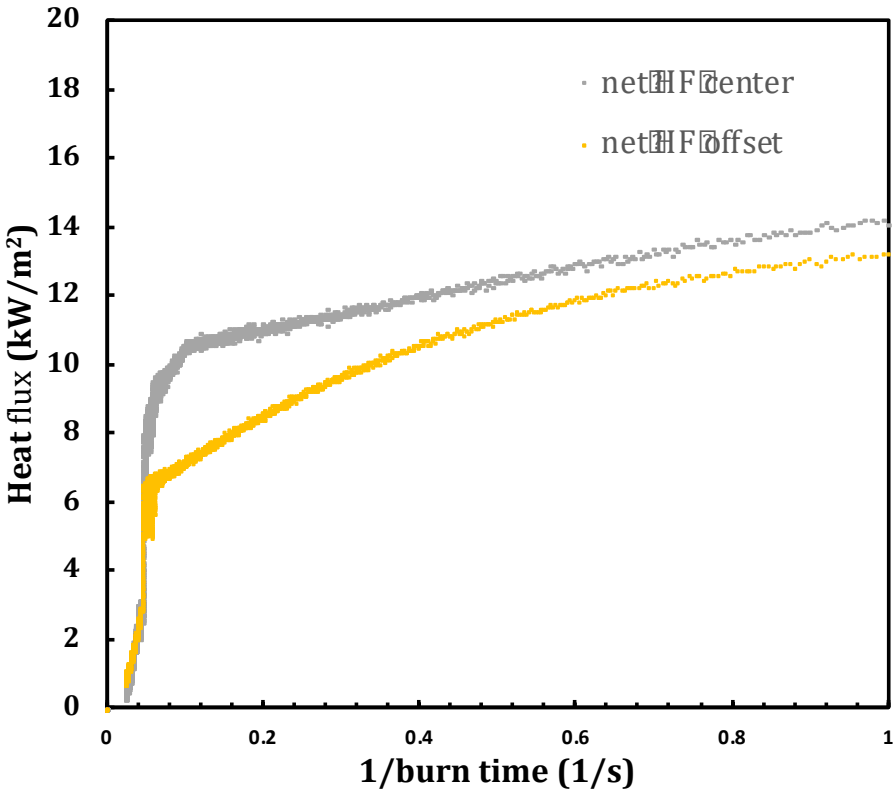


Figure 17. Transient heat flux vs. inverse of the burn time for 20279B4 with X_{O2} = 37.95%, p = 14.7 psia and mass flux of 2.46 g/m²s.

Our preliminary analysis on the heat flux data led to the flammability map (heats of gasification, *L*, vs. fuel mixture mass flux) shown in Figure 18 which contains all of the round 2 tests including the self-extinguished cases.

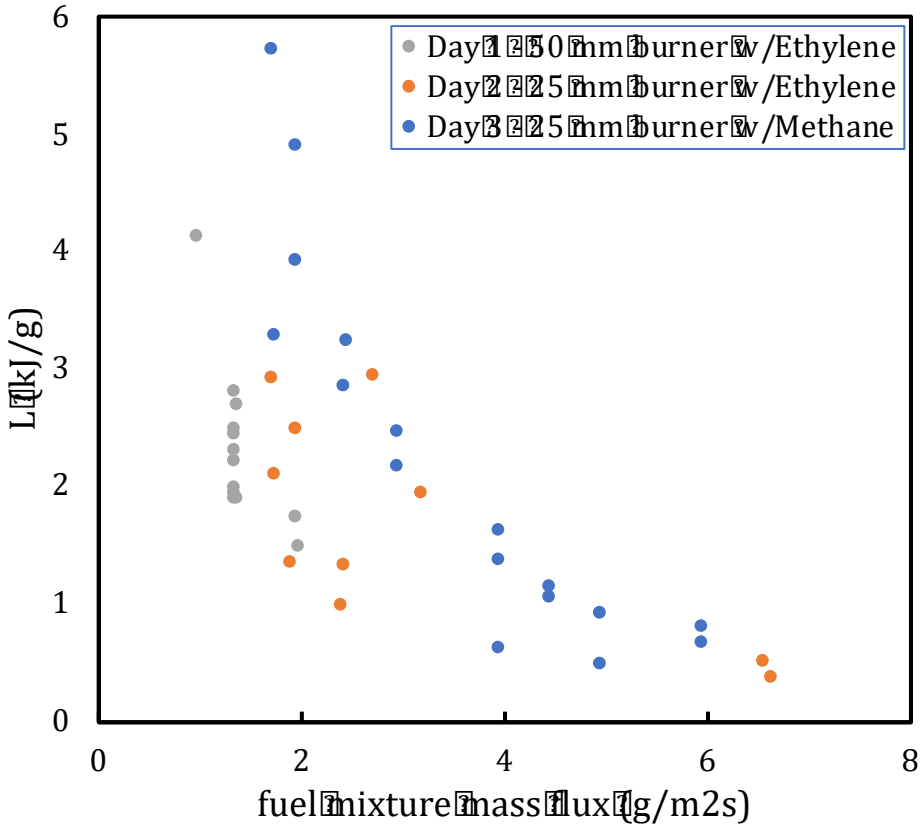


Figure 18. Preliminary flammability map based on the arithmetic average of the two heat flux sensors data.

Radiation data was limited to the tests taken with the smaller burner. The offset radiometer measurements are used to obtain the preliminary radiation fraction for the available data which are shown against fuel mixture mass flux in Figure 19.

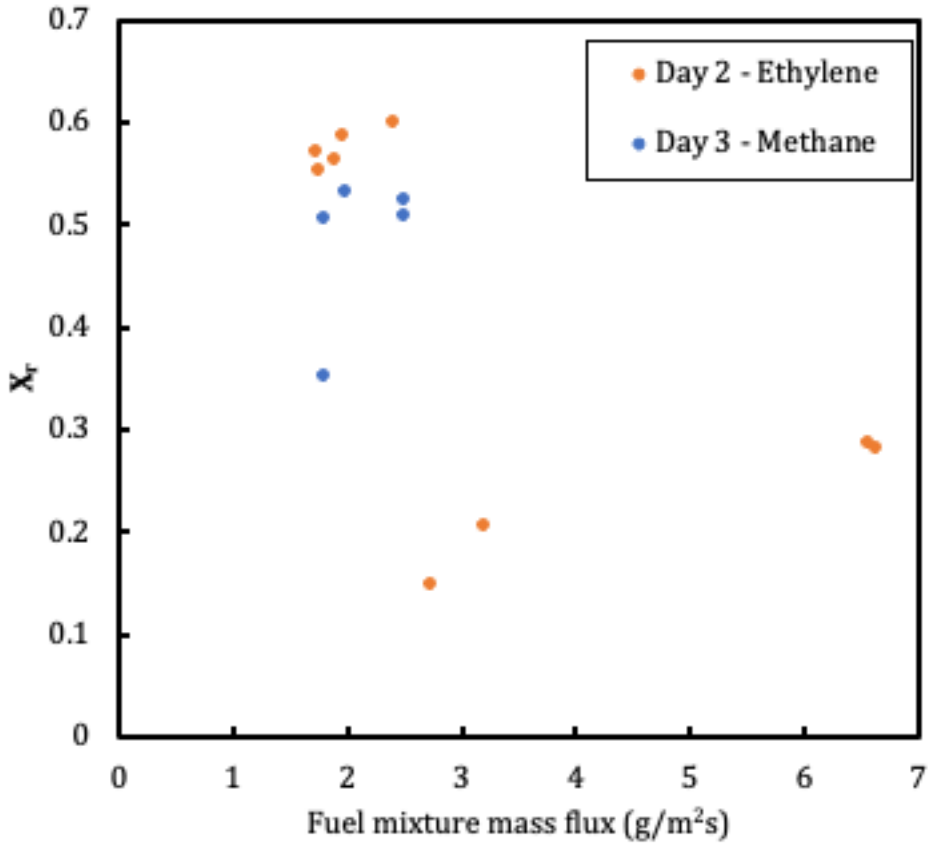


Figure 19. Radiation fraction vs. fuel mixture mass flux

Early ISS results have been submitted for publication, where the first is currently in press:

- A. Snegirev, E. Kuznetsov, E. Markus, P. Dehghani, and P.B. Sunderland, "Transient dynamics of radiative extinction in low-momentum microgravity diffusion flames," Proceedings of the Combustion Institute. https://doi.org/10.1016/j.proci.2020.06.110
- P. Dehghani, P.B. Sunderland, J.G. Quintiere, and J.L. deRis, "Burning in Microgravity: Experimental Results and Analysis," *Combustion and Flame*. (submitted)
- P. Dehghani and J.G. Quintiere, "Burning in Microgravity: Theoretical Predictions," *Combustion and Flame*. (submitted)

Early ISS results were presented with papers at the following conferences:

- Sergey Pichugin, Oleg Artemiev, Sergey Prokopiev, Oleg Kononenko, Oleg Novitsky, and Sergey Frolov, "Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience," XIII International Scientific and Practical Conference 'Manned Flights in Space,' Gagarin Cosmonaut Training Centre, Star City, Russia, 13-15 Nov. 2019.
- P. Dehghani, E. Auth, C. Cui, D.P. Stocker, J.L. de Ris, P.B. Sunderland, and J.G. Quintiere, Microgravity Observations of a Burning Emulator to Investigate Material Flammability, Spring Technical Meeting of the Eastern States Section of The Combustion Institute, Columbia, SC, 8-11 March 2020.

ISS results, without a paper, were orally presented as follows:

- James G. Quintiere and Peter B. Sunderland, "Burning Rate Emulator (BRE): Material Burning in Microgravity," ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019.
- P. Dehghani and P.B. Sunderland, "Burning Emulation in Microgravity - Experimental Results and Analysis," 36th Annual Meeting of the American Society for Gravitational and Space Research, 5-6 Nov. 2020.

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project, its hardware, and the value of combustion research can be found at: https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Videos related to BRE and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

The results of this study will have a substantial impact on the methodology used by NASA to assess the fire safety of materials in the future. [From BRE PI Jim Quintiere]

References

1. NASA-STD-6001B. Flammability, Offgassing, and Compatibility Requirements and Test Procedures. (2011). https://standards.nasa.gov/standard/nasa/nasa-std-6001

Appendix E. 30-Day Operational Accomplishments Report for ISS Increment 64

25 June 2021

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increment 64

ACME Advanced Combustion via Microgravity Experiments
BRE Burning Rate Emulator

BRE is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation was carried out in two separate rounds, where the first and second rounds were respectively conducted during Increments 58-59 and 63-64. While this report was prepared for Increment 64, it comprehensively addresses the operations in all four increments.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for BRE and ACME information.

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Co-Investigator (Co-I):	Prof. Peter B. Sunderland, U. of Maryland, College Park, MD
Co-Investigator (Co-I):	Dr. John L. de Ris, FM Global (retired)
Collaborator:	Prof. Alexander Yu. Snegirev, Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia
Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Uday G. Hegde, Case Western Reserve University, Cleveland, OH

I. Introduction

The Burning Rate Emulator (BRE) experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. BRE is focused on spacecraft fire prevention. More specifically, BRE's goals are to improve our fundamental understanding of materials flammability, such as extinction behavior and the conditions needed for sustained combustion, and to assess the relevance of existing flammability test methods for low and partial-gravity environments. A flat porous burner fed with gaseous fuel simulates the burning of solid and liquid fuels, where measurements are made of the thermal feedback (i.e., to the burner) upon which the vaporization of condensed-phase fuels is dependent. A small number of gaseous fuels (including mixtures with inert gases) are used to simulate the burning of fuels such as paper, plastic, and

alcohol by matching properties including the heats of combustion and gasification, the surface temperature, and smoke point. The experiment's results can establish fuel properties that directly relate to fire hazards in spacecraft, and if effective in microgravity could provide an efficient, safe, and comprehensive technique for assessing flammability and selecting fire-resistant spacecraft materials.

BRE's specific experimental objectives are as follows, where it must be emphasized that NASA does not currently plan to carry out the desired tests for objective B.

J. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21 % oxygen; 10.2 psia, 26.5 %; and 8.2 psia, 34 % oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels.
K. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired).
L. Consider the conditions of Objectives 1 – 2 for extinction limits.

The observations will enable quantification of the microgravity behavior of condensed fuels (i.e., liquids and non-charring plastics).

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flame, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

BRE and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the operations center at the NASA Glenn Research Center. The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it could alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, gas flow rate(s) can be adjusted after quasi-steady data has been collected. But testing has not been conducted in a 'manual' mode thus far for BRE or ACME in general.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

The first round of BRE tests began in February 2019, during Increment 58, when a total of 45 flames were ignited. That round was completed in April 2019, during Increment 59, when an additional 17 flames were ignited. BRE's second-round flame testing began in Sept. 2020, during Increment 63, in which 47 flames were ignited. The second and final round of tests was completed in Jan. 2021, during Increment 64, when another 85 flames were ignited. As a result, 62 and 132 flames were respectively ignited in the first and second rounds of testing for a total of 194 flames.

The 'interruption' in the BRE testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome. However, the break was longer than had been planned because of combination of hardware issues and limited crew availability. As a result, BRE's second round was rescheduled to occur after both rounds of the preceding ACME experiment, Structure and Response of Spherical Diffusion Flames (s-Flame), was completed, where that occurred in July 2020. A factor in that decision was the delayed start of commercial crew launches and understanding that crew availability would be limited until such crew launches entered an operational, rather than demonstration, phase.

All first-round tests were conducted with the 25-mm diameter BRE burner while burning ethylene (C2H4) - sometimes diluted with nitrogen. The tests were carried out at a range of chamber pressures and oxygen concentrations, generally representative of possible crew environments in future spacecraft or habitats. More specifically, testing was especially focused on atmospheric conditions at (1) 14.7 psia, 21% oxygen; (2) 10.2 psia, 26.5% oxygen; and (3) 8.2 psia, 34% oxygen. In all cases, the atmosphere was diluted with

nitrogen and any combustion products (e.g., water vapor and carbon dioxide) from preceding tests. An exception is tests conducted after the chamber atmosphere was vented and refilled, such as the first test of the day. But during tests days, the atmosphere was typically refreshed once or twice between sets of tests.
Second round testing was carried out including both BRE burner diameters, i.e., 25 and 50 mm, and on some days with methane (CH4) – again sometimes diluted with nitrogen. Therefore, the second round of tests featured a broader range of conditions than investigated during the first round. However, in contrast to the first round most of the round 2 tests were carried out at roughly 14.7 psia, although pressure has a secondary effect on flames. Meanwhile, the tests in this round were carried out at various oxygen concentrations as in the first round but were more often conducted with ~40% oxygen, where oxygen of course has a strong effect on combustion.

BRE does not have living subjects like a life science experiment. The tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C2H4), is normally highly sooting because of its carbon double bond. Tests are conducted with pure fuel and fuel mixed with nitrogen, where the fuel and nitrogen were premixed on the ground prior to launch for some cases. But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations.

There are no samples (e.g., of soot) collected for BRE, where all ISS results are electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for BRE during Increments 58 and 59 (the first round of tests) and Increment 63 and 64 (the second round) is found in Table 1. The table includes a limited overview of the test conditions. For completeness, a row was included for testing on 1 April 2019 (GMT 091) even though it was quickly discovered that the hot-wire igniter tip had failed at the end of the preceding test day. As a result, flames were not ignited until a few days later after astronaut Nick Hague had replaced the igniter tip.

Table 1. BRE log of flames ignited in Increments 58-59 (round 1) and 63-64 (round 2)

Test Day[1]	Year	Month	Date	Burner Size[2]	Fuel bottle	Oxygen conc.[3]	Flames ignited[4]
First round of tests, with the 25-mm burner, from Feb.-April 2019							
19039	2019	Feb.	8	25-mm	100% C2H4	21%, 26.5%, 40%	10
19051			20			26.5%, 34%	8
19056			25		50% C2H4	21%, 26.5%, 34%, 40%	8
19063		March	4			26.5%, 40%	6
19066			7			34%	6
19070			11			26.5%, 34%	7
19087			28		100% C2H4	26.5%, 34%	8
19091		April	1			21%	0
19094			5			21%, 26.5%, 34%	9
Second round of tests, with the 25-and 50-mm burners, from Sept. 2020 to Jan. 2021							
20247	2020	Sept.	3	50-mm	100% C2H4	25%, 34%, 40%	13
20261			17	25-mm		26.5%, 34%	13
20279		Oct.	5		100% CH4	25%, 34%, 40%	21
20339		Dec.	4			25%, 40%	18
20346			11		50% C2H4		12
20352			17		100% C2H4	34%, 40%	10
20358			23			25%, 34%, 40%	17
21020	2021	Jan.	20			21%, 26.5%, 34%	15
21028			28	50-mm		25%, 34%, 40%	13

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Burner size** refers to the diameter of the BRE burner's perforated circular outlet.
[3]**Oxygen conc.** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations for tests that ignited are listed. *The chamber pressures were also varied but are not identified in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

There were no significant anomalies hindering BRE operations during Increments 59 and 63, but there was one in Increment 58 during the preparations for BRE testing (that was mistakenly not addressed in the 30-day report including that increment). It was discovered on Feb. 4 (GMT 035) that the pressure was low in a check gas bottle for the CIR's Gas Chromatograph Instrumentation Package (GCIP). The bottle had

been on the space station for 9 years, and the low pressure was attributed to a nominal leak rate. The check gas was for calibration of the GCIP measurements and not needed for regular operations which proceeded unhindered. The bottle was replaced on June 13 to enable the necessary calibration.

The anomaly in Increment 64 was similar in that there was a slow leak of the GCIP's argon carrier gas. As a result, the bottle valves for all of the instrument's gas bottles were closed on Dec. 29, 2020 (GMT 364) making those measurements unavailable for the BRE's last three test days.

2. Any other difficulties in obtaining scientific data

BRE's primary problem was that the sensors for measuring the thermal radiation often saturated. Although the measurements were not required to meet experiment success, they were considered sufficiently critical that BRE was equipped with its own radiometer array, featuring three thermopile detectors. Estimates had been made of the signal levels based on 5-second drop tests since the hot gases tend to blanket the flame in the absence of buoyancy and thereby remain within the view of the radiometers. As a result, the signal levels are strongly dependent on gravity. The estimates were not part of the science requirements and may not have been sufficiently tracked in the hardware development. Furthermore, the ISS tests lasted as long as 5 minutes, which is over four times the envisioned BRE test duration and 60 times the duration of the drop tests. With the fixed amplification of the radiometer signals, the data was frequently lost - other than the knowledge that the limits were exceeded. An engineering review failed to identify an acceptable solution and BRE testing continued unchanged during the first round, i.e., during increments 58 and 59.

> ***From the BRE PI Jim Quintiere:***
> The radiometry array proved problematic as its sensors were mostly out of range and quickly became saturated. However, for some of the small size flames, these data were complete. Those data will be vital to assessing the role of flame and combustion product radiation.

Saturation of the radiometers was partially mitigated in the round 2 testing, during increments 63 and 64, through use of ACME's standard array instead of the one developed specifically for BRE.

Ignition was not always successful. As previously discussed, it was discovered on 1 April 2019 (GMT 091) that the hot-wire igniter tip had failed on the preceding test day, preventing further flame testing until it could be replaced. While such circumstances are uncommon for ACME operations, BRE tests were sometimes conducted under conditions, e.g., for igniter activation and deactivation, which led to momentary flashes without sustained ignition. Given that unplanned crew time is required to replace failed igniter tips, the BRE ops team limited - to varying degrees - the igniter voltage and the time that it was energized within the flame. The latter was

known from earlier research in the Combustion Integrated Rack (CIR) to degrade the hot-wire igniter tip and hasten its demise. It is notable that the flashes of flame (across the burner surface) sometimes continued briefly after the igniter was deactivated and retracted. In one case on the first day of testing, there was excess fuel and a 'fireball' was ignited which promptly quenched. Ignition also sometimes failed due to a software limitation where the fuelFlowIgnitionDeltaMs parameter which specifies the relative timing of the fuel flow and igniter activation must be evenly divisible by 100 (ms). This limitation wasn't initially understood and the igniter didn't energize as a result.

From the BRE PI Jim Quintiere:
It was realized that the igniter for the burner gases could only be applied for up to 4 seconds, on insertion and retraction. This produced an unanticipated problem, as the time for the burner gases to reach the igniter could be tens of seconds depending on the flow rate. Also, after the first test, the burner would contain that fuel mixture mixed with the chamber gas. Thus the burner would need to be flushed with nitrogen before the second fuel mixture was initiated.

Intense communication with the NASA team proceeded the day before the first test. The gas volume of the burner along with the feed pipe volume was supplied by NASA. Accordingly ground experiments at UMD were conducted with an identical NASA BRE burner along with its piping volume from the flow valve. These experiments and theory were used to develop an algorithm for determining the relative timing of the igniter and gas flow activation that will result in ignition.

Day 1 [of the first round] was conducted with rough estimates made on the fly, and fortunately nearly all of the tests led to a satisfactory ignition. By Day 2 an algorithm was established and successfully used for the remainder of the testing. This algorithm will need to be refined and applied for the larger 50 mm diameter burner that will be used in Round 2.

A document on the development of the algorithm and its refinement for the larger burner [was] prepared and … shared with the NASA team to facilitate [ignition in] round 2.

The second day of round-1 flame testing was delayed to allow for the development of a new ignition procedure, where concerns arose that the mixture of gas flowing from the burner wasn't sufficiently well understood, i.e., in terms of concentrations. This concern had to do with the burner's large internal volume relative to that of the burners used for the other ACME experiments. To be clear, this issue had to do with the analysis of results rather than the achievement of ignition. Re-planning was also necessary early in BRE testing because the test durations were often much longer than had earlier been envisioned by the investigators. While these factors didn't directly interfere with the acquisition of test data, they did slow progress.

While it rarely occurred, BRE tests sometimes automatically aborted (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time associated with the failed test attempt. Such aborts typically occur because of communication issues with an Image Processing and Storage Unit (IPSU).

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel and nitrogen have unfortunately not always performed adequately, where the flow rates have not always been appropriately steady. The resulting effects on the flame behavior are tentatively assumed to be minor, but this still needs to be assessed.

While it didn't prevent the acquisition of scientific data, the intended igniter arm, which establishes the relative positioning of the igniter tip and burner, was not found for the round-2 hardware set-up. Therefore, an alternate arm, which positioned the igniter further from the burner than intended, was used for the 3 Sept. 2020 testing. Fortunately, the flames were still reliably ignited and the proper igniter arm was found and installed prior to the subsequent test day.

However, it was determined that the 3 Sept. 2020 science objectives were not accomplished because the fuel didn't always transition to the intended test flow rate. As a result of concerns about the ignition success because of the use of the alternate igniter arm, the decision was made to repeatedly ignite the flame at the same fuel flow rate and then subsequently adjust it to the desired test flow rate. The first attempt was successful, but the transition time was shortened to save fuel, etc. which unfortunately prevented the flow transition. In other words, the transition time was too short where that wasn't discovered until review of the data days later. As a result, the tests were generally conducted at the same fuel flow rate, where that was meant to be a variable. As a result, the data currently doesn't allow a full assessment of the effect of the burner size on the flammability. Therefore, a make-up test day in Increment 64 has been planned to address this shortcoming.

During testing on 23 Dec. 2020, it was discovered that the lowest of the 14-micron silicon carbide fibers on the Thin Filament Pyrometry (TFP) fiber array had broken. After some discussion, testing continued that day. Given that the translating array had five ceramic fibers, the loss of one was not a significant science loss. But it was a concern for crew safety. Recovery plans were developed and enacted on January 18 to remove the broken fragments from the harp-like array. Unfortunately, the next fiber was broken during the delicate operations where the fibers are only spaced 5 mm apart. As a result, the last two BRE test days were conducted with only three rather than five fibers in the TFP array. ACME does have alternate arrays, but also has months of remaining testing, so the decision was made to continue testing with only 3 fibers.

C. Completeness/Quality of Data

The 194 flames ignited for BRE significantly exceeds the 73 tests which were specified as required in the ACME Integrated Science Requirements Document (ISRD). As such, more than 2.5 times as many flames were ignited than had been planned.

It should be understood that a complete set of image data is not necessarily available for all tests. Recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done to maximize the scientific return of each day of testing within the project constraints.

> ***From BRE Co-I Peter Sunderland:***
> The video and image data of the testing produced satisfactory data to display the nature of the burning sequence from ignition to self-extinction or fuel termination. However, the image quality is less than was hoped for. The ACME ops camera provides video, but only offers 0.3 megapixels of resolution, and for many flames it is too zoomed out. [As a clarification, the ops camera has a fixed lens and its primary purpose is not flame measurements.] The GigE camera has poor sensitivity in the blue, and after the ignition transient all our flames have been blue. This camera only offers 1.4 megapixels of resolution and requires about 1 s exposures for blue flames, which is far too slow to be considered video. The images provided by both cameras are, according to UMD marketing staff, at too low resolution to be used in press releases. In round 2 we will increase the gain on the GigE camera, an option we were unaware of in round 1.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

See the discussion in III.D regarding fouling or degradation of the paint on the BRE burner's heat flux sensors.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of such effects and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). Note that although the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS) it was not utilized for BRE (or any ACME) testing because of the crew time needed to enable and disable the isolation before and after testing.

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and BRE's first round is considered to be successful despite the frequent saturation of the radiometer data discussed in section III.B.2. Meanwhile, the second round is tentatively considered to be highly successful, where final assessment will depend on further review and analysis of the data.

D. Status of Data Analysis (submitted by BRE PI Jim Quintiere and doctoral student Parham Dehghani; lightly edited by Dennis Stocker)

All measured transient raw data (e.g., heat flux sensors, thermocouples, radiometers, mass flow meters and chamber pressure sensor) have been converted to engineering units and plotted using the calibrations provided in ACME's Software Interface Design Description (ACME-DOC-010 Rev. D).

All of the round 1 and 2 tests have had the heat flux sensor data reduced using their original calibrations. As presented below, the data is being processed to address the effect of the combustion product water condensation on the sensor surfaces.

The primary measurements are used, by two techniques, to obtain the net heat flux from the flame to the burner surface. In the steady state, this net heat flux divided by the mass flow rate per unit area of the fuel mixture gives the heat of gasification of the emulated fuel. The key hypothesis is that the heat of combustion, smoke point, and heat of gasification of the fuel mixture will represent a real solid fuel (or liquid fuel absorbed in an inert matrix). Thus, achieved steady burning would represent the potential for real solid fuels to achieve steady burning in microgravity. Other than for a candle flame, this has not been shown before.

The first technique uses two thermopile heat flux gages embedded in the burner surface. One gage is at the center the other is offset from the center (r = 9.5 mm). As the sensors are not cooled, but rise with temperature as the flame grows, water from its combustion products can condense and then evaporate from the sensor. This occurs, as the dew point for combustion products is about 60-70°C, and the sensor begins at ambient temperature and can achieve temperatures over 100°C. Thus, condensation can occur at temperatures below the dew point that lead to added heat to the sensor. Above the dew point, the condensed water will evaporate and reduce the heat to the sensor. When the sensor is dry and above 100°C, it will only measure the heat flux from the flame gases as intended.

We have duplicated this condensation effect in 1-g laboratory experiments with the duplicative NASA BRE burner. Fig. 1 illustrates the condensation and evaporation effect in the ISS tests. The offset sensor in red shows a clear jump increase when the water fully evaporates. More analysis is needed to properly correct for this effect. Fortunately, by the end of the test shown here (and for all of the other data), this water effect is gone.

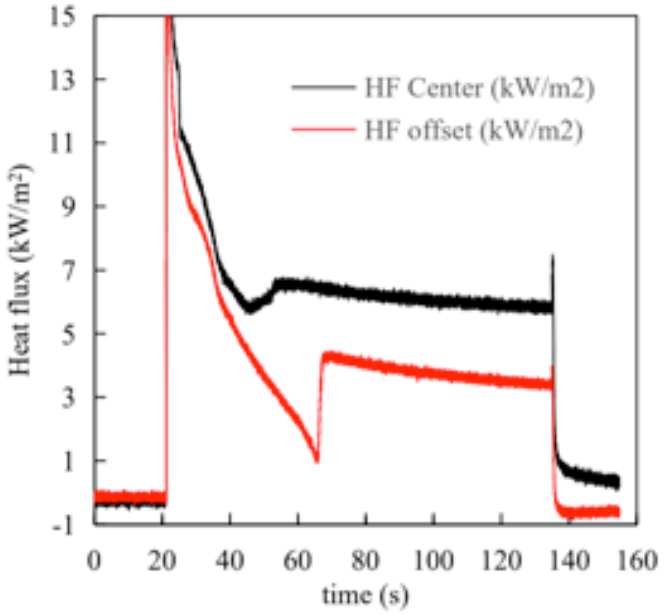


Figure 1. Typical net heat flux of the heat flux sensors.

The second effect on the heat flux sensors is that the heat may have fouled their paint. This could then affect their calibration. Figure 2 shows an image of the ISS burner after round 1 test were concluded. It can be seen that the paint on the sensors is degraded. This was the original manufacturer's paint, instead of that used for the copper top. The copper paint appears fine. We worked with the heat flux gage manufacturer to explore the effect of heat damage on the calibration of the sensors. The results showed that the calibration was stable and didn't change during the process of heating and cooling.



Figure 2. Paint degradation on heat flux sensors.

The second technique to determine the heat flux is the copper calorimeter forming the top of the burner. The temperature rise of the copper is key to that determination.

Figure 3 shows that typically the copper temperature is nearly uniform and smooth in time. It does not appear that condensation or evaporation is having any effect on its temperature behavior. So, we believe we are justified in using this measurement to predict the overall average heat flux from the flame to the burner.

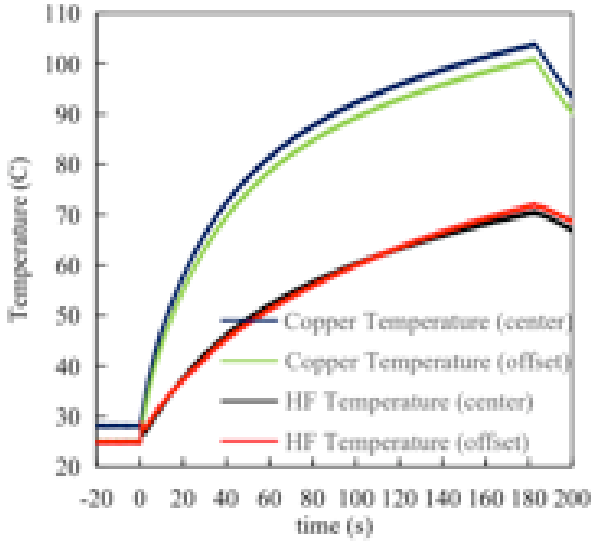


Figure 3. Copper and heat flux sensor temperatures in test 19070D1.

Recently it has come to our attention that the actual thermocouple and wiring types were as summarized in Table 2, below. We used type K conversion coefficients for south (i.e., center copper) and far south (i.e., offset copper) thermocouples and type T for heat flux transducers respectively from NIST data set provided in this link:

https://srdata.nist.gov/its90/download/allcoeff.tab

Table 2. Thermocouple and wiring types for BRE burner.

BRE burner position	Thermocouple type	Wiring type
Offset heat flux transducer	T	K (mismatched)
Center heat flux transducer	T	B (mismatched)
South thermocouple	K	K (matched)
Far south thermocouple	K	B (mismatched)

One sees that only the south (i.e., center copper) thermocouple has consistent type K wiring throughout. This latter thermocouple recorded an initial temperature of 28 deg. Celsius from which we can infer that initial ambient temperature must be 28 deg. C applicable to all thermocouples.

Table 2 shows the center transducer thermocouple was mismatched with a type B extension wire, and the offset transducer was mismatched with a type K extension wire.

Meanwhile the far south (i.e., offset) copper plate thermocouple was of type K, but was mismatched with type B extension wiring.

Ambient temperature before ignition is applicable to all thermocouples that can be obtained by using the igniter side and relay side thermocouples. This thermocouple and its extension wiring are of type K. The BRE burner thermocouples output before test starts must match igniter side's temperature.

According to the *Intermediate Metal Law* the extension wires do not affect the voltages since they are located at a constant temperature throughout their length. Therefore, each thermocouple voltage is converted to temperature based on their thermocouple type (shown in Table 2).

Since the south thermocouple output is the only one that shows higher temperature compared to other three and the igniter side temperature, it is concluded that this thermocouple is receiving a constant extra voltage of 0.127 mV. Therefore, this spurious voltage should be subtracted from the south thermocouple voltage. The resulting temperatures are shown in Figure 4.

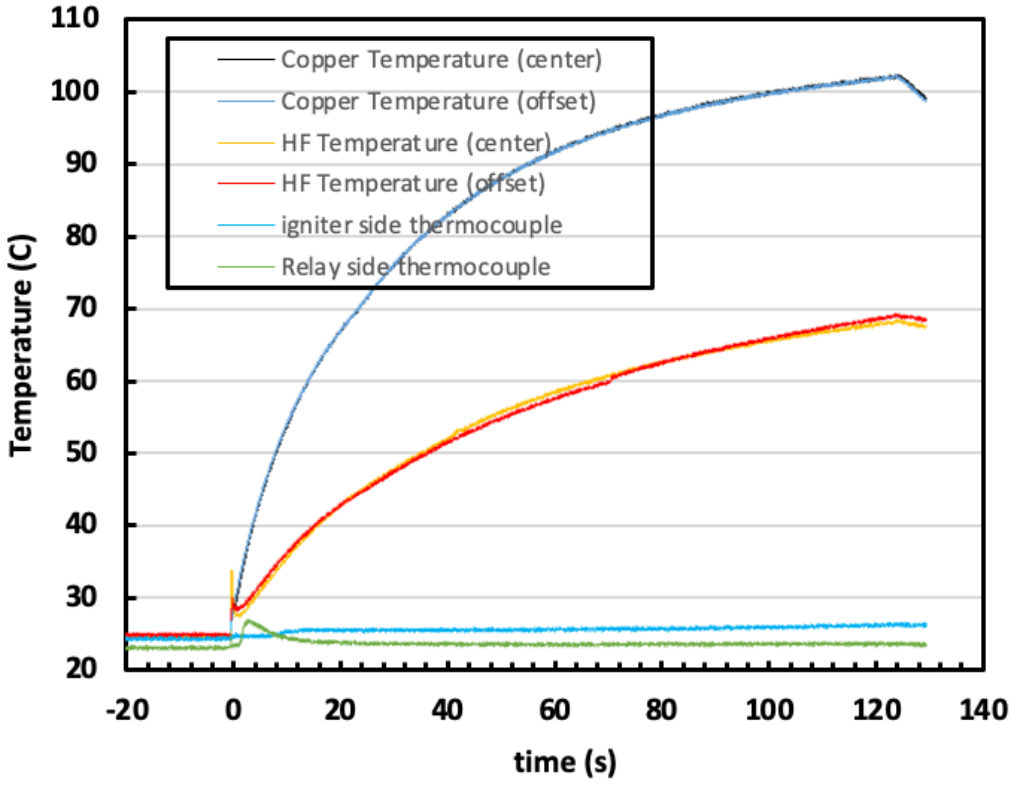


Figure 4. Temperature measurements after modifications made for the heat flux transducers and far south thermocouple.

The calorimeter required a new calibration to account for heat being absorbed through the bevel edge of copper plate. So, control surfaces are now adjusted (i.e., the edge and

burner back surface area, A_{back}) and then optimized for new set of calibration constants which are specific to the burner used in ISS tests (i.e., mc_{Cu}, h_{back}, h_{rod}, and e_0). The calorimeter represents the average heat flux to the surface and is being used to obtain heats of gasification. Figure 5 shows the transient plots of test 19066D5. The test conditions for the plots used in this report are presented in Table 3. We found out that there is significant heat absorption by the bevel edge of the burner that induces error in the calorimeter heat flux. A theoretical analysis was employed to estimate the radiative, convective and re-radiation heat flux to and from the top surface and the bevel edge of the burner which helped in estimating the error induced in the calorimeter data.

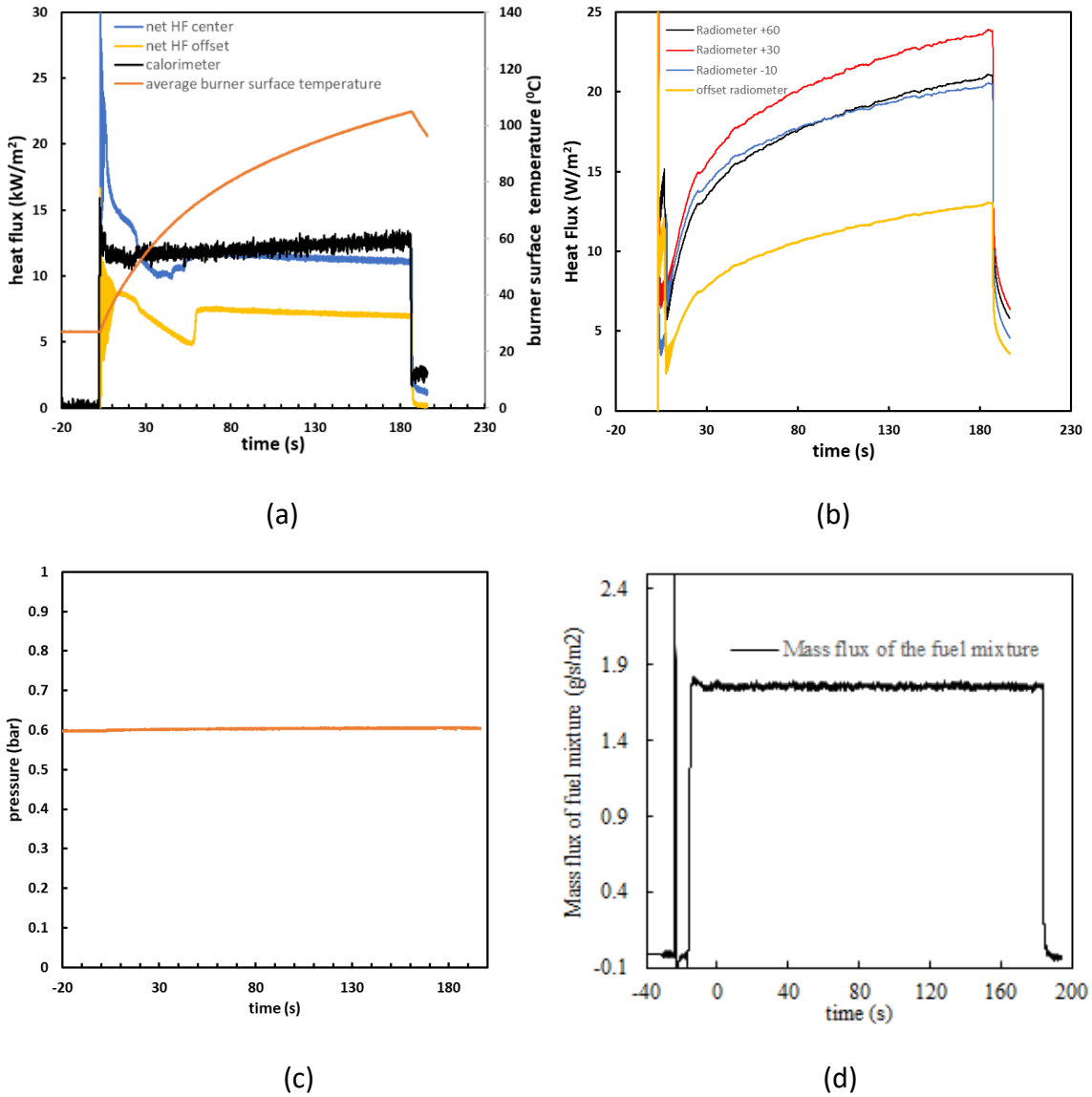


Figure 5. Average burner surface temperature and measured heat flux to the burner surface calorimeter (a), radiometer measurements (b), measured chamber pressure (c), and measured burner mass flux (d).

Table 3. Nominal test conditions of 19066D5 and 19070L1.

Test Day (in 2019)	Test point	Burner dia. (mm)	Oxygen conc.	Pressure (bar)	Fuel conc.	Mass flux (g/m^2s)
Mar. 7	19066D5	25	34%	0.56	50%	1.76
Mar. 11	19070L1		27%	0.7		1.74

Our analysis of the radiometer sensor data has indicated its consistent ability to measure the flame overall radiation heat loss fraction. The method to determine the radiative heat flux component to the burner is more complex, and it is based on the geometrical details of the radiometers and burner surface relative distance and orientation. Also, the burner surface will re-radiate heat that impinges the radiometer and it needs to be corrected. All of the available radiation data are processed for the first round of testing and an empirical radiation correlation with flame height was found.

The video images of all the flames have been compiled and are being quantified in terms of flame shapes. Also, a summary booklet of all of the key findings along with heat flux and surface temperature data for each test was compiled.

Alexander Snegirev, our Russian partner, has attempted a numerical simulation of at least one of the ISS BRE tests. That work is still in progress. It should be remarked his student won the Student Prize for Best Presentation at the 9th International Seminar on Fire and Explosion Hazards held in St. Petersburg, Russia in April 2019 for simulating our 5-s BRE drop tests.

E. Preliminary Research Findings **(submitted by BRE PI Jim Quintiere and doctoral student Parham Dehghani; lightly edited by Dennis Stocker)**

The burner has shown that quasi-steady burning is possible for real solids with a flat surface of 25 mm diameter. Up to now most flammability studies in microgravity have focused on flame spread with a wind for concurrent spread and at least a self-induced wind for opposed flow spread. We already knew that this would be possible from normal-gravity research. The burning of liquids, as in a wick of a candle flame and small droplets, has been shown to produce steady flames. However, many droplet flames were found to extinguish owing to radiative cooling of the flame. We see this as well in the extended ISS microgravity tests. However, this extinction was predominately seen for nominal ambient conditions of 21% oxygen and 14.7 psia. At conditions of 27% oxygen, 10.2 psia and 34% oxygen, 8.2 psia, the flames generally persisted for 3 minutes or more. These are all atmospheric conditions that NASA is considering for future spacecraft

Figure 6 shows a typical test of ignition and flame growth. Although extinction occurred at 45 s with the flame still growing, the heat flux was nearly steady by 30 s. Therefore, a heat of gasification (L) could be estimated. This was done by using the arithmetic average of the center and offset heat flux sensor as the net average heat flux.

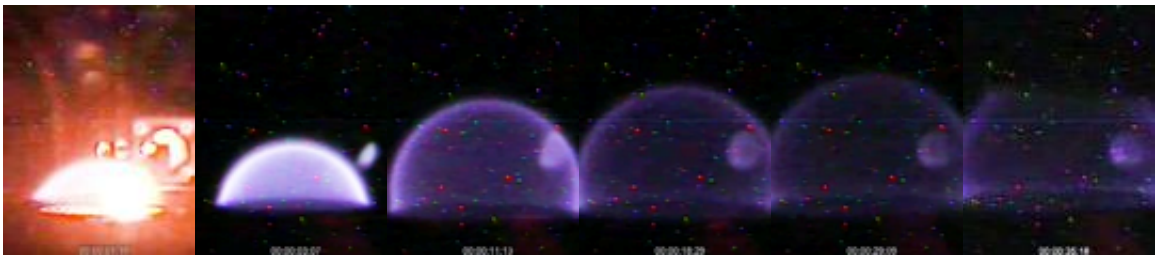


Figure 6. Flame growth from ignition to self-extinction in test 19056F1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 4.55 g/m^{2}s, 27% O_2, 10.2 psia, L = 0.66 kJ/g, extinction at 45 s.

Figures 7 through 10 illustrate results for nominal atmospheric conditions of 26.5% oxygen and 0.7 atm (10.2 psia), and 34% oxygen and 0.56 atm (8.2 psia), respectively with two figures each. Both use 50% mixtures of nitrogen and ethylene giving a fuel mixture heat of combustion of 24 kJ/g. Both of these tests reached quasi-steady conditions and were terminated by shutting off the fuel. The plots of transient heat flux shown by Figures 8 and 10, show the condensation effect removed after 80 s and 40 s respectively and arithmetic average of the two heat flux gauge measurements at the end of each test is recorded as steady surface heat flux which are used to obtain heats of gasification.



3 s 10 s 52 s 91 s 125 s 155 s 178 s

Figure 7. Flame growth from ignition to steady state in test 19070K1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 2.3 g/m^{2}s, 26.5% O_2, 10.2 psia, L = 3.0 kJ/g, steady.

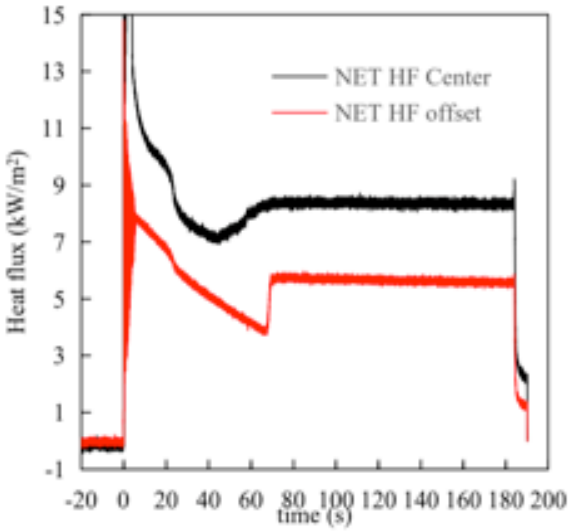


Figure 8. Measured heat flux in test 19070K1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 2.3 g/m^{2}s, 26.5% O2, 10.2 psia, L = 3.0 kJ/g, steady.

In Figures 9 and 10 the flame reaches a quasi-steady state in about 80 seconds as seen by the flame shape and the heat flux data. This emulation corresponds to a material with a heat of combustion of about 24 kJ/g and a heat of gasification of about 3 kJ/g. As such it could represent PMMA as a material solid. This is a profound result indicating a common plastic could steadily burn under these NASA proposed atmospheric conditions.



Figure 9. Flame growth from ignition to steady state in test 19070D1. The conditions are ethylene/N_2, Δh_c = 24 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Figure 9 shows steady results for a NASA proposed nominal atmosphere of 34% oxygen and 0.56 atm. After about 100 s the flame appears to be steady. The heat of gasification here is estimated as 2.42 kJ/g.

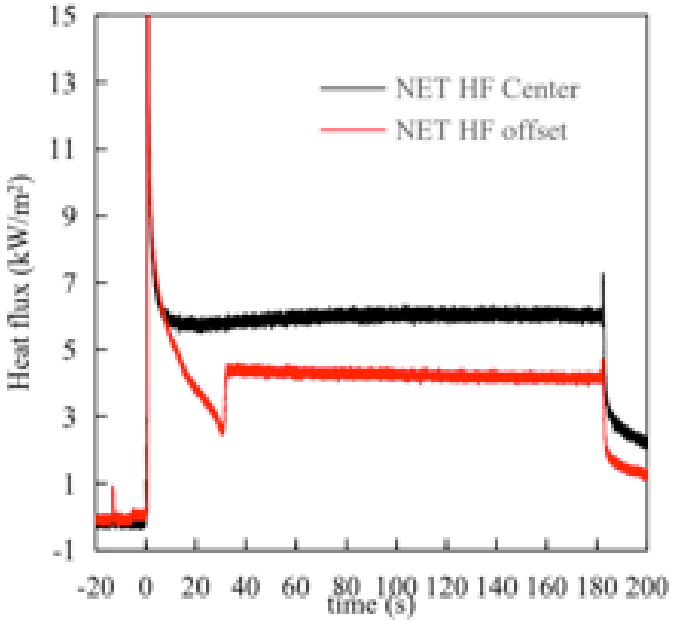


Figure 10. Measured heat flux in test 19070D1. The conditions are ethylene/N_2, Δh_c = 22 kJ/g, 4.95 g/m^{2}s, 34% O2, 8.2 psia, L = 2.42 kJ/g, steady.

Radiative loss fraction from the flames are obtained using the radiometer's measurements. Unfortunately, the data set is limited as BRE sensors were saturated in many tests. The +60 mm and the offset radiometers were used to obtain the radiative loss fraction as these two sensors provided data in multiple cases while the other two radiometers were saturated. The transient radiative loss fraction for 19066D5 and 19070L1 are shown in Figure 11 for both radiometers, which show good agreement with each other. Table 3 provides information about the test conditions. As BRE radiometers were saturated for many tests of round 1, the BRE radiometer array was replaced with

the standard ACME array to reduce the likelihood of sensor saturation. The offset radiometer data has been analyzed and the results are presented in Fig. 20.

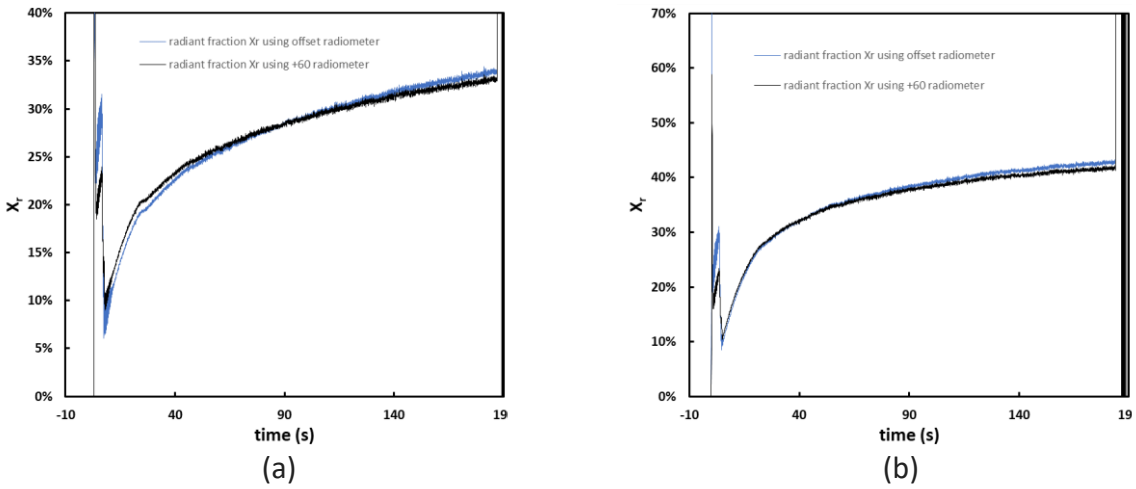


Figure 11. Transient radiative loss fraction for tests 19066D5 (a) and 19070L1 (b).

Flame heights at the end of the tests are measured from the image data. The radiative loss fraction is correlated with flame height and fuel mixture mass flux. The results from first round tests are shown in Figure 12.

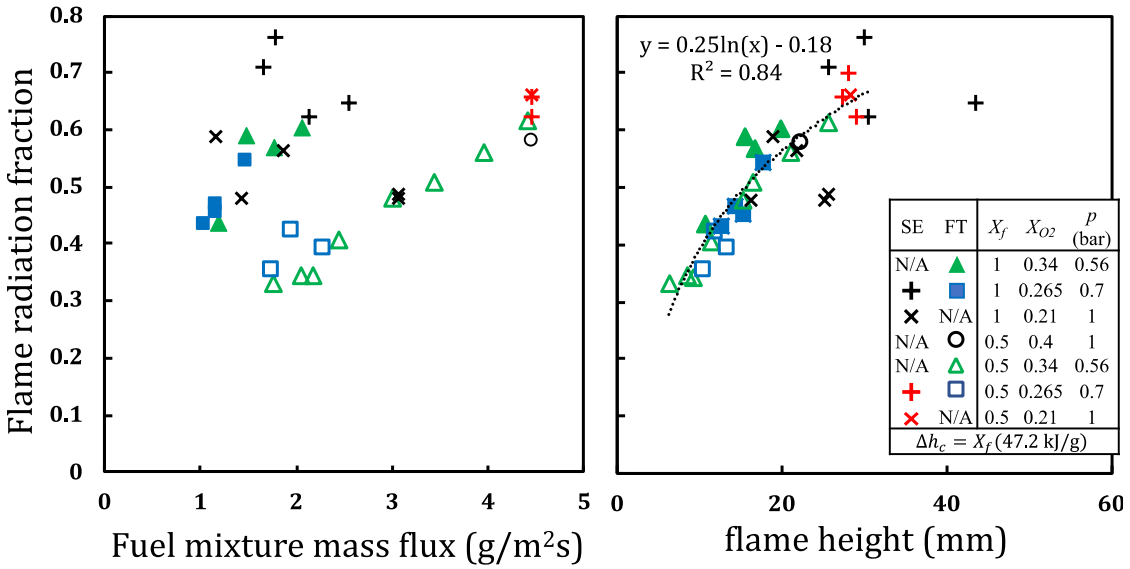


Figure 12. Flame radiant fraction vs. mass flux (a) and flame height (b) for all steady end-point results.

A simple analysis, based on the geometrical view factor of the surface with respect to the flame sheet, allowed us to estimate the fraction of the total available heat that impinges surface as seen in Figure 13. Also, the analytical model (the ellipsoidal model) based on classical diffusion theory was used to estimate the convective heat flux to the surface

which is shown in Figure 14. Adding up the convective and radiative heat transport plus re-radiation from the surface results in a total average heat flux that when divided by fuel mixture mass flux yields the theoretical heat of gasification.

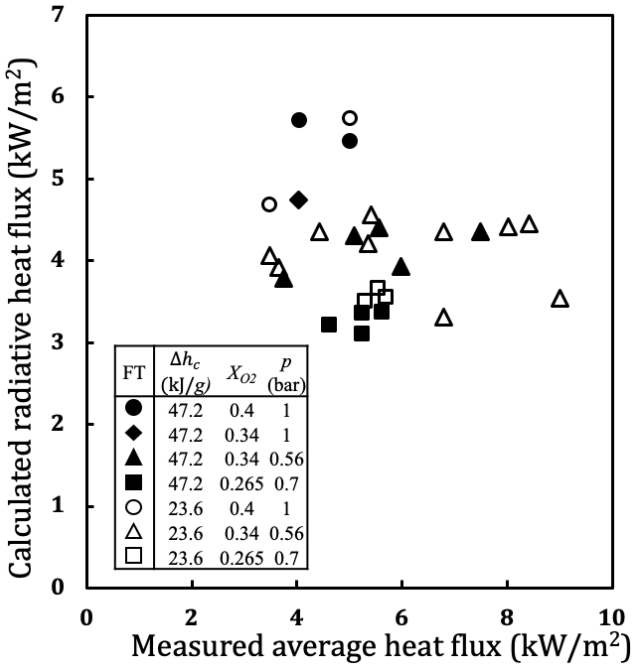


Figure 13. Radiative heat flux on the burner surface obtained by the View Factor method for nominal ambient conditions used for round 1 BRE tests on the ISS.

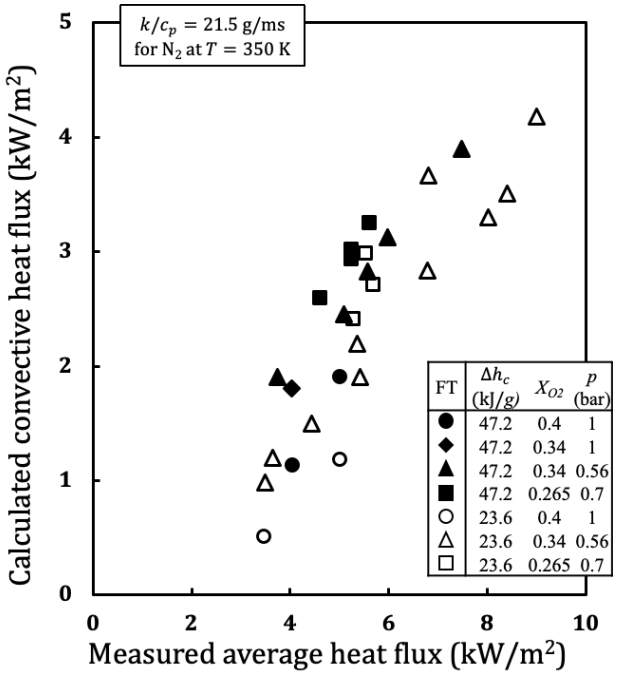


Figure 14. Convective heat flux on the burner surface obtained by the View Factor method for nominal ambient conditions used for round 1 BRE tests on the ISS.

The experimental flammability maps are shown in Figures 15 and 16. Both plots indicate a decreasing, inverse asymptotic-like curve that approaches zero, for heat of gasification with mass flux. It is surprising that the data cluster about a nearly singular trend. However, both plots show that the heat of gasification is up to 20 % higher for the lower heat of combustion 23.6 kJ/g. Our tests in the University of Maryland lab have shown, with the BRE in 1-g that similar results for propane-nitrogen mixtures have the opposite behavior with heats of gasification decreasing with heat of combustion for the same mass flux. But again, the data mostly cluster about one trend line except for very low heats of combustion. However, the corresponding 1-g heat of combustion data shift far to the right and down from the microgravity data of Figures 15 and 16. For example, the 1-g data would indicate ($\dot{m}$", L) points of about 30 g/m^{2}s, 2 kJ/g and 10, 0.5. Moreover, our previous work in a 5- s drop facility suggested similar data trends but with an apparent vertical asymptote at about 3 g/m^{2}s, rather than 1 as shown here. Obviously, the current data at durations of minutes are more accurate than those limited to 5 s. However, it has been shown that the ISS heat flux data even at self-extinction are theoretically nearly steady. Figures 15 and 16 include the heats of gasification derived from the self-extinguished endpoint data as well. Remarkably, the self-extinguished not-so-steady data follow the same trend as the steady data.

Let us interpret Figures 15 and 16 in terms of the flammability of real materials in microgravity. Typical melting plastics have heats of gasification ranging from about 2 to 3 kJ/g, with effective heats of gasification for charring plastics above 4 kJ/g. Some selected literature values for materials with heats of combustion in the BRE range are listed in Table 4. The accuracy of the calorimeter heat flux is found to be less than the two sensors, thus, we relied on the arithmetic average of the two heat flux sensors data for obtaining the flammability map. However, same results are shown considering the calorimeter data. Figure 15 would suggest that polyethylene (PE) and polypropylene (PP) would burn in microgravity at about 3 g/m^{2}s. Nylon and PMMA would burn at about 2.5 g/m^{2}s, and the phenolic glass fiber composite would burn at about 1 g/m^{2}s.

Table 4. Materials close to BRE heats of combustion and corresponding heats of gasification

Material	LHV, kJ/g	L, kJ/g
PE	43.4	1.9
PP	44	2.0
Nylon	28.8	1.5, 2.4
PMMA	24.2	1.6, 2.3
Phenolic glass fibers	22.0	7.3

Round 2 tests were finished with 196 successful ignitions. The measured heat flux by the transducers were used and the end point data of the long-duration tests (i.e., the tests in which the fuel flow was terminated as opposed to the tests that self-extinguished) were used to obtain heats of gasification. The heats of gasification of these data from round 2 testing as a function of fuel mixture mass flux are shown in Fig. 19. One important fire

safety consideration, based on the findings of this study, is that solid materials with heats of combustion in the range 23.6 to 47.2 kJ/g and heats of gasification from about 1 to 10 kJ/g can burn in microgravity. The NASA alternative flammability Test 2 [1] using the Cone Calorimeter can measure these material properties. As this study develops a more complete understanding, NASA may wish to put more emphasis into Test 2.

Moreover, the flammability map based on our theoretical analysis is shown in Figure 17. It shows similar asymptotic trend and clearly indicates lower pressure and higher oxygen concentration as favorable conditions for long duration burning.

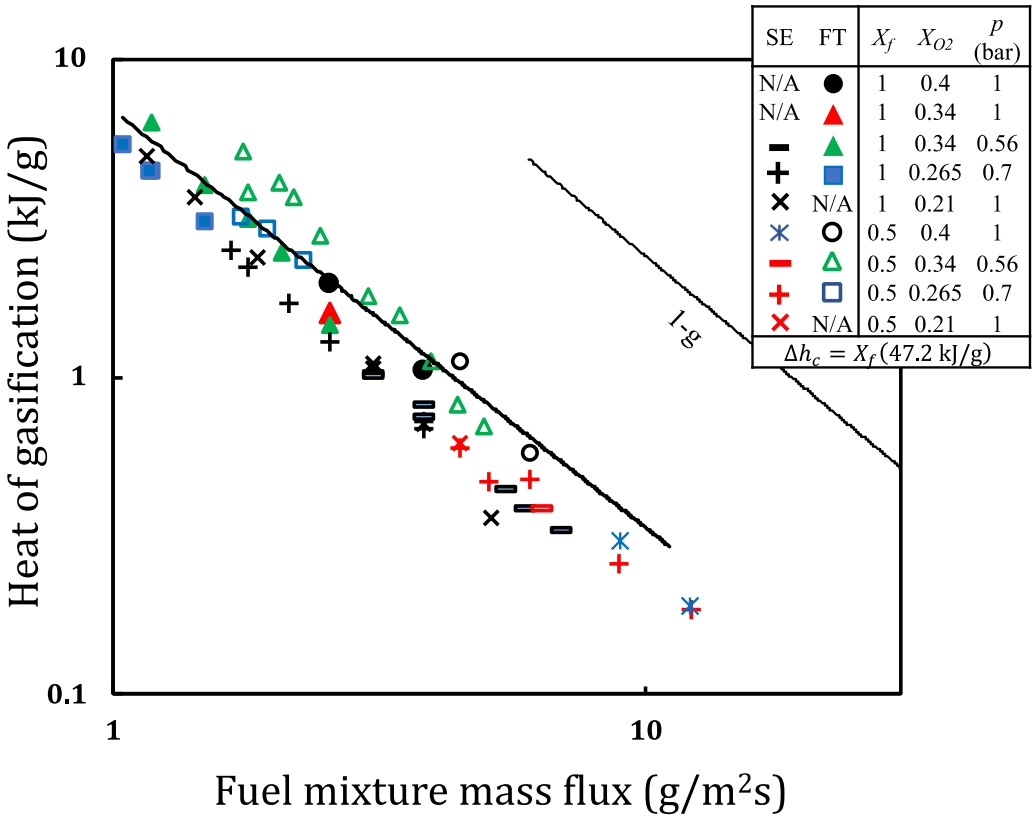


Figure 15. Round 1 result for the emulation of 25-mm flat materials using the arithmetic average of the two local heat flux measurements.

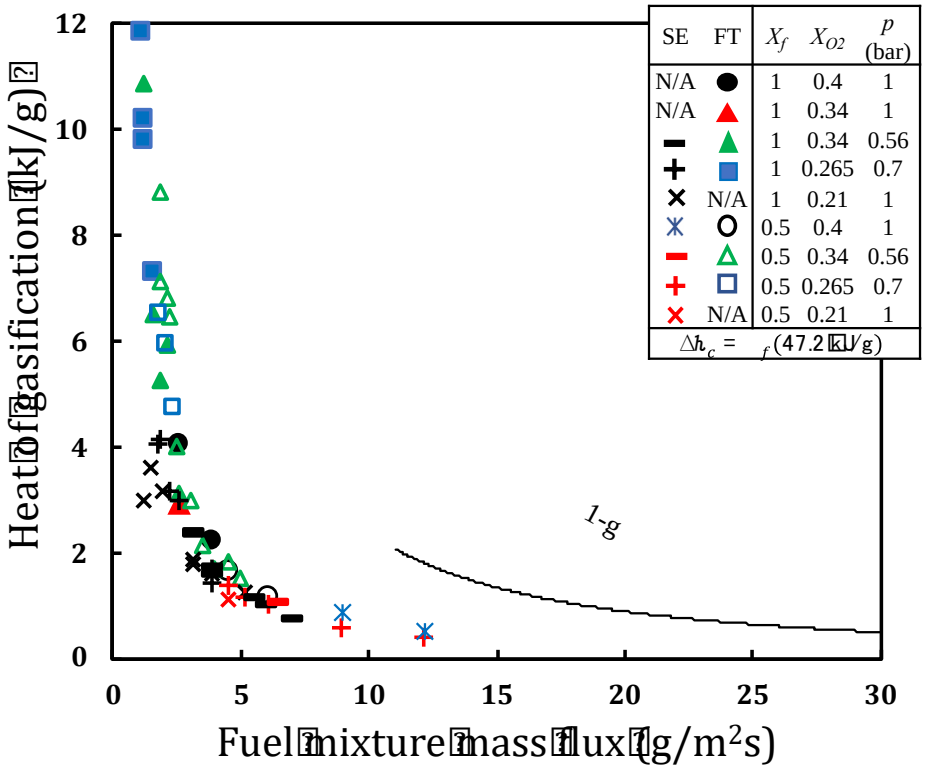


Figure 16. Heats of gasification calculated from the corrected calorimeter measurements vs. fuel mixture mass flux.

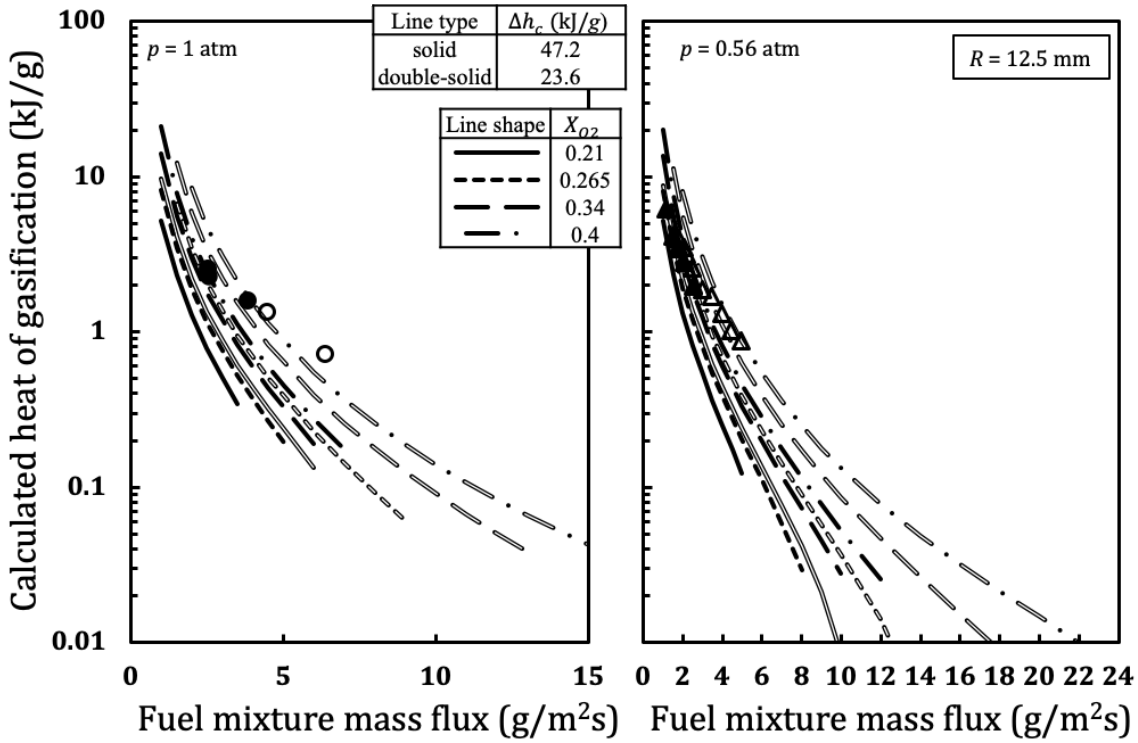


Figure 17. Calculated heats of gasification vs. mass flux for different oxygen mole fractions and p = 1 atm (left) p = 0.56 atm (right).

The imaging data from ops camera and ACME camera are used to create videos and image sequences for all of the tests. Preliminary analysis on the first three days of the second round of ISS testing data was performed. Pure methane was the fuel used for testing on the third day at 3 different nominal ambient conditions. It was observed that methane was hard to ignite compared to ethylene and flames tend to self-extinguish faster even at a high oxygen concentration.

A new method is used to examine the asymptotic behavior of the heat flux data. The inverse of the burn-time was used as the abscissa and the transient data from the two heat flux sensors are shown in Figure 18 for one of the tests.

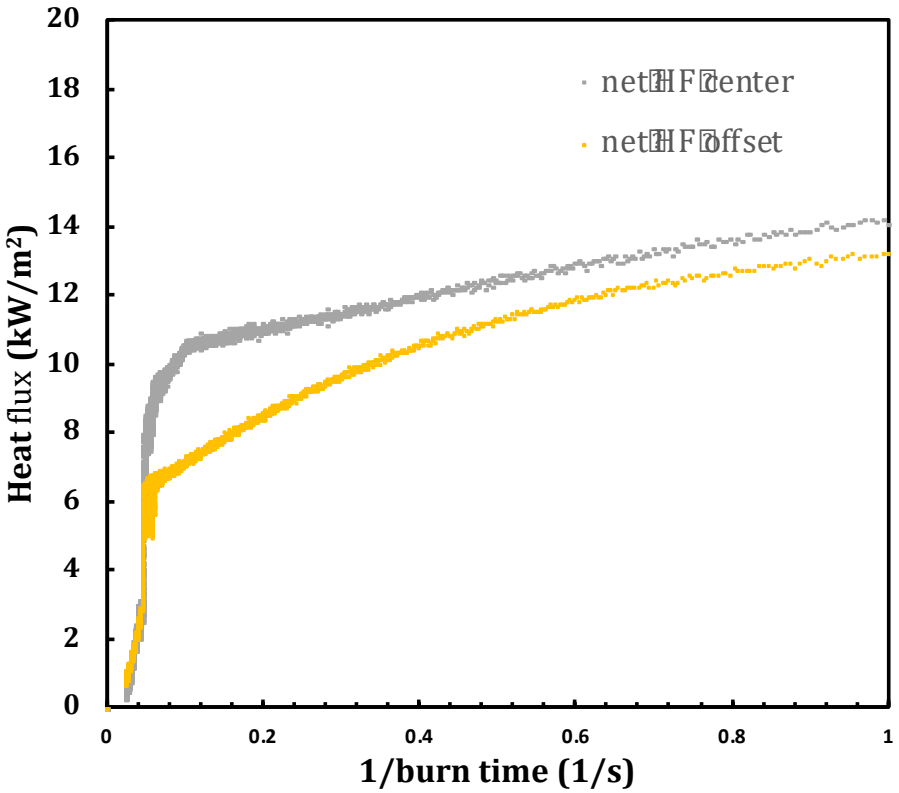


Figure 28. Transient heat flux vs. inverse of the burn time for 20279B4 with X_{O2} = 37.95%, p = 14.7 psia and mass flux of 2.46 g/m²s.

Our preliminary analysis on the heat flux data led to the flammability map (heats of gasification, L, vs. fuel mixture mass flux) shown in Figure 19 which contains all of the round 2 tests with quasi-steady data [2].

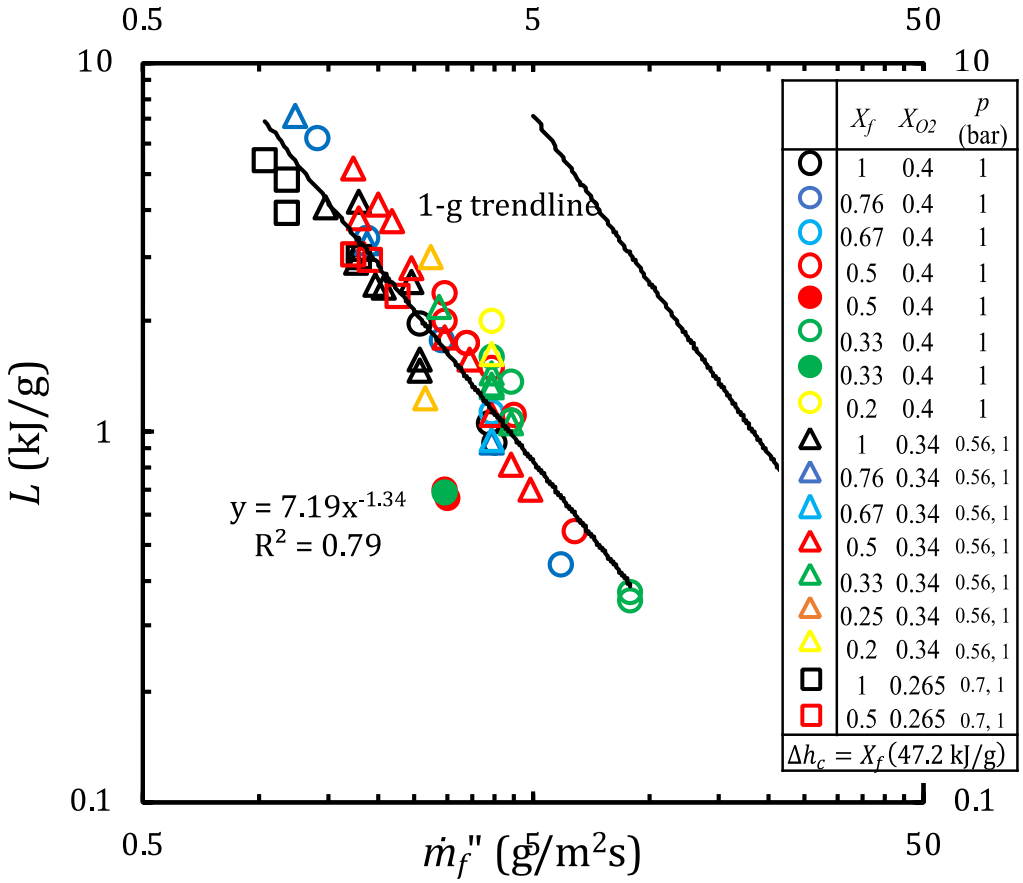


Figure 19. Preliminary flammability map based on the arithmetic average of the two heat flux sensors data.

Radiation data was limited to the tests taken with the smaller burner. The offset radiometer measurements are used to obtain the preliminary radiation fraction for the available data which are shown against fuel mixture mass flux in Figure 20 [2].

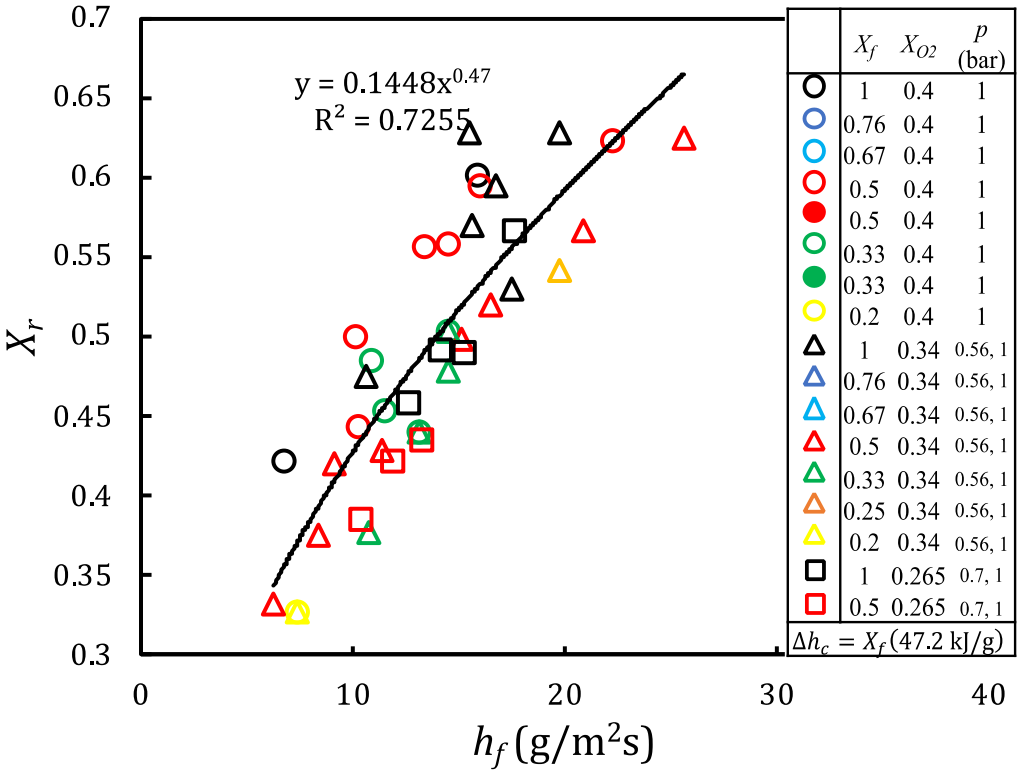


Figure 20. Radiation fraction vs. fuel mixture mass flux

Thin Filament Pyrometry was used during many tests of round 2. The technique of converting data to peak flame temperature at fiber locations was previously developed and published in ref. [3]. Four flames were analyzed to obtain the transient flame temperature [4].

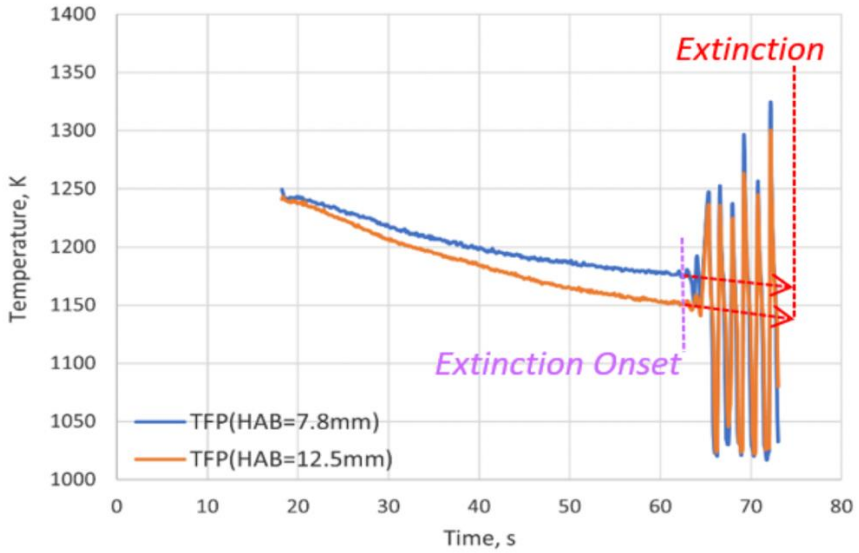


Figure 21. TFP measurements for test 21020D1

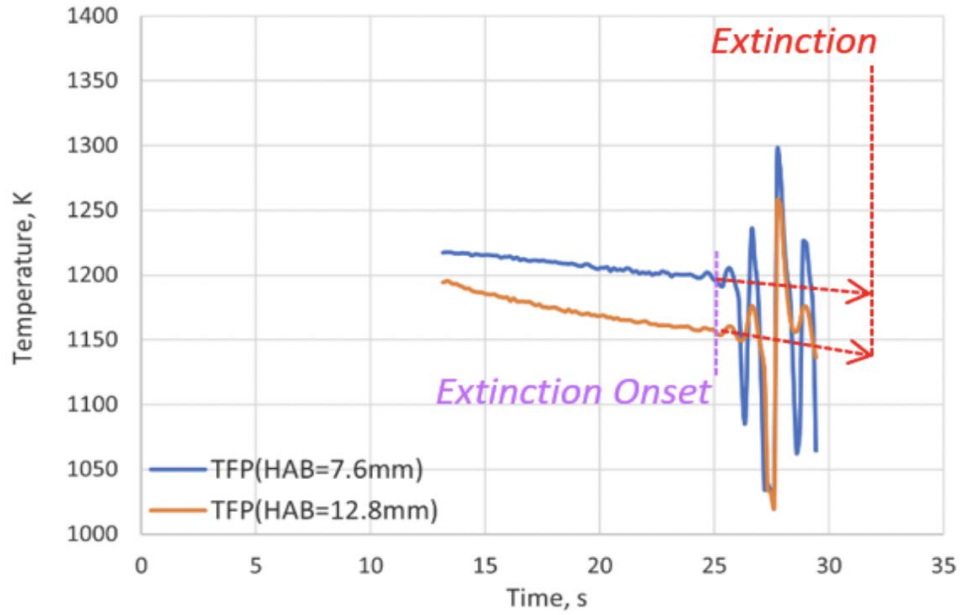


Figure 22. TFP measurements for test 20358G2

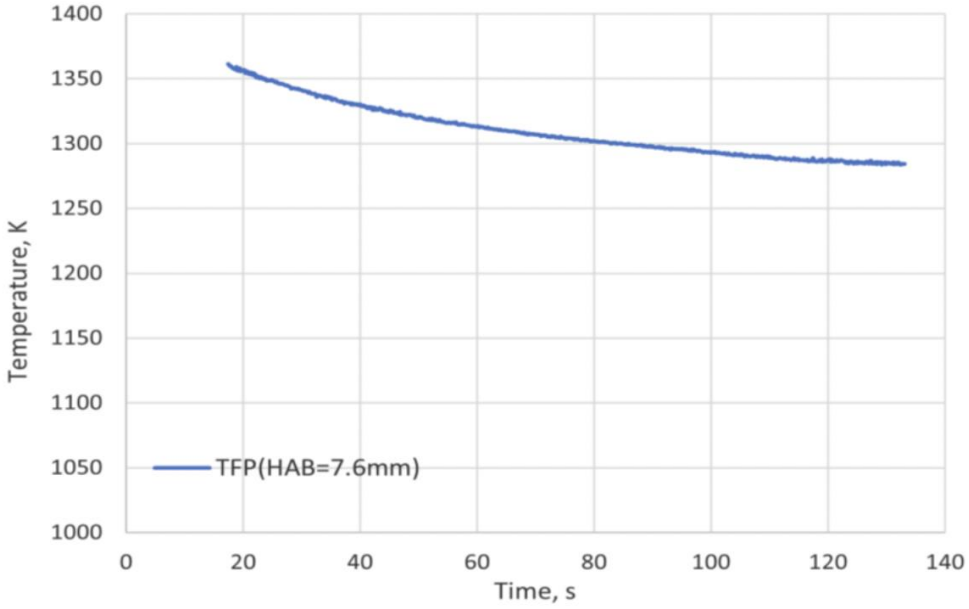


Figure 23. TFP measurements for test 21020A1

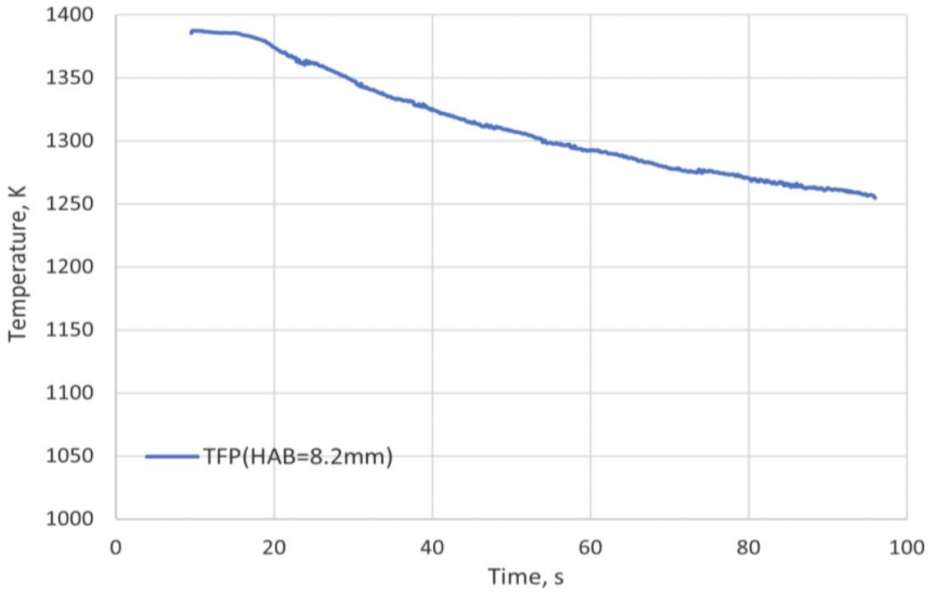


Figure 24. TFP measurements for test 20346A5

Measured oscillation onset and extinction times of all of the BRE flames were analyzed which resulted in a modeling of these timings as a function of Reynolds number and ambient oxygen mole fraction. The empirical coefficient (i.e., A) and exponents were found by fitting the modeled vs. measured data [5]. The results are shown in Figs. 25 and 26.

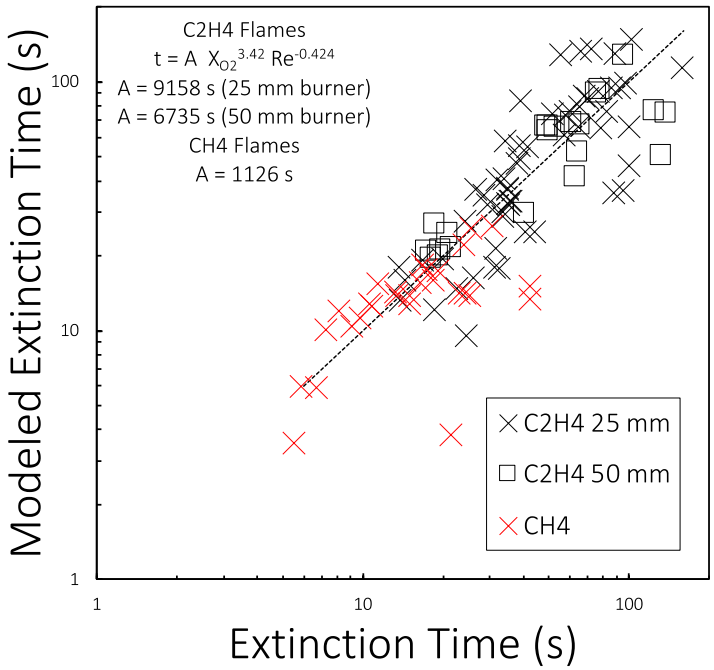


Figure 25. TFP measurements for test 20346A5

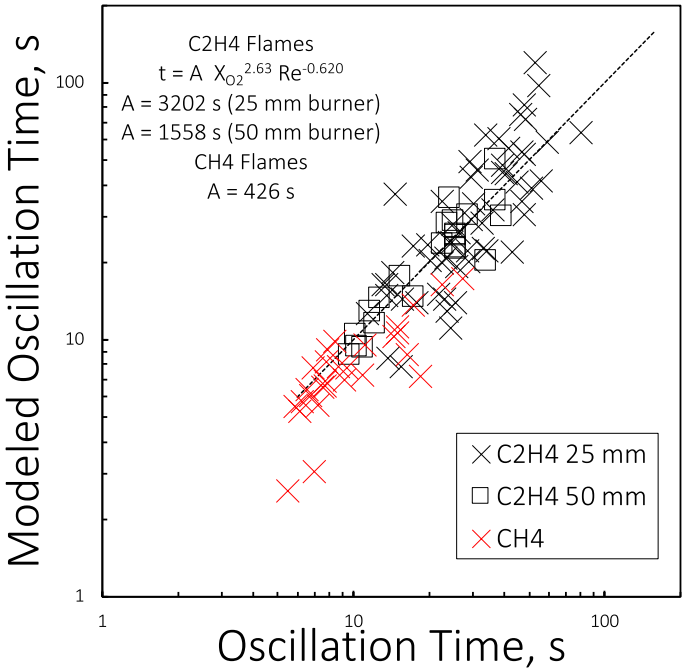


Figure 26. TFP measurements for test 20346A5

ISS results have been submitted for publication, where the first has been published:

- A. Snegirev, E. Kuznetsov, E. Markus, P. Dehghani, and P.B. Sunderland, "Transient dynamics of radiative extinction in low-momentum microgravity diffusion flames," Proceedings of the Combustion Institute, Volume 38, Issue 3, 2021, pp. 4815-4823. https://doi.org/10.1016/j.proci.2020.06.110
- P. Dehghani, P.B. Sunderland, J.G. Quintiere, and J.L. de Ris, "Burning in Microgravity: Experimental Results and Analysis," *Combustion and Flame*. (accepted)
- P. Dehghani and J.G. Quintiere, "Burning in Microgravity: Theoretical Predictions," *Combustion and Flame*. (submitted)

ISS results can be found in these two master's theses:

- A.E. Wright, "Using a Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station," M.S. thesis, 2021.
- Venzon, R., 2021, "Thin-Filament Pyrometry and Analysis of Extinction OnsetTemperature for Ethylene Flames on the Burning Emulator in the International Space Station," University of Edinburgh.

ISS results were presented with papers at the following conferences:

- Sergey Pichugin, Oleg Artemiev, Sergey Prokopiev, Oleg Kononenko, Oleg Novitsky, and Sergey Frolov, "Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience," XIII International Scientific and Practical Conference: Manned Flights in Space, Gagarin Cosmonaut Training Centre, Star City, Russia, 13-15 Nov. 2019.
- P. Dehghani, E. Auth, C. Cui, D.P. Stocker, J.L. de Ris, P.B. Sunderland, and J.G. Quintiere, Microgravity Observations of a Burning Emulator to Investigate Material Flammability,

Spring Technical Meeting of the Eastern States Section of The Combustion Institute, Columbia, SC, 8-11 March 2020.

- P. Dehghani, A. Wright, J.L. de Ris, and P.B. Sunderland, "Burning rate emulations aboard the International Space Station," 12th U.S. National Combustion Meeting, Texas A&M University, College Station, Texas (but virtual), 24-26 May 2021.
- S. Pichugin, D. Samsonov, S. Frolov, O. Korobeinichev, and A. Shmakov, "Flames Investigations at ISS," Cosmonautics of XXI Century Conference, TSNIIMash, Moscow, Russia, 12-14 May 2021.

ISS results were presented without papers, via oral presentations or posters, as follows:

- James G. Quintiere and Peter B. Sunderland, "Burning Rate Emulator (BRE): Material Burning in Microgravity," ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019.
- P. Dehghani and P.B. Sunderland, "Burning Emulation in Microgravity - Experimental Results and Analysis," 36th Annual Meeting of the American Society for Gravitational and Space Research, 5-6 Nov. 2020.
- P. Dehghani, Anna Wright, J.L. de Ris, P.B. Sunderland, "Pool Fire Emulations in Microgravity: Experimental Results and Analysis," poster, 13th International Symposium on Fire Safety Science, 26-30 April 2021.
- A. Wright, P. Dehghani, P. Sunderland, and J. de Ris, "Using A Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station," poster, 12th U.S. National Combustion Meeting, Texas A&M University, College Station, Texas (but virtual), 24-26 May 2021.
- P. Dehghani, A. Wright, J.L. de Ris, and P.B. Sunderland, "Burning emulations aboard the International Space Station," poster, 12th U.S. National Combustion Meeting, Texas A&M University, College Station, Texas (but virtual), 24-26 May 2021.

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Videos related to BRE and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

The results of this study will have a substantial impact on the methodology used by NASA to assess the fire safety of materials in the future. [From BRE PI Jim Quintiere]

References

1. NASA-STD-6001B. Flammability, Offgassing, and Compatibility Requirements and Test Procedures (2011). https://standards.nasa.gov/standard/nasa/nasa-std-6001
2. P. Dehghani, A. Wright, J.L. de Ris, and P.B. Sunderland, "Burning emulations aboard the International Space Station," poster, 12th U.S. National Combustion Meeting, Texas A&M University, College Station, Texas (but virtual), 24-26 May 2021.
3. P.H. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, and R.L. Axelbaum, "First Observations of Long Duration Microgravity Spherical Diffusion Flames Aboard the International Space Station," *Combustion and Flame,* 229 (2021) 111373. https://doi.org/10.1016/j.combustflame.2021.02.019
4. Venzon, R., 2021, "Thin-Filament Pyrometry and Analysis of Extinction OnsetTemperature for Ethylene Flames on the Burning Emulator in the International Space Station," University of Edinburgh.
5. A. Wright, P. Dehghani, P. Sunderland, and J. de Ris, "Using A Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station," poster, 12th U.S. National Combustion Meeting, Texas A&M University, College Station, Texas (but virtual), 24-26 May 2021.

Appendix F. Operational Timeline

This summary timeline of BRE operations was prepared by Dennis Stocker, where the red text indicates the key days when flames were ignited. The dates for crew activities were based on plans, where it is possible that the dates for some simple tasks, e.g., hardware gather or stowage, may have shifted slightly.

Date	GMT	Activity
BRE round 1, limited to C_2H_4 and the 25-mm burner		
2018		
Dec. 3	*337*	*Launch of Kononenko, McClain, and Saint-Jacques*
2019		
Jan. 14	014	LLL-UV camera reconfigured by Saint-Jacques
Jan. 16	016	Hardware gathered for CIR optics bench, etc. by Saint-Jacques
Jan. 17	017	CIR optics bench reconfigured by Saint-Jacques
Jan. 18	018	Hardware stowed by Saint-Jacques
Jan. 30	030	Hardware gathered for chamber insert reconfiguration by Saint-Jacques
Jan. 31	031	Reconfiguration of ACME chamber insert (part 1) by Saint-Jacques
Feb. 1	032	Reconfiguration of ACME chamber insert (part 2) by Saint-Jacques
Feb. 2	033	Hardware stowed by Saint-Jacques
Feb. 4	035	Bottle swap by Kononenko (assisted by McClain)
Feb. 6	037	Hardware check-out, etc. in preparation for BRE
Feb. 8	**039**	**BRE tests using 25-mm burner and 100% C2H4**
Feb. 20	**051**	**BRE tests using 25-mm burner and 100% C2H4**
Feb. 22	053	Bottle swaps by Kononenko
Feb. 25	**056**	**BRE tests using 25-mm burner and 50% C2H4**
Feb. 26	057	Swap of fuel Mass Flow Controller by Saint-Jacques
Feb. 28	059	MFC testing & tuning
March 4	**063**	**BRE tests using 25-mm burner and 50% C2H4**
March 5	064	Bottle swap by Kononenko
March 7	**066**	**BRE tests using 25-mm burner and 50% C2H4**
March 11	**070**	**BRE tests using 25-mm burner and 50% C2H4**
March 13	072	Bottle swap by Kononenko
March 14	*073*	*Launch of Ovchinin, Hague, and Koch*
March 18-19	077-078	FOMA calibration by Kononenko
March 26	085	Bottle swap by Kononenko
March 28	**087**	**BRE tests using 25-mm burner and 100% C2H4**
April 1	091	BRE tests deferred because of the discovery of an igniter failure
April 3	093	Igniter tip replacement by Hague
April 4	**094**	**BRE tests using 25-mm burner and 100% C2H4**
June 24	*175*	*Landing of Kononenko, McClain, and Saint-Jacques*
Oct. 3	*276*	*Landing of Ovchinin and Hague*
2020		
BRE round 2, including C_2H_4, CH_4, and both burner sizes		
April 9	*100*	*Launch of Cassidy, Ivanishin, and Vagner*
July 22	204	MFC testing & tuning
Aug. 26	239	Hardware gather, bottle swaps, etc. by Cassidy
Aug. 27	240	Hardware gather and ACME insert reconfigured by Cassidy

Aug. 28	241	Hardware check-out, etc. for BRE; hardware stowed by Cassidy
Sept. 3	**247**	**BRE tests using 50-mm burner and 100% C2H4**
Sept. 14	258	ACME insert reconfigured for 25-mm burner testing by Cassidy
Sept. 15	259	Hardware stowed by Chris Cassidy
Sept. 17	**261**	**BRE tests using 25-mm burner and 100% C2H4**
Sept. 22	266	Bottle swaps by Ivanishin
Sept. 28	272	Bottle swap by Ivanishin
Oct. 5	**279**	**BRE tests using 25-mm burner and 100% CH4**
Oct. 14	*288*	*Launch of Ryzhikov, Kud-Sverchkov, and Rubins*
Oct. 21	*295*	*Landing of Cassidy, Ivanishin, and Vagner*
Nov. 15	*320*	*Launch of Hopkins, Glover, Noguchi, and Walker*
Dec. 3	338	Bottle swap by Kud-Sverchkov (assisted by Hopkins)
Dec. 4	**339**	**BRE tests using 25-mm burner and 100% CH4**
Dec. 10	345	Bottle swaps by Kud-Sverchkov
Dec. 11	**346**	**BRE tests using 25-mm burner and 50% C2H4**
Dec. 15	350	Bottle swaps by Kud-Sverchkov
Dec. 17	**352**	**BRE tests using 25-mm burner and 100% C2H4**
Dec. 21	356	Bottle swap by Kud-Sverchkov
Dec. 23	**358**	**BRE tests using 25-mm burner and 100% C2H4**
Dec. 29	364	Bottle swap by Kud-Sverchkov

2021

Jan. 7	007	Imaging of damaged Thin Filament Pyrometry (TFP) fiber array
Jan. 13	013	Imaging of damaged Thin Filament Pyrometry (TFP) fiber array, etc.
Jan. 18	018	Broken TFP fibers removed and igniter tip replaced by Glover
Jan. 20	**020**	**BRE tests using 25-mm burner and 100% C2H4**
Jan. 21	021	Bottle swaps by Rubins
Jan. 25	025	ACME insert reconfigured for 50-mm burner testing by Rubins
Jan. 26	026	Hardware stowed by Rubins
Jan. 28	**028**	**BRE tests using 50-mm burner and 100% C2H4**
April 17	*107*	*Landing of Ryzhikov, Kud-Sverchkov, and Rubins*
May 2	*122*	*Landing of Hopkins, Glover, Noguchi, and Walker*

Appendix G. Astronauts and Cosmonauts who Conducted ISS Operations for BRE

This list was prepared by Dennis Stocker from his log of ACME operations.

Ten cosmonauts and astronauts participated in ISS operations directly supporting the BRE investigation, where they are listed with the dates of their activity (at least as planned). They are listed by the number of days in which they conducted supporting on-orbit activities and alphabetically. The crew activity on each of those dates is very briefly summarized in the Appendix F. Additional information about the crew activity can be found in some of the subsequent appendices.

The CIR and ACME hardware was initially set up for BRE testing by Canadian astronaut David Saint-Jacques, while U.S. astronaut Chris Cassidy reconfigured the hardware for BRE's second round of testing after intervening testing for other experiments. Meanwhile, the gas bottles were most often replaced by Russian cosmonauts Oleg Kononenko and Sergey Kud-Sverchkov.

Crew member	Year	Day(s) of year	# days
Saint-Jacques, David (CSA)	2019	014, 016-018, 030-033	8
Kononenko, Oleg (RSA)	2019	035, 053, 064, 072, 077-078, 085	7
Cassidy, Chris (NASA)	2020	239-241, 258-259	5
Kud-Sverchkov, Sergey (RSA)	2020	338, 345, 350, 356, 364	5
Rubins, Kate (NASA)	2021	021, 025-026	3
Ivanishin, Anatoly (RSA)	2020	266, 272	2
Glover, Victor (NASA)	2021	018	1
Hague, Nick (NASA)	2019	093	1
Hopkins, Mike (NASA)	2020	338 (with Kud-Sverchkov)	1
McClain, Anne (NASA)	2019	035 (with Kononeko)	1

Together, the 8 men and 2 women represent three space agencies and three nations, namely Canada, Russia, and the United States.

Appendix H. ISS Operations per the Payload Weekly Science Summary

This section is from Dennis Stocker's log of ACME operations.

BRE round 1, featuring tests with ethylene and the 25-mm burner

2019

Week 16: January 11-17

David: On GMT 014, thank you for replacing the imaging modules in the CIR LLL-UV Imaging Package. Then on GMT 018, great job configuring the back of the CIR Optics Bench to support future ACME experiments. We appreciate all your hard work and diligence during the many intricate tasks!

Week 18: January 25-31

David: On GMT 031 & 032, great job configuring the ACME Chamber Insert to support the next ACME experiment, BRE (Burning Rate Emulator). We appreciate your attention to detail during the many complicated tasks! We especially appreciate you taking all the closeout photos of the newly installed hardware. We will check out and calibrate the newly installed hardware next week and perform the first test points late next week.

Week 19: February 1-7

Oleg & Anne: Great job replacing the CIR Manifold & GC Bottles this week! We these new bottles installed, we look forward to starting the next ACME experiment, BRE (Burning Rate Emulator).

Week 20: February 8-14

On GMT 039, we performed our first ACME BRE (Burning Rate Emulator) test points. Unlike the other ACME experiments, BRE is focused on improving spacecraft fire safety rather than practical terrestrial combustion. More specifically, BRE uses gaseous fuels to determine the flammability of liquids and solids in microgravity by selecting test conditions to match key parameters. We performed nine successful tests burning ethylene on the 25-mm diameter burner, where the burner's embedded sensors measure the heat flux that would normally vaporize a liquid or solid fuel. The tests were conducted under a range of oxygen concentrations and pressures, including some atmospheres that could be used for future crew vehicles or habitats.

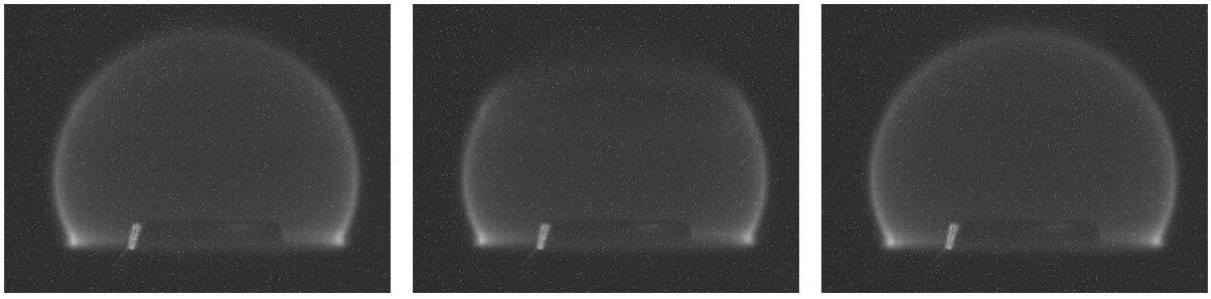


Figure 1. Sequential Monochrome Images of the First BRE Flame Stabilized on the Cylindrical Burner

Figure 1 shows a sequence of three images from BRE's first on-orbit test. The first image shows that the flame grew to become nominally hemispherical (over the burner outlet, which can also be seen). The second image shows a hole in the top of the flame due to radiative heat loss. However, the third image

shows that the flame recovered, where the hole closed. In this latter phase of the test, measurements of chemiluminescence (from OH*) clearly reveal that the flame was oscillating at approximately 1 Hz. These oscillations (the opening and closing of the flame's top) were seen to grow in magnitude and are associated with the flame's approach to extinction. In the first test, this lasted for an extended time, but in other tests at lower oxygen concentrations the oscillations sometimes lasted 10 seconds or less before the flame quenched. BRE seeks to understand the flame extinction and the conditions under which the flames become unsteady and finally extinguish.

Week 21: February 15-21

Testing for the Burning Rate Emulator (BRE) experiment resumed on GMT 051 following a revision of the ignition procedure. Eight successful tests were conducted burning pure ethylene, which is naturally a sooty fuel because of its double bond, but the soot produced during ignition quickly vanished leaving a pale blue flame, where the flame temperatures were insufficient to generate soot. These flames grow to be nominally hemispherical, then become quasi-steady although weakening, and finally become unsteady and extinguish when radiative heat losses become too great. The experiment's goal is to improve our understanding of material flammability in microgravity and thereby guide the selection of materials use in spacecraft.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

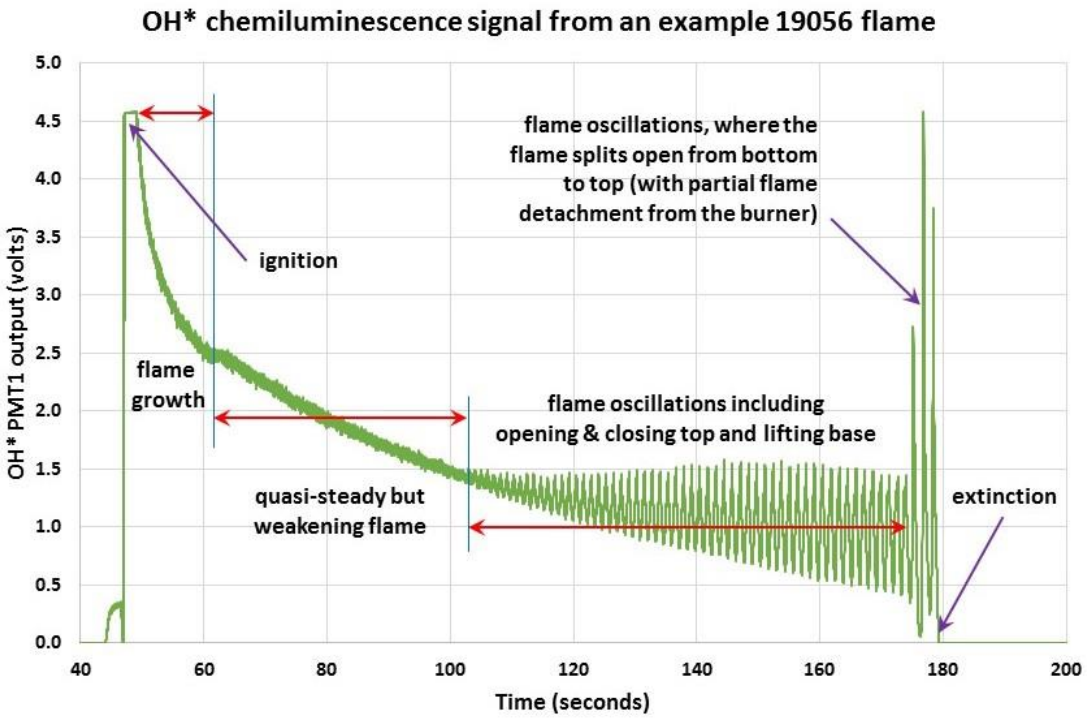
The video from test 19039G1 is typical of the BRE tests conducted thus far, where the flame grows to become nominally hemispherical, during which time the soot produced at ignition vanishes. The flame then becomes quasi-steady, e.g., in terms of the flame shape and size, although it can be seen that the flame weakens with time – becoming dimmer and more diffuse. Oxygen transport to the flame is inhibited because it is blanketed by an ever-thickening cloud of carbon dioxide and water vapor. Eventually, radiative heat loss leads the top of the flame (on the video's left) to locally extinguish, leaving a hole in the flame. But the fuel and oxygen, which are normally separated by the flame, mix and the flame quickly reestablishes, filling the hole. A pattern of oscillation begins, which also features periodic lifting of the flame base from the cylindrical burner (on the right of the video). The oscillations grow in strength, where the flame becomes momentarily just a ring. At the end, before complete extinction, the flame also partially detaches from the burner, leaving a crack in the flame's side extending from the base to its top.

The durations of at least the quasi-steady and unsteady periods are strongly dependent on the atmosphere in the chamber, where both the oxygen concentration and pressure have been varied. As one might expect, longer durations are observed at higher oxygen concentrations.

Note that the brighter circle that can be seen in upper middle of the video is due to reflection from an optical sensor, which is not seen in the other three camera views.

Week 22: February 22-28

On GMT 056, we performed ACME BRE (Burning Rate Emulator) test points with a 50% ethylene/50% nitrogen mixture. The flames with 50% ethylene may be steadier than the 100% ethylene flames, which require more oxygen at the same total flow rate. The 50% ethylene flames still exhibited flame growth, a quasi-steady period, and oscillations leading to extinction. The flames also quickly became soot free (i.e., all blue flame) after ignition.



Ignition, extinction, and the three main phases of the BRE flame's life can be seen in this example plot of the OH* chemiluminescence measured by one of the three PhotoMultipiler Tube (PMT) modules on the ACME Chamber Insert. OH* is an excited molecule that loses energy by emitting a photon of light at a specific wavelength (in the ultraviolet). In combustion of most fuels, there is oxygen, hydrogen, and lots of energy making OH* measurements a good indicator of the flame.

This data shows that the flame is weakening during the quasi-steady period when the flame size and shape is nominally constant. The OH* measurement also clearly shows the oscillations that precede extinction and their growing strength. The flame's oscillations (e.g., with the opening and closing of the top) are at a nominally constant frequency that is a little less than 1 Hz. The huge OH* oscillations are the flame's death throes when it splits open from bottom to top and partially detaches from the burner.

Week 23: March 1-7

On GMT 064, Oleg Kononenko replaced the CIR Manifold #2 Bottle that was consumed during test points on GMT 063. For these ACME BRE (Burning Rate Emulator) test points, we used higher fuel flow rates with the same fuel (50% ethylene/50% nitrogen) as the test points on GMT 056. We performed a total of six tests, where the first—in an atmosphere of 40% oxygen—was notable in that the flame didn't extinguish during the five-minute test even though it was oscillating in the last three minutes. As expected, the flames in subsequent tests with higher flow rates grew to be significantly larger in size, and with their higher oxygen consumption, they extinguished in significantly shorter times. We observed dramatic oscillations with the large flames. Furthermore, these large oscillating flames became asymmetric; whereas, smaller BRE flames have been largely axisymmetric even when oscillating.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

The test shown in this video is the first ACME test—of all the experiments thus far—to have such a high fuel flow rate. Initially, the flame behavior is similar to previous BRE tests. The flame is very sooty at ignition, due to ethylene's double bond, where you can see the glowing soot particles pass through the flame. But the soot quickly fades away leaving a dim blue flame, where blue is the natural color of hydrocarbon flames. The flame expands to reach enough oxygen to consume the fuel. Meanwhile, the flame is blanketed by a thickening cloud of carbon dioxide and water vapor inhibiting the oxygen's diffusion to the flame. With the poor oxygen transport, the flame weakens, and radiative heat loss leaves a hole in the flame. But oxygen then mixes with the fuel, causing the flame to quickly reestablish and the hole closes.

With this flame's large size, the edge of the flame opening can be seen to curve upward and outward—somewhat like a vase—where that is the fuel-lean rim of the flame edge. The opening and closing grows in strength as in previous tests, but the flame size is apparently so large that asymmetries begin. The flame eventually becomes mostly quenched with just a ring at the base and a highly dynamic spiral extending upward. As in previous tests, the flame partially detaches from the burner (right side of video), but in this case, the flame surprisingly persists for roughly 20 seconds despite the large-scale detachment. While careful analysis will be challenging, this flame is visually dramatic with its swirling pulsations and powerful death throes.

Week 24/01: March 8-14

On GMT 072, Oleg Kononenko replaced the CIR Manifold #4 Bottle to support upcoming ACME BRE test points with pure ethylene fuel, and he installed the CIR Manifold #1 Bottle to support the upcoming CIR FOMA Calibration.

On GMT 066 & 070, we performed the remaining ACME BRE (Burning Rate Emulator) test points using diluted ethylene fuel (50% ethylene/50% nitrogen). These tests were conducted with low fuel flow rates under atmospheric conditions that could be used for future crew exploration vehicles (or habitats), i.e., with elevated oxygen concentrations (between 26% and 36%) and reduced pressures. Compared to prior tests, these flames were steady and didn't extinguish during the allotted test durations.

Even though these flames appeared steady, measurements of OH* and CH* chemiluminescence revealed that they were slowly weakening as each test progressed. Chemiluminescence occurs when excited molecules at a high energy state drop to a lower energy state by emitting a photon of a certain wavelength. CH* chemiluminescence is why hydrocarbon flames are blue; meanwhile the OH* chemiluminescence is in the ultraviolet.

Although the flames were weakening, the thermal radiation from each flame slowly increased because of the lack of buoyancy, which, in gravity, removes the hot combustion gases from the vicinity of a flame. Instead, these flames became blanketed by a thickening cloud of hot and radiating combustion products, namely carbon dioxide and water vapor. This accumulation of products inhibited the transport of oxygen to the flame. Yet, with the low flow of diluted fuel, the oxygen diffusion was enough to maintain these flames during each test's duration. If the test durations weren't limited, these flames would have eventually gone out, but it could have taken many minutes for that to happen.

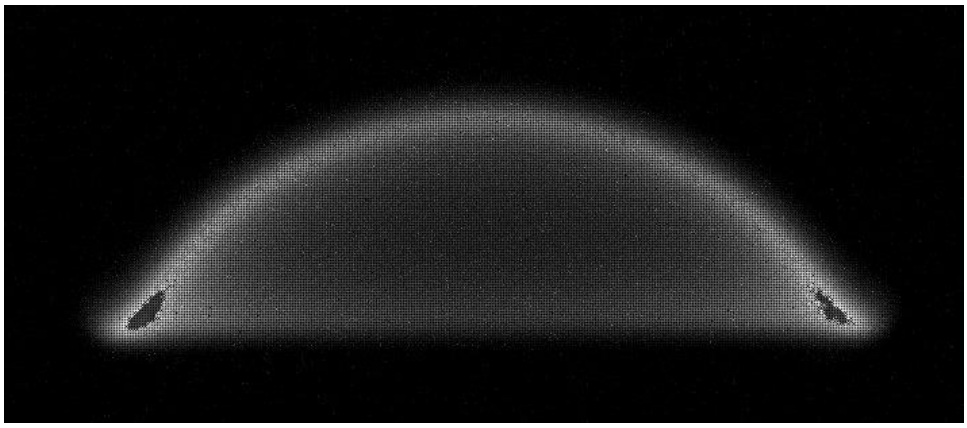


Figure 1. Sequential Monochrome Images of the First BRE Flame Stabilized on the Cylindrical Burner

Figure 1 shows a flame near the end of a test conducted with 50% ethylene/50% nitrogen in an atmosphere of approximately 27.5% oxygen at 0.7 bar. The dim blue flame did not appear to visibly change following ignition. (The dark spots at the base are a result of image processing.)

Increment 59-60

Week 2: March 15-21

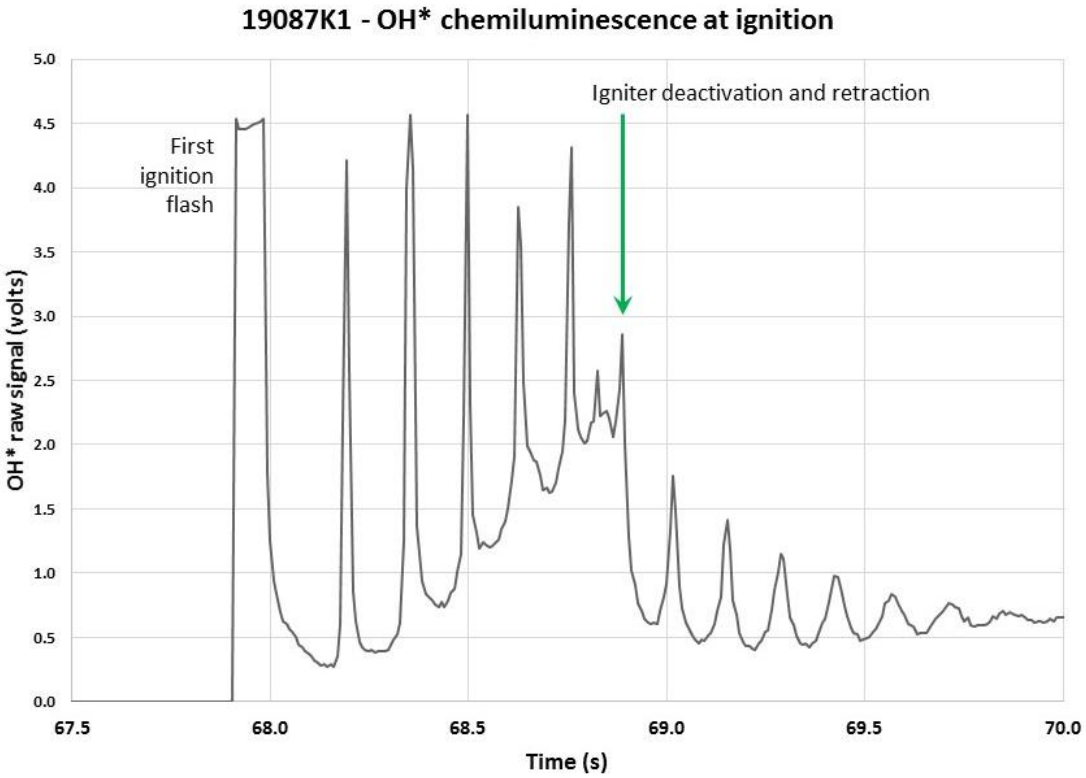
On GMT 077 and 078 in support of the CIR FOMA Calibration, Oleg Kononenko closed and opened valves in the CIR's Fuel/Oxidizer Management Assembly (FOMA) and switched the settings on the CIR Valve Timers. This calibration needs to be completed yearly to ensure that all the pressure transducers in the CIR are accurate.

Week 3: March 22-28

On GMT 085, Oleg Kononenko replaced the CIR Manifold #2 Bottle, so we could continue performing ACME BRE (Burning Rate Emulator) test points. He also closed the valve on the CIR Manifold #1 Bottle since this bottle will not be used until a future ACME experiment.

Week 4: March 29 – April 4

On GMT 087, ACME BRE (Burning Rate Emulator) test points were performed using pure ethylene fuel in atmospheric conditions that might be used for future crew exploration vehicles or habitats. This flammability testing has focused on flame conditions at low fuel flow rates, where ignition wasn't always achieved. Nonetheless, eight successful tests were conducted this day. During the day's final test point, the igniter tip burned out. On GMT 093, Nick Hague replaced the burnt-out igniter tip.



This plot illustrates the challenge faced in igniting some BRE tests. The plot is of OH* chemiluminescence, which is measured by a Photomultiplier Tube (PMT) during the ignition period. Chemiluminescence occurs when a molecule drops from a higher to a lower energy level by emitting a photon of a specific wavelength. In this case, the excited OH* emits light in ultraviolet (at 310 nm). While this light is not visible, it is a good indicator of the flame's strength or intensity. This plot shows how the flame repeatedly flashed across the burner. The flame didn't stabilize until after the igniter was retracted. Unfortunately, by that time the igniter tip had already burned out.

Week 5: April 5-11

On GMT 094, the final test points were performed for the first round of testing for ACME BRE (Burning Rate Emulator). These were the first BRE test points to use very fine ceramic fibers in the flame to acquire temperature measurements from the color of the glowing fibers. During the past two months, over 60 flames were ignited in this first round of testing to improve our understanding of microgravity flammability.

This week, Christina Koch configured the ACME Chamber Insert to support the next ACME experiment, Flame Design. Also to support Flame Design, Oleg Kononenko replaced the bottles on CIR Manifold #2, CIR Manifold #4, and the CIR GC Helium Manifold. Flame Design is focused on understanding the soot processes of flames. Test points will start in a few weeks.

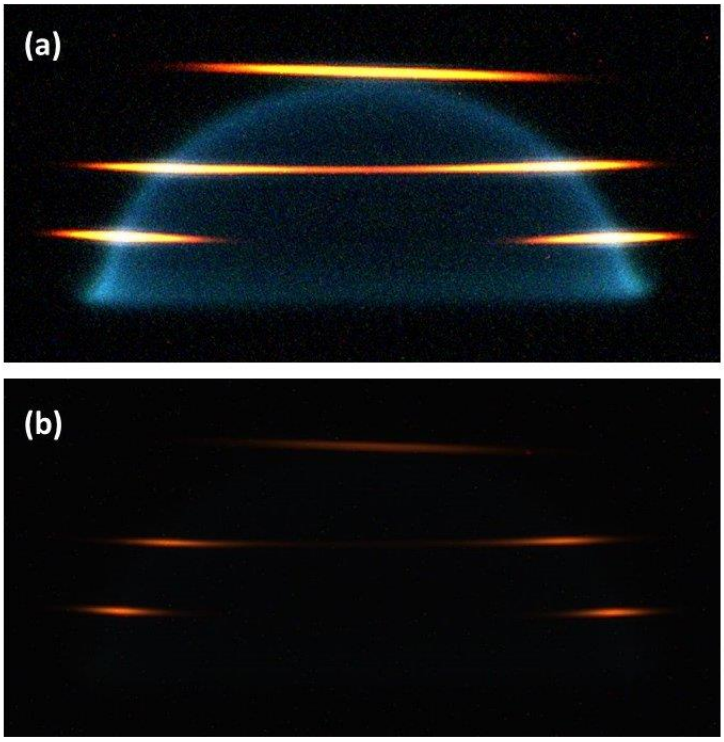


Figure 1. Sample preliminary images from a BRE test, where both were processed from the same raw image file. Image (a) reveals the relative positions of the glowing ceramic fibers, while (b) is adjusted for determination of the temperature along the fibers.

Figure 1 displays sample images, in which Thin Filament Pyrometry (TFP) was used to allow optical determination of the temperature along the ceramic fibers where they are hot enough to visibly glow. The harp-like array of 14-micron SiC fibers was translated throughout the flame allowing measurement of the entire temperature field. The temperature is determined by comparison of the red, blue, and green color planes of the images, which were taken with a digital camera characterized for quantitative measurements. The color of the fiber's glow is indicative of the temperature, where yellow is hotter than orange, which in turn is hotter than red. (The blue in the flame results from the flame chemistry rather than the fibers' broadband thermal radiation.)

BRE round 2, featuring tests with ethylene & methane and the 25-mm & 50-mm burners

2020

Week 20: August 21-27

On GMT 239, Chris Cassidy replaced the bottles and lubricated the QDs on the CIR GC Helium Manifold, the CIR GC Check Gas Manifold, the CIR Manifold #2 and the CIR Manifold #4. On GMT 240, Chris Cassidy configured the ACME Chamber Insert and the CIR Combustion Chamber. These tasks prepared the CIR and ACME for supporting the next ACME experiment, BRE Part 2 (Burning Rate Emulator). BRE studies material flammability with the goal of improving spacecraft fire safety.

Week 22: September 4-10

On GMT 247, we began the second round of testing for ACME BRE (Burning Rate Emulator) with 13 flames ignited using ethylene fuel. While the first round of testing only used the 25-mm diameter burner, this round of testing began with the 50-mm diameter burner to better understand the effect of burner size on flame behavior. Even though these flames burned for a long time, they were unsteady and extremely dynamic. Since BRE's research focus is on quasi-steady flame behavior, the remaining tests will use the 25-mm burner.

In many of these 50-mm burner tests, heat loss caused the top of the flame to locally quench—creating a hole that allowed for increased mixing of the fuel and oxygen. The flame was able to reestablish, closing the hole in its top. These oscillations occurred repeatedly and grew in strength such that the flame became a ring instead of a dome. Later in the burn, the flame often partially detached from the burner, creating a hole in the side of the flame that would also open and close repeatedly. In some cases, this behavior transitioned to a more constant hole, where the partial flame would swirl around the burner's rim as if it were "chasing its tail."

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This clip from a test point video shows the flame dynamics observed in the latter portion of the test. This test is burning 100% ethylene fuel in a combustion chamber environment of about 32% oxygen/68% nitrogen. The flame, which in microgravity would normally be dome-shaped, has largely quenched where only a portion is burning. In this phase of the burn, the flame appears to have mostly detached from the burner. The flame swirls around the burner's rim, where the leading edge is bright because of the fuel/oxidizer mixing that occurs where the flame has quenched. While the nature of the oscillations changed over time, they made up about 70% of the flame's life. In fact, this flame was still oscillating when the fuel flow was shut off (at a preset time), and the flame went out.

Week 23: September 11 - 17

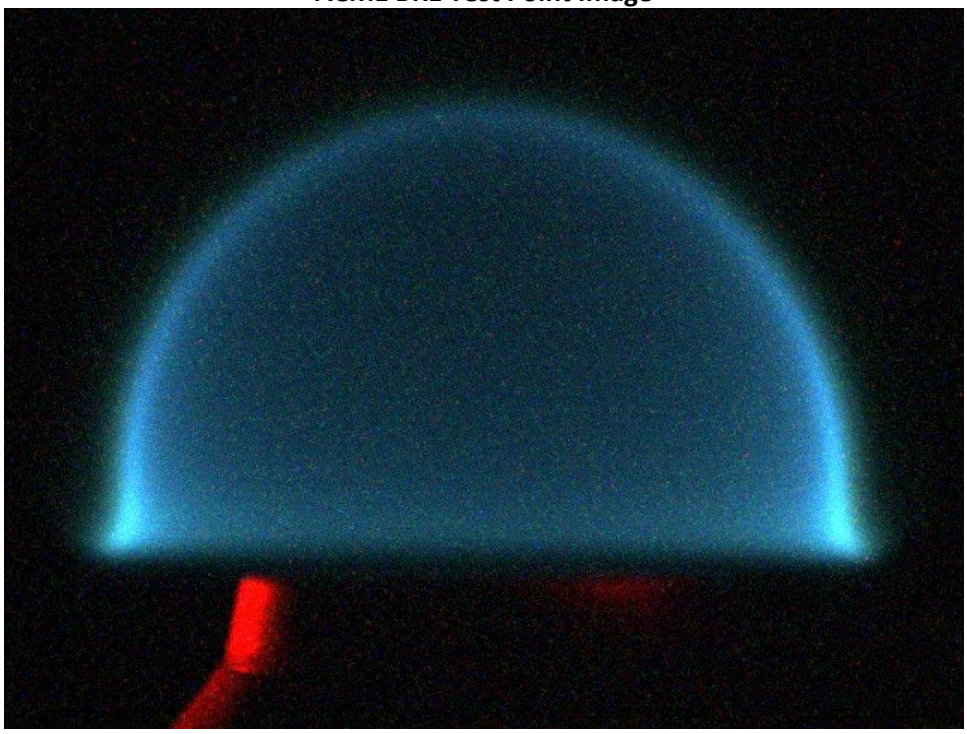
In support of ACME BRE (Burning Rate Emulator) test points, Chris Cassidy replaced the burner, igniter arm and tip, and two mass flow controllers on the ACME Chamber Insert on GMT 258. This burner allows for steadier flames, and the mass flow controllers allow for slower nitrogen and fuel flow rates.

Week 24: September 18 - 24

On GMT 261, 13 ethylene flames were ignited for ACME BRE (Burning Rate Emulator) with its primary burner, which has a 25-mm diameter outlet. Today's flames were steadier than those with the 50-mm burner used on the previous test day. In fact, one of today's tests appears to be the most stable BRE flame thus far, including the 62 BRE flames ignited in early 2019. The stability of flames in a no-breeze environment has important fire safety implications. The current approach to spacecraft fire is to stop ventilation so that the flames will quench, but the BRE results indicate that there are conditions under which flames will continue to burn in still air.

On GMT 266, Anatoly Ivanishin replaced the bottles at CIR Manifold #2 and CIR Manifold #4. An issue with the CIR Manifold #4 bottle valve delayed testing this week. From crew photos and video, we determined that the valve is already fully open, which we confirmed with powered ops on GMT 267. The CIR Bottle Valve Cover was cross-threaded when last installed, which caused it to bind on the valve. When rotating the valve cover to remove it, the valve also rotated. We will be able to resume test points next week!

ACME BRE Test Point Image



This image of a dome-shaped flame is from late in a test in which the blue flame remained quasi-steady (i.e., changes are occurring at such a slow rate that the flame can be considered constant). The burner is illuminated from the left in red. The test was conducted in an atmosphere of about 30% oxygen (with a balance of nitrogen and gases resulting from preceding tests) with a reduced pressure of roughly 0.6 atm. Future crew exploration may feature vehicles and habitats at elevated oxygen concentrations and reduced pressures to facilitate ExtraVehicular Activity (EVA). BRE includes testing under such conditions because of the increased fire hazard with elevated oxygen concentrations.

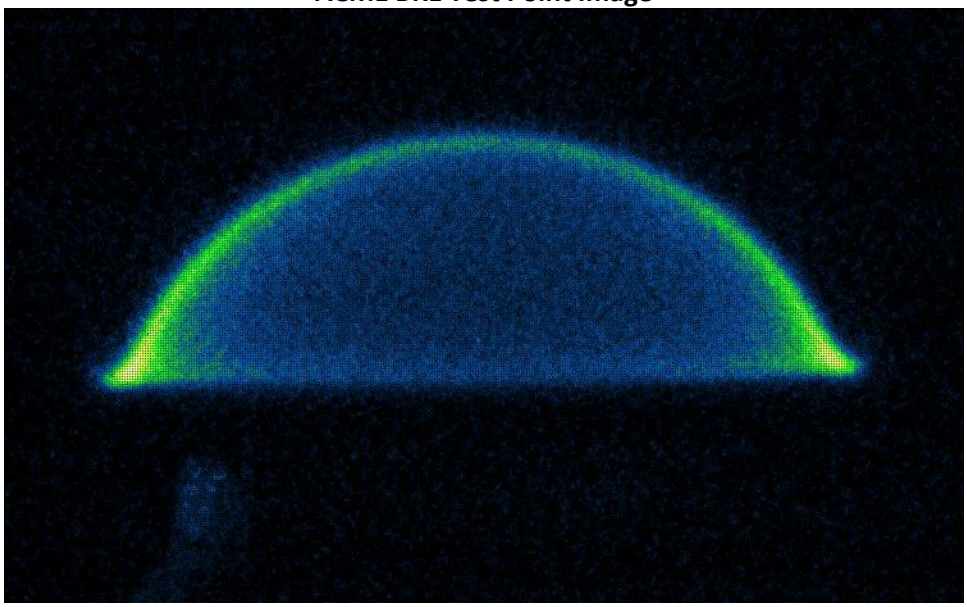
Week 25: September 25 - October 1

In support of ACME BRE (Burning Rate Emulator) test points, Anatoly Ivanishin replaced the CIR GC Argon Bottle and opened the bottle valve at CIR Manifold #2 on GMT 272. He also took some very nice, detailed photos of the damaged CIR Bottle Valve Cover.

Week 26: October 2 - 8

On GMT 279, a record 21 flames were successfully ignited for the ACME BRE (Burning Rate Emulator) experiment using methane as the fuel, which was a first for this investigation. Methane is the primary component of natural gas and is a good contrast to ethylene, the fuel used in BRE's preceding test days. Ethylene (C2H4) is naturally sooty in normal-gravity flames because of its C=C double bond; whereas, methane (CH4) has no carbon bonds. With a goal of improved spacecraft fire safety, BRE simulates the combustion of solid and liquid materials by matching key parameters in tests conducted with gaseous fuels. Testing with different gaseous fuels provides an understanding of flammability for a broader range of materials.

ACME BRE Test Point Image



This preliminary false-colored image of a dome-shaped flame is from late in a test in which the blue flame remained quasi-steady through most of the test duration. Quasi-steady describes when changes are occurring at such a slow rate that the flame can be considered constant. This photo was taken shortly before the flame oscillated for a few seconds and then self-extinguished. While the flame's top often locally quenches due to radiative heat loss, in this example the flame's circular rim (at its base) has pulled away from the burner due to conductive heat loss. The curvature of the flame's rim away from, rather than toward, the burner's edge (which can be faintly seen on the left) is evidence of the flame-burner separation and the approach of the subsequent flame instability.

Increment 64

Week 7: November 27 - December 3

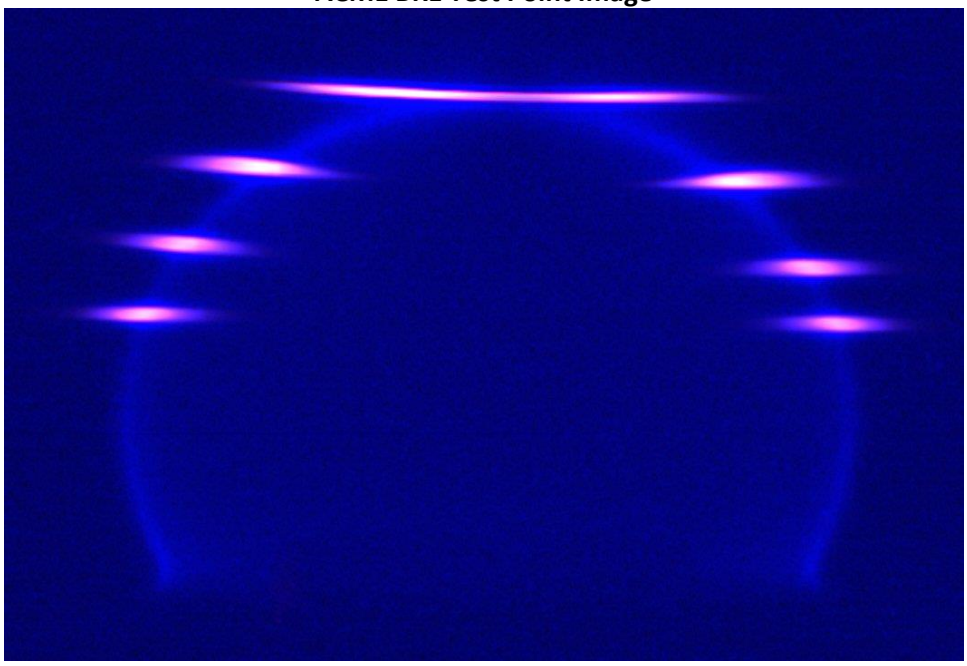
In support of ACME BRE (Burning Rate Emulator) test points, Sergey Kud-Sverchkov and Mike Hopkins replaced the bottle at CIR Manifold #2 on GMT 338.

Week 8: December 4 - 10

On GMT 339, 18 flames were ignited for the ACME BRE (Burning Rate Emulator) experiment. In this experiment, the University of Maryland's Dept. of Fire Protection Engineering studies microgravity flammability in quiescent (i.e., no-breeze) atmospheres. These tests were conducted using methane, which is the primary component of natural gas, as the fuel. By innovatively matching key properties, BRE uses gaseous fuels to study the flammability of solid and liquid materials. These tests were conducted in oxygen concentrations of up to 40% with a goal of investigating fire risks in atmospheres being considered for future spacecraft and habitats.

On GMT 345, Sergey Kud-Sverchkov replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support upcoming ACME BRE test points using diluted ethylene fuel (50% ethylene/50% nitrogen).

ACME BRE Test Point Image



This preliminary false-colored image of a dome-shaped flame shows an array of 14-micron ceramic fibers making temperature measurements. The four visible fibers pass through or near the flame and glow because of the high temperature. This flame, which is non-premixed, is like a bubble in that the methane fuel is inside the bubble and the oxygen is outside the bubble. The flame shell, like the film of a bubble, is a thin sheet where the methane and oxygen come together and react, producing carbon dioxide, water vapor, light, and heat. The fibers are only visible where they are hot, near the flame shell, which is why the centers of the three lower fibers are not visible.

The temperature profile along each fiber is optically determined from its varying hue through a technique known as Thin Filament Pyrometry (TFP). In this image, the blue has been enhanced so that the soot-free flame is clearly visible along with the much brighter fibers, which naturally transition from red to orange to yellow as the temperature increases.

Week 9: December 11 - 17

On GMT 346, 12 flames were ignited for the ACME BRE (Burning Rate Emulator) experiment. To investigate the microgravity flammability of solid materials, these tests burned nitrogen-diluted ethylene fuel (50% ethylene/50% nitrogen). Generally, the flames continued to burn through their test duration, where some exhibited partial detachment from the burner's rim or partial quenching—creating a hole in the flame top. Other flames were quasi-steady, which means changes were occurring at such a slow rate that the flame could be considered constant. The results from these tests may guide material selection for future spacecraft.

On GMT 350, Sergey Kud-Sverchkov replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support upcoming ACME BRE test points using pure ethylene fuel.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This clip from a test point video shows the flame dynamics observed in the latter portion of the test. Early in the test, the array of five ceramic fibers was translated into the flame to determine

temperature measurements from the color of the glowing fibers. In the second half of the test, the flame begins to locally quench due to radiative heat loss. The hole allows the fuel and oxygen, which are normally separated by the flame sheet, to mix. This mixing enhances flammability and causes the flame to reestablish and the hole to close. This oscillatory behavior occurs repeatedly at roughly 1 Hz while the opening in the flame becomes successively greater. This flame would have self-extinguished but was put out when the fuel was shut off at a preset time.

Week 10: December 18 - 24

On GMT 352, ten nitrogen-diluted ethylene flames were ignited for the ACME BRE (Burning Rate Emulator) experiment. Ethylene's (C2H4) C=C double bond can create sooty flames. These tests were conducted in oxygen concentrations ranging between 32% to 40%. BRE's goal is to improve our understanding of material flammability in microgravity and thereby guide the selection of materials used in spacecraft.

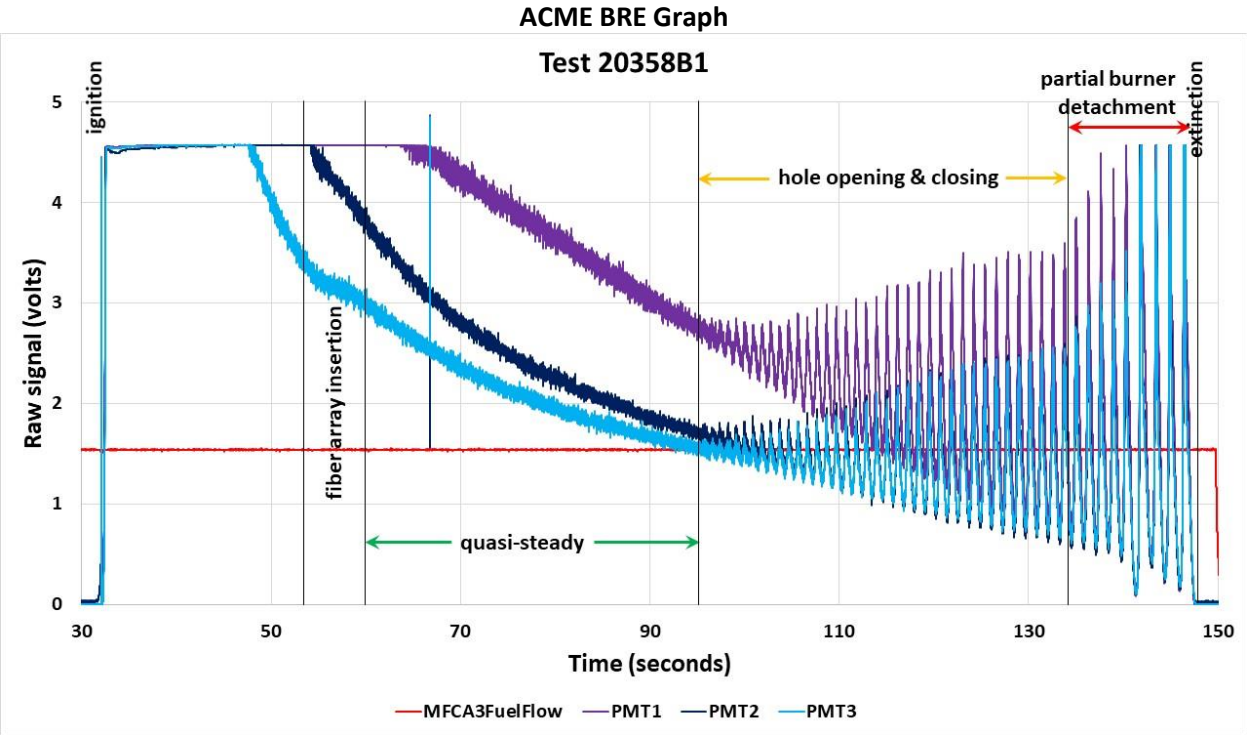
On GMT 356, Sergey Kud-Sverchkov replaced the gas bottle at CIR Manifold #2 to continue supporting ACME BRE test points.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

Upon ignition, the ethylene flame in this video clip exhibits a lot of soot, which is bright whitish-yellow in appearance. On Earth, soot is normally consumed as it passes through the flame sheet, but in this test point the soot particles are large enough to escape the flame. In microgravity, soot growth can be enhanced because it stays within the flame longer due to the absence of buoyancy. Despite this enhanced residence time, the soot fades away quickly as the burn continues. High temperatures are required to create the soot and microgravity conditions can lead to lower flame temperatures. There is no buoyancy to bring in fresh oxygen and the combustion products blanket these low-momentum flames. The carbon dioxide, water vapor, and soot contribute to radiative heat loss. This video clip ends as the flame becomes mostly blue and soot-free.

Week 11: December 25 - 31

On GMT 358, 17 ethylene flames were ignited for the ACME BRE (Burning Rate Emulator) experiment. These tests were conducted in oxygen concentrations ranging from 24% to 40%, and the ethylene fuel was generally diluted with nitrogen. At the higher oxygen concentrations, the flames were generally quasi-steady and continued to burn until the fuel was shut off. In contrast, at the lower oxygen concentrations, the flames all self-extinguished. On GMT 364, Sergey Kud-Sverchkov replaced the gas bottle at CIR Manifold #2 to support continuation of ACME BRE test points.



This graph shows as a function of time, the fuel flow (red) and three optical flame measurements from a test in which pure ethylene was burned. The optical flame measurements were made with Photomultiplier Tube (PMT) modules of OH* chemiluminescence (purple), CH* chemiluminescence (light blue), and broadband emissions in the visible and near UV spectrum (dark blue). OH* and CH* are transient molecules which, in the process of chemiluminescence, emit a photon at specific wavelengths when they drop to a lower energy level. The CH* chemiluminescence is the principal source of the blue seen in hydrocarbon flames, while OH* is in the ultraviolet and is used in ACME to indicate ignition and extinction.

The initially sooty flame was ignited at 32 seconds, after which the hot-wire igniter was retracted. At about 60 seconds, after the ceramic fiber array was inserted, the flame became quasi-steady. But the PMT signals weakened with time, where the low-momentum flame became increasingly blanketed by carbon dioxide and water vapor in the absence of buoyancy.

At about 95 seconds, the increasing radiative loss caused the flame's top to quench, creating a hole in the flame. This hole allowed the fuel and oxygen, which are normally separated by the flame, to directly mix. This mixing enhanced flammability, and the flame quickly reestablished and closed the hole, but then again, it locally quenched. This oscillatory behavior happened repeatedly at roughly 1 Hz with the flame opening growing with time until the flame became just a ring. We saw this phenomenon in the first round of BRE testing performed on ISS in 2019. A paper about computational modeling of it is now in press.

At about 135 seconds, the flame began to partially detach from the rim of the burner and then reattach until it completely extinguished at 148 seconds.

2021

Week 14: January 15 - 21

In support of ACME BRE (Burning Rate Emulator) test points, Victor Glover removed the broken fiber and replaced the igniter tip on the ACME Chamber Insert on GMT 018. On GMT 020, 15 ethylene flames were ignited for the ACME BRE experiment. The array of ceramic fibers (now with three instead of five functional fibers) was successfully used to determine the temperature distributions where the fibers were hot enough to visibly glow. Tests were conducted in atmospheres being considered for future spacecraft and habitats, where that included elevated oxygen concentrations and reduced pressures which can facilitate EVA preparation. At normal atmospheric conditions (e.g., ~21% oxygen by volume), the flames all self-extinguished. In contrast, three of the five tests conducted at 30-35% oxygen and 0.57 bar (i.e., 8.2 psia) featured quasi-steady flames, which continued to burn through the test duration. While the oxygen concentration dominates the flame behavior, this day's tests will enable a better understanding of the role of the atmospheric pressure on microgravity flammability. On GMT 021, Kate Rubins replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support continuation of ACME BRE test points.

Week 15: January 22 - 28

In support of ACME BRE (Burning Rate Emulator) test points, Kate Rubins replaced the burner in the ACME Chamber Insert on GMT 025. In contrast to the 25-mm burner that was removed, this burner is 50 mm in diameter and allows for larger flames. Tests are being conducted with this larger burner to investigate the dependence of the microgravity flammability on flame size.

Week 16: January 29 – February 4

On GMT 028, the final test points were performed for the ACME BRE (Burning Rate Emulator) experiment led by researchers from the University of Maryland and Russia's Peter the Great St. Petersburg Polytechnic University. In the study of microgravity flammability, 13 flames were ignited with the 50-mm diameter burner at oxygen concentrations from 40% to 23% in 1 atm. This larger burner allows for larger flames compared to the 25-mm burner that was used for previous tests. Generally, these larger flames tended to self-extinguish due to radiative heat loss, which scales with the flame size. In contrast, the smaller flames on the 25-mm burner were generally quasi-steady. BRE studies material flammability with the goal of improving spacecraft fire safety.

On GMT 034, Kate Rubins and Victor Glover configured the ACME Chamber Insert to support the next ACME experiment, CLD Flame Part 3 (Coflow Laminar Diffusion). The general goal of CLD Flame is to extend the range of flame conditions that can be accurately predicted, especially under conditions where the flame is sooty or could become unstable. The results could lead to the improved design of practical combustion devices on Earth.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This clip from a test point video shows a flame's oscillatory "death throes" leading ultimately to the flame's extinction (not shown in the clip). Initially, radiative heat loss causes the top of the flame (on the video's left) to locally quench, creating a hole in the flame. That opening allows the fuel (from the burner) and oxygen (in the chamber) to directly mix, which increases the flammability and causes the flame to reestablish. But, due to heat loss, the flame partially quenches again, and the opening grows with each oscillation cycle. Later, the flame also begins to detach from the burner's rim, leaving an opening at the flame's base (on the video's right). Here too the flame reestablishes but then opens again—the opening growing until it extends from the

burner to the flame's top. At this point in the flame's powerful death throes, it begins swirling around the burner's rim. This behavior is notably evident in the glowing ceramic fibers, which can be seen to glow on alternating ends when heated by the swirling flame. To gain a better understanding of this extinction phenomena, the local temperatures of the flame will be derived from the glowing fibers.

Appendix I. Crew Activity per Space Station Science Highlights

The subsequent text and images are from the weekly highlights prepared by the Lead Increment Scientist and made available via e-mail or online via www.nasa.gov/mission_pages/station/research/.

Note that hyperlinks were generally not included in the excerpts below, but they typically go to web pages accessible via https://www.nasa.gov/mission_pages/station/research/experiments/explorer/. Those pages are secondary sources of information for the ACME project, for which the primary web site is https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

BRE round 1, featuring tests with ethylene and the 25-mm burner

Week of January 28, 2019

Other work was performed on these investigations: …

- The Advanced Combustion Microgravity Experiment (ACME) investigation is a set of five independent studies of gaseous flames to be conducted in the Combustion Integration Rack (CIR).

Week of February 4, 2019

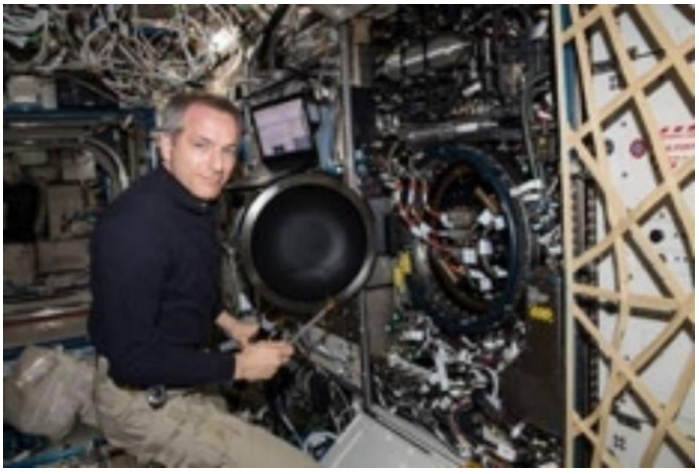
Other work was performed on these investigations and facilities:

- CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

Week of February 25, 2019

Studying gaseous flames and flammability

Following replacement of an ACME fuel flow controller, which allows observation of different flow conditions, the crew documented its final configuration with photographs. The ACME series performed in the Combustion Integrated Rack (CIR) includes five independent studies of gaseous flames. The current experiment, Burning Rate Emulator (BRE), focuses on spacecraft fire prevention, specifically improving understanding of materials flammability and assessing existing flammability test methods in low and partial-gravity environments.



Astronaut David Saint-Jacques of the Canadian Space Agency is inside the U.S. Destiny laboratory module working on the Combustion Integrated Rack. Saint-Jacques replaced fuel flow controllers inside the device

for the Advanced Combustion via Microgravity Experiments which are a set of five independent studies of gaseous flames. *Credits: NASA*

Week of March 11, 2019

Other investigations on which the crew performed work: …

- The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Week of March 18, 2019

Other investigations on which the crew performed work: …

- The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Week of March 25, 2019

Other investigations on which the crew performed work: …

- The Combustion Integrated Rack (CIR**)** includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Week of April 8, 2019

Other investigations on which the crew performed work:

- The Combustion Integrated Rack (CIR**)** includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.



NASA astronaut Nick Hague replaced an igniter in the Advanced Combustion via Microgravity Experiment (ACME), a set of six studies of gaseous flames conducted in the Combustion Integrated Rack (CIR). The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity. *Credits: NASA*

The caption was initially faulty; the corrected version above was provided in the 'Week of April 22.'

BRE round 2, featuring tests with ethylene & methane and the 25-mm & 50-mm burners

Week of August 31, 2020

Space Station Science Image of the Week: CIR

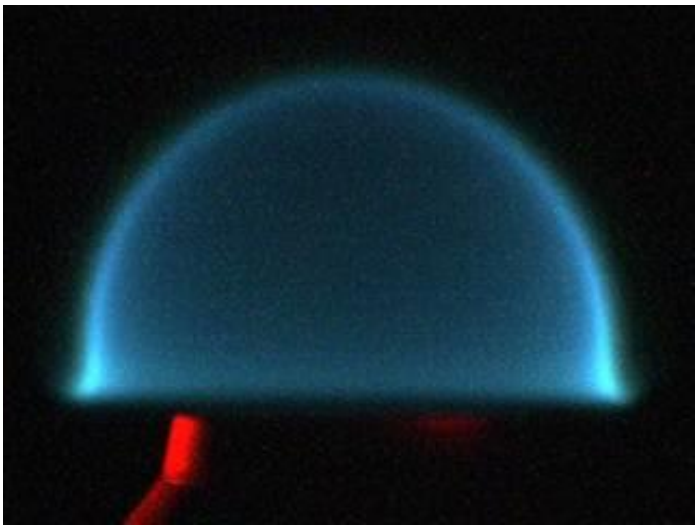


NASA astronaut Chris Cassidy changed out the experiments in the Combustion Integrated Rack (CIR) this past week. These investigations could help us learn about ***fire prevention in space***.

Week of September 28, 2020

Testing fire safety in space

The Burning Rate Emulator (BRE) fire safety study aims at improving our fundamental understanding of the flammability of certain materials and assesses whether existing flammability test methods are appropriate for low and partial-gravity environments. BRE is conducted in the Combustion Integrated Rack (CIR) as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The crew exchanged a gas chromatograph bottle for conducting the second part of BRE.



This image shows a dome-shaped flame and the burner illuminated in red on the left during a BRE test in an atmosphere of about 30% oxygen. Future crew exploration missions may feature vehicles and habitats at elevated oxygen concentrations, and these conditions present increased fire hazard.
Credits: NASA

Week of November 30, 2020

Other investigations on which the crew performed work:

- ACME is a set of six independent studies of gaseous flames. ACME's goals are to advance fuel efficiency and reduce pollutant production in practical combustion on Earth, and to improve spacecraft fire prevention.

Week of December 7, 2020

Other investigations on which the crew performed work:

- BRE focuses on fire prevention in spacecraft, examining burning conditions and the flammability of materials in microgravity. BRE is part of ACME, a set of six independent studies of gaseous flames intended to advance fuel efficiency and reduce pollutant production in practical combustion on Earth, and to improve spacecraft fire prevention.

Weeks of December 21 and 28, 2020

Other investigations on which the crew performed work:

- BRE focuses on fire prevention in spacecraft, examining burning conditions and the flammability of materials in microgravity. BRE is part of ACME, a set of six independent studies of gaseous flames intended to advance fuel efficiency and reduce pollutant production in practical combustion on Earth, and to improve spacecraft fire prevention.

Week of January 4, 2021

Other investigations on which the crew performed work:

- BRE focuses on fire prevention in spacecraft, examining burning conditions and the flammability of materials in microgravity. BRE is part of ACME, a set of six independent studies of gaseous flames intended to advance fuel efficiency and reduce pollutant production in practical combustion on Earth and to improve spacecraft fire prevention.

Week of January 11, 2021

Other investigations on which the crew performed work:

- ACME is a set of six independent studies of gaseous flames. ACME's goals are to advance fuel efficiency and reduce pollutant production in practical combustion on Earth, and to improve spacecraft fire prevention.

Week of January 25, 2021

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

NASA astronaut Kate Rubins sets up the Combustion Integration Rack for BRE, one of six ACME investigations into combustion and fire safety in space.
Credits: NASA

Other investigations on which the crew performed work:

- BRE, part of the set of Advanced Combustion via Microgravity Experiments (ACME), simulates the flammability of solid and liquid materials by burning gaseous fuels under conditions corresponding to those materials.

Appendix J. Crew Activity per the ISS Daily Summary Report

*The subsequent text was prepared by Dennis Stocker, where the source information is available from https://blogs.nasa.gov/stationreport. Text was added identifying the investigation's two rounds, but BRE operations listed under the headings "*All activities are complete unless otherwise noted*" or "Completed Task List Activities" were not included in this report. Some days of crew activity were only listed in those ways and as a result this section does not comprehensively document crew activity in support of the BRE experiment.*

BRE round 1, featuring tests with ethylene and the 25-mm burner

2019 January 14: GMT 014

Combustion Integrated Rack (CIR) imaging reconfiguration: Today the crew replaced a filter module and relay optics module. This is part of the reconfiguration to support the upcoming Burning Rate Emulator (BRE) series of experiments. These reconfiguration activities are expected to continue over the next two to three weeks. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2019 January 18: GMT 018

Combustion Integrated Rack (CIR) Hardware Reconfiguration: As a continuation of the ongoing CIR reconfiguration for the Advanced Combustion via Microgravity Experiments (ACME) BRE (Burning Rate Emulator) investigation, the crew replaced the relay optics module in the CIR HiBMS (High Bit depth Multi Spectral) Imaging Package, installed the CIR LLL-UV (Low Light Level Ultraviolet) Imaging Package, and installed the FCF IPSU (Fluids and Combustion Facility Image Processing and Storage Unit). The BRE experiment is focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2019 January 31: GMT 031

Advanced Combustion via Microgravity Experiments (ACME): A crewmember replaced hardware on the ACME unit by installing a new mesh power supply and closeout panel. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2019 February 1: GMT 032

The ISS Experience: Today a crewmember setup the ISS Experience hardware in the US Lab in order to record portions of the CIR ACME reconfiguration and chamber insertion into the CIR rack. ISS Experience creates a virtual reality film documenting daily life aboard the space station. The 8 to 10 minute videos created from footage taken during the six-month investigation cover different aspects of crew life, execution of science aboard the station, and the international partnerships involved. The ISS Experience uses a Z-CAM V1 Pro Cinematic VR (Virtual Reality) 360-degree camera with nine 190° fisheye lenses.

Advanced Combustion via Microgravity Experiments (ACME): Today a crew completed the part 2 of the ACME hardware reconfiguration. The ACME experiment series being performed in the CIR includes five

independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2019 February 4: GMT 035

Combustion Integrated Rack (CIR): Today crewmembers performed a Manifold #2 high percentage O2 Bottle and Gas Chromatograph (CG) bottle changeout on the front Optics Bench. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2019 February 19: GMT 050

Advanced Combustion via Microgravity Experiments (ACME): Today the crew removed ACME hardware items from storage and consolidated them with other ACME hardware. The ACME experiment series being performed in the CIR includes five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2019 February 22: GMT 053

CIR (Combustion Integrated Rack): Today a crewmember performed O2 bottle exchanges in the CIR rack facility Manifold 2 and 4 in preparation for upcoming test runs. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2019 February 26: GMT 057

Advanced Combustion via Microgravity Experiments (ACME) Controller Replacement: The crew replaced one of the ACME fuel flow controllers. This was a nominally planned activity and allows different flow conditions to be observed. Following the completion of the activities, the crew photo documented the final configuration of ACME Chamber Insert. The ACME series being performed in the Combustion Integrated Rack includes five independent studies of gaseous flames, with the Burning Rate Emulator (BRE) experiment currently active. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2019 March 5: GMT 064

Combustion Integrated Rack (CIR): Today a crewmember replaced the Manifold #2 O2/N2 bottle in preparation for upcoming CIR runs. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2019 March 13: GMT 072

Combustion Integrated Rack (CIR) Manifold Bottle Replacement: The crew removed a 50% ethylene fuel bottle and replaced it with a 100% ethylene bottle. They also installed a 100% helium bottle. This is in support of the ongoing BRE (Burning Rate Emulator) experiment, which is part of the ACME (Advanced Combustion via Microgravity Experiments) suite of experiments. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental

understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2019 March 18: GMT 077

Combustion Integrated Rack (CIR) Fuel Oxidizer Management Assembly (FOMA): The crew performed the FOMA calibration preparatory activities which consisted of closing and opening the appropriate valves on manifold bottles and the GIP (gas interface panel). The FOMA calibration is a periodic activity required to support CIR science operations. CIR provides sustained, systematic microgravity combustion research and it houses hardware capable of performing combustion experiments to further research of combustion in microgravity.

2019 March 26: GMT 085

CIR (Combustion Integrated Rack): The crew swapped out the CIR Manifold 02/N2 bottles to support upcoming CIR Burning Rate Emulator runs. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2019 April 3: GMT 093

CIR (Combustion Integrated Rack): The crew replaced a burned out Advanced Combustion via Microgravity Experiment (ACME) igniter for the Spherical Burner. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

BRE round 2, featuring tests with ethylene & methane and the 25-mm & 50-mm burners

2020 August 26: GMT 239

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew replaced the Gas Chromatograph (GC) helium bottle, GC check gas bottle, and manifold bottles 2 and 4. During these bottle exchanges they also performed quick-disconnect (QD) lubrication to prevent the potential for any future leaks at the QD interfaces. The bottle exchanges are in preparation for the upcoming BRE part 2 (Burning Rate Emulator), which is one of the ACME suite of five experiments. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2020 August 27: GMT 240

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew performed the steps necessary to reconfigure the CIR/ACME hardware components from the completed s-Flame part 2 experiment to BRE part 2 (Burning Rate Emulator), which is one of the ACME suite of five experiments. Although the crew was not able to locate and install igniter arm 2, the currently-installed igniter arm 1 was left in the system and will be assessed for acceptable alignment and function with the burner head during checkout. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2020 September 14: GMT 258

Burning Rate Emulator (BRE): The crew reconfigured the ACME hardware in preparation for BRE experiment runs. This includes swapping in Igniter Arm #2. The Burning Rate Emulator (BRE) investigation is conducted in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials. This technique has been demonstrated for a wide variety of materials in normal-gravity and could provide an efficient way to screen and select fire-resistant materials for use in spacecraft, if the technique is similarly effective in microgravity.

2020 September 22: GMT 266

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME)/BRE-2: The crew exchanged a 40% oxygen/60% nitrogen bottle at the manifold 2 location for a new bottle with the same composition. They also replaced the ethylene manifold 4 bottle with a methane bottle, but were not able to open the bottle valve. CIR powered ops for today have been cancelled while the ground team discusses the issue. This was performed in support of the continuing BRE part 2 (Burning Rate Emulator) investigation, which is one of the ACME suite of five gas combustion experiments. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2020 September 28: GMT 272

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME)/BRE-2: The crew exchanged a GC (gas chromatograph) argon bottle and opened the valve on the 40% O2/60% N2 bottle on manifold 2. This was performed in support of the continuing BRE part 2 (Burning Rate Emulator) investigation, which is one of the ACME suite of five gas combustion experiments. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2020 December 3: GMT 338

Burning Rate Emulator (BRE): The crew performed O2/N2 gas bottle changes in preparation for upcoming BRE runs. The BRE investigation is conducted in the Combustion Integrated Rack (CIR) on the ISS, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials. This technique has been demonstrated for a wide variety of materials in normal-gravity and could provide an efficient way to screen and select fire-resistant materials for use in spacecraft, if the technique is similarly effective in microgravity.

2020 December 10: GMT 345

BRE (Burning Rate Emulator): The crew performed O2/N2 gas bottle changes and C2H4 fuel bottle changes in preparation for upcoming BRE runs. The Burning Rate Emulator (BRE) investigation is conducted in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials. This technique has been demonstrated for a wide variety

of materials in normal-gravity and could provide an efficient way to screen and select fire-resistant materials for use in spacecraft, if the technique is similarly effective in microgravity.

2020 December 15: GMT 350

CIR/ACME (Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments)/BRE-2: The crew performed a manifold bottle exchange to allow the continuation of the BRE-part 2 series of experiment burns. The fuel currently installed in the rack is ethylene (CH_2=CH_2). This was performed in support of the continuing BRE part 2 (Burning Rate Emulator) investigation, which is one of the ACME suite of five gas combustion experiments. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2020 December 21: GMT 356

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME)/Burning Rate Emulator (BRE-2): The crew replaced the current 40% O2 / 60% N2 manifold bottle with a full bottle of the same composition. This was in support of the continuing BRE part 2 investigation, which is one of the ACME suite of five gas combustion experiments. The BRE experiment runs are focused on spacecraft fire prevention. More specifically, BRE's objective is to improve our fundamental understanding of materials flammability and to assess the relevance of existing flammability test methods for low and partial-gravity environments.

2020 December 29: GMT 364

Burning Rate Emulator (BRE): The crew completed the activities needed to replace the current 40% O2 / 60% N2 manifold bottle with a full bottle of the same composition. The Burning Rate Emulator (BRE) investigation is conducted in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials. This technique has been demonstrated for a wide variety of materials in normal-gravity and could provide an efficient way to screen and select fire-resistant materials for use in spacecraft, if the technique is similarly effective in microgravity.

2021 January 18: GMT 018

CIR/ACME hardware replace: The crew opened the doors on the CIR, removed the ACME insert, and removed 2 broken silicon carbide (SiC) fiber from the 5-fiber array inside the ACME system. These thin SiC fibers are used to determine temperature of areas in and around the flames generated by the ACME experiments. The crew also replaced an igniter tip used to ignite the gas mixtures in the ACME insert. This will allow the continuation of the BRE-part 2 experiment. The Burning Rate Emulator (BRE) investigation is conducted in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials.

ISS Experience: The crew set up the ISS Experience hardware and captured the CIR/ACME hardware activities performed today. The International Space Station Experience (The ISS Experience) is a cinematic virtual reality (VR) series documenting life and research aboard the space station. Filmed over multiple months, the immersive VR series documents different crew activities – from science conducted aboard

the station to preparation for a spacewalk. The ISS Experience uses a Z-CAM V1 Pro Cinematic Virtual Reality (VR) 360-degree camera with nine 190° fisheye lenses.

2021 January 21: GMT 021

CIR/ACME – The crew replaced a 40% O_2/60% N_2 gas bottle and a 100% CH_2=CH_2 gas bottle to allow the continuation of the BRE part 2 investigation. The Burning Rate Emulator (BRE) investigation is conducted in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials. This technique has been demonstrated for a wide variety of materials in normal-gravity and could provide an efficient way to screen and select fire-resistant materials for use in spacecraft, if the technique is similarly effective in microgravity.

2021 January 25: GMT 025

Burning Rate Emulator (BRE): The crew configured the ACME Chamber Insert for the 9th day of BRE-2 test points. This involved lubrication of the water Quick Disconnect interfaces and a replacement of the Heat Flux burner. The Burning Rate Emulator (BRE) investigation is conducted in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this fire safety study, the flammability of solid and liquid materials is simulated by burning gaseous fuels under key conditions corresponding to the solid and liquid materials. This technique has been demonstrated for a wide variety of materials in normal-gravity and could provide an efficient way to screen and select fire-resistant materials for use in spacecraft, if the technique is similarly effective in microgravity.

2021 January 26: GMT 026

CIR-Combustion Integrated Rack – Fluids and Combustion Facility: The crew stowed CIR hardware for return. The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Appendix K. Crew Activity per the NASA Space Station Blog

This section consists of excerpts from https://blogs.nasa.gov/spacestation/. Hyperlinks were generally not included, but clarifications have sometimes been added in red.

BRE round 1, featuring tests with ethylene and the 25-mm burner

2019 January 14 - Dragon Back on Earth as Crew Revs Up Station Science

Flight Engineer David Saint-Jacques of the Canadian Space Agency opened up the Combustion Integrated Rack and replaced optics gear inside the flame and soot research device.

2019 January 18 - Tech Work and Life Science Ahead of Orbital Boost Today

Satellite and combustion technology are being worked on today aboard the International Space Station. … Canadian Space Agency astronaut David Saint-Jacques opened up the Combustion Integrated Rack and configured hardware inside the flame and soot research device. The work is being done ahead of operations for the Advanced Combustion in Microgravity Experiments that encompass a set of five independent gaseous flames studies.

2019 January 30 - Crew Works CubeSats, Life Science and Configures Physics Hardware

Saint-Jacques … later swapped parts in the Combustion Integrated Rack that permits safe research into fuel and flames aboard the orbital lab.

2019 January 31 - Fiery Research Work and CubeSats Deployed Today

The Expedition 58 crew set up a variety of combustion research hardware today to look at what happens to high temperatures, fuels and flames in space. … Flight Engineer David Saint-Jacques opened the Combustion Integrated Rack inside the U.S. Destiny lab module again today for more maintenance work. The Canadian astronaut replaced a control unit and a radiometer inside the fuel and flame research device.

2019 February 1 - Crew Wraps Up Biomedical Studies; Films Station in Virtual Reality

The Combustion Integrated Rack received more attention today as Saint-Jacques replaced hardware in the fuel and flame research platform.

2019 February 19 - Experimental Fuel Hardware, Astrophysics and Life Science Fill Crew Day

The Expedition 58 crew … worked on a variety of life science, astrophysics and combustion science gear. … McClain … later cleaned up gear supporting gaseous flame studies inside the Combustion Integrated Rack.

2019 March 5 - Crew Resumes Normal Ops Before Crew Dragon Leaves Friday

Commander Oleg Kononenko replaced fuel bottles used during experiment operations inside the Combustion Integrated Rack.

2019 March 13 - **Housekeeping and Maintenance Punctuate Last Full Day of Expedition 58**
Expedition Commander Oleg Kononenko of Roscosmos replaced fuel bottles on the Combustion Integrated Rack, which allows the crew members to conduct fluids and combustion studies in microgravity.

BRE round 2, featuring tests with ethylene & methane and the 25-mm & 50-mm burners

2020 September 14 - **Crew Sets Up Advanced Science Gear, Loads Cargo Ship With Trash**
Today, Commander Chris Cassidy … swapped components supporting safe combustion investigations. ... Next, the veteran NASA astronaut replaced a burner and igniter tip inside the Combustion Integrated Rack that supports fuel and flame research.

2020 September 22 - **Crew Takes on Eye Doc and Plumber Roles as Station Avoids Debris**
The Combustion Integrated Rack's fuel bottles were replaced today to support ongoing research into flames and fuels. Ivanishin serviced the science rack that provides insights helping scientists and engineers improve fire safety and fuel performance for both space and Earth systems.

2020 September 28 - **Weather Delays U.S. Cargo Mission; Crew Back to Work After Leak Test**
Veteran cosmonaut Anatoly Ivanishin once again opened up the Combustion Integrated Rack and replaced fuel bottles to support fuel and flames studies inside the research device.

2020 December 3 - **Crew Immersed in Space Science as Cargo Dragon Nears Launch**
Flight Engineers Michael Hopkins and Sergey Kud-Sverchkov swapped fuel bottles inside the Combustion Integrated Rack that enables safe investigations of flames and fuels on the station.

2020 December 10 - **Cancer and Heart Studies, Spacesuit Swaps Aboard Station Today**
Flight Engineer Sergey Kud-Sverchkov swapped fuel bottles inside the Combustion Integrated Rack that enables safe research into fuel and flame studies.

2020 December 15 - **More Heart Research as Next Cygnus Cargo Mission Booked**
Flight Engineer Sergey Kud-Sverchkov swapped fuel bottles inside the Combustion Integrated Rack research facility before configuring life support and communications hardware.

2021 January 21 - **Station Boosts Orbit During Research and Spacewalk Preps**
DNA, time perception and combustion investigations filled the research schedule aboard the International Space Station today. … Biologist and NASA Flight Engineer Kate Rubins … replaced fuel bottles inside the Combustion Integrated Rack to maintain safe fuel and flame studies aboard the orbiting lab.

Appendix L. ACME Operations per @ISS_Research on Twitter

These tweets (from https://twitter.com/ISS_Research/) were from the period of BRE's second round operations, i.e., from July 2020 to Jan. 2021. However, some of the tweets feature images or video clips from other ACME investigations.

2020 July 23: GMT 205

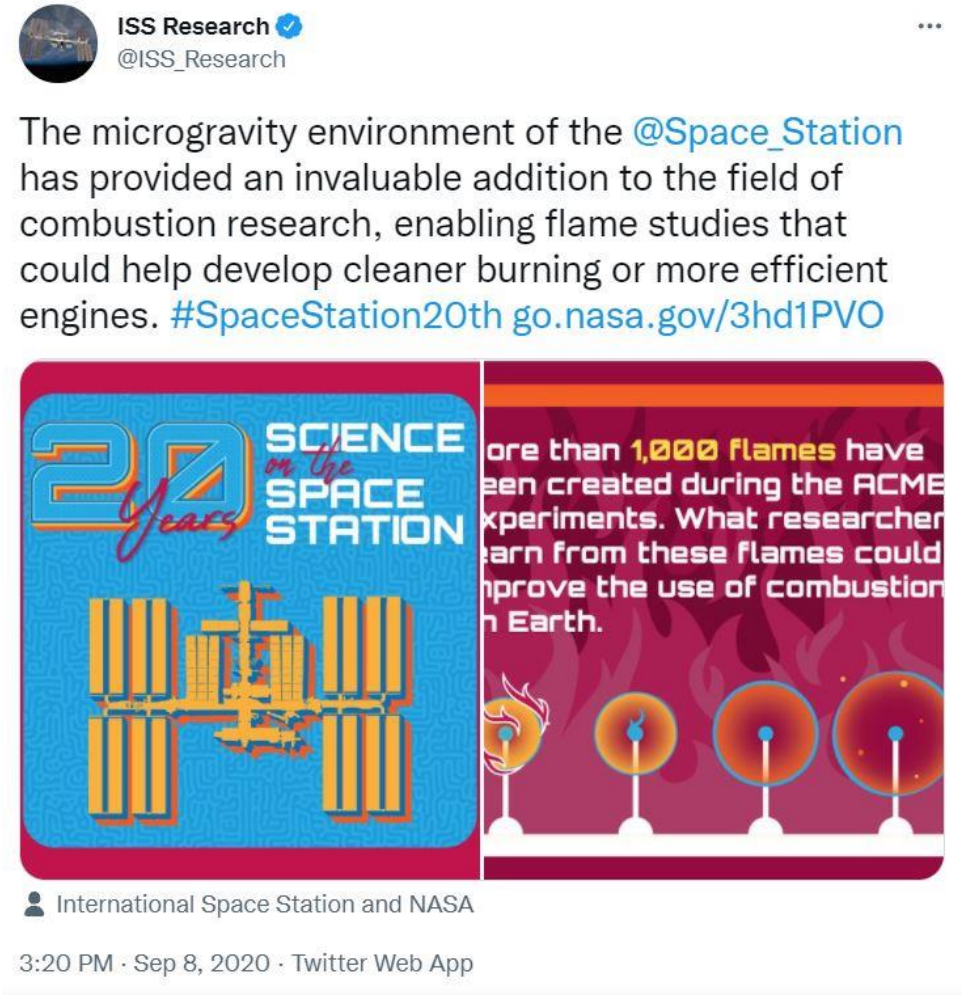


This ACME/Flame Design experiment video clip can be viewed at and downloaded from https://images.nasa.gov/details-jsc2019m000710_Designing-Flames-Aboard-ISS-MP4.html.

2020 August 27: GMT 240



2020 Sept. 8: GMT 252



2020 Dec. 31: GMT 366



2021 Jan. 22: GMT 022

ISS Research
@ISS_Research

💠Testing technologies for space debris removal
🔥Studying fire safety
🩸Testing a device for in-flight measurement of white blood cell counts

The @Space_Station crew worked on these investigations and more this past week. Learn more: go.nasa.gov/396He4K



4:57 PM · Jan 22, 2021 · Twitter Web App

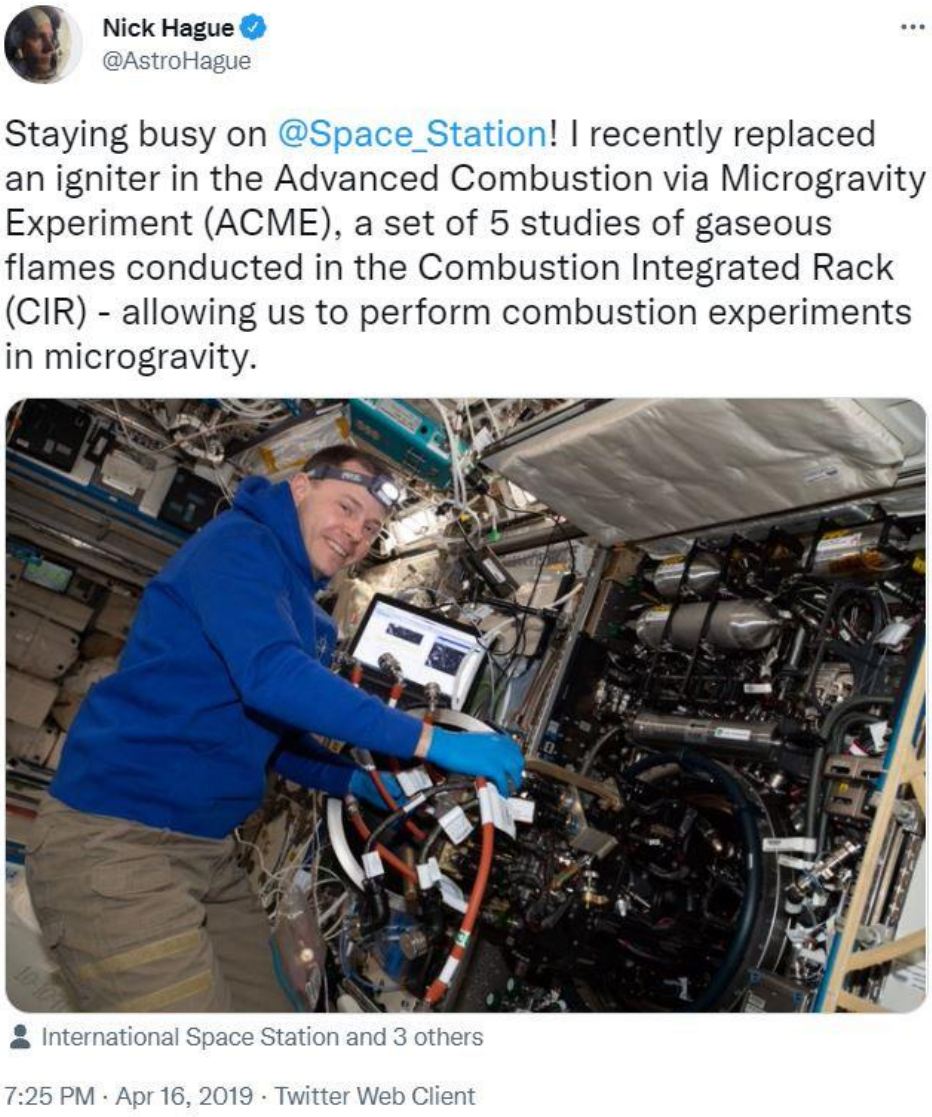
Appendix M. ACME Operations per Astronauts on Twitter

These tweets are from ISS crew members who conducted operations in support of the BRE experiment.

BRE round 1, featuring tests with ethylene and the 25-mm burner

Nick Hague (@AstroHague) on Twitter, https://twitter.com/astrohague

2019 April 16: GMT 106



BRE round 2, featuring tests with ethylene & methane and the 25-mm & 50-mm burners

2020 Sept. 1: GMT 245



Appendix N. Summary of the ISS data collection

The BRE data is all digital and includes both image and non-image data, where this is true for the other ACME experiments. To be clear, there was no collection of soot or other physical samples that were subsequently transported to the ground.

The BRE image data includes images from the following cameras:

1. ACME DataCam, used with a clear fused-silica 'filter' and various zoom and focus settings (but those settings were consistent within each individual test)
2. CFI intensified camera, typically used with a CH* filter (and likely always or nearly always for BRE)
3. High-Bitdepth Multispectral (HiBMs) camera, equipped with a red filter corresponding to the opposing CIR illumination package, which was sometimes left off

The raw DataCam images are in a Halcon Iconic Object (hobj) format, while the images from the CFI, and HiBMs cameras are provided in a Tagged Image File (tif) format. In contrast, the analog video was recorded on the ground in a Transport Stream (ts) format.

The key non-imaging data from a test is a comma-delimited file that is described in detail in the Software Interface Design Description (ACME-DOC-010 Rev. D).

The raw data is currently archived at NASA and work is underway to make it available publicly through NASA's Physical Science Informatics System (PSI).

Appendix O. Publications, presentations, awards, and outreach

Papers, etc. listed in italics are general to ACME rather than specific to the BRE experiment.

O.1 Publications and Presentations with ISS Results

Refereed Journal Articles

3. P. Dehghani, J.G. Quintiere, *Theoretical Analysis and Predictions of Burning in Microgravity Using a Burning Emulator*, Combustion and Flame, 233 (2021) 111572. https://doi.org/10.1016/j.combustflame.2021.111572
2. P. Dehghani, P.B. Sunderland, J.G. Quintiere, J.L. deRis, *Burning in Microgravity: Experimental Results and Analysis*, Combustion and Flame, 228 (2021) 315-330. https://doi.org/10.1016/j.combustflame.2021.01.035
1. A. Snegirev, E. Kuznetsov, E. Markus, P. Dehghani, P.B. Sunderland, *Transient dynamics of radiative extinction in low-momentum microgravity diffusion flames*, Proceedings of the Combustion Institute, Volume 38, Issue 3, 2021, pp. 4815-4823. https://doi.org/10.1016/j.proci.2020.06.110

Conference Proceedings, Abstracts, Posters, Etc.

12. Peter B. Sunderland, Transformative Combustion Research in the ACME Suite of ISS Experiments, 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, Maryland, 3-6 Nov. 2021.

11. Dennis P. Stocker, Recent Research on Flames of Gaseous Fuel Aboard the International Space Station, Conference of the Japanese Society of Microgravity Application (JASMAC-33), virtual, 13-15 Oct. 2021, 11 pp.

10. P. Dehghani, A. Wright, J.L. deRis, P.B. Sunderland, *Burning Rate Emulations Aboard the International Space Station*, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.
9. P. Dehghani, A. Wright, J.L. de Ris, and P.B. Sunderland, *Burning emulations aboard the International Space Station*, Work In Progress poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.

8. Dennis P. Stocker, Advanced Combustion via Microgravity Experiments on the International Space Station, Work in Progress Poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.

7. A. Wright, P. Dehghani, P.B. Sunderland, and J.L. de Ris, *Using a Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station*, Work In Progress poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.

6. S. Pichugin, D. Samsonov, S. Frolov, O. Korobeinichev, A. Shmakov, Flames Investigations at ISS, Cosmonautics of XXI Century Conference, TSNIIMash, Moscow, Russia, 12-14 May 2021.

5. P. Dehghani, A. Wright, J.L. deRis, and P.B. Sunderland, *Pool Fire Emulations Aboard the International Space Station*, Poster, 13th International Symposium on Fire Safety Science, virtual, 26-30 April 2021.
4. P. Dehghani, P.B. Sunderland, *Burning Emulation in Microgravity - Experimental Results and Analysis*, 36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), virtual, 5-6 Nov. 2020.
3. P. Dehghani, E. Auth, C. Cui, D.P. Stocker, J.L. deRis, P.B. Sunderland, J.G. Quintiere, *Microgravity Observations of a Burning Emulator to Investigate Material Flammability*, Eastern States Section of the Combustion Institute, Columbia SC, 8-11 March 2020, 6 pp.

2. *Sergey Pichugin, Oleg Artemiev, Sergey Prokopiev, Oleg Kononenko, Oleg Novitsky, and Sergey Frolov, Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience, XIII International Scientific and Practical Conference 'Manned Flights in Space,' Gagarin Cosmonaut Training Centre, Star City, Russia, 13-15 Nov. 2019.*
1. James G. Quintiere, Peter B. Sunderland, *Burning Rate Emulator (BRE): Material Burning in Microgravity*, ISS Increments 61 & 62 Science Symposium, virtual, 17-19 Sept. 2019.

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2. Venzon, Ryan, *Thin-Filament Pyrometry and Analysis of Extinction Onset Temperatures for Ethylene Flames on the Burning Rate Emulator in the International Space Station*, M.S. Thesis, University of Edinburgh, Dept. of Civil and Environmental Engineering, 2021.
1. Wright, Anna, *Using a Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station*, M.S. Thesis, University of Maryland, Dept. of Fire Protection Engineering, 47 pp., 2021. https://doi.org/10.13016/dc26-dbds

O.2 Publications and Presentations (comprehensive list)

Papers, etc. in this section which include ISS results are enclosed in borders like this.

Refereed Journal Articles

13. P. Dehghani, J.G. Quintiere, *Theoretical Analysis and Predictions of Burning in Microgravity Using a Burning Emulator*, Combustion and Flame, 233 (2021) 111572. https://doi.org/10.1016/j.combustflame.2021.111572
12. P. Dehghani, P.B. Sunderland, J.G. Quintiere, J.L. deRis, *Burning in Microgravity: Experimental Results and Analysis*, Combustion and Flame, 228 (2021) 315-330. https://doi.org/10.1016/j.combustflame.2021.01.035
11. A. Snegirev, E. Kuznetsov, E. Markus, P. Dehghani, P.B. Sunderland, *Transient dynamics of radiative extinction in low-momentum microgravity diffusion flames*, Proceedings of the Combustion Institute, Volume 38, Issue 3, 2021, pp. 4815-4823. https://doi.org/10.1016/j.proci.2020.06.110
10. E.A. Kuznetsov, A.Y. Snegirev, E.S. Markus, *Radiative Extinction of Laminar Diffusion Flame above the Flat Porous Burner in Microgravity*, Combustion, Explosion, and Shock Waves, 56 (2020), 394–411. https://doi.org/10.1134/S0010508220040036
9. E. Auth, J.G. Quintiere, P.B. Sunderland, *Emulation of Condensed Fuel Flames with Gaseous Fuels Supplied Through a Porous Copper Calorimeter*, Fire and Materials, 44 (2020), 935-942. https://doi.org/10.1002/fam.2896
8. A. Markan, H.R. Baum, P.B. Sunderland, J.G. Quintiere, J.L. deRis, *Transient Ellipsoidal Combustion Model for a Porous Burner in Microgravity*, Combustion and Flame, 212 (2020) 93-106. https://doi.org/10.1016/j.combustflame.2019.09.030
7. F. Vermina Plathner, J.G. Quintiere, P. van Hees, *Analysis of extinction and sustained ignition*, Fire Safety Journal 105 (2019) 51–61. https://doi.org/10.1016/j.firesaf.2019.02.003
6. A. Markan, P.B. Sunderland, J.G. Quintiere, J.L. de Ris, H.R. Baum, *Measuring Heat Flux to a Porous Burner in Microgravity*, Proceedings of the Combustion Institute 37 (2019) 4137-4144. https://doi.org/10.1016/j.proci.2018.05.006

5. A. Markan, P.B. Sunderland , J.G. Quintiere, J.L. de Ris, D.P. Stocker, and H.R. Baum, *A Burning Rate Emulator (BRE) for Study of Condensed Fuel Burning in Microgravity*, Combustion and Flame 192 (2018) 272-282. https://doi.org/10.1016/j.combustflame.2018.01.044
4. F.V. Lundström, P.B. Sunderland., J.G. Quintiere, P. van Hees, J.L de Ris, *Study of Ignition and Extinction of Small-Scale Fires in Experiments with an Emulating Gas Burner*, Fire Safety Journal 87 (2017) 18-24. https://doi.org/10.1016/j.firesaf.2016.11.003
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1. Y. Zhang, M.J. Bustamante, M.J. Gollner, P.B. Sunderland, J.G. Quintiere, *Burning on Flat Wicks at Various Orientations*, Journal of Fire Sciences 32 (2014) 52-71. https://doi.org/10.1177%2F0734904113495650

Conference Proceedings, Abstracts, Posters, Etc.

33. Peter B. Sunderland, Transformative Combustion Research in the ACME Suite of ISS Experiments, 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore MD, 3-6 Nov. 2021.
32. Dennis P. Stocker, Recent Research on Flames of Gaseous Fuel Aboard the International Space Station, Conference of the Japanese Society of Microgravity Application (JASMAC-33), virtual, 13-15 Oct. 2021, 11 pp.
31. P. Dehghani, A. Wright, J.L. deRis, P.B. Sunderland, *Burning Rate Emulations Aboard the International Space Station*, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021, 10 pp.
30. P. Dehghani, A. Wright, J.L. de Ris, and P.B. Sunderland, *Burning emulations aboard the International Space Station*, Work In Progress poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.
29. Dennis P. Stocker, Advanced Combustion via Microgravity Experiments on the International Space Station, Work in Progress Poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.
28. A. Wright, P. Dehghani, P.B. Sunderland, and J.L. de Ris, *Using a Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station*, Work In Progress poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.
27. S. Pichugin, D. Samsonov, S. Frolov, O. Korobeinichev, A. Shmakov, Flames Investigations at ISS, Cosmonautics of XXI Century Conference, TSNIIMash, Moscow, Russia, 12-14 May 2021.
26. P. Dehghani, A. Wright, J.L. deRis, and P.B. Sunderland, *Pool Fire Emulations Aboard the International Space Station*, Poster, 13th International Symposium on Fire Safety Science, virtual, 26-30 April 2021.
25. P. Dehghani, P.B. Sunderland, *Burning Emulation in Microgravity - Experimental Results and Analysis*, Abstract, 36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), virtual, 5-6 Nov. 2020.

24. P. Dehghani, E. Auth, C. Cui, D.P. Stocker, J.L. deRis, P.B. Sunderland, J.G. Quintiere, *Microgravity Observations of a Burning Emulator to Investigate Material Flammability*, Eastern States Section of the Combustion Institute, Columbia SC, 8-11 March 2020, 6 pp.
23. Sergey Pichugin, Oleg Artemiev, Sergey Prokopiev, Oleg Kononenko, Oleg Novitsky, and Sergey Frolov, Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience, XIII International Scientific and Practical Conference 'Manned Flights in Space,' Gagarin Cosmonaut Training Centre, Star City, Russia, 13-15 Nov. 2019.
22. James G. Quintiere, Peter B. Sunderland, *Burning Rate Emulator (BRE): Material Burning in Microgravity*, ISS Increments 61 & 62 Science Symposium, virtual, 17-19 Sept. 2019.
21. A. Markan, H.R. Baum, P.B. Sunderland, J.G. Quintiere, J.L. de Ris, *Transient Ellipsoidal Combustion Model for the Burning Rate Emulator*, 34th Annual Meeting of the American Society for Space and Gravitational Research, Abstract, Bethesda MD, 31 Oct. - 3 Nov. 2018.
20. A. Markan, H.R. Baum, P.B. Sunderland, J.G. Quintiere, J.L. de Ris, *Ellipsoidal Combustion Model for a Porous Burner in Microgravity*, Poster, 37th International Symposium on Combustion, Dublin, Ireland, 29 July - 3 Aug. 2018.
19. A. Markan, H.R. Baum, P.B. Sunderland, J.G. Quintiere, J.L. de Ris, *Transient Ellipsoidal Flames in Microgravity*, Eastern States Section of the Combustion Institute, State College PA, 4-7 March 2018, 6 pp.
18. A. Markan, P.B. Sunderland, J.G. Quintiere, J.L. de Ris, D.P. Stocker, H.R. Baum, *Unsteady Analysis and an Alternative Heat Flux Measurement Technique for the Burning Rate Emulator*, Abstract, 33rd Annual Meeting of the American Society for Space and Gravitational Research, Seattle WA, 25-28 Oct. 2017.
17. J.G. Quintiere, A. Markan, P B. Sunderland, H.R. Baum, J. deRis, D.P. Stocker, F.V. Lundström, *A Gas Burner Emulator for Condensed-Phase Burning*, Ninth International Seminar on Flame Structure, Novosibirsk, Russia, 10-14 July 2017.
16. A. Markan, J. G. Quintiere, P. B. Sunderland, J. L. de Ris, D. Stocker, *Emulation of Condensed Fuel Flames using a Burning Rate Emulator (BRE) in Microgravity*, U.S. National Combustion Meeting, College Park MD, 23-26 April 2017, 6 pp.
15. J.G. Quintiere, A. Markan, P.B. Sunderland, J. deRis, D.P. Stocker, *Burning Rate Emulator (BRE) for Study in Microgravity*, Poster, 32nd Annual Meeting of the American Society for Space and Gravitational Research, Cleveland OH, 26-29 Oct. 2016.
14. J.G. Quintiere, A. Markan, P.B. Sunderland, J. deRis, D.P. Stocker, *Development of Spaceflight BRE Burners*, Abstract, 32nd Annual Meeting of the American Society for Space and Gravitational Research, Cleveland OH, 26-29 Oct. 2016.
13. F. Lundström, P.B. Sunderland, J. Quintiere, P. van Hees, J. de Ris, *Using a Porous Bed Gas Burner for Studying Ignition and Extinction of Condensed-Phase Material*, Proceedings of the Eighth International Seminar on Fire and Explosion Hazards, Hefei, China, 25-28 April 2016, in press.
12. J.G. Quintiere, A. Markan, P.B. Sunderland, J. deRis, D.P. Stocker, *Microgravity Burning Using a Gas Burner Emulator*, Eastern States Section of the Combustion Institute, Princeton NJ, 13-16 March 2016, 6 pp.
11. A. Markan, P.B. Sunderland, J.G. Quintiere, J. DeRis, D.P. Stocker, *A Burning Rate Emulator (BRE) for Study in Microgravity*, Abstract, 31st Annual Meeting of the American Society for Space and Gravitational Research, Alexandria VA, 11-14 Nov. 2015.
10. Y. Zhang, M. Kim, P.B. Sunderland, J.G. Quintiere, J. de Ris, *A Burner to Emulate Condensed Phase Fuels*, Ninth Mediterranean Combustion Symposium, Rhodes, Greece, 7-11 June 2015, 12 pp.

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8. Y. Zhang, M. Kim, P.B. Sunderland, J.G. Quintiere, J. deRis, D.P. Stocker, F. Takahashi, P.V. Ferkul, *A Burning Rate Emulator for Microgravity Studies*, Poster, International Symposium on Combustion, San Francisco CA, 3-8 Aug. 2014.
7. M.J. Bustamante, Y. Zhang, K.T. Dotson, P.B. Sunderland, J.G. Quintiere, *Burning on Flat Wicks at Various Orientations*, Proceedings of the Seventh International Seminar on Fire and Explosion Hazards, D. Bradley, G. Makhviladze, V. Molkov, P. Sunderland, F. Tamanini, Eds., Providence RI, 5-10 May 2013, 10 pp.
6. Y. Zhang, M.J. Bustamante, K.T. Dotson, P.B. Sunderland, J.G. Quintiere, *A Gas Burner to Emulate Condensed-Phase Burning*, Proceedings of the Seventh International Seminar on Fire and Explosion Hazards, D. Bradley, G. Makhviladze, V. Molkov, P. Sunderland, F. Tamanini, Eds., Providence RI, 5-10 May 2013, 9 pp.
5. Y. Zhang, M.J. Bustamante, P.B. Sunderland, J.G. Quintiere, P. Ferkul, *A Burning Rate Emulator for Microgravity Fire Safety*, Poster, Seventh International Seminar on Fire and Explosion Hazards, Providence RI, 5-10 May 2013.
4. Y. Zhang, M.J. Bustamante, P.B. Sunderland, J.G. Quintiere, P. Ferkul, *A Burning Rate Emulator for Study in Microgravity*, Eighth U.S. National Combustion Meeting, Salt Lake City UT, 19-22 May 2013, 14 pp.
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2. M.J. Bustamante, K.T. Dotson, Y. Zhang, P.B. Sunderland, J.G. Quintiere, *Laminar Burning of Flat Diffusion Flames at Various Orientations*, Central States Section of the Combustion Institute, Dayton OH, 22-24 April 2012, 15 pp.
1. K.T. Dotson, M.J. Bustamante, P.B. Sunderland, J.G. Quintiere, *Laminar Burning on Flat Wicks at Various Orientations*, 7th U.S. National Combustion Meeting, Atlanta GA, 20-23 March 2011, 5 pp.

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6. Venzon, Ryan, *Thin-Filament Pyrometry and Analysis of Extinction Onset Temperatures for Ethylene Flames on the Burning Rate Emulator in the International Space Station*, M.S. Thesis, University of Edinburgh, Dept. of Civil and Environmental Engineering, 2021.
5. Wright, Anna, *Using a Burning Rate Emulator to Analyze Flame Extinction Time on the International Space Station*, M.S. Thesis, University of Maryland, Dept. of Fire Protection Engineering, 47 pp., 2021. https://doi.org/10.13016/dc26-dbds
4. Auth, Eric, *Using a Burning Rate Emulator (BRE) to Emulate Condensed Fuels and Study Pool Fire Behavior in 1G*, M.S. Thesis, University of Maryland, Dept. of Fire Protection Engineering, 92 pp., Aug. 2019. https://doi.org/10.13016/3gfb-yhqp
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O.3 Awards

University of Maryland graduate student Michael Bustamante was selected as a NASA Space Technology Research Fellow for 2011/2012 for work on the BRE project, a distinct honor described at https://fpe.umd.edu/release/fire-protection-experts-available-to-the-media.

For his work on BRE, Akshit Markan was awarded the University of Maryland Graduate School's Outstanding Graduate Assistant Award for 2017. Markan's recognition for his outstanding work is further described at https://fpe.umd.edu/news/story/fpe-phd-candidate-wins-2017-outstanding-graduate-assistant-award. It explains, for example, that only the top 2% of Graduate Assistants were selected for this honor.

In 2021, BRE Co-Investigator Peter Sunderland was inducted as a Fellow in the Combustion Institute for his "significant contributions to understanding laminar flame dynamics and soot formation and oxidation."

O.4 Outreach

NASA Project Scientists Dennis Stocker created the Space Flames Facebook page (www.facebook.com/space.flames) and Flickr site (https://www.flickr.com/photos/space-flames), where some entries are specific to BRE. He continues to administer these sites, where most content has been about the ACME project of which BRE is a part. Meanwhile, Stocker has made educational and general information is available via https://tinyurl.com/space-flames. As an educational example, a simple board game related to the ACME's spherical flames experiments, which can be downloaded. As another example, an ACME information sheet providing an overview and recent status can also be downloaded through that site. Such flyers have been distributed at meetings of The Combustion Institute and the American Society for Gravitational and Space Research (ASGSR). Other documents, including the BRE Science Requirements Document (SRD), are available for download from https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/downloads/.

The University of Maryland promoted awareness of the BRE investigation online, where that has included precursor ground-based research as well as the ISS testing, as can be seen via these links:

- Bustamante Wins Prestigious NASA Fellowship
- FPEs Bustamone Goes Zero-G
- FPE Ph.D. Candidate Wins 2017 Outstanding Graduate Assistant Award
- UMD Experiments Launched into Orbit on SpaceX Capsule
- Fires in Microgravity
- UMD Fire Researchers Ignite the First of Two Space Station Experiments
- A Brilliant, Yet Humble Mind: FPE Research Professor Takes His Leave
- FPE's BRE Experiment on the ISS Continues
- Unlocking the Mysteries of Fire ... in Space
- Fire Protection Experts Available to the Media
- Sunderland Gives Invited Talk at ASGSR 2021 Symposium

Meanwhile, NASA promoted awareness of ISS combustion research with the video 'The Power of Studying Combustion on the ISS' which was posted on March 13, 2018.

Awareness of ISS combustion research, including the ACME project, was promoted on Oct. 15, 2018 by The Combustion Institute in their online news article, "International Space Station (ISS) Combustion Experiments."

Stocker is involved in a broad range of NASA educational outreach to students and teachers, Scouts of both genders, and the general public. His primary outreach focus is the fundamentals of microgravity (for which resources are available via http://tinyurl.com/SEECmicrogravity) and human spaceflight. But he periodically addresses other topics, e.g., where he sometimes speaks on 'NASA and the Science of Superman' – for example at the Superman Celebration in Metropolis, IL. In this regard, his NASA outreach is carried out locally, regionally, nationally, and online. For example, he regularly presents on microgravity at Space Center Houston's Space Exploration Educators Conference (SEEC). He has exhibited the last five National Scout Jamborees, four national Boy Scout conferences, the last three national Girl Scout conventions, and the 2019 World Scout Jamboree (which was held in the U.S.). Stocker has been a key lead for the agency in NASA's participation in these Scouting events, where as an example the latter outreach is visually described at www.nasa.gov/feature/stem/world-scout-jamboree.html.